

APPENDIX OF THE GENERAL CONDITIONS

Astana city

14.03.2023

CONTRACT AGREEMENT dated 14.03.2023

Work stages

Work stages		Date of completion (day-month-year)
0	Effective Date of EPC Contract	14 mar 2023
1	Design – P-Stage	14 nov 2023
2	Design – W-Stage	14 jul 2024
3	Manufacture & Transportation	31 jan 2025
4	Civil and Erection Completion	23 jul 2025
5	Mechanical completion	23 aug 2025
6	Act of acceptance of the object into operation	23 aug 2025
7	Test & Commissioning	23 aug 2026
8	Completion whole of the Works	23 aug 2026

Note:

Pursuant to Sub-Clause 8.2 [Time of Completion] of General Conditions, Date of Completion, deadlines for Key Dates and other deadlines may be changed upon mutual agreement of the Parties based on a positive conclusion from the State expertise.

SIGNED

In the name of and on behalf of the Employer:

In the name of «CCGT Turkistan» LLP

Kusainov Askar Aitbolatovich
General Director acting on the basis of the Charter

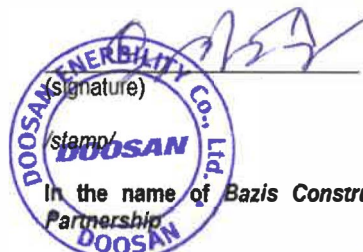


SIGNED

In the name of and on behalf of the Contractor:

In the name of Doosan Enerbility Co., Ltd.

Iljoon Won
Deputy General Manager acting on the basis of Power of Attorney dated 9 March 2023



In the name of Bazis Construction Limited Liability Partnership

Uali Bauyrzhan Kudaibergenuly
President, acting on the basis of the Charter



APPENDIX TO THE CONTRACT

Astana city

14.03.2023

CONTRACT AGREEMENT dated 14.03.2023

Works determined in Foreign currency,

Works determined in Local currency

Category		Price (Exclude VAT)	Price (Exclude VAT)	Total Price after Conversion to KZT
		KZT	USD	KZT
Offshore part of the scope of work	Design/Engineering and design work		11 773 856	5 376 649 000
	Material supply of Contractors		773 149 899	353 066 632 800
	Construction and installation works			
	Transportation			
	Total	0	784 923 755	358 443 281 800
Onshore part of the scope of work	Design/engineering work	9 787 666 500		9 787 666 500
	Equipment of Contractors	60 935 357 960		60 935 357 960
	Construction and installation works	48 758 069 579		48 758 069 579
	Transportation			
	Test & Commissioning			
	Total	119 481 094 039	0	119 481 094 039
Sum total		119 481 094 039	784 923 755	477 924 375 839

Note:

※ Payment Currency of Offshore part of the scope of work shall be US Dollar.

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In the name of «CCGT Turkistan» LLP

Kusainov Askar Aitbolatovich
General Director acting on the basis of the Charter

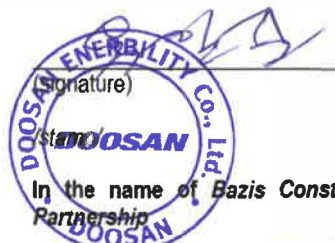


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APPENDIX OF THE GENERAL CONDITIONS

Astana city

14.03.2023

CONTRACT AGREEMENT dated 14.03.2023

Main stage of the project and payment by project stages

Stage	Expected Month	For delivery at the end of the stage	Category	Offshore / non-offshore	Supporting documentation	% of the cost for the section	% of the cost for the each section	Amount (cost of work)	Advance	Assessments	Deductions	Amount of payment
I Engineering and design work												
1	15	Approval by State Expertise (positive Conclusion), Design complete	E	Offshore	Approved Design Document	1,12%	35,33%					
2	15	Approval by State Expertise (positive Conclusion) for Bazis scope of work	E	Onshore	Approved Design Document	2,05%	64,67%					
Total for Engineering and Design work						3,17%	100,00%					
II Procurement												
1	1	Advance Payment	P	Offshore	N/A	40,00%	46,17%					
2	5	Gas Turbine & Generator Purchase Order Issue	P	Offshore	Manufacturer's Confirmation	0,00%	0,00%					

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Stage	Expected Month	For delivery at the end of the stage	Category	Offshore / non-offshore	Supporting documentation	% of the cost for the section	% of the cost for the each section	Amount (cost of work)	Advance	Assessments	Deductions	Amount of payment
3	6	Heat Recovery Steam Generator Purchase Order Issue	P	Offshore	Manufacturer's Confirmation	0,00%	0,00%					
4	6	Steam Turbine & Generator Purchase Order Issue	P	Offshore	Manufacturer's Confirmation	0,00%	0,00%					
5	6	Air Cooled Condenser Purchase Order Issue	P	Offshore	Manufacturer's Confirmation	0,00%	0,00%					
6	7	GIS (Gas Insulated SWGR) Purchase Order Issue	P	Offshore	Manufacturer's Confirmation	0,00%	0,00%					
7	8	Closed Cooling Water System Purchase Order Issue (Pump)	P	Offshore	Manufacturer's Confirmation	0,00%	0,00%					
8	8	Fuel Gas Supply System Purchase Order	P	Offshore	Manufacturer's Confirmation	0,00%	0,00%					
9	8	Aux. Boiler System Purchase Order Issue	P	Offshore	Manufacturer's Confirmation	0,00%	0,00%					
10	10	DCS (distribution control system)	P	Offshore	Manufacturer's Confirmation	0,00%	0,00%					

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Stage	Expected Month	For delivery at the end of the stage	Category	Offshore / non-offshore	Supporting documentation	% of the cost for the section	% of the cost for the each section	Amount (cost of work)	Advance	Assessments	Deductions	Amount of payment
		Purchase Order Issue										
11	11	Pipe Rack Purchase Order Issue	P	Offshore	Manufacturer's Confirmation	0,00%	0,00%					
12	11	MV / LV Switchgear Purchase Order Issue	P	Offshore	Manufacturer's Confirmation	0,00%	0,00%					
13	12	EDG (emergency diesel generator) Purchase Order Issue	P	Offshore	Manufacturer's Confirmation	0,00%	0,00%					
14	14	DC / UPS / Battery Purchase Order Issue	P	Offshore	Manufacturer's Confirmation	0,90%	1,04%					
15	15	GT Step Up Transformer Purchase Order Issue	P	Offshore	Manufacturer's Confirmation	0,80%	0,92%					
16	15	Aux. Boiler Purchase Order Issue	P	Offshore	Manufacturer's Confirmation	0,80%	0,92%					
17	15	Compressed Air System Purchase Order Issue	P	Offshore	Manufacturer's Confirmation	1,40%	1,62%					

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Stage	Expected Month	For delivery at the end of the stage	Category	Offshore / non-offshore	Supporting documentation	% of the cost for the section	% of the cost for the each section	Amount (cost of work)	Advance	Assessments	Deductions	Amount of payment
18	15	Water Treatment System Purchase Order Issue	P	Offshore	Manufacturer's Confirmation	1,20%	1,39%					
19	15	Waste Water Treatment System Purchase Order Issue	P	Offshore	Manufacturer's Confirmation	0,54%	0,62%					
20	15	Fire Fighting Pump Station/Fire Fighting System Purchase Order Issue	P	Offshore	Manufacturer's Confirmation	0,00%	0,00%					
21	16	Embedded Materials of Gas Turbine, Generator #11 Delivery at Site	P	Offshore	Arrival Notice	5,25%	6,06%					
22	17	Embedded Materials of Gas Turbine, Generator #12 Delivery at Site	P	Offshore	Arrival Notice	6,00%	6,93%					
23	18	Embedded Materials of HRSG#11 Delivery at Site	P	Offshore	Arrival Notice	6,00%	6,93%					

Stage	Expected Month	For delivery at the end of the stage	Category	Offshore / non-offshore	Supporting documentation	% of the cost for the section	% of the cost for the each section	Amount (cost of work)	Advance	Assessments	Deductions	Amount of payment
24	19	Gas Turbine, Generator #11 FOB	P	Offshore	Shipping Notice	6,00%	6,93%					
25	19	Gas Turbine, Generator #12 FOB	P	Offshore	Shipping Notice	0,00%	0,00%					
26	19	Fire Fighting Pump Station/Fire Fighting System FOB (Piping)	P	Offshore	Shipping Notice	1,35%	1,56%					
27	20	GT Step Up Transformer FOB	P	Offshore	Shipping Notice	3,00%	3,46%					
28	20	MV / LV Switchgear FOB	P	Offshore	Shipping Notice	0,00%	0,00%					
29	21	Fuel Gas Supply System FOB	P	Offshore	Shipping Notice	2,25%	2,60%					
30	21	HVAC Purchase Order Issue	P	Offshore	Manufacturer's Confirmation	1,10%	1,27%					
31	22	Heat Recovery Steam Generator #11, #12 FOB (Pressure Parts)	P	Offshore	Shipping Notice	2,25%	2,60%					
32	22	Closed Cooling Water System FOB (Pump)	P	Offshore	Shipping Notice	0,00%	0,00%					

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Stage	Expected Month	For delivery at the end of the stage	Category	Offshore / non-offshore	Supporting documentation	% of the cost for the section	% of the cost for the each section	Amount (cost of work)	Advance	Assessments	Deductions	Amount of payment
33	22	DCS (distribution control system) FOB	P	Offshore	Shipping Notice	0,00%	0,00%					
34	22	Waste Water Treatment System FOB	P	Offshore	Shipping Notice	0,50%	0,58%					
35	22	Water Treatment System FOB	P	Offshore	Shipping Notice	0,50%	0,58%					
36	22	Pipe Rack ex-work	P	Offshore	Notice for shipping ready	0,00%	0,00%					
37	23	GIS (Gas Insulated SWGR) FOB	P	Offshore	Shipping Notice	2,25%	2,60%					
38	23	DC / UPS / Battery FOB	P	Offshore	Shipping Notice	0,00%	0,00%					
39	25	Gas Turbine, Generator #21, #22 FOB	P	Offshore	Shipping Notice	2,25%	2,60%					
40	28	EDG (emergency diesel generator) FOB	P	Offshore	Shipping Notice	1,63%	1,88%					
41	30	HVAC Ex work	P	Offshore	Notice for shipping ready	0,66%	0,76%					
Total for Procurement						86,63%	100,00%					
III Construction and installation works												
1	15	Site Temp. Work (SI, Test piling, Site	C	Onshore	Completion Notice (ACT)	0,85%	8,33%					

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Stage	Expected Month	For delivery at the end of the stage	Category	Offshore / non-offshore	Supporting documentation	% of the cost for the section	% of the cost for the each section	Amount (cost of work)	Advance	Assessments	Deductions	Amount of payment
		Leveling, Camp)										
2	20	Main Electrical Building Foundation Finish	C	Onshore	Completion Notice (ACT)	0,85%	8,33%					
3	25	Main Electrical Building Structure Finish	C	Onshore	Completion Notice (ACT)	0,48%	4,71%					
4	25	Steam Turbine Area (#10, #20) Piling and Foundation Work Finish	C	Onshore	Completion Notice (ACT)	0,37%	3,63%					
5	26	Gas Turbine Area (#11~#12)Piling and Foundation and Foundation Finish (For Major Component) Work Finish	C	Onshore	Completion Notice (ACT)	1,70%	16,67%					
6	27	Heat Recovery Steam Generator Area (#11~#22) Piling and Foundation Work Finish	C	Onshore	Completion Notice (ACT)	0,85%	8,33%					

Stage	Expected Month	For delivery at the end of the stage	Category	Offshore / non-offshore	Supporting documentation	% of the cost for the section	% of the cost for the each section	Amount (cost of work)	Advance	Assessments	Deductions	Amount of payment
7	29	Pipe Rack installation finish	C	Onshore	Completion Notice (ACT)	0,34%	3,33%					
8	34	Main Electrical Building FF, Elec, I&C installation Finish	C	Onshore	Completion Notice (ACT)	0,34%	3,33%					
9	34	GT Step Up Transformer installation finish	C	Onshore	Completion Notice (ACT)	0,34%	3,33%					
10	35	GIS installation finish	C	Onshore	Completion Notice (ACT)	0,34%	3,33%					
11	36	EDG (emergency diesel generator) installation Finish	C	Onshore	Completion Notice (ACT)	0,34%	3,33%					
12	36	Closed Cooling Water System installation finish	C	Onshore	Completion Notice (ACT)	0,34%	3,33%					
13	36	Compressed Air System installation finish	C	Onshore	Completion Notice (ACT)	0,34%	3,33%					
14	36	Water Treatment System	C	Onshore	Completion Notice (ACT)	0,34%	3,33%					

Stage	Expected Month	For delivery at the end of the stage	Category	Offshore / non-offshore	Supporting documentation	% of the cost for the section	% of the cost for the each section	Amount (cost of work)	Advance	Assessments	Deductions	Amount of payment
		installation finish										
15	36	Aux. Boiler System installation finish	C	Onshore	Completion Notice (ACT)	0,34%	3,33%					
16	36	Water Treatment System installation finish	C	Onshore	Completion Notice (ACT)	0,34%	3,33%					
17	36	Aux. Boiler System installation finish	C	Onshore	Completion Notice (ACT)	0,34%	3,33%					
18	38	Fuel Gas Supply System installation finish	C	Onshore	Completion Notice (ACT)	0,34%	3,33%					
19	39	Mechanical Completion	C	Onshore	Mechanical Completion certification	0,51%	5,00%					
20	52	Reliability Run Test of Combined Cycle Block #2 (GTG#21, #22, STG#20) Finish	C	Onshore	Project Taking Over Certification	0,51%	5,00%					
Total for Construction and installation works						10,20%	100,00%					
Grand Total						100,00%						

Remark: The scope of work is not fully specified. The scope of work, types of work, equipment supply (onshore/offshore) and delivery, including financing, are preliminary and is a subject for adjustment after the release of project documentation with a positive conclusion of the State Expertise.

SIGNED

In the name of and on behalf of the Employer:

In the name of «CCGT Turkistan» LLP

Kusainov Askar Aitbolatovich
General Director acting on the basis of the Charter

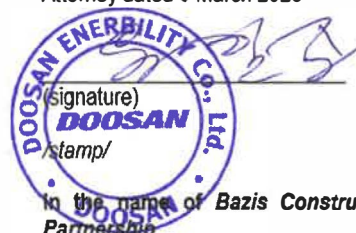


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In the name of and on behalf of the Contractor:

In the name of Doosan Enerbility Co., Ltd.

Iijoon Won
Deputy General Manager acting on the basis of Power of Attorney dated 9 March 2023



In the name of Bazis Construction Limited Liability Partnership

Uali Bauyrzhan Kudaibergenov
President, acting on the basis of the Charter



APPENDIX TO EMPLOYER'S REQUIREMENTS

Astana city

14.03.2023

CONTRACT AGREEMENT dated *14.03.2023*

Dislocation of the project team

To ensure practical solutions to issues and tasks, the contractor must determine the permanent dislocation of the project team staff at the location of the construction of the facility.



APPENDIX TO EMPLOYER'S REQUIREMENTS

Astana city

14.03.2023

CONTRACT AGREEMENT dated 14.03.2023

Contractor's key personnel

The Contractor shall assign key project personnel in accordance with the quality requirements of the project and the Employer's quality expectations.

No	NAME	Position	Mailing Address Companies	Email address	Phone
1	Gihun Bek	Project Manager	155, Jeongjail-ro, Bundang-gu, Seongnam-si, Gyeonggi-do, Republic of Korea	gihun.bek@doosan.com	+82-31-5179-2041
2	Buzgulov Utegen Kurmanbaevich	Site Manager	Astana, Amanzholov street 26, block A, room 200	u.buzgulov@basis.kz	
3	Minwouk Cho	Procurement Manager	155, Jeongjail-ro, Bundang-gu, Seongnam-si, Gyeonggi-do, Republic of Korea	minwouk.cho@doosan.com	+82-31-5179-1351
4	Hyun Hur	Engineering Manager	155, Jeongjail-ro, Bundang-gu, Seongnam-si, Gyeonggi-do, Republic of Korea	hyun.hur@doosan.com	+82-31-5179-1437
5	Bobrov Egor Rudolfovich	All division Lead construction manager & design Lead Engineer	Astana, Amanzholov street 26, block A, room 200	y.bobrov@basis.kz	+7 707 712 0119
6	Taewoog Kang	QA/QC Manager	155, Jeongjail-ro, Bundang-gu, Seongnam-si, Gyeonggi-do, Republic of Korea	taewoog.kang@doosan.com	+82-31-5179-1081
7	Chunkee Park	Project Schedule Manager	155, Jeongjail-ro, Bundang-gu, Seongnam-si, Gyeonggi-do, Republic of Korea	chunkee.park@doosan.com	+82-31-5179-1943

※ This above personnel is the candidate of each division and will be assigned after notice of commencement of construction.

Personnel must be full-time employees of the Contractor and must be fully qualified, competent, trained and experienced in their respective fields and positions.

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APPENDIX TO EMPLOYER'S REQUIREMENTS

Astana city

14.03.2023

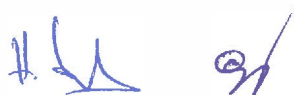
CONTRACT AGREEMENT of 14.03.2023

Employer's key personnel

The Employer must assign key project personnel to the project's quality requirements and quality expectations.

No	NAME	Position	Company's Mailing Address	Email address	Phone
1	Musataev Niyaz Gazizovich	Managing Director for Operational Affairs	17/10 Syganak street, Astana, Republic of Kazakhstan	n.mussatayev@sk-pgu.kz	+7 701 908 5801
2	Nurmukanov Daniyar Nurlanovich	Managing Director for Administrative and Legal Affairs	17/10 Syganak street, Astana, Republic of Kazakhstan	d.nurmukanov@sk-pgu.kz	+7 702 888 6226
3	Erbolganov Abzal Turysbekovich	Managing Director for Economics and Finance	17/10 Syganak street, Astana, Republic of Kazakhstan	a.yerbolganov@sk-pgu.kz	+7 701 777 7240
4	Tanbetov Almat Zhailibayevich	Director of the Contract Administration Department	17/10 Syganak street, Astana, Republic of Kazakhstan	a.tanbetov@sk-pgu.kz	+7 701 723 7997
5	Alen Amirkhan Anuvarbekovich	Head of Security Service	17/10 Syganak street, Astana, Republic of Kazakhstan	a.alen@sk-pgu.kz	+7 705 633 7742

Personnel must be full-time employees of the Employer and must be fully qualified, competent, trained and experienced in their respective fields of work and in their respective positions.



APPENDIX TO EMPLOYER'S REQUIREMENTS

Astana city

14.03.2023

CONTRACT AGREEMENT dated 14.03.2023

Materials and Plant required to perform the work specified in the Employer's Requirements: construction materials, construction equipment and tools, power plant equipment, fuel and lubricants, chemical reagents are provided by the Contractor.

Materials and Plant	Details	Country of import	Price/Currency

The parties agreed that this application will be completed after passing the State examination.



APPENDIX TO EMPLOYER'S REQUIREMENTS

Health, Safety and Environment protection

Astana city

14.03.2023

CONTRACT AGREEMENT dated 14.03.2023

Conditions on liability for violation of health, safety and environmental requirements, recommended for incorporation in the contract, providing for the performance of works / rendering of services

1. The Contractor/Subcontractor shall be liable for violations of the applicable requirements of the laws of RK on occupational health and safety (hereinafter, the "OHS"), the applicable requirements of the Employer in the field of OHS, with which the Contractor/Subcontractor was familiarized in writing, including payment of penalties (fines) and compensation for damages, committed during the performance of contract work by the Contractor's personnel or Subcontractor's personnel in relation to which it acts as the Employer.

2. If the Employer has been held liable by the state authorized body for the above-mentioned violations due to the failure of the Contractor's employees or the Subcontractors involved, the latter shall compensate the Employer for all losses caused by it, including, but not limited to fines, penalties, court fees, etc.

3. If there is a fault of the Contractor/Subcontractor for fires, accidents, and incidents that occurred during the performance of contract work, the Contractor/Subcontractor shall compensate the Employer for the losses caused by these circumstances.

4. Terms of liability for violation of OHS requirements recommended for inclusion in contracts providing for the performance of works / provision of services may be replaced by the results of the evaluation of the effectiveness of the Employer and contractors / subcontractors in the field of OHS with the model of prevention and avoidance of violations and accidents.

5. For failure to perform and/or improper performance of obligations under the contract in terms of the Contractor/Subcontractor's compliance with OHS requirements identified by the Employer's representative, the Contractor/Subcontractor shall pay a forfeit to the Employer in the following amount:

No.	Nature of violations of OHS requirements	Amount of penalty (fine) for each violation, KZT, thousand
1	Detection of a Contractor/Subcontractor employee at the work site while intoxicated by alcohol, drugs or toxic substances, bringing or carrying on the work site of substances causing alcohol, drugs or toxic intoxication (except when the above facts are revealed directly by the Contractor/Subcontractor representative with a written notice to the Employer within 24 hours of the detection and removal of the guilty person from the work)	500, as well as termination of the contract with the employee
2	Failure to inform the Employer within 24 hours (concealment of information) about accidents, fires, incidents, occupational injuries, violations of the technological process that occurred	500

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	during the performance of contract work, or notification with a delay of more than 24 hours from the detection of the incident	
3	Contractor/Subcontractor carrying out hazardous work without the required work permit	500
4	Disabling or tampering with interlocks and other safety devices on operating Contractor or Employer equipment without proper authorization	200
5	Smoking by Contractor/Subcontractor employees on the Employer's premises outside designated areas	200
6	Use by Contractor/Subcontractor employees on the Employer's territory of an open fire outside places designated for this purpose, if this is not stipulated by the permit, including the burning of garbage	200
7	Involvement by the Contractor in the performance of contract work of a Subcontractor without appropriate approval of the Employer	1% of the cost of the work under the Contract
8	Discovery at the Employer's facilities of the Contractor/Subcontractor employees performing contract work without proper qualification and certification, concluded labor relations between the employee and the employer	200
9	Damage to equipment, pipelines, power lines, cable communications occurred on the Employer's territory through the fault of the Contractor/Subcontractor when performing contract work without the Employer's approval, in addition to other payments related to direct and indirect losses of the Employer	500
10	Failure to comply with the applicable requirements of regulatory legal acts of the Republic of Kazakhstan in the field of industrial safety, operating at hazardous production facilities of the working site	200
11	Absence of a responsible person (works supervisor) at the place of work with increased danger, performed according to the permit	200
12	Performance of production operations by an employee: - without undergoing induction training, in-service training (primary, repeated, targeted); - who did not pass a knowledge test in a timely manner; - if the employee does not have a certificate for the right to perform special work at the workplace.	200
13	Failure to comply with occupational safety regulations when carrying out electrical and gas welding and firing work	200
14	Failure to comply with safety requirements when working at height, climbing (not using safety harnesses, barriers, etc.)	500
15	Failure to comply with the fire safety regulations of Kazakhstan during the works at the Employer's worksite	200
16	Failure to eliminate within the established timeframe and previously identified/recorded violations of OHS requirements (for each violation)	200
17	Failure to comply with the requirements to remove from the work site materials and equipment not used in the course of contract work, scrap metal upon completion of contract work, or individual phases of work	500

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18	Failure to comply with the Employer's requirements to eliminate violations of OHS requirements	200
19	The Contractor/Subcontractor's contractual relationship with the worker performing contract work at the worksite has not been formalized	200

6. The Employer may give notice to the Contractor to terminate the subcontract in the following cases:

- Fatal, serious and group accidents, as well as accidents and fires caused by the fault of the Subcontractor;
- Repeated violations by the Contractor/Subcontractor employees of OHS requirements, failure by the Subcontractor to eliminate deficiencies identified in the course of OHS monitoring at work sites within the established timeframe;
- The Employer shall not be liable for injury, injury or death of any Contractor/Subcontractor employee or third party engaged by the Contractor through no fault of the Employer, or for their violation of health, safety, industrial and fire safety and industrial hygiene requirements during the performance of contract work on the Employer's premises.

APPENDIX TO EMPLOYER'S REQUIREMENTS

Health, Safety and Environment protection

Astana city

14.03.2023

CONTRACT AGREEMENT dated 14.03.2023

Model form for issuing remarks
Health, Safety, Security and
environmental protection
PGU Turkestan LLP

Name of contractor

INSTRUCTIONS

**on taking measures to eliminate the revealed violations
on safety, occupational safety and environmental protection
dated "___" "___" 20___ No. ___**

To _____

Name and surname, position

In accordance with subparagraph 3) of paragraph 4 of Article 202 of the Labor Code of the Republic of Kazakhstan dated November 23, 2015, I request that you eliminate the following violations of health, safety and environmental protection requirements:

No.	Identified violations and requirements for their elimination	Place of discovery	Deadline for elimination	Notes

Report on the implementation of the Instructions after the deadline specified therein to submit a written report to the Service of Health, Safety and Environment protection of PGU Turkestan LLP.

The instruction was provided by:

Signature *Full name*

I received the instruction:

Signature *Full name.*

APPENDIX TO EMPLOYER'S REQUIREMENTS

Astana city

14.03.2023

CONTRACT AGREEMENT dated 14.03.2023

Checklist

* - significant discrepancies

In terms of OHS				
No.	Name of criteria	available	Not available	note
1*	Provided diagnosis, testing, inspection of equipment and technical devices (provide information on the availability of specialized laboratories and technical means of the Contractor, in their absence, copies of contracts with specialized organizations)	☐	☐	☐
2*	Employees are fully provided with protective clothing, footwear, and other personal protective equipment	☐	☐	☐
3*	Requirement to wear safety glasses at all times while on the job site	☐	☐	☐
4*	When working, use only a safety harness with two straps	☐	☐	☐
5*	Availability of a contract for the removal for recycling and disposal of all types of hazardous and non-hazardous waste	☐	☐	☐
6*	Availability of registers to account for all types of waste	☐	☐	☐
7*	Availability of containers for separate waste disposal	☐	☐	☐
8*	Availability of full-time employees to carry out inspections of compliance with OHS requirements	☐	☐	☐
9	Availability of a system for recording incidents without disability (micro-injuries, requests for medical help, incidents without consequences, etc.). The statistics of accidents for the last calendar year is provided.	☐	☐	☐
10	Availability of a system for the collection, recording and analysis of indicators of injuries, accidents and incidents, periodic reporting in the field of OHS (if available, statistics for the past 12 months are provided)	☐	☐	☐

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11	Whether an OHS risk assessment is performed for contract work (if available, a description and map of critical risks and risk management methods are provided)	☐	☐	☐
12	Presence of measures to protect the environment (activities and relevant permits are provided in terms of environmental protection)	☐	☐	☐
13	There is a system of industrial control in the field of occupational health and safety and environmental protection (copies of the Regulations on industrial control are provided)	☐	☐	☐
14	Availability of the organization's logo on special clothing and PPE	☐	☐	☐
15	Availability of a system for prompt notification of incidents (copies of current procedures and notification schemes are provided)	☐	☐	☐
16	Existence of an internal procedure for organizing and conducting internal investigation of accidents (a copy of the current procedure and scheme is provided)	☐	☐	☐
17	Availability of a developed plan for production and consumption waste management	☐	☐	☐
In terms of transport support				
No.	Name of criterion	available	Not available	note
1*	The presence of special signs, route maps on vehicles and special equipment performing dangerous work (lifting and unloading work, transportation of dangerous goods, etc.)	☐	☐	☐
2*	Ensuring the safety of vehicles for the transport of passengers and goods (data on the passage of state technical inspection is provided)	☐	☐	☐
3*	Availability of first aid kits, serviceable fire extinguishers, emergency signs)	☐	☐	☐
4	Whether control inspections of vehicles are organized before they leave for the route before the start of work	☐	☐	☐
In terms of Personnel and Health and Living Conditions				
1*	List of workers involved in the work (provide copies of identity cards, permits to hire foreign labor, registration)	☐	☐	☐
2*	Contractor's personnel (including managers and specialists) have received the necessary OHS training (copies of documents confirming certification and training are provided)	☐	☐	☐
3*	Availability of qualified personnel with experience to perform work at hazardous production facilities, devices (provide copies of diplomas, certificates, lists, orders, etc.)	☐	☐	☐
4	Training and additional OHS training (training programs, certification and work permit documents are provided)	☐	☐	☐
5*	Training and proper qualification of drivers	☐	☐	☐

6*	Availability of facilities for employee accommodation, or a contract to provide accommodation in accordance with sanitary requirements	☐	☐	☐
7*	Availability of facilities for catering to employees or a contract to provide food in accordance with sanitary requirements	☐	☐	☐
8*	Availability of certificates of medical examination and admission to work for catering staff	☐	☐	☐
9*	Availability of medical peerage for pre-shift medical examinations, or a contract for the provision of medical examination services	☐	☐	☐
10*	Whether emergency medical care is organized on the Employer's territory (for facilities remote from medical institutions, the medical station and ambulance must be available)	☐	☐	☐
In terms of material and technical equipment, IMS and financial stability				
1*	Production equipment, machinery and mechanisms are in a satisfactory condition that meets the requirements of the current regulations of Kazakhstan (a list of equipment, machinery and mechanisms used in the performance of Contract work, passports for equipment, permission to operate, technical certification is provided)	☐	☐	☐
2*	The Contractor has the necessary authorization documentation for the right to carry out activities, if it is provided by the current regulatory legal acts of Kazakhstan.	☐	☐	☐
3*	Constant telephone communication with the place of work and the Employer's dispatching service	☐	☐	☐
4*	Availability of energy personnel or a contract with organizations for the maintenance of electrical equipment	☐	☐	☐
5*	Availability of technological regulations and a schedule of technical inspections and preventive maintenance work on the equipment, machinery and mechanisms used	☐	☐	☐
6*	Whether transportation of work crews from the workplace to the housing and back is provided	☐	☐	☐
7*	Availability of a contract of insurance of employees against accidents at work when performing work at the production facilities of the Employer	☐	☐	☐
8*	Availability of a civil liability insurance contract for owners of facilities whose activities are related to the danger of causing harm to third parties (for facilities declared industrial safety)	☐	☐	☐
9	Availability of an OHS policy (written copy)	☐	☐	☐
10	Internal standards and procedures have been developed, which establish and regulate the order of work (supporting documents are provided as a general list)	☐	☐	☐
11	Is there a system of internal audits in the field of quality	☐	☐	☐
12	International certificates in the field of occupational safety and health, environmental protection	☐	☐	☐

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Occupational injuries, accidents (for the last 12 months)	
Average headcount of the Contractor's personnel	
Number of actually worked man/hours	
Accidents/number of casualties, Including:	
• Fatal	
• Group, number of victims	
• Road Accidents	
• Fires	
• Accidents	
• Incidents related to the incapacity of the injured person	

Responsible person of the Employer:

Name, signature

Contractor/Subcontractor representative:

Name, signature

APPENDIX TO EMPLOYER'S REQUIREMENTS

Astana city

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CONTRACT AGREEMENT dated 14.03.2023

**List
of "stop points" of violations, in case of which the work must be stopped until the non-compliance is corrected**

No.	Name of non-conformity
1	Failure to comply with the conditions of the issued permit/permit for work of increased danger
2	Conducting work without appropriate personal protective equipment (special clothing, safety helmet with special chinstrap, respirators, life belts and ropes, ropes, signal vests)
3	Work in the absence of appropriate certificates (qualification, fire-technical minimum, admittance to work in the field of occupational safety and health, etc.)
4	Absence of a responsible person at the place where hot work, work at height, climbing work is carried out
5	Lack of primary fire extinguishing equipment at the place of fire works
6	Performing work without proper authorization documents (work permit, high-risk work permit)
7	Smoking on the territory of the Employer's facilities outside designated areas
8	Violation during slinging of cargo, use of defective slings
9	Presence of oil contamination of the site of the ointment works
10	Operation of electrical equipment without earthing
11	Working with defective, uncertified and homemade equipment
12	Operation/connection of equipment without a plug
13	Being an employee under the influence of alcohol or narcotics
14	Violation of the order of briefing or presence of the Contractor's employee on the Employer's territory absent in the lists provided to the Employer
15	Incentive to work
16	Defective vehicles, equipment, machinery with leaks of fuel and lubricants
17	Failure of drivers, machinists to pass pre-shift medical examinations
18	Violations during installation/operation of scaffolding, ladders and equipment



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CONTRACT AGREEMENT dated 14.03.2023

Evaluation sheet of the contractor's activities in the field of industrial safety and occupational safety and health

1. Name and address of contractor _____

2. Type of activity/subject of the contract _____

3. Object _____

4. The actual term of work under the contract _____
5. Accident Statistics

Accidents (number of victims)	
Microtrauma	
A minor accident at work	
Serious accident at work	
Group accident at work	
Fatal industrial accident	
Road accident	
Fires	
Accidents	
Total number of injuries (persons) with disability - X1	
Number of man-hours worked - Y1	
Lost Time Injury Frequency Rate - LTIR= $X1 * 1,000,000 / Y1$	

6. Results of control and preventive work/Audits

No.	Indicators	Points			
		Not applicable	3	2	1
1	Number of OHS inconsistencies identified in the course of inspections, units				
1.1	Number of detected non-compliances in OHS training and testing of personnel, units	☐	0 ☐	1-2 ☐	3≤ ☐
1.2	Number of detected discrepancies in the maintenance of OHS documentation, units.	☐	0-2 ☐	3-4 ☐	5≤ ☐

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1.3	Number of instances of employees not using PPE, units.	☐	0 ☐	1-2 ☐	3≤ ☐
1.4	The number of identified non-compliances with the requirements of the Labor Code of RK, units.	☐	0-2 ☐	3-4 ☐	5≤ ☐
1.5	Number of non-compliances identified in the field of industrial safety, units.	☐	0-2 ☐	3-4 ☐	5≤ ☐
1.6	The number of detected discrepancies in the maintenance of documents on traffic safety, units.	☐	0-2 ☐	3-4 ☐	5≤ ☐
1.7	The number of detected non-compliances in the field of traffic safety (equipment of transport), units.	☐	0-2 ☐	3-4 ☐	5≤ ☐
1.8	Failure to comply with environmental protection measures	☐	0 ☐	1 ☐	2≤ ☐
1.9	The presence of the Contractor's employee and vehicles on the territory of the Employer's work sites without permits	☐	0 ☐	1-3 ☐	4≤ ☐
1.10	The fact of presence of the Contractor/Subcontractor employees at the Employer's work sites in a state of alcoholic, narcotic intoxication (their analogues), including the use of substances causing a state of alcoholic, narcotic intoxication (their analogues) during the working day	☐	0 ☐	1 ☐	2≤ ☐
1.11	The fact of carrying into the territory of the Employer substances causing a state of alcoholic, narcotic intoxication (their analogues)	☐	0 ☐	1 ☐	2≤ ☐
1.12	Failure of the Contractor's employees to ensure the safety of official documents	☐	0 ☐	1 ☐	2≤ ☐
1.13	Number of repeated non-compliances in the field of occupational safety and health, units.	☐	0 ☐	1-2 ☐	3≤ ☐

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APPENDIX TO EMPLOYER'S REQUIREMENTS

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Standard requirements for the quality of maintenance of the working site of the capital construction facility

1. The Contractor undertakes to comply with all applicable requirements of the Employer in the field of quality of maintenance of work sites of capital construction facilities set forth in this Appendix to the Standard, with which the responsible representative of the Employer has made the Contractor aware.

2. These requirements do not apply to the conduct of work:

- In underground conditions (except for the Employer's infrastructure on the surface);
- In the shops in operation;
- On the construction of linear objects, including power lines, communications, trunk (external) pipelines for various purposes and other similar facilities;
- On construction of facilities located in the right-of-way (right-of-way) of railroads and highways and other transport routes located outside the Employer's work site.

3. The Contractor undertakes to ensure the implementation of the necessary measures to ensure the following quality requirements for the maintenance of the work site.

3.1 Information board (construction passport)

At the entrance (entrance) to the construction site, in the visibility area, there shall be an information board with the following information: the name of the reconstruction/construction facility, the name of the Employer and Contractor, telephone numbers, positions and names of the responsible workmen, the planned start and end dates of reconstruction/construction (the recommended format of the Information Board in Appendix 8 to this Standard).

The size, possible design features of the Information Board and methods of its installation shall be determined by the Contractor depending on the peculiarities of the work site.

3.2 Scheme of safe movement of vehicles and routes for pedestrians on the working site of the capital construction project

At the entrance to the construction site, in the visibility area, a "Traffic diagram of vehicles and pedestrian routes on the working site of the capital construction project" shall be installed, containing:

- permitted routes for vehicle and pedestrian traffic on the work site;
- road signs regulating traffic;
- places where vehicles and pedestrians cross each other and hazardous areas;
- permitted places for parking and washing of vehicles;
- the permitted maximum speed of vehicles;
- the place of exit from the work site.

The recommended format of "Schemes of safe vehicle traffic and pedestrian routes on the working site of the capital construction facility" is shown in Appendix 9 to this Standard.

The size, possible design features of the “Traffic and pedestrian routes on the working site of the capital construction facility” and methods of its installation shall be determined by the Contractor depending on the peculiarities of the working site.

3.3 Contractor's construction headquarters

The Contractor's construction headquarters shall be in the form of a container-type modular building, unless otherwise specified in the construction organization project and/or the Employer's requirements. The size/size and number of floors of the building shall be determined by the Contractor based on his need for the number of workplaces for managers and specialists engaged by the Contractor to organize and support the contract work.

The following rooms shall be arranged in the Contractor's construction headquarters:

- Meeting room (intended use - production meetings, briefings, staff meetings);
- Rooms / “open spaces” for the work of managers and specialists of the Contractor;
- Restrooms;
- Rooms for the archive of documents;
- Warehouse inventory, clothing and personal protective equipment;
- The rooms for meals (provided that such rooms are not provided by the Employer).

3.3.1 Meeting room

Must be provided with the following information materials posted on the stands:

- Information plate with contact information of key employees of the Employer, Contractor, Subcontractors (full name, position, contact information (phone number, e-mail));
 - Information plate with the name of the person responsible for fire safety;
 - The master plan of the building (graphic part with explications);
 - Construction plan, graphic part with explications for all stages / stages of construction / reconstruction);
 - Up-to-date schedule of works on construction sites with the main milestones and updated fact on the reporting date;
 - Staff mobilization schedule and a schedule of the need for machines and mechanisms;
- At the request of the Employer must be equipped with equipment for audio/video conferences. The offices/“open spaces” for the work of the Contractor's managers and specialists shall be equipped with communication equipment, computers with Internet access, and printing devices.

3.4 Contractor's camp

The Contractor/Subcontractor camp must be equipped in accordance with the construction organization project, current requirements of sanitary regulations and fire safety rules of Kazakhstan.

Prior to commencing work on the equipment of the camp, the Contractor, in order to coordinate with the Employer the possibility of bringing mobile living quarters to the worksite, shall present the mobile living quarters to the Employer's representative to assess and inspect the overall technical condition, compliance with sanitary and fire safety standards.

Prior to the placement of Contractor/Subcontractor personnel in the camp, the Contractor shall develop detailed plans of the camp with the layout, types of structures, their dimensions, dimensions of rooms within them, connections to utilities, approaches and entrances.

The camp area and all living quarters within it should be decorated with appropriate inscriptions and signs (Contractor/Subcontractor name, purpose of the rooms, responsible persons for fire safety, evacuation routes, etc.).

The typical composition of the premises of the camp:

- The room for the OHS classes;
- Checkrooms;
- Room for rest and warming of workers;
- Shower room, washroom;

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- Room for drying clothes and shoes;
- Dining Room;
- Restrooms.

Typical elements of the improvement of sites and areas within the residential campus:

- Shoe washing/cleaning points (near the entrances to all buildings of the camp);⁴
- Awnings for recreation;
- Smoking areas;
- Information boards;
- Fire Shields.

In order to organize the management of the camp, the Contractor shall appoint and inform the Client of the Commandant who will organize and perform the administrative management of the camp and the construction headquarters.

3.5 Control of compliance with the requirements for the maintenance of the construction headquarters and the camp

In the course of weekly inspections of the work site, the Employer's representative shall make a tour of the camp in order to monitor the Contractor/Subcontractor's compliance with the requirements for the maintenance of premises and adjacent areas, compliance with sanitary and occupational safety and health regulations.

If violations of these rules are found, the Employer's representative shall draw up and issue a notice to correct the non-compliance within a reasonable period of time.

3.6 Parking place for construction and special equipment

3.6.1 Requirements for covering the parking space for construction and special equipment

The parking area for construction and special equipment shall be arranged with the use of construction materials that exclude erosion of the roadbed and formation of dirt/dust, regardless of possible climatic conditions - crushed stone, grades and fractions used in road construction, or precast or monolithic reinforced concrete. The choice of materials for parking areas should be based on the availability/acquisition and use of materials in the construction area.

3.6.2 Requirements for fencing of parking places for construction and special equipment

The construction of the fence, including the type, choice of materials and geometric parameters of the fence are provided in the Construction Management Project.

3.6.3 Rules for the maintenance of parking places for construction and special equipment

All maintenance work, including replacement of technical fluids and repairs, is prohibited in the areas designated for parking of the equipment;

It is prohibited to place equipment that is not involved in construction, as well as equipment that is in a state of disrepair, in the parking areas.

3.7 Open areas for temporary storage of materials and equipment

Sites for temporary storage of materials and equipment shall be located in accordance with the requirements of the Construction Management Plan/Stroygen Plan. The sites shall have an even, level, leveled surface providing for natural drainage of surface water. Equipment shall be stored in areas enclosed by temporary fencing. Storage of materials and equipment shall protect them from weather conditions and any damage. Contractor/Subcontractors shall minimize such open storage of materials and equipment.

3.8 Installation of storage containers for household and construction waste station

Close to each construction site, a site for installation of containers for temporary collection of household and construction waste shall be allocated and equipped in accordance with the requirements of the sanitary rules of RK. The number of points for installation of containers for household and construction waste shall be regulated by the total volume of garbage generated and the standards of the sanitary rules of the Republic of Kazakhstan in force.

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3.9 Toilets

Toilets must be located in close proximity (no further than 100 m) to each of the work sites. If there are no fixed equipped toilets at the work sites, it is necessary to ensure installation of toilets with sealed tanks of "Bio-toilet" type. It is necessary to conclude a contract for bio-toilet maintenance (pumping waste from toilet tanks, filling with a special liquid that prevents unpleasant odor, washing the tanks and the bio-toilet cabin)

It is prohibited to locate and install toilets with cesspools within the working areas.

3.10 Areas of work

At each work site, an information sign with the name of the site, the name of the responsible subcontractor, the name of the responsible person and contact information must be installed. At each site should be installed appropriate signs and information signs of the staging posts, temporary storage of materials, places of storage of hazardous substances, gas cylinders, etc.

All areas shall be fenced in accordance with the applicable construction standards and regulations of Kazakhstan. Places of work fenced with signal tape, signal fencing shall be inspected daily by the Contractor/Subcontractor for integrity and the need for immediate restoration.

In order to maintain cleanliness and orderliness, the Contractor shall make regular rounds and inspect the Subcontractors' work areas. In the event of irregularities and non-conformities from the requirements, take the measures necessary to rectify them unconditionally within a reasonable period of time.

3.11 Smoking areas

There must be places for smoking:

- Located in close proximity, at a safe distance from each of the work areas outside the work area;
- Equipped with canopies of non-combustible materials, benches and garbage cans;
- Marked by appropriate signs and information boards, primary fire extinguishing equipment;

Smoking outside organized places **is prohibited**.

3.12 Requirements for the equipment used in the production of works

Machines and mechanisms must be properly registered with the supervisory authorities of Kazakhstan.

Machines and mechanisms used at the construction site must be kept clean and in good condition, with no traces of dripping liquids and oils. Washing points for machines and mechanisms

Washing points of machines and mechanisms should be located in accordance with the Project of the organization of construction.

Washing of machines and mechanisms outside equipped areas designated for washing **is prohibited**.

3.13 Requirements for cleanliness and organization of cleaning of the work site

The work site area, including the territory of the workers' camps, passageways, passageways, sites for storage and consolidated assembly of structures and elements, workplaces, as well as the surrounding area must be kept clean and tidy. To this end:

- During the warm season, roads and driveways should be cleaned as needed with a water washer, and process roads are regularly sprayed with water or bituminous emulsions for dust suppression;
- During the cold season, roads and passages are cleaned with snow removal equipment at a frequency that ensures its serviceability;
- Construction debris, household waste and snow are removed from the work site in a timely manner.

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It is not allowed to bury construction debris and waste in the ground, as well as their incineration directly on the work site.

Industrial and domestic wastewater generated at the work site must be treated and decontaminated in accordance with the procedure provided for in the Construction Management Project.

3.14 Requirement for lighting of the work site area

Illumination of the construction site should meet the standards not lower than those stipulated by the international standard GOST 12.1.046-2014 "Construction. Illumination standards of construction sites. Lighting of work areas should allow for work around the clock".

In addition to the requirements stipulated in the Construction Management Project, there must be lighting:

- Entrance group, including separate lighting sources for information boards, traffic patterns;
- The area in front of the construction headquarters;
- The territory of the household camp;
- Areas of storage and warehousing, unloading and reloading of materials and equipment (security lighting).

3.15 Requirements for signs and information signs

Recommended forms of signs and information signs to provide a working site in Appendix 11 of the Standard.

3.16 Requirements for means of remote video monitoring of the work site

In order to be able to monitor the construction of the facility remotely, construction sites are equipped with web-cameras. The cameras are mounted on artificial structures at a height that provides the fullest possible view of the construction site.

The images from the cameras should be displayed on a separate monitor installed in the Contractor's construction headquarters and the Employer's project office premises.

3.17 Requirements for the organization of the route around the work site by official delegations

In order to provide the officials with full and high-quality information on the current status of construction, the route around the site by official delegations (hereinafter, the "**Route**") shall be developed. The Route is graphically mapped on the Construction Plan.

Unless otherwise specified, the route must begin and end at the construction headquarters.

The bypass route should include the completed, most responsible and significant sections of the current stage of construction.

If necessary, and if the terrain or artificial structures permit, an observation platform is arranged by constructing fenced platforms with the location of basic information about the progress of construction and the characteristics of visible objects.

The itinerary should be updated monthly or by the specific date of the event.

Roads and walkways included in the route should be as far away as possible from places of exposure to hazards.

APPENDIX TO EMPLOYER'S REQUIREMENTS

regarding temporary networks

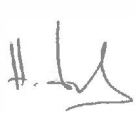
Astana city

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During the period of the Work (performance of the Work), the Contractor shall be subject to the following obligations:

- 1) Obtaining temporary technical conditions for engineering networks to ensure and carry out works at the Site;
- 2) erection of temporary buildings and structures, materials and equipment, including construction and installation work, as well as covering all costs at the Contractor's expense associated with obtaining permits for the necessary communications to them;
- 3) organization of infrastructure on the construction site and adjacent territory, namely, the construction of temporary roads, fences, wheel wash point, providing transport services, lighting masts, flagpoles and checkpoints, etc;
- 4) further operation of temporary utility networks, buildings and facilities, including cleaning services, security services, garbage collection, and other expenses;
- 5) reimbursement of utilities, security, garbage collection, internet, cleaning and other services;
- 6) implementation of environmental protection measures, ensuring cleanliness at the construction site and adjacent areas;
- 7) ensuring the implementation of the necessary fire protection measures at the construction site, as well as measures for the safe production of works;
- 8) compliance with environmental standards at the construction site.



APPENDIX TO EMPLOYER'S REQUIREMENTS

Specification of Works (Results of Works)

Astana city

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“Work” in accordance with clause 1.1.79 of the Contract means the Permanent works and Temporary works, including but not limited to the following: project works, engineering and exploration works, Equipment and Materials supply, responsible storage, material and technical support, transportation, logistics services, customs clearance, loading and unloading according to the Employer's Requirements, operations, training, mechanical readiness, construction and installation work, preparation for commissioning, development of project on commissioning and coordination plan, commissioning of the Equipment and Permanent works, supervision of installation, supply of consumables and spare parts, as well as any additional, auxiliary and other related works and services necessary and/or expedient for successful preparation for commissioning, successful commissioning, commissioning of Permanent works on a turnkey basis and elimination of any defects, as more fully described in the Employer's Requirements and the Design, as well as all other works and services performed by the Contractor under the Contract (even if it is not directly enshrined there), including auxiliary works or connected to the Design, or anything of the above-mentioned if applicable.



APPENDIX TO EMPLOYER'S REQUIREMENTS

Technical Maintenance and Repair Methodology

Astana city

14.03.2023

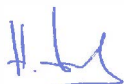
CONTRACT AGREEMENT dated 14.03.2023

When transferring facilities from the Contractor to the Employer, in order to maintain correct accounting in accordance with IFRS, as well as to implement the Employer's asset management system, the Contractor must provide the Employer with a register of all transferred facilities (assets) that meets the following requirements:

1. all objects (assets) must be classified in accordance with IFRS into fixed assets, intangible assets and inventories and their groups, according to the Employer's accounting policy, or general definitions of IFRS in case of its absence.
2. all items of fixed assets must be divided into components in accordance with the criteria agreed with the Employer or the basic requirements of IFRS.

At a minimum, components whose original cost is significant in relation to the total original cost of the entire item of fixed assets and having different useful lives and/or depreciation method should be identified.

3. with respect to all the Contractor's fixed assets (parent and components) and intangible assets must specify:
 - Name in accordance with passport data
 - Unique identifier (inventory number) applied to the object with a unique identifier (inventory number) of the parent fixed asset for components
 - Cost in KZT (with VAT, without VAT) (in case of cost in foreign currency, the rules and date of recalculation in KZT shall be agreed upon with the Employer or correspond to IFRS 21)
 - Standard terms of useful service
 - Year of issue (construction)
 - Date of acceptance for accounting/transfer to the Employer
 - Manufacturer
 - Model
 - Location on site
 - Scheduled repair/inspection date
 - Generation rates
 - Reference to datasheet (another document)
 - Other parameters as agreed with the Employer



APPENDIX TO EMPLOYER'S REQUIREMENTS

Methodology for local content calculation

Astana city

14.03.2023

CONTRACT AGREEMENT dated 14.03.2023

Methodology for calculating the share of local content of the Works (results of the Works)

Calculation of the share of local content in the contract for the performance of works, is made by the formula:

$$BL_{p/y} = 100\% * \frac{\sum_{j=1}^m ((CA_j - CT_j - CCA_j) * R_j + \sum_{i=1}^n (CT_i * M_i))}{S}$$

Where:

m - the total number of j-contracts concluded for the purpose of work (service), including the contract between the customer and the supplier, the contracts between the supplier and the subcontractor (co-contractor);

j - serial number of the contract concluded for the purpose of work (service);

CA_j - value of the j-contract;

CT_j is the total cost of goods purchased by the supplier or subcontractor to fulfill the j-contract;

CCA_j - the total value of subcontracts (co-contractor) contracts concluded in order to execute the j-contract;

R_j - the share of the payroll of Kazakhstani personnel in the total payroll of the supplier or subcontractor (co-contractor), performing the j-contract;

n is the total number of items purchased by the supplier or subcontractor (co-contractor) in order to perform the j-contract;

i is the serial number of the goods purchased by the supplier or subcontractor in order to perform the j-contract;

CT_i - the cost of the i-commodity;

M_i - the share of intra-country value in the commodity specified in the certificate of origin of the form "ST-KZ";

In the absence of a certificate of origin form "ST-KZ", unless otherwise stipulated by paragraph 10 of the Uniform Methodology, M_i = 0;

S is the total value of the contract.

7. R_j - the share of the payroll of Kazakhstani personnel in the total payroll of the supplier or subcontractor, performing the j contract, is calculated by the following formula:

$$R_j = SFRK/SF$$

Where:

Salary Foundation of RK (SFRK) is the payroll of the Kazakhstani personnel of the supplier or subcontractor performing the j contract for the period of the j-contract;

Salary Foundation (SF) is the total payroll of the supplier's or subcontractor's employees performing the j contract for the period of the j contract.

Branches (representative offices) of legal entities are not suppliers or subcontractors.

Provided that a party to the j-contract is a branch (representative office) of a legal entity, when calculating the coefficient R_j , indicate the total number of employees of such a legal entity.

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APPENDIX TO EMPLOYER'S REQUIREMENTS

SCHEDULE OF RATES AND PRICES

Astana city

14.03.2023

CONTRACT AGREEMENT dated 14.03.2023

The parties have agreed that this appendix will be completed after passing the State Expertise.



APPENDIX TO EMPLOYER'S REQUIREMENTS

Protection construction

Astana city

14.03.2023

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In order to ensure the protection of working personnel, the Contractor, as part of the Works, must provide for the construction of a civil defense protective structure according to local regulations of the Republic of Kazakhstan, ensuring safety from weapons of destruction and from radiation.



APPENDIX TO EMPLOYER'S REQUIREMENTS

Guaranteed performance

Astana city

14.03.2023

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Net Electrical power (installed/disposable)	1050 MW at ISO condition
Electric power consumption for own needs	Up to 6%
Gross Efficiency (electrical/thermal)	Not less than 53%
Time of delivery of nominal power from cold/uncooled/hot condition	Hot (within 8 hrs after turbine trip): 210 min. Cold: 300 min. Warm (uncooled): Between 210 min. to 300 min. Will be considered after providing technical and operational characteristics from the OEM/sub-supplier.
CO concentrations in exhaust flue gases (maximum per day)	Not greater than 100 mg/nm ³ at 100% Plant Load. In accordance with the regulations of the Republic of Kazakhstan, a possible tightening up to 30 mg/Nm ³ .
Sound pressure level at the border of the Sanitary Protection Zone (SPZ)	Not greater than 55 decibels (A)

※ Fuel for performance Guarantee is main gas specified in Employer's Requirements.



APPENDIX TO EMPLOYER'S REQUIREMENTS

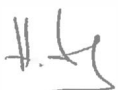
Automated monitoring system

Astana city

14.03.2023

CONTRACT AGREEMENT dated 14.03.2023

The presence of a system for monitoring engineering structures in the form of an intelligent system, which is a hardware and software complex that monitors load-bearing and critical structures of buildings and structures. The task of the monitoring system for engineering load-bearing structures is the timely warning and prevention of emergency situations arising from the fact that the load-bearing structures of an object are subjected to critical changes.



APPENDIX TO EMPLOYER'S REQUIREMENTS

Building management systems

Astana city

14.03.2023

CONTRACT AGREEMENT dated 14.03.2023

It is necessary to provide for the presence of a Building Management System as a solution for the most efficient operation of buildings and saving energy, including the integration of several building engineering systems, in order to effectively and economically manage buildings. Building automation should include metering, control and supervision solutions, as well as energy efficiency solutions that help save as much as possible on the amount of energy consumed per year, while also providing high comfort.

The main systems, including but not limited to, to be managed are:

Ventilation system. Maintaining the air in the building at a certain level of temperature and humidity is provided by an intelligent system;

Heating system. Thereby to the preliminary settings, it is possible to automatically turn on in the weather and calendar modes, as well as maintain the set temperature with a minimum consumption of resources;

Cooling system. The launch of centralized heating occurs depending on the time of year and the air temperature outside the window. The system operates in an economical mode while maintaining the set temperature and without unnecessary energy consumption;

Water and gas supply system. It implies automatic filling of tanks, water and gas supply, as well as monitoring of water and gas pipes. Issuance of a command to actuate shutoff valves is performed automatically in case of water leakage or gas leakage;

Lighting and power supply system. Uninterrupted power supply to the building is ensured through the introduction of a "smart" system. Programmed operating modes provide maximum energy savings. Security system that includes various alerts, authorized access and video surveillance.

Among the platform modules, it is also necessary to provide for the presence of the following modules:

- system core (database server, web server, IoT engine, LoraWAN server, MQTT broker, VoIP server, API server);
- database of users, premises, personal accounts, services, etc.;
- unified communications module (VoIP, SMS, Telegram, e-mail, chat, video conferencing) and integration with SIP;
- management of assets, applications, maintenance and repair;
- digital twin based on 3D BIM model;
- module of integration with engineering systems;
- module for integration with external systems;
- business process development module;
- integration module with ASKUE;
- monitoring module, summary screens and analytics;

- analytics and machine learning module;
- blockchain data security module;
- center of accidents and notifications;
- mobile applications for users (employees, visitors), managers and top managers, engineers.



APPENDIX TO EMPLOYER'S REQUIREMENTS

REQUIREMENTS FOR PACKAGING, LABELING, SHIPPING, TRANSHANDING AND UNLOADING

Astana city

14.03.2023

CONTRACT AGREEMENT dated 14.03.2023

The Contractor is responsible for the committed violations of the requirements applicable in the field of international forwarding in the performance of contract work.

The Contractor must instruct the relevant responsible persons of his organization, his subcontractors on all requirements applicable in the field of international forwarding when performing contract work. Any requirements for the transport of cargo and related activities contained in the technical specification take precedence over the following minimum requirements.

PACKAGING AND LABELING REQUIREMENTS

A. Packing list(s)

1. The packing list(s) must contain the following information:

1.1 Name and address of the Supplier

1.2 Name and address of the Buyer and Consignee

1.3 Place of delivery

1.4 Packing list number

1.5 Packing list date

1.6 Number of the Supply Agreement and/or Shipping Order

1.7 Material code, product name, unit of measurement, quantity by position. The data must exactly match the Order Details or Shipping Order

1.8 Number of packages (eg 5 boxes, 3 containers, 6 bundles, 2 loose items) and total number of packages. (For example, 1 out of 5, 2 out of 5, etc.)

1.9 Gross and net weight in kilograms per package.

1.10 Total gross and net weight in kilograms of the entire consignment.

1.11 Dimensions in centimeters and package volume measurements in cubic meters.

1.12 Measurements of the volume of the entire cargo in cubic meters.

1.13 Container number and type, where applicable.

2. For all containerized cargo, the Supplier must provide a "container contents list" with a detailed list of the contents of the container. A list of the contents of the container must be enclosed in a sealed bag affixed to the inside of the left side door of the container. The second list of the contents of the container must be enclosed in a sealed waterproof bag affixed to the outside of the container's right-hand door. The third copy of the list of contents of the container should be sent as a separate package to the final recipient on the spot. All listings must include the container number, shipment identification number and vessel/transport name.

3. The packing list must be stamped by the Supplier's company and signed. Two original packing lists must be enclosed with each shipment. One copy inside the package, one copy in a waterproof envelope attached to the outside of the package, with the following documents:

- The consignment note of the Supplier;

- Original commercial invoice stamped and signed by the Supplier;

- Dangerous goods declaration or material safety data sheet, if applicable;
- Certificate of origin, if applicable;
- Quality certificate, if applicable.

b. Packing instructions

The Contractor is responsible for providing adequate protection and/or packaging for the safe transport and arrival of the goods in an undamaged condition at their final destination. All packaging must comply with the Export Packaging Standard.

The cost of replacing or repairing any unsound or unsuitable packaging, or resulting from loss or damage to the contents, will be borne by the Contractor.

1. Packaging must comply with international standards and be suitable for the mode of transportation (air, sea, rail, road and inland waterway) and must be appropriate and sufficient to prevent theft, withstand repeated unloading, loading and storage for an indefinite period in various conditions;
2. Packaged goods must be protected inside the case from horizontally occurring forces (tilting, movement, strapping, etc.) in such a way that they can withstand such a load without any damage. Protection can be provided by bolting to the skid or base, fixing to the base with a frame, a gasket between the device and the enclosure wall, etc.;

In the manufacture of stands, bundles, boxes, containers, partial packages, etc., the packaging method must be appropriate for the weight and characteristics of the goods.

The use of cardboard boxes or soft wood fiber boards as external protection is unacceptable.

Packages must be consecutively numbered from 1 upwards for each Purchase Order (e.g. 1 of 3, 2 of 3, etc.)

3. Where possible, good quality wooden cases/boxes or similar packaging should be used;
4. Packages weighing five (5) tons or more must be marked with sling points and clearly marked on both sides with the center of gravity for careful handling of the package;
5. Items that do not need significant mechanical or climatic protection should be packed in a box or pallets (eg heat exchangers, vessels, etc.). Machined or screw-cut surfaces and all moving internal parts of the machine must be coated with an easily removable anti-corrosion coating. All fittings, valves, products, etc., smaller than 100 mm must be properly packaged in boxes with clear indications of which Purchase Order item numbers they refer to. All fittings, valves, etc. larger than 100 mm must be properly packed in boxes or pallets;
6. All hatches, nozzles and similar openings in equipment and machinery, etc., must be closed with wooden or metal plugs and securely bolted using suitable gasket material. Threaded connections must be plugged or insulated to provide a waterproof seam. In addition, the edges of all seams must be sealed with waterproof tape;
7. Rotors or electric motors and rotating parts fitted with anti-friction bearings must be secured against movement by clearly visible external clamps to prevent any internal damage;
8. All equipment must be completely dry prior to packing and loading to prevent damage due to freezing and corrosion;
9. The Contractor is responsible for proper inner packing and wrapping of items subject to moisture damage (eg tools, precision instruments, electric motors, etc.). Such protection necessarily involves sealing and embedding moisture-absorbing materials, as the case may be;
10. Packages containing fragile items must be packed with extreme care to avoid breakage in strong wooden crates and be screen printed on all sides with "CAUTION FRAGILE" in English and Russian. Where crates are not to be turned over, the Supplier shall provide a clear indication that they are in the correct position. For example: "TOP";
11. All timber used in the packing process must be properly handled before or after cargo preparation. Timber must be clearly ISPM15 marked or have an appropriate fumigation certificate;

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12. Transportation of all goods with a weight of one piece over 15 kg must be carried out on pallets and / or in containers. Packaging of all goods requiring a crane for unloading must ensure readiness for unloading by a crane (using a substrate (bars), towels, etc.);
13. All dangerous goods supplied for air transport must be supplied in appropriate UN approved packaging. For assistance in determining appropriate packaging, please refer to the IATA Regulation or contact a professional packer who has been trained and certified to IATA standards. Delivery of dangerous goods - all hazardous materials must be clearly marked;
14. The Contractor shall be responsible for packing and pre-slinging necessary to ensure safe and efficient unloading. The Contractor shall supply all lifting and lifting means necessary for the proper unloading of the goods;
15. The Contractor shall ensure that the goods are adequately protected from damage and deterioration during transportation and short-term storage, taking into account the conditions and conditions at the site or other place that is the point of delivery and in the territories through which the materials will be transported, including climate, roads, and shipping instructions. All packages must be appropriately designed for transport on uneven roads, taking into account the appropriate handling. When packing, labeling and transporting materials, the Contractor shall comply with international and national laws applicable to the transport of materials, as well as health, safety and environmental protection;
16. The Supplier shall be liable for any damage, loss or destruction of any materials supplied under the contract or any consequential loss, damage or destruction of any other materials or property resulting from improper packaging or transportation by the Supplier. Expenses incurred as a result of the Contractor's failure to comply with the packaging, labeling and transportation procedures shall be reimbursed by the Contractor. All goods must be packaged, labeled and transported in accordance with the requirements for the transport of hazardous materials (where applicable) and best practice in this area.

C. Marking Instructions

All boxes, bags and boxes are marked as indicated below. On bundles, the marking must be embossed on a metal or similar tag and securely attached to the bundle.

1. If more than one package or box, groupage or bulky item is sent under the same item number (stock), EACH box must be numbered and display the TOTAL number of packages shipped (for example, Box 1 of 5, Box 2 of 5, etc.)
2. The contractor must provide a list of all transported goods, with a detailed list of contents. For item(s) supplied as a kit, kit, etc., an app is required. Give an additional list of the components of a given kit, kit, etc.
3. Goods (s) entering the warehouse as a set must be packed separately from each other (each set must be packed separately).
4. Markings must be written in capital letters not less than 20 mm high and applied either with paint or indelible ink. This marking should, if possible, be applied at three different points on each case, package, bundle, etc.
5. Each individual piece of welded/structural steel must have a metal stamping with an identifying mounting mark.
6. Spare parts boxes must be marked "SPARE PARTS".
7. Goods (which are cable and wire products) must have a factory marking (metric), which must be applied directly along the entire length of the Goods, indicating linear length signs (meters). In case of non-compliance with this requirement, the Customer has the right to refuse to accept the Goods. Return and / or replacement of the Goods that do not meet the labeling requirements specified in this clause are carried out solely by the forces and at the expense of the Contractor in full.

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8. In each case, the box, crate and container, as well as the individual materials, must be clearly labeled as specified in the contract. The seller must ensure that labels containing details of the contract number and date, quantity and description of the materials are attached to the materials or the bill of lading/packing list to accompany the materials to their destination.

9. Shipments of harmful/dangerous materials require the Contractor to apply appropriate/standard warning labels for the commercial transport of hazardous/dangerous goods. Each consignment must be accompanied by a Material Safety Data Sheet (MSDS).

The marking on the package must contain at least the following data:

Delivery Contract Number (if applicable)

Product code / Marking number

Country of origin

Loading unit no. from

Gross weight t.....kg

Net weightkg

Dimensions L.....W.....H.....(cm)

Special requirements for movement

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APPENDIX TO EMPLOYER'S REQUIREMENTS

SHIPMENT REQUIREMENTS

Astana city

14.03.2023

CONTRACT AGREEMENT dated 14.03.2023

The Contractor shall at least comply with the following requirements:

The Contractor is obliged to obtain GREENLIGHT for the Customer's incoming shipments prior to shipment of the goods(s).

The Contractor must fill out a checklist of the established form and obtain approval of the checklist completed by the Contractor from the Customer for incoming cargo.

The pallet/box/crate must be accompanied by a packing list indicating the number of the Purchase Agreement and the Shipping Order, if applicable, which must be included in each piece. The packing list must indicate the commodity items that are in this particular box / box, indicating the quantity of the product (s).

Package/box must not be damaged

Item(s) must be placed on a pallet

Item(s) such as: teeth, comb knives, etc. (heavy metal goods) must be placed on a pallet in one tier

Oil and lubricants should be placed in one tier on a pallet on vehicles

SIGNED

In the name of and on behalf of the Employer:

In the name of «CCGT Turkistan» LLP

Kusainov Askar Atbolatovich
General Director acting on the basis of the Charter

(signature)
/stamp/



SIGNED

In the name of and on behalf of the Contractor:
Consortium composed of:

In the name of Doosan Enerbility Co. Ltd.
(Consortium Leader)

Iljoon Won
Deputy General Manager acting on the basis of Power of Attorney dated 9 March 2023

(signature)
/stamp/



In the name of Bazis Construction Limited Liability Partnership

Uali Bauyrzhan Kudaibergenuly
President, acting on the basis of the Charter

(signature)
/stamp/



EMPLOYER REQUIREMENTS

Astana city

14 March 2023

CONTRACT AGREEMENT dated 14 March 2023

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Section 1. INTRODUCTION

In this technical specification, Potential supplier/the Contractor means an individual engaged in entrepreneurial activities, a legal entity (with the exception of state institutions, unless otherwise provided for them by the laws of the Republic of Kazakhstan), a temporary association of legal entities (consortium) that claims to conclude a Contract on procurement of complex works on surveys, design, logistical support and «turnkey» construction (hereinafter referred to as the Contract) with «CCGT Turkistan» LLP (hereinafter referred to as the Customer).

The Contractor undertakes to carry out complex work on design, logistical support and construction on a «turnkey» basis under the project «Construction of a power plant on the basis of CCGT (combined cycle gas turbine) with a capacity of 1000 MW in the Turkestan region. Main site», which must be performed in accordance with the terms of the Contract, this document and other conditions with access to full design capacity (hereinafter referred to as the Works referred to in the Contract as the «Facility / Works»).

All terms in this document should be interpreted in accordance with the definitions of such terms provided for in the Contract (in the Contract Agreement and General Conditions), unless otherwise specified in this technical specification and its appendices.

The work shall consist of the design, procurement, construction, inspection, testing, the process of pre-commissioning works (PCW) to bring it up to production capacity with the development of a project for PCW, commissioning, all and any actions necessary to deliver to the Customer a complete, safe, functional, operable and maintained power plant.

Any design, material, method or equipment provided by the Contractor shall be fit for purpose and meet all requirements of the Contract.

No deficiencies, errors or omissions in this document with respect to any design, material, method, equipment or specification contained therein shall release the Contractor from his obligation to perform the Work under the Contract.

The Contractor shall be deemed to have included in the Contract Price all installations and equipment, whether or not specifically specified herein, that are necessary to form a complete Facility, fit in all respects for its intended use.

Works include, but are not limited to, design, obtaining permits and approvals, obtaining all required expertise certificates, manufacturing, construction, supply, in-process testing, packaging for export, shipping and unloading, transportation, delivery to the Facility, unloading at the Facility, storage on and off the Facility, full construction, connection to existing infrastructure (network) or construction of new infrastructure, conservation, putting into operation and commissioning, performance and reliability tests upon completion of works, including tools and equipment for maintenance and operation, spare parts, instruction / training of the Customer's personnel and liabilities for the defect liability period for equipment required for a fully functioning Facility.

The Contractor shall provide all equipment, scope of work and services, as specified, in order to make the Facility fully completed and functional for safe, reliable, serviceable and suitable for commercial operation under all conditions and relations.

The Contractor shall provide full details of the labor resources he offers during construction periods to be included in the defined scope of work.

All materials and equipment must be new and of the best quality for use under the conditions and fluctuations in temperature and pressure that will be encountered in operation, without excessive corrosion, change in shape, aging or fatigue, and without undue stress or deformation in any part that may affect efficiency and reliability of a plant that is in all respects suitable for its intended purpose. No repairs will be permitted without the written permission of the Customer. Complete information on the properties and the thermal, chemical and mechanical treatments used must be provided.

Particular care must be taken to eliminate the possibility of corrosion due to galvanic effects. Design, material selection and all installation methods should be such as to minimize these effects.

All parts of the installation that come into contact with potable water, industrial water or waste water, such as pumps, pipes, heat exchangers, measuring instruments, etc., must be designed in such a way as to prevent contamination and corrosion. The same applies to individual plant equipment exposed to humid gases below the dew point.

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All materials to be used in the Project must be accompanied by the necessary certificates confirming their required quality in accordance with relevant international and local standards. These certificates must be presented to the Customer for verification.

All materials, installations and equipment must be suitable for the purposes of the Facility. Equipment must be selected to ensure high reliability and availability of the entire Facility with low levels of downtime and minimal maintenance. There must be permanent, safe access to all parts of the installation for inspections, operation and maintenance.

The construction of a power plant on the basis of CCGT with a capacity of 1000 MW in the Turkestan region includes all of the following works that are described or implied in this Technical Specification, as well as all matters not taken into account in it, which may be considered necessary for completion of works, further stable, safe, reliable and efficient exploitation, whether or not specified in this document.

1.1. Technical documentation of the Tender proposal of the Contractor

To evaluate the technical part of the Tender offer, the Contractor must provide the following documentation:

- Work schedule;
- Receiving LTSA offers from manufacturers of the main critical equipment, within 12 and 27 years from the expiration date of the warranty period. The Participant is obliged to conclude service and maintenance agreements with the Customer during its warranty period - 2 years;
- The Supply Specification (delivery set) of the energy source that is clarified/specified by the Contractor (a complete list of supplied equipment, utilities, communication systems, inventory, materials, products, structures, works, services, indicating types, main technical characteristics, weight, and so on);
- Preliminary Master Plan within (the allocated by the Customer) construction site, proposed by the Contractor;
- Updated title list of buildings and structures;
- Layout drawings of the placement of energy source equipment in buildings and structures;
- A general PI diagram of the Main body of the energy source, a schematic diagram of the output of electric power of the energy source including Switchgears of own needs (RUSN), a block diagram of the automation of the energy source;
- Drawings of the general type of the proposed equipment (with overall, installation, connection dimensions, weight) and specifications;
- Aerodynamic, hydraulic and thermal calculation of energy installations (Gas Turbine, HRSG, Steam Turbine) in various modes (10%, 20%, 30%, 50%, 75%, 100%) and at different outdoor temperatures, taking into account the height of the construction site above sea level;
- Correlation curves in order to adjust the test results to the standard conditions (according to ISO), local conditions and to the nominal conditions with respect to the turbine gas temperature;
- Dependence of the nominal base and peak power and efficiency of the gas turbine, the flow rate and temperature of the gases behind the turbines on the ambient air temperature;
- Dependences of fuel consumption (CCGT efficiency, gas consumption and outdoor temperature) on electrical load;
- Formulae and diagrams showing the change in recommended operating hours between mandatory inspections/major repairs, depending on the number of emergency shutdowns and startups, operating hours at base load and continuous operation at peak load;
- Technical description of all the equipment of the energy source with indication of the characteristics of energy efficiency;
- Correlation diagrams of the energy source parameters against external conditions;
- Schedules of energy source start-ups at various states;
- Noise characteristics of the main equipment;
- Environmental indicators of energy source emissions;
- Conditions for the implementation of technological protections, locks, automatic reserve input (AVR), technological and emergency alarms



- Information on the frequency and duration of inspections and repairs;
- A list of activities carried out by the Contractor as part of compensation for seismic impacts on the power plant;
- Power and quality characteristics of energy sources to meet their own needs during the construction of a power plant;
- A document indicating deviations and/or non-compliance with the requirements of the Tender Documentation. The absence of deviations implies confirmation of the requirements of the Tender Documentation

1.2. General information

The name of the project - «Construction of a power plant on the basis of CCGT (combined cycle gas turbine) with a capacity of 1000 MW in the Turkestan region. Main site» (hereinafter - the Facility).

The location of the facility - the Republic of Kazakhstan, Turkestan region, Tolebi district.

Customer – «CCGT Turkistan» LLP.

The source of financing - own funds.

CCGT product range - electrical energy.

The main goal of the Project is to ensure the balance reliability of the energy system of the Republic of Kazakhstan, cover irregular fluctuations in the electrical load and imbalance in the power flow resulting from the uneven supply of electricity produced by renewable energy sources, increase the reliability and safety of the power plant through the use of modern technologies for the production of electrical energy.

Climatic conditions of the location site

The climatic conditions of the construction area of the Facility shall be accepted in accordance with CII PK 2.04-01-2017* (SP RK 2.04-01-2017*) with a change in design temperatures and duration of the heating period (Table 1.2.1).

Table 1.2.1.

Name of indicator	Value	Rationale
Climatic subregion	IVA	СП РК 2.04-01-2017* (SP RK 2.04-01-2017*)
Humidity zone	dry	МСН 2.04-03-2004, прил. 3 (МСН 2.04-03-2004, app. 3)
Estimated outdoor temperature, °C:		
absolute minimum	-30.3	СП РК 2.04-01-2017* (SP RK 2.04-01-2017*)
absolute maximum	+44.2	СП РК 2.04-01-2017* (SP RK 2.04-01-2017*)
The coldest day with a probability of 0.92	16.9	СП РК 2.04-01-2017* (SP RK 2.04-01-2017*)
The temperature of the coldest five-day period with a security of 0.92, °C	14.3	СП РК 2.04-01-2017* (SP RK 2.04-01-2017*)
Average annual outdoor air temperature, °C	+12.6	СП РК 2.04-01-2017* (SP RK 2.04-01-2017*)
Average monthly outdoor temperature of the coldest month, January, °C	-1.5	СП РК 2.04-01-2017* (SP RK 2.04-01-2017*)

Name of indicator	Value	Rationale
Average monthly outdoor air temperature of the hottest month, July, °C	+26.4	СП РК 2.04-01-2017* (SP RK 2.04-01-2017*)
Average maximum outdoor air temperature of the hottest month, July, °C	+33.5	СП РК 2.04-01-2017* (SP RK 2.04-01-2017*)
Temperature under normal conditions for GT, CCGT, °C	+15	ГОСТ Р 52200-2004* (GOST R 52200-2004*)
Temperature of the beginning of the heating period, °C	+8	СП РК 2.04-01-2017* (SP RK 2.04-01-2017*)
Duration of the heating period ≤ 8 °C, days.	136	СП РК 2.04-01-2017* (SP RK 2.04-01-2017*)
Average number of days with minimum air temperature equal to or below ≤ -25 °C	0	СП РК 2.04-01-2017* (SP RK 2.04-01-2017*)
Average number of days with maximum air temperature equal to or higher than $\geq +25$ °C	141.7	СП РК 2.04-01-2017* (SP RK 2.04-01-2017*)
Average number of days with maximum air temperature equal to or higher than $\geq +30$ °C	87.9	СП РК 2.04-01-2017* (SP RK 2.04-01-2017*)
The average number of days with a maximum air temperature equal to or higher than $\geq +34$ °C	37.2	СП РК 2.04-01-2017* (SP RK 2.04-01-2017*)
Barometer height above sea level, m	695	СП РК 2.04-01-2017* (SP RK 2.04-01-2017*)
Relative humidity, year, %	57	СП РК 2.04-01-2017* (SP RK 2.04-01-2017*)
Atmospheric pressure at the height of the barometer, monthly average for July/January, hPa	937.5/951.4/946.517	СП РК 2.04-01-2017* (SP RK 2.04-01-2017*)
Atmospheric pressure at the height of the barometer, average for the year, hPa	946.517	СП РК 2.04-01-2017* (SP RK 2.04-01-2017*)
Normative value of snow cover weight (II region), kPa	1.5	НТП РК 01-01-3.1 (4.1) - 2017 (NTP RK 01-01-3.1 (4.1) - 2017)
Standard value of wind pressure (III region), kPa	1.0	СП РК 2.04-01-2017* (SP RK 2.04-01-2017*)
Seismicity of the construction area, points	8 (category I)	СП РК 2.03-30-2017* (SP RK 2.03-30-2017*)

СП РК 2.04-01-2017 «Строительная климатология» (SP RK 2.04-01-2017 «Construction climatology») No. 312-NK dated December 20, 2017 was introduced by the Order of the Committee for Construction, Housing and Communal Services of the Ministry for Investments and Development of the Republic of Kazakhstan.

The main equipment of the Facility includes: a gas turbine (GT) with an electric generator, a heat recovery steam generators (HRSG), a steam turbine (ST) with an electric generator, a bypass and a chimney.

The main fuel is fuel gas, emergency (back-up) fuel is diesel fuel.

1.3. Enterprise capacity

The power of a maneuverable Power plant is determined by the requirements for installed and control power, the characteristics of industrially developed gas turbine equipment.

The defining factors are: high maneuverability with a large control range and maintaining the efficiency of the equipment in the control range while reducing power to a technical minimum.

For the Facility, due to the significant influence of the outside air temperature on the GT power, the nominal and maximum power values and the limits of the control range change significantly with the change in the outside air temperature.

The indicators of installed capacity and maneuverability for each type of power equipment are determined in the Project by the technical characteristics of the equipment and are specified during the annual certification of power units.

The installed electrical power of condensing combined cycle plants is determined as the sum of the capacities of steam and gas turbines at an average annual temperature, air humidity and the corresponding temperature of the cooling water.

The determination of the average annual generation of electrical energy was made on the basis of simulated graphs of the dependence of electrical power on the ambient temperature (Figure 1.3.1). In doing so, it is accepted:

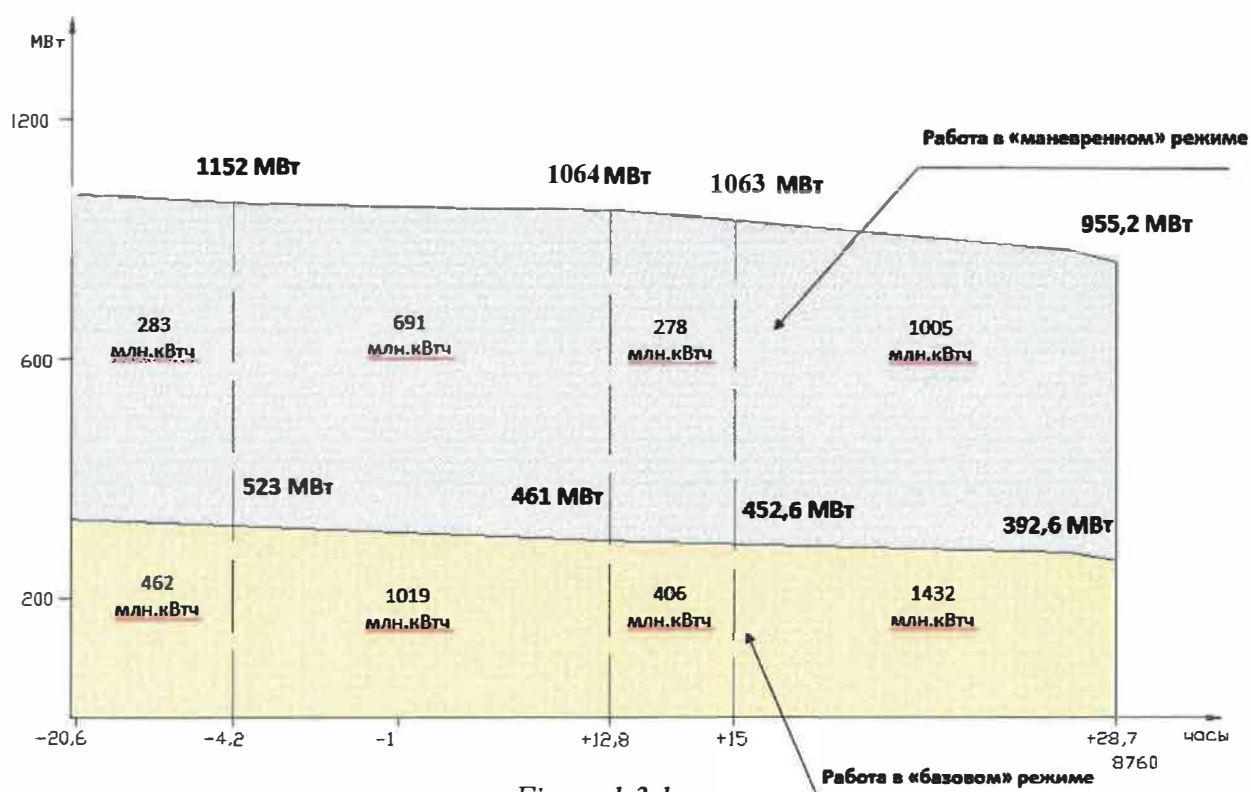


Figure 1.3.1.

- «base» mode of operation is taken below the technical minimum.
- within the regulation range, the coefficient of influence of the participation of equipment in the regulation of daily schedules of electrical loads (0.5) was applied to the volume of electricity generation.

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1.4. Providing the enterprise with resources

1.4.1. Fuel

The main fuel that provides production activities for the generation of electricity of the designed Facility, is fuel (natural and associated) gas provided by «QazaqGaz» JSC from two sources:

- Main gas pipeline (MGP) «Beineu-Bozoi-Shymkent» (main);
- Main gas pipeline (MGP) «BGR-TBA» (reserve).

To provide the projected power plant with fuel gas, the construction of the linear part of the gas pipeline-outlet with AGDS was considered, which ensures the transportation and supply of natural gas with the declared volume of consumption.

The equipment of the external gas supply system includes the following units:

- security crane units (SCU);
- cleaning device launch unit (CDLU), cleaning device receiving unit (CDRU);
- automated gas distribution station (AGDS);
- gas pipeline.

Power supply of the site AGDS – «CCGT», CDRU, CDLU, SCU is carried out from the projected OTLPW-10 kV (overhead transmission lines with protected wires) for auxiliary needs of the CCGT.

Steel pipes of the gas pipeline provide for measures to protect engineering networks from corrosion. The external gas supply complex is provided with communication and signaling systems: fiber optic communication, industrial and technological USW (ultra-short waves) radio communication, satellite communication, IP telephony, burglar alarm, fire alarm, video surveillance.

The complex of all measures to ensure the reliable operation of all units of the gas supply system makes it possible to ensure uninterrupted supply of fuel gas from the main gas pipeline to the site of the designed combined-cycle plant with the required parameters.

ГОСТ 305-82 (GOST 305-82) diesel fuel is provided as an emergency fuel for gas turbines.

For the storage of a 3 -day supply of fuel, a diesel fuel warehouse is provided, the total capacity is determined by the project. A diesel fuel pumping station is provided for receiving and pumping diesel fuel.

To replenish the emergency reserve of diesel fuel, it is transported by road and/or rail to the sites.

Table 1.4.1.1. Characteristics of natural gas - MGP «Beineu-Bozoi-Shymkent» (according to the Passport for gas No. 312 dated November 21, 2022, GMS «Akbulak»)

No.	Name of indicators	ND of measurement methods	Requirements ST RK 1666-2007	Actual value
1	Component composition (% mol)	ГОСТ 31371.7-2008 (GOST 31371.7-2008)	Not normalized	
	Methane CH ₄			86.8
	Ethane C ₂ H ₆			8.3
	Propane C ₃ H ₈			2.11
	Iso-butane i - C ₄ H ₁₀			0.210
	Nor-butane n - C ₄ H ₁₀			0.164
	Neo-pentane neo- C ₅ H ₁₂			0.000
	Iso-pentane i - C ₅ H ₁₂			0.009
	Nor-pentane n - C ₅ H ₁₂			0.006
	n-Hexanes C ₆ H ₁₄			0.018
	Heptanes C ₇ H ₁₆			0.001
	Octanes C ₈ H ₁₈			-
	Nitrogen N ₂			2.28
	Carbon dioxide CO ₂ no more		2.5	0.104
	Volume fraction oxygen O ₂ % max		0.02 /0.1	0.008
	Mass concentration of hydrogen sulfide, g/m ³ no more (Iodometry) from 21.11.2022	ГОСТ 22387.2-97	0.007/0.006	0.001

	Mass concentration of mercaptan sulfur, g/m ³ no more (Iodometry) from 11/21/2022	(GOST 22387.2-97)	0.016/ 0.014	0.012
2	Density, kg / m ³ at 20 °C and 760 mm Hg (calculated)	ГОСТ 31369/ 17310 (GOST 31369 / 17310)	Not normalized	0.7618
3	Density, kg / m ³ at 20 °C and 760 mm Hg (pycnometric) from	ГОСТ 17310 (GOST 17310)		-
4	Moisture dew point, °C at P _{gas} = 7.0 MPa	CT PK 53763-2011 (ST RK 53763-2011)	from 01.05 to 30.09. (-10)/(0) from 01.10 to 30.04. (-10)/(-5)	-13.3
5	Hydrocarbon dew point, P _{gas} = 2.7 MPa	CT PK 53762-2011 (ST RK 53762-2011)	from 01.05 to 30.09. (-2)/(0) from 01.10 to 30.04. (-2)/(-2)	-13.8
6	Lowest volumetric calorific value, MJ/m ³ , at 20 °C, 101.325 kPa, not less	ГОСТ 31369-2008 (GOST 31369-2008)	31.80	36.23
7	Range of values of the Wobbe number MJ/m ³	ГОСТ 31369-2008 (GOST 31369-2008)	Not normalized	50.37
8	Mass concentration of mechanical impurities, g/m ³ , no more than from 08.11.2022	ГОСТ 22387.4-77 (GOST 22387.4-77)	0.001	0.0001

* The actual values of gas temperature and pressure at the moment of measurement of DPT are indicated.

Table 1.4.1.2. Characteristics of natural gas - MGP «BGR-TBA» (according to the Passport for gas dated November 14, 2022 GDS «Shymkent-1»)

No.	Name of indicators	ND of measurement methods	Requirements ST RK 1666-2007	Actual value
1	Component composition (% mol)			
	Methane CH ₄	ГОСТ 31371.7-2008 (GOST 31371.7-2008)	Not normalized	95.30
	Ethane C ₂ H ₆			2.36
	Propane C ₃ H ₈			0.472
	Iso-butane i-C ₄ H ₁₀			0.037
	Nor-butane n-C ₄ H ₁₀			0.041
	Neo-pentane neo-C ₅ H ₁₂			0.000
	Iso-pentane i-C ₅ H ₁₂			0.011
	Nor-pentane n-C ₅ H ₁₂			0.009
	n-Hexanes C ₆ H ₁₄			0.019
	Heptanes C ₇ H ₁₆			0.003
	Octanes C ₈ H ₁₈			-
	Nitrogen N ₂			1.10
	Carbon dioxide CO ₂ no more		2.5	0.63
	Volume fraction oxygen O ₂ %, no more		0.02/0.1	0.005
	Mass concentration of hydrogen sulfide, g/m ³ no more (Iodometry)	ГОСТ 22387.2-97 (GOST 22387.2-97)	0.007/0.006	-
	Mass concentration of mercaptan sulfur, g/m ³ no more (Iodometry)		0.016/0.014	-
2	Density, kg/m ³ at 20 °C and 760 mm Hg (estimated)	ГОСТ 31369/17310 (GOST 31369/17310)	Not normalized	0.7030
3	Density, kg/m ³ at 20 °C and 760 mm Hg (pycnometric) from	ГОСТ 17310 (GOST 17310)		-
4	Moisture dew point, °C at P _{gas} = 7.0 MPa	CT PK 53763-2011 (ST RK 53763-2011)	from 01.05. to 30.09. (-10)/(0) from 01.10. to 30.04. (-10)/(-5)	-
5	Dew point for hydrocarbons, P _t = 2.7 MPa	CT PK 53762-2011 (ST RK 53762-2011)	from 01.05. to 30.09. (-2)/(0) from 01.10. to 30.04. (-2)/(-2)	-
6	The volumetric heat of combustion is lower, MJ / m ³ , at 20 °C, 101.325 kPa, not less	ГОСТ 31369-2008 (GOST 31369-2008)	31.80	33.82
7	Range of values of the Wobbe number MJ/m ³	ГОСТ 31369-2008 (GOST 31369-2008)	Not standardized	49.07

8	Mass concentration of mechanical impurities, g/m ³ , not more than	ГОСТ 22387.4-77 (GOST 22387.4-77)	0.001	-
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* The actual values of gas temperature and pressure at the moment of measurement of DPT are indicated.

If necessary, it is required to foresee and implement measures aimed at eliminating corrosion, depending on the component composition of natural gas.

1.4.2. Water supply and sewerage

1.4.2.1. Water supply

The source of water supply for the projected power plant is determined from the Badam reservoir (main), the water intake point is 3.9 km from the site and the groundwater of the Badam-Sairam groundwater deposit (reserve), located in the middle reaches of the river. Badam, 1.5 km southeast of Shymkent.

The Badam reservoir is located in the Tolebi district of the Turkestan region, 55 km southeast of the city of Shymkent and 17 km southeast of the village of Pervomayskoye, in the upper reaches of the river Badam.

Source of water supply: r. Badam

Reservoir type: Bulk

Purpose: Irrigation

Year of commissioning: 1974

NHL (m): 649.4

Total volume (million m³): 61.5

Usable volume (million m³): 59.0

Dead volume (million m³): 2.5

Mirror area (km²): 4.75

Length (km): 4.0

Width (km): 1.5

Table 1.4.2.1.1. Characteristics of technical water from the Badam reservoir Sampling point No. 1 - at the water outlet from the Badam reservoir (according to Test Report No. 140 dated November 29, 2022)

Product (object)name	Name of the determined indicators	ND for test methods	MPC	Test results
Sampling point No. 1-At the drainage from the Badam Reservoir	Hydrogen indicator (pH)	ГОСТ 26449.1-85 п. 4 (GOST 26449.1-85 p.4)	-	7,0
	Total iron, mg/dm ³	ГОСТ 26449.1-85 п. 16 (GOST 26449.1-85 p. 16)	-	0,01
	Total hardness, mg/dm ³	ГОСТ 26449.1-85 п. 10 (GOST 26449.1-85 p. 10)	-	5,8
	Permanganate oxidizability	ГОСТ 26449.1-85 п. 5 (GOST 26449.1-85 p. 5)	-	1,6
	Ammonium nitrogen	СТ РК ИСО 5664-2006 (ST RK ISO 5664-2006)	-	0
	Nitrites	СТ РК 1963-2010 (ST RK 1963-2010)	-	0,012
	Sulfates, mg /dm ³	СТ РК 1015-2000 (ST RK 1015-2000)	-	104,16
	Phosphates	ГОСТ 26449.1-85 п. 14.2 (GOST 26449.1-85 p. 14.2)	-	0
	Fluorites	СТ РК ИСО 10359-1-2008 (ST RK ISO 10359-1-2008)	-	0,3
	Chlorides, mg/dm ³	СТ РК ИСО 9297-2008 (ST RK ISO 9297-2008)	-	0
	Nitrates, mg/dm ³	СТ РК 2730-2015 (ST RK 2730-2015)	-	2,0
	Sulfides	СТ РК 2275-2013	-	0

	(ST RK 2275-2013)		
Dry residue	ГОСТ 26449.1-85 п. 3 (GOST 26449.1-85 p. 3)	-	3,4
Hydrogen sulfide	ГОСТ 26449.1-85 п. 3 (GOST 26449.1-85 p. 3)	-	0
Calcium, mg/dm ³	ГОСТ 26449.1-85 (GOST 26449.1-85)	-	56,0
Magnesium, mg/dm ³	ГОСТ 26449.1-85 (GOST 26449.1-85)	-	36,6
Total alkalinity	ГОСТ 26449.1-85 п. 6 (GOST 26449.1-85 p. 6)	-	0
Hydrocarbon, mg / dm ³	ГОСТ 26449.1-85 п. 7 (GOST 26449.1-85 p. 7)	-	292,8
Sodium, mg/dm ³	ГОСТ 26449.1-85 п. 17 (GOST 26449.1-85 p. 17)	-	32,2
Silicon and its compounds	ГОСТ 26449.1-85 п. 22 (GOST 26449.1-85 p. 22)	-	0
Carbonate, mg/dm ³	ГОСТ 26449.1-85 п. 7 (GOST 26449.1-85 p. 7)	-	0
Chromium	СТ РК 1511-2006 (ST RK 1511-2006)	-	0
Copper	СТ РК 1998-2010 (ST RK 1998-2010)	-	0
Cadmium	СТ РК 1998-2010 (ST RK 1998-2010)	-	0
Lead	СТ РК 1998-2010 (ST RK 1998-2010)	-	0
Aluminum	СТ РК ИСО 1956-2010 (ST RK ISO 1956-2010)	-	0
Boron	СТ РК ГОСТ Р 51210-2003 (ST RK GOST R 51210-2003)	-	0
Manganese	СТ РК Р 52180-2010 (ST RK R 52180-2010)	-	0,2
Petroleum	ГОСТ 26449.1-85 п. 26 (GOST 26449.1-85 p. 26)	-	0,01

Table 1.4.2.1.2.2. Characteristics of industrial water from the Badam reservoir Sampling point No. 2 - on the western side of the dam of the Badam Reservoir (according to Test Report No. 141 of 29.11.2022)

Product (object)name	Name of the determined indicators	ND for test methods	MPC	Test results
Sampling point No.2 - on the western side of the dam Водохранилище of the Badam Reservoir	Hydrogen index (pH)	ГОСТ 26449.1-85 п. 4 (GOST 26449.1-85 p. 4)	-	7,0
	Total iron, mg/dm ³	ГОСТ 26449.1-85 п. 16 (GOST 26449.1-85 p. 16)	-	0
	Total hardness, mg/dm ³	ГОСТ 26449.1-85 п. 10 (GOST 26449.1-85 p. 10)	-	5,7
	Permanganate oxidizability	ГОСТ 26449.1-85 п. 5 (GOST 26449.1-85 p. 5)	-	1,6
	Ammonium nitrogen	СТ РК ИСО 5664-2006 (ST RK ISO 5664-2006)	-	0
	Nitrites	СТ РК 1963-2010 (ST RK 1963-2010)	-	0,01
	Sulfates, mg /dm ³	СТ РК 1015-2000 (ST RK 1015-2000)	-	104,1
	Phosphates	ГОСТ 26449.1-85 п. 14.2 (GOST 26449.1-85 p. 14.2)	-	0
	Fluorites	СТ РК ИСО 10359-1-2008 (ST RK ISO 10359-1-2008)	-	0,28
	Chlorides, mg/dm ³	СТ РК ИСО 9297-2008 (ST RK ISO 9297-2008)	-	0
	Nitrates, mg/dm ³	СТ РК 2730-2015 (ST RK 2730-2015)	-	2,0

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Sulfides	CT PK 2275-2013 (ST RK 2275-2013)	-	0
Dry residue	ГОСТ 26449.1-85 п. 3 (GOST 26449.1-85 p. 3)	-	3,6
Hydrogen sulfide	ГОСТ 26449.1-85 п. 3 (GOST 26449.1-85 p. 3)	-	0
Calcium, mg /dm ³	ГОСТ 26449.1-85 (GOST 26449.1-85)	-	56,0
Magnesium, mg/dm ³	ГОСТ 26449.1-85 (GOST 26449.1-85)	-	34,1
Total alkalinity	ГОСТ 26449.1-85 п. 6 (GOST 26449.1-85 p. 6)	-	0
Hydrocarbonate, mg/dm ³	ГОСТ 26449.1-85 п. 7 (GOST 26449.1-85 p. 7)	-	291,1
Sodium, mg/dm ³	ГОСТ 26449.1-85 п. 17 (GOST 26449.1-85 p. 17)	-	31,6
Silicon and its compounds	ГОСТ 26449.1-85 п. 22 (GOST 26449.1-85 p. 22)	-	0
Carbonate, mg / dm ³	ГОСТ 26449.1-85 п. 7 (GOST 26449.1-85 p. 7)	-	0
Chromium	CT PK 1511-2006 (ST RK 1511-2006)	-	0
Copper	CT PK 1998-2010 (ST RK 1998-2010)	-	0
Cadmium	CT PK 1998-2010 (ST RK 1998-2010)	-	0
Lead	CT PK 1998-2010 (ST RK 1998-2010)	-	0
Aluminum	CT PK ИСО 1956-2010 (ST RK ISO 1956-2010)	-	0
Boron	CT PK ГОСТ Р 51210-2003 (ST RK GOST R 51210-2003)	-	0
Manganese	CT PK Р 52180-2010 (ST RK R 52180-2010)	-	0,18
Petroleum	ГОСТ 26449.1-85 п. 26 GOST 26449.1-85 p. 26	-	0,01

The selection of water samples from the Badam reservoir is given. For a rational water treatment scheme, it is necessary to conduct additional analyzes during the annual period.

Due to the high demand for water resources in this region and the further increase in water consumption for irrigation of agricultural land, there is a shortage of free volumes in the reservoir.

In this connection, the Badam-Sairam groundwater deposit was determined as a reserve source of water for the Facility. Located in the middle reaches of the r. Badam, 1.5 km southeast of Shymkent.

Groundwater is fresh, with a mineralization of 0.76-0.9 g/l, sulfate-bicarbonate, sodium-calcium.

Water intake facilities are envisaged with a capacity of 16.80 thousand m³/day, 4900.0 thousand m³/year from a network of artesian wells, water storage tanks and a pumping station.

At the design stage, it is necessary to re-evaluate groundwater reserves.

The following water supply systems are considered at the power plant site:

- technical water supply;
- household and drinking;
- production and fire fighting.

At the projected site of the power plant, water is used for household and drinking needs, fire fighting, watering green spaces and roads.

The maximum consumption of initial water of the designed power plant will be 700.0 m³/hour, 4900.0 thousand m³/year, including:

- 525.0 m³/h - at the boiler feeding WTP;
- 120.0 m³/hour - for drinking water treatment;
- 48.5 m³/hour - to replenish the fire-fighting water supply in the tanks in case of fire;
- 6.5 m³/hour - for watering green spaces and roads.

Technical water supply

The total consumption of cooling water in the cooling system of steam turbine condensers and auxiliary equipment of the Facility, according to the considered options, is 56,000.0 m³/h. To cool the circulating water, you will need:

- evaporative cooling towers with an irrigation area of 3×2100 m², while the frequency of irrigation in the condensing mode will be 8.9 m³/m²·hour;
 - circulation pumping station with a capacity of 56,000.0 m³/h with a switching chamber and a suction chamber;
 - circulation conduits with a diameter of 1600÷2400 mm made of steel pipes with stiffening rings.
- To recharge the circulating system, water will be required in the amount of 1100.0 m³/hour, 7700.0 thousand m³/year.

An evaporation field of about 150 ha will be required to utilize the circulatory system blowdown water.

Taking into account the shortage of water resources in the region under consideration, a dry condenser was considered for cooling the installed equipment. When installing a dry condenser, water is not required to feed the cooling system.

Household and drinking water supply

The estimated consumption of drinking water will be 75.0 m³/hour, 300.0 m³/day.

On the site of the power plant, circular networks of household and drinking water supply from polyethylene pipes are provided with the installation of shut-off valves in wells.

For drinking water supply, the use of existing water supply networks in Shymkent is considered.

Fire water supply

Estimated fire extinguishing consumption is the total consumption for external and internal fire extinguishing for the main building 162 l/s, including 100 l/s for external fire extinguishing, 62 l/s for internal fire extinguishing with a design head of 70.0 wcm (water column meters).

The water consumption for external fire extinguishing of the projected main building is determined depending on the volume of 1030.7 thousand m³, the degree of fire resistance - II, fire hazard category B1 in accordance with Table 1 of Appendix 5 to the Technical regulation «General requirements for fire safety», for internal fire extinguishing - according to table 2 of CII PK 4.01-101-2012 «Внутренний водопровод и канализация зданий и сооружений» (SP RK 4.01-101-2012 «Internal water supply and sewerage of buildings and structures»).

To provide external and internal fire extinguishing at the site of the power plant, a fire extinguishing pumping station and fire-fighting water tanks are provided.

The pumping station provides for the installation of two groups of pumps:

- two fire pumps (working, standby) with a capacity of 580-600 m³/h, head 80 wcm;
- two pumps for supplying water to the drinking water treatment plant and boiler feeding water treatment plant, with a capacity of 650.0 m³/h, a head of 40.0 wcm.

The capacity of the tanks is taken from the condition of the duration of the fire extinguishing of the main building for 3 hours, the water supply for the boiler make-up water treatment plant and the drinking water treatment plant.

The estimated volume of tanks is 2400 m³, 2 tanks of 1200 m³ are accepted.

The maximum period for the restoration of the fire volume of water, in accordance with the requirement of СНиП РК 4.01-02-2001 (SNiP RK 4.01-02-2001), should be no more than 36 hours. The estimated water consumption during the restoration of the fire-fighting volume will be 48.5 m³/hour.

Fire protection networks at the site are made of ring, steel pipes in accordance with the «Instructions for the design of fire protection of energy enterprises» RD 34.4.49.101-87, with a diameter of 150÷300 mm. Providing external fire extinguishing from the fire water supply network is provided from fire hydrants installed in wells every 50-70 m. Hydrants, according to СНиП РК 4.01-02-2001 (SNiP RK 4.01-02-2001), are located along the highway at a distance of no more than 2.5 m from the edge of the carriageway. In accordance with ГОСТ 12.4.009-83 (GOST 12.4.009-83), the installation of fluorescent indicators is provided for at the locations of fire hydrants.

1.4.2.2. Sewerage

The following water disposal systems are provided at the power plant site:

- household sewerage;
- sewerage of industrial effluents;
- sewerage of emergency drain of transformer oil.

Due to the small amount of precipitation and the significant evaporation capacity of the area, as well as taking into account the soils that make up the site - sands, an open rainwater collection system with subsequent evaporation is provided at the power plant site.

Household sewerage

Effluent from sanitary appliances of the designed buildings is diverted to the domestic sewerage network. The volume of domestic wastewater disposal is equal to the volume of water consumption and is 75.0 m³/hour, 300.0 m³/day.

Household wastewater flows by gravity into a sewage pumping station with a capacity of 75.0 m³/hour, factory-made, with subsequent supply to treatment facilities. A station for deep biological treatment of domestic wastewater is provided at the site of the power plant. Wastewater treatment plant of prefabricated modular design, with a capacity of 300 m³ per day. Treatment facilities operate in automatic mode. The treatment facilities include a block of biological treatment, post-treatment of wastewater on the filter, disinfection and sludge treatment.

Water after treatment at the plant must meet the quality requirements for discharge into fishery water bodies.

Treated domestic wastewater after treatment facilities, depending on the quality of treated water, can be used for irrigation of green spaces and roads. Issues related to the use of purified water should be resolved with the participation of sanitary protection authorities. If the water quality does not meet sanitary requirements, the treated effluents are sent to the evaporation site.

Gravity networks of household sewerage are provided from polyethylene pipes.

Sewerage of industrial effluents

Industrial effluents at the power plant site include saline effluents from boiler feed water treatment plant with a flow rate of 183.0 m³/h, 1281.0 thousand m³/year, drinking water treatment water treatment plant with a flow rate of 45.0 m³/h, 315.0 thousand m³/year, m³/year, blowdown of boilers, washing of equipment (periodically), oily effluents after treatment, with a flow rate of 40.0 m³/hour, 280.0 thousand m³/year.

The total amount of industrial effluents is 268.0 m³/hour, 1876.0 thousand m³/year.

Industrial effluents are sent to the industrial effluent collection tank, from where they are periodically pumped out to the evaporation site by pumps installed in the industrial effluent pumping station. Factory-made pumping station for industrial wastewater, with a capacity of 270.0 m³/hour.

Gravity networks at the site, the pressure pipeline pumping wastewater to the evaporation site is made of polyethylene pipes.

Evaporation pad

For the disposal of industrial effluents, an evaporation site is provided, located 1.5 km from the power plant.

The total amount of effluents entering the evaporation site is 1876.0 thousand m³/year.

The surface area of the evaporation platform was calculated taking into account the total amount of wastewater of 1876.0 thousand m³/year, the evaporative capacity of the area - 1500 mm/year, the annual rainfall of 300 mm/year and the filtration of the impervious screen made of polymeric materials.

The estimated surface area of the evaporation platform will be 150 ha.

The evaporation site is an earthen container, in a half-ditch-half-fill, with enclosing dams made of local soil.

The bottom and slopes of the evaporation platform, taking into account engineering and geological conditions and in accordance with the requirements of СНиП РК 1.04-14-2003, «Полигоны по обезвреживанию и захоронению токсичных промышленных отходов» (SNiP RK 1.04-14-2003,

«Landfills for the neutralization and disposal of toxic industrial waste»), are covered with an impervious screen made of a geomembrane of the following design:

- protective layer 20 cm of crushed stone;
- protective layer 80 cm of sand;
- geomembrane 1.0 mm thick;
- planned and etched base.

The protective layer of the crushed stone impervious screen is adopted to preserve the protective layer of the sand impervious screen from wind erosion.

To control the effect of the evaporation site on the composition and level of groundwater, observation (regime) wells are provided along the perimeter of the evaporation field dams.

Sewerage of emergency drain of transformer oil

Transformer oil emergency drain sewerage is designed for emergency draining of oil, fire extinguishing water and atmospheric water from transformer pits under transformers to special oil tanks.

Emergency oil drains from outdoor installations of transformers are diverted to underground tanks located in 220 kV and 500 kV switchgear (SG).

The sewerage network of emergency oil drains is made of cast-iron pipes.

In order to prevent the spread of fire through the sewerage networks for emergency draining of transformer oil, they are provided with wells with a hydraulic seal.

1.4.3. Demand for heat and electricity

The need of the projected Facility for fuel is determined by the capacity of the installed mechanisms for the production and distribution of products to consumers, as well as the capacity of current collectors that provide appropriate working conditions and safety in production.

The need of the projected Facility for heat energy for heating, ventilation and hot water supply is determined according to the temperature schedule based on calculations of the heat demand of each building and structure, depending on its purpose, operating conditions and the heating and ventilation systems used, taking into account the needs of the designed buildings and structures.

The demand for thermal and electric energy of consumers and for own needs of CCGT is shown in Table 1.4.3.1.

Table 1.4.3.1.

Name	Value
Electricity generation, million kWh	Determined by the adopted version of the CCGT
Electricity consumption for own needs, million kWh	Until 6%
Annual supply of electricity from tires, million kWh	Determined by the adopted version of the CCGT

1.4.4. The need for reagents

The operating costs of the main reagents of the WTP heat recovery steam generators, auxiliary steam boilers are determined at the design stages.

Chemicals and consumables, including oils for GT, if necessary, flushing fluids and antifreeze, a set of filters for various purposes intended for the operation of the Facility, for commissioning, maintenance and commissioning, testing, for the entire warranty period of operation for all equipment Facility.

Provide warehouses for the storage of chemicals and consumables, in accordance with the rules for storage and warehousing, as well as an express laboratory.

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1.4.5. Provision of labor resources

The need for labor resources for the construction period is determined at subsequent design stages in the section «Construction organization project» taking into account the duration of construction and the timing of commissioning.

1.5. General plan and transport

1.5.1. Location of the site

The Turkestan region is one of the largest regions of the republic.

The regional center is the city of Turkestan.

The preliminary selection of the location site was based on the availability of the following criteria: water resources, gas supply, connection to the electrical network, accessible transport and other infrastructure.

The territory of the site is located in the Tolebi district of the Turkestan region. Administrative center - city Lenger . The Facility is located on the western shore of the Badam reservoir, 24 km east of the center of Shymkent along the Lenger highway, 6 km south of the highway near the Mayatas quarter of Shymkent, 20 km west of Lenger. From the Mayatas quarter, 2 km asphalt road to the recreation area, then to the object along the reservoir - a dirt road (about 3 km).

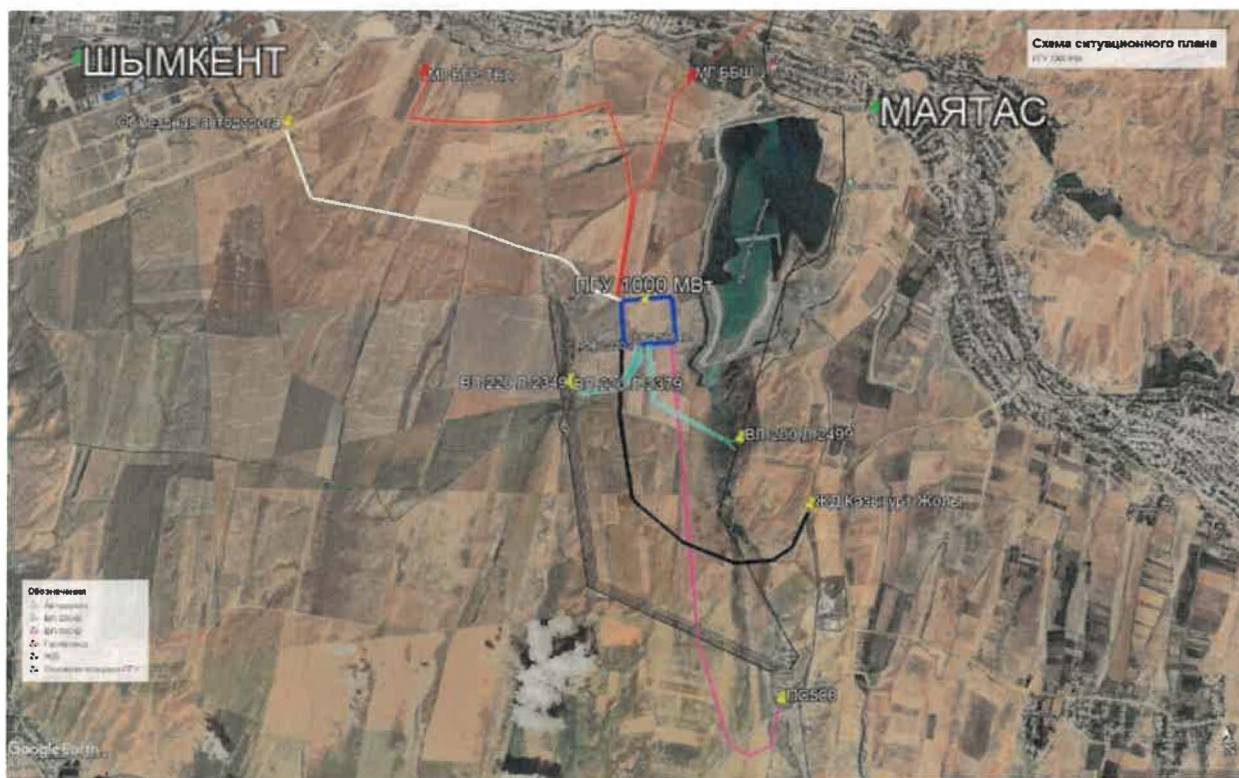
Site distance from:

- roads - 6 km;

- Kazykurt railway station (in a straight line) - 12 km.

The site is a rugged terrain, complicated by agricultural land (arable land).

Figure 1.5.1.1. Situation plan diagram



1.5.2. General plan

The site for the location of the power plant within the boundaries of the allotment is free from development and is 100 hectares.

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Figure 1.5.2.1. General plan

When laying out the general plan, fire breaks, communication corridors, technological communications, driveways and entrances to the main buildings and structures were taken into account.

The Facility includes the following buildings and structures:

- CCGT main building with bypass chimneys;
- Auxiliary electric boiler for own needs;
- Installation of transformers;
- SG 500 kV;
- SG 220 kV;
- Relay board with MCP;
- Dry coolers;
- Gas treatment plant;
- Air compressor room;
- Initial water storage tanks with a pumping station;
- Water treatment for make-up of HRSG and auxiliary systems
- Diesel fuel warehouse with pumping station;
- Administrative building;
- Checkpoint;
- Service building;
- Canteen;
- Joint auxiliary building;
- Fire-fighting water tanks with a pumping station;
- Cleaning households. household drains;
- Fire station.

Vertical layout

Elevations on the site range from 674 m to 712 m. The relief of the site is rugged terrain. The territory will be leveled on average up to the level of 695 m with observance of slopes ensuring water

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runoff from the site. The volume of earthworks is preliminary 2 340 000 m³ (Movement of earth from excavation to embankment within the allotted area).

1.5.3. Transport

Highways

The access road adjoins the Southern bypass road of Shymkent (from the Lenger highway to the regional road KH-15). Estimated KP33.00 3.3 km of the route. The length of the road 6 km. The main entrance to the site is expected from the east side.

The road to the fire station runs along the fence also on the north side. Access road categories not lower than IIIB. To provide technological connections and fire-prevention entrances on the site, a network of intra-site motor roads of category IIIB and IVB is provided.

Railway

The railway is adjacent to the private tracks of the branch owner «Kazygurt Zholy» LLP in the area of the «Karyernaya» station. The preliminary length of the railway is 6 km. The main entrance to the site is expected from the western side. To provide technological connections at the site, a network of on-site railways is provided.

Section 2. SELECTION OF THE MAIN EQUIPMENT

2.1. Power plant functional specification

These requirements are the basic criteria of the power plant, and not a detailed description of all works, services, equipment, systems, products and materials that the Contractor must perform as part of the construction of the Facility. The Contractor's arguments that some services, works, equipment, systems are not listed in the tender documentation and therefore they are not included in the Contractor's delivery set will not be accepted by the Customer during the implementation of the Project due to the fact that at the stage of preparing the tender offer the Contractor should everything necessary for a safe, efficient, environmentally friendly, cost-effective, convenient operation by the Customer has been taken into account. In the event that the Contractor is not clear about any requirements for the power plant, the Contractor is obliged to fulfill a formal request as part of the tender procedure to obtain an appropriate clarification.

This document includes the Customer's requirements, which must be carefully analyzed and considered by Potential suppliers (Contractor) when preparing a tender offer. The technical solutions proposed by the Contractor must ensure the exploration, design, construction and commissioning of a power plant that provides electric energy in full in an automated mode of operation and in the entire range of regulation.

According to this specification, the power plant must be complete, new and modern; work on the basis of well-established technologies; safe and economical; with high availability and low rates of forced decommissioning under all site environmental conditions. The equipment must be selected in such a way that the power plant has high reliability and a slight decrease in operating efficiency over its entire service life. The power plant must be constructed in such a way that it can operate over the entire range of ambient temperatures, within the limits specified in this specification, and can also operate continuously with a minimum of operator intervention and with a minimum of maintenance. The layout of the general plan of the power plant should be carried out so that the power plant fits on the allocated site for construction.

The Contractor shall comply with all the requirements listed in this document, unless such implementation, for some reason, jeopardizes safety, reliability, compliance with the mandatory rules, norms and standards of the Republic of Kazakhstan, or unless such implementation of the requirements of this technical specification worsens economic performance of the power plant due to deviation from the standard and time-tested practices of the Contractor or the Equipment manufacturer. In any case, the Contractor shall work in accordance with the objectives of this technical specification. All deviations from this specification shall be clearly indicated by the Contractor in his tender offer. Unless the deviations are explicitly listed and agreed with the Customer, the Contractor shall be deemed to have

undertaken to comply with the requirements of this technical specification. Agreed deviations from these requirements are listed and this list is attached to the Contract subsequently concluded as a separate Appendix, canceling the agreed positions of this technical requirement.

The power plant must be designed, manufactured, mounted in such a way as to ensure reliable operation of the equipment in all respects in the specified site conditions in the basic mode of operation of the power plant with various loads in the control range under the specified environmental conditions, while demonstrating a high availability and high reliability .

The power plant and its implementation must satisfy all requirements in terms of ensuring the functional, operational, as well as environmental requirements and restrictions specified in these technical requirements.

The power plant must be designed in such a way as to meet the requirements of internationally recognized standards and comply with the current legislation of the Republic of Kazakhstan, including norms, rules and standards.

The technical documentation of the power plant and its implementation must meet all the requirements of climatic conditions prevailing at the construction site, take into account the impact on the equipment of climatic conditions (for example, rain, temperature, ice wall thickness, relative humidity, floods, storms, sudden snowfalls, earthquakes), which should be taken into account in the technical documentation, namely in the choice of measures to protect against adverse weather conditions, the choice of materials, coatings, drainage systems, as well as in the organization of repair and maintenance of power plant equipment in the specified conditions.

2.2. Description of the power plant

The main strategy for the construction of a power plant based on a CCGT unit with a capacity of 1000 MW in the Turkestan region is to ensure the balance reliability of the energy system of the Republic of Kazakhstan, cover irregular fluctuations in the electrical load and power flow imbalances resulting from uneven power output from renewable energy sources, increase the reliability and safety of the power plant through the use of modern technologies for the production of electrical energy. All technical solutions must take into account the seismic hazard of the region.

It is planned to build a new power plant based on modern types of CCGT, which will meet the requirements of high maneuverability with a large control range and maintaining the efficiency of equipment operation in the control range while reducing power to a technical minimum. Implementation is provided on the territory of the Turkestan region.

In order to optimize the construction time of the power plant, it is planned to carry out parallel work on the main CCGT site and external infrastructure facilities (power generation of the electric power system, water intake system, access road, railway line and gas supply).

This Technical specification provides for works related to the construction of the main site.

2.3. Operating mode of the power plant

The power plant will be used as a source of electrical energy. The basic mode of operation of the power plant must ensure the issuance of electrical energy in accordance with the balance of electrical energy agreed by the Customer.

The Contractor must also indicate in his proposal the maximum possible output of electricity to the network in each of the modes of operation of the power plant. The characteristics of the auxiliary equipment of the power units should allow the operation of the GT, ST based on the maximum characteristics of the GT, ST.

The operating mode required for the power plant is at least 350 starts/year. All requirements listed in ST RK ISO 14661-2011 apply to equipment supplied by the Contractor. The interval of inspections and repairs of high-temperature parts of the gas path is at least 8000 hours. The resource before the first overhaul and between major repairs is at least 24000 hours. The interval between maintenance is at least 1500 hours.

The power plant must ensure stable operation in the event of short circuits on the 220-500 kV overhead lines extending from the buses of the CCGT Main site. The duration of such work is set by the settings for the operation of relay protection and automation (RPA) without disconnecting the power plant from the network.



It is necessary to provide the ability to control reactive power ($Q=\text{const}$) at the point of common connection. The reactive power control system must be designed in such a way that the reactive power setpoint is controlled at the point of common connection. The set point must be adjusted in steps of 1 MVar within the technical limits, according to the P/Q curve valid at the point of common connection. The error limit of reactive power measurement at the point of common connection must be higher than ± 1 MVar.

The second reactive power control option ($U=\text{const}$) must provide voltage regulation at the point of common connection. The control system must be arranged in such a way as to use the entire reactive power reserve within the technical limits according to the P/Q curve in order to achieve the voltage setting. The voltage regulation dead zone is ± 1 kV at the point of common connection. The setting must be adjusted in steps of 1 kV. Voltage regulation systems must be designed in such a way that simultaneous voltage regulation does not cause instability in operation (voltage fluctuations, control conflict).

2.4. Functional requirements for the power plant:

- high energy efficiency;
- high reliability and operational readiness for work;
- high degree of automation;
- the level of emissions into the atmosphere in accordance with the norms of the Republic of Kazakhstan;
- low cost of maintenance;
- suitability for repair at the installation site in accordance with the regulations established by the equipment manufacturer;
- the total electric power of the power plant on the high voltage side (excluding the power plant's own needs) must be at least 1000 MW, taking into account climatic conditions, geographical location, ground level above sea level, seismic activity level and other characteristics of the construction site;
- electric installed capacity of each power unit - to be determined by the project.
- the rate of output of the available power (taking into account the work on the bypass chimney and taking into account seasonal, regime and repair restrictions) of the power plant to the network from the moment the «Start» command is received is at least 10 MW/min, regardless of the thermal state of the unit (s);
- power plant control ranges must correspond to the lower limit within the technical and technological minimums;
- the scheme for connecting the units to the electrical network, as well as the features of the operation of the auxiliary equipment of the power plant should not lead to a limitation in the output of power from more than one power unit during maintenance;
- decrease in the rated power of the units during their operation during the overhaul period should not exceed 4%, and the decrease in electrical EF should not exceed 2% (relative). Ensuring the possibility of carrying out all types of inspections (repairs) at the place of operation of the equipment. The increase in specific fuel consumption should not exceed 2% during the overhaul period. The restoration of the specified indicators after the overhaul should be ensured.
- capacity limitation depending on the weather conditions typical for the construction site no more than 15% of the installed capacity. Data on weather conditions in accordance with CII PK 2.04-01-2017* «СТРОИТЕЛЬНАЯ КЛИМАТОЛОГИЯ» (SP RK 2.04-01-2017* «CONSTRUCTION CLIMATOLOGY»).
- long-term operation of the power plant at a frequency in the electrical network from 49 Hz to 51 Hz should be ensured. When the frequency drops to 47 Hz, the operating time of the power plant is at least 120 minutes. When the frequency increases to 51.5 Hz, the operating time of the power plant is at least 90 minutes.
- an automatic voltage regulation system should be provided to maintain the specified voltage level at the point of connection of the power plant to the power system.
- it should be possible to start the units from emergency and system automation devices (including the automatic frequency and power control system (AFPCS)).



- the automated process control system (APCS) should ensure the operation of the power plant (main and auxiliary equipment) in automatic mode, without the participation of permanent maintenance personnel during start-ups, shutdowns and work under load.

The Contractor integrates and completes power output equipment depending on the type and power of a single unit and the existing site conditions.

2.5. Performance indicators

Calculate guaranteed performance and operation indicators in accordance with BAT, including net electrical EF, net total fuel use, and analysis of the effectiveness of the selected scheme by the Contractor.

Performance requirements

Type of combustion chamber	Net Electrical EF (%) New Chamber		Total fuel use (%)
	Existing Requirements	Possible tightening of requirements	
Open cycle gas turbine			
Open cycle gas turbine, ≥ 50 MW	>33	36–41.5	missing
Combined cycle gas turbine			
CCGT 50–600 MW	>40	53–58.5	missing

2.6. Technical requirements for regulating electric power

Newly commissioned generating units with a maneuverable generation mode: must comply with the Requirements of the «Rules for organizing and conducting auctions for the construction of newly commissioned generating plants with a maneuverable generation mode» (Order of the Ministry of Energy of the Republic of Kazakhstan No. 161 dated April 30, 2021), the requirements of the Terms of reference.

The available volume of regulating electric power, which will be involved in the secondary regulation of frequency and power flows, should provide:

- in the case of parallel operation of the UES of RK with the UES of Russia - maintaining the total power flows (balance of power flows) for external communications with frequency correction specified by the national dispatch center of the system operator of Kazakhstan;

- adjusting electric power is involved in automatic secondary regulation of frequency and power;

- the volume of regulating electric power must correspond to the range of regulating electric power of CCGT, $\pm 50\%$ of P^* . MW;

- the rate of change in the volume of regulating electric power of generating installations for loading or unloading (corresponding to the regulation range) - at least 10 MW/min.;

- the involvement of the regulating electric power in the automatic regulation of frequency and power is provided by the power plant without time and duration restrictions for the entire specified value of the regulating electric power;

- it is possible to start and put into operation generating installations in the absence of voltage in the external electrical network;

- generating installations allow the required number of starts per year (at least 350 starts/year);

- the activation of the regulating electric power for loading and unloading is carried out both promptly, by the commands of the dispatching personnel, and automatically from the automatic frequency and power control system of the system operator.

The requirements for the control range for the newly introduced control electric power of condensing combined-cycle plants that are not involved in heat supply are given in Table 2.6.1.

Table 2.6.1.

No.	Type of power plant of the energy producing organization	Adjustment electric power range, MW	The rate of change in the volume of regulating electric power, MW/minute	Note
1.	Gas turbine power plants	+/- 50% of P_{rab} .	at least 10	***
2.	Combined-cycle power plants	+/- 50% of P^* .	at least 10	***
3.	Combined-cycle power plants involved in heat supply	+/- 20% of R^* .	at least 10	**
4.	Gas piston power plants	+/- 50% of P_{rab} .	at least 10	***
5.	Hydroelectric power plants, including counter-regulating	+/- 20% of R^* .	at least 10	**

* - P - variable base, in MW, the value of which is determined for each hour of the day according to the following formula:

$$P = P_{rab} - \max(P_{tech.min.}; P_{technol.min.}), \text{ where:}$$

P_{rab} - the value of the operating electric power of the generation of the electric station, MW;

$P_{tech.min.}$ - the value of the technical minimum of the power plant of the energy-producing organization, technological and technical minimums, in MW;

$P_{technol.min.}$ - the value of the technological minimum, technological and technical minima, MW;

$\max(P_{tech.min.}; P_{technol.min.})$ is the largest of the values $P_{tech.min.}$ и $P_{technol.min.}$, MW.

The values P_{Σ} , P_{rab} , $P_{technol.min.}$, $P_{tech.min.}$ are determined according to the Power Market Rules.

The use of regulating electric power is carried out for loading within the limits of working electric power, for unloading to the highest of the values of the $P_{tech.min.}$ и $P_{technol.min}$ power plant of the energy-producing organization.

The sharply continental climate of the Republic of Kazakhstan and, accordingly, low temperatures in winter and high temperatures in summer, require taking into account the range of ambient temperatures at which the operation of the equipment is planned, temperature minimum/maximum °C, atmospheric pressure at the height of a particular site, on which the power of gas turbines depends.

With an increase in temperature in summer mode, the power of the GT is significantly reduced, so when switching from +15C to +30C, the power drop can be ~ 11-12% of the nominal one. In this case, the GT will not be able to produce the required power declared during certification.

An increase in the level of the site and, as a result, a decrease in atmospheric pressure leads to a decrease in the power of the GT.

An increase in temperature and an increase in altitude above sea level have a cumulative effect on the power of the GT.

The electric power control range of the power plant (+50% $P = P_{rab}$ - $P_{min.}$) is determined in accordance with the requirements for the regulating electric power of the electric power plant on gas with a maneuverable generation mode (Appendix 1. Order of the MERK No. 161 dated April 30, 2021).

The values of operating power, technical and technological minimum are determined according to "Rules of organization and functioning of the electric power market".

The range of regulating electric power is determined for loading within the operating electric power, for unloading - up to the highest of the technical or technological minimum values.

The value of the certified working electric power of the power plant (P_{rab}) is determined by the technical characteristics of the CCGT equipment during the annual certification of the power of the power plant, taking into account the conditions of the construction site (barometric pressure, current outdoor temperature).

In order to ensure the specified control range, the average hourly power of a CCGT with a maneuverable generation mode is allowed no more than 0.5 of the control power.

The technical minimum also includes an environmental minimum that meets the requirements for NOx emissions at the lowest possible power.

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The use of the CCGT control range, taking into account the technical minimum, allows the use of industrial gas turbines, instead of aircraft derivatives, the use of a standard maintenance schedule and repair periods without increasing the equivalent operating hours with frequent starts. The assessment of the cost of the entire life cycle of a maneuverable power plant includes both capital costs and service costs, taking into account the requirements for the number of starts.

For the industrial gas turbines under consideration, the requirement of at least 350 starts/year can only be achieved by significantly reducing the overhaul period and the service life of the plants, which is inappropriate. The optimal solution for the tasks of the widest regulation is the aircraft-derived GT, operating in a simple cycle, if necessary.

Industrial grade GT have different wear rates in base mode and in cyclic mode.

Thermal mechanical fatigue is the life limiting factor for peak duty units, while creep, oxidation and corrosion are life limiting factors for base duty units.

The maintenance interval is determined by which criterion limit is reached first.

The normal start-up cycle of a CCGT unit from a cold state to a full load of 100% is approximately 4 hours.

The normal start-up cycle of a CCGT unit from a hot state, in which less than 8 hours have passed from the moment of shutdown to the moment of start-up, until reaching 100% load, is approximately 2.5 hours. At the same time, a set of approximately 80% of the power takes about 30-40 minutes. With the use of modern methods, this time can be significantly reduced. The more powerful the equipment, the longer it will start.

Fast start in a simple cycle from any state applies only to aircraft-derived GT and some industrial GT up to 100 MW. It is these GT in a simple cycle that are most often used in world practice to solve the problems of ensuring network flexibility and emergency backup of renewable energy sources.

2.7. Technical characteristics of the maneuverable power plant

The main criteria for choosing an option for the construction of a maneuverable power plant are:

- Providing electric power at the level of 1000 MW during the year;
- Ensuring the maneuverability of the operation of power units due to the widest range of regulation;
- Ensuring the efficiency of the power plant due to the high EF of gas turbine units, and the steam-gas cycle as a whole, a smaller decrease in the EF of GT in the control range of the power unit;
- Minimal decrease in power plant capacity during downtime of power units for maintenance;
- Ensuring the reliability of the power plant by installing industrially developed equipment.

The Shymkent region is characterized by hot summers with a maximum temperature of $\geq 25^{\circ}\text{C}$ 141.7 days, which requires the power plant to be provided both in winter and in summer.

The selected technology should ensure the operation of power units in the control range without falling below the technical minimum, with shutdowns mainly only for maintenance.

The range of regulation depends on the season, the power of the GT is reduced in the summer and can be increased by installing evaporative cooling systems, optionally supplied with the GT.

When choosing the main equipment, it is necessary to be guided by the characteristics of power units with a higher EF and the lowest cost.

At the same time, it should be taken into account that the withdrawal of one of the power units for maintenance or its emergency shutdown causes a loss of power of the power plant, the replacement of power is problematic and depends on the availability of reserve capacities in the power system.

Due to the fact that a single shutdown of generation with a capacity of more than 300 MW without the action of the anti-emergency automatics (AEA) in the South Zone will lead to a violation of stability, the maximum power of one generator of the Facility should be up to 300 MW.

It is possible to use generators with a unit capacity of up to 370 MW, provided that a balancing action is organized to disconnect consumers in the amount of 100 MW from emergency automatics, triggered by the fact of an emergency shutdown of the generator, which should be provided for in the corresponding project.

The advantages of building a variant of GT, HRSG, ST are determined by the following parameters:

- Expected attestation power at the average January temperature (-1.5°C) ~ MW;

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- Expected regulation range (expected base load minus the technical minimum) (-1.5 °C) during operation of the GT ~MW, when one GT is stopped, the range is reduced to ~MW;
- Possibility of bringing each unit of generating equipment into repair while maintaining the remaining units of generating equipment in operation.

Provide the layout of the main building of power units.

Provide the main characteristics of the power plant equipment according to the proposed option for the characteristic outdoor temperature regimes, according to the Table 2.7.1.

Table 2.7.1. Technical characteristics of CCGT with maneuverable generation mode
(-14.3 °C, -1.5 °C, +12.6 °C, +15 °C, +26.4 °C, +30 °C, +33.5 °C, +40 °C)

Quantity x CCGT	Quantity x CCGT based on GT			
<i>CCGT power, MW</i>	GT 100%	Nmin 2(3)xGT	Nmin 1xGT	Range of reg. Nmin 1, 2(3)xGT
ISO				
-14.3 °C				
-1.5 °C				
+12.6 °C				
+15 °C				
+26.4 °C				
+30 °C				
+33.5 °C				
+40 °C				
<i>Number of CCGT</i>				
<i>Power 1xCCGT, MW</i>				
ISO				
-14.3 °C				
-1.5 °C				
+12.6 °C				
+15 °C				
+26.4 °C				
+30 °C				
+33.5 °C				
+40 °C				
<i>GT power, MW</i>				
ISO				
-14.3 °C				
-1.5 °C				
+12.6 °C				
+15 °C				
+26.4 °C				
+30 °C				
+33.5 °C				
+40 °C				
<i>ST power, MW</i>				
ISO				
-14.3 °C				
-1.5 °C				

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Quantity x CCGT	Quantity x CCGT based on GT			
<i>CCGT power, MW</i>	GT 100%	Nmin 2(3)xGT	Nmin 1xGT	Range of reg. Nmin 1, 2(3)xGT
+12.6 °C				
+15 °C				
+26.4 °C				
+30 °C				
+33.5 °C				
+40 °C				
<i>EF of GT, %</i>				
ISO				
-14.3 °C				
-1.5 °C				
+12.6 °C				
+15 °C				
+26.4 °C				
+30 °C				
+33.5 °C				
+40 °C				
<i>EF of CCGT, %</i>				
ISO				
-14.3 °C				
-1.5 °C				
+12.6 °C				
+15 °C				
+26.4 °C				
+30 °C				
+33.5 °C				
+40 °C				

Provide the range of power control of GTs of the CCGT plant with a minimum load in the form of a diagram.

Provide a schematic diagram of the GT with ST.

To start up the power units, it is planned to install low-pressure gas-fired steam boilers.

For feeding HRSGs, preparing demineralized water for injection into the combustion chambers of GT when operating on diesel fuel, feeding closed cooling circuits of GT, and preparing drinking water, water treatment is provided. Cleaning of polluted effluents is provided for at treatment facilities.

Treatment facilities are provided for the treatment of oily industrial effluents from the Facility.

Discharge of water treatment effluents, blowdown of HRSGs, after treatment facilities is carried out to the evaporation field.

A system for the treatment of domestic wastewater is provided at the site of the Facility.

The site provides for the construction of a gas treatment facility with control, purification and measurement units.

To supply diesel fuel to the GT used as an emergency one, a fuel facility for liquid fuel is being built on the site with a storage of containers determined by the project, designed for 3-day operation of the power plant. Fuel delivery is provided by vehicles according to an agreed schedule, taking into account emergency interruptions in gas supply. Also, to deliver diesel fuel to the site, the construction of an access railway and a railway station on the site will be required.

As source water:

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- for production needs, the Badam reservoir (main) and underground artesian water (reserve) are used;
- for drinking water supply, the use of existing water supply networks in Shymkent city is considered.

The shortage of water resources in the region and the need to build our own underground water intake at the site necessitates the use of «dry» systems and will require a comparison of options at the next design stage.

In the cooling system for condensers and auxiliary equipment, the use of «dry» condensers, «dry» fan cooling towers and «wet» cooling towers for the steam turbine condenser is being considered.

Electric heating is provided for all premises of the industrial site of the power plant.

The removal of blowdown water and other waste water will require the construction of an evaporation field.

Heat recovery steam generator (HRSG)

The HRSG is a drum boiler with natural circulation in high and low pressure evaporative circuits, single-shell, horizontal profile, suspended from its own frame through intermediate metal structures, with a deaerator built into the LP drum. The framework is executed on high-strength bolts.

To ensure the operation of the GT in open cycle mode (when the HRSG is off), the HRSG is equipped with a bypass system including:

- bypass valve (divertor);
- shut-off valve guillotine type;
- a bypass chimney with stairs, platforms, built-in silencer, light protection and lightning protection.

The HRSG is designed to operate on combustion products coming from the GT. The main and reserve fuel for GT is dry natural gas. Emergency fuel - diesel.

The main HRSG equipment is installed in the building.

For access and ease of maintenance of equipment and fittings, stairs and platforms with galvanized flooring are provided.

The maximum gas temperature at the inlet to the HRSG is 580 °C (to be specified after receiving data on the GT).

The HRSG and flues are made gas-tight, based on the excess gas pressure of 3.5 kPa. The way to ensure gas tightness is sheathing and sealing.

The heating surfaces of the HRSG are made of pipes with external slotted and continuous spiral-band finning. The location of the heat exchange pipes of the heating surfaces is vertical.

The HRSG design provides complete drainage of heating surfaces and pipelines, as well as the possibility of pre-start, operational, chemical and water flushing, conservation.

The HRSG layout provides for places for tracing cable routes, impulse pipes, and also provides the ability to install the necessary measuring devices, fittings, fittings, bosses and other selected devices for CMI, automation and protection in places convenient for maintenance.

The wall material of the high-pressure drum is selected from the condition of the minimum start-up time of the HRSG from a cold state.

The HRSG has manual devices for sampling water and steam, made of stainless steel.

The HRSG design provides for the possibility of maintaining it in a hot standby.

The HRSG is supplied in blocks. The design of the HRSG provides installation both with supply blocks and with mounting blocks with their enlargement at the construction site.

The design of the HRSG and the delivery blockiness of its units was developed taking into account the technology of manufacturing, transportation, installation, repair and operation.

The supply of HRSG is carried out in prefabricated blocks of maximum overall dimensions and maximum factory completeness of parts, based on the conditions of loading and unloading operations, transportation and storage. Coefficient of delivery blockiness is not less than 80% of the calculated mass of HRSG.

The operating range of the load change of the HRSG corresponds to the GT load range of 100÷50%. The load change is achieved by changing the fuel and air consumption in the GT. This changes the flow rate and temperature of the gases at the inlet to the HRSG.

The HRSG allows operation at sliding parameters of high-pressure and low-pressure steam, determined by the flow rate and temperature of gases entering the HRSG from the GT.

The operation of the GT through the disabled HRSG is not provided.

The HRSG, working under pressure, is supplied in prefabricated blocks after factory tests, regulated by regulatory documents for this type of product.

Water and steam quality requirements

Condensate quality indicators at the inlet to the condensate gas heater must comply with the requirements of CTO 70238424.27.100.013-2009 (STO 70238424.27.100.013-2009) for HRSGs of a power plant with CCGT power units.

Steam quality standards - in accordance with CTO 70238424.27.100.013-2009 (STO 70238424.27.100.013-2009) for HRSGs of power plants with CCGT power units.

Water-chemical regime - in accordance with CTO 70238424.27.100.013- 2009 (STO 70238424.27.100.013-2009) for HRSGs of a power plant with CCGT power units.

Requirements for the automated control and monitoring system of the HRSG (ACMS of the HRSG)

The automated control and monitoring system of the HRSG is a subsystem of the APCS of the CCGT. The start-up of the GT and HRSG is carried out according to an agreed algorithm.

When developing technical documentation for automated process control systems, the requirements set forth in the following documents shall be taken into account to the extent necessary:

- РД 153-34.1-35.104-2001 «Методические указания по объёму технологических измерений, сигнализации и автоматического регулирования на тепловых электростанциях с ПГУ, оснащенных АСУ ТП» (RD 153-34.1-35.104-2001 «Guidelines for the scope of technological measurements, signaling and automatic control at thermal power plants with CCGT equipped with APCS »);

- РД 153-34.1-35.143-00 «Объём и технические условия на выполнение технологических защит теплоэнергетического оборудования газотурбинных установок ТЭЦ» (RD 153-34.1-35.143-00 «Scope and technical conditions for the implementation of technological protection of heat and power equipment of gas turbine units of TPP»).

The boiler supplier provides the terms of reference for automatic control of the HRSG from the APCS, including: PI-diagrams, a list of measurements, a list of auxiliary mechanisms, a list of shut-off and control valves, control conditions, conditions and algorithms for protection and blocking, including the values of settings, protections, interlocks, alarms, ARI, automatic control algorithms, step-by-step control algorithms for the main and auxiliary equipment of the boiler.

Optionally, a HRSG can be equipped with a local ACMS.

All shut-off and control valves included in the scope of delivery of the boiler, participating in the algorithms of automatic regulation, remote control, protection and blocking, are equipped with electric drives.

All protections leading to the shutdown of the boiler are carried out according to the «2 out of 3» scheme with separate independent inserts of parameters control points for each protection measurement channel (inserts for the selection of pressure sensors, temperature sensor sleeves, etc.).

Reliability requirements

The design of the HRSG and its elements, as well as the materials used, comply with the requirements of TPTC 032/2013 (TRTS 032/2013).

Estimated resource of HRSG elements operating under pressure with the temperature of the corresponding creep area, at least 200,000 hours for pipes of heating surfaces and outlet headers of a high-pressure superheater;

Estimated service life of the HRSG - 40 years.

Mean time between failures - not less than 8000 hours.

The service life between overhauls - not less than 8 years.

Availability ratio - not less than 98%.

Security Requirements

The HRSG design meets the requirements of TRTS 032/2013.

Environmental requirements

The amount of nitrogen oxide emissions downstream of the HRSG is determined by their concentration downstream of the gas turbine. The HRSG does not provide for measures to reduce these emissions.

The equivalent sound pressure level from the HRSG flues at a distance of 1 m from the casing and 1.5 meters from the floor does not exceed 80 dBA.

Maneuverability requirements

The HRSG allows a total number of starts-stops for the entire service life of at least: 300 from a cold state - 1700 from a warm state - 8000 from a hot state.

The HRSG design provides the possibility of combined start-up of the GT and HRSG.

The HRSG ensures compliance with the requirements for the maneuverability of a combined cycle power plant.

To ensure explosion and fire safety during the operation of gas turbine power plants, gas-air paths, including gas ducts and a HRSG, must be ventilated (ventilated) for at least 5 minutes before ignition of the GT burners. In order to ensure a quick start-up of the GT, it is allowed to ventilate the gas-air path in advance during the shutdown process during the GT rotor coastdown by turning on the starter. In this case, the GT must be equipped with a system that excludes the penetration of fuel gas into the gas-air path when the GT is in standby, by locking the gas pipeline with air at a pressure higher than the pressure of the fuel gas in the gas pipeline. The control of the state «GT and gas ducts are purged» when the GT is in reserve should be carried out by the pressure difference between the shut-off air and fuel gas in the gas pipeline before the shut-off valve.

Technical documentation

List of technical documentation supplied together with the HRSG:

- The HRSG operation manual;
- List of documentation and installation drawings for installation;
- Auxiliary equipment operation manuals;
- Passport of the HRSG (with applications in accordance with TPTC 032/2013 (TRTS 032/2013));
- Declaration of conformity or certificate of conformity HRSG TPTC 032/2013 (TRTS 032/2013);
- Declaration of conformity or certificate of conformity of TRTS auxiliary equipment;
- Security justification.

Repair suitability

Maintainability of heating surfaces, collectors and pipelines.

The heating surfaces of the HRSG are made in the form of vertical packages suspended from the ceiling metal structures of the HRSG using metal rods.

Between the rows of packages, openings of at least 1400 mm are organized, designed for inspection and repair of heating surfaces. In the upper and lower parts of the HRSG, at the level of the section headers, horizontal enclosing sheets are installed, forming with the ceiling and bottom of the boiler, respectively, the upper and lower «warm boxes». In «warm boxes» repair of connecting manifolds and pipelines is provided, as well as disconnection of replaceable sections of heating surfaces from inlet and outlet pipes and enclosing structures.

Maintenance personnel have access to the «warm boxes» and to the gas duct to the heating surfaces through rectangular manholes in the shields of the gas-tight casing of the HRSG. Maintenance personnel access to heating surfaces is organized using inventory rack scaffolding. For inspection and repair of the inner lining and insulation of the diffuser, as well as for inspection and repair of the flue to the chimney, manholes are arranged on the side walls for access by maintenance personnel and the delivery of inventory scaffolding elements and other repair materials.

Manholes are also arranged for inspection and repair of the silencer, shut-off valve and flue compensator behind the HRSG. Methods of control and testing at all stages of the manufacture of HRSG elements provide a low probability of defects in pipes and welded joints during normal operation of the HRSG.



The list of electrical consumers of the supplied equipment:

- Pipe fittings and shut-off valve of the chimney 30 kW;
- Units (1 working, 1 standby) - 2x1100 kW;
- CGH recirculation pumps (1 working, 1 standby) 2x30 kW;
- Drainage pump for pumping water from the BDT (1 working, 1 standby) - 2x15 kW;
- Diverter sealing air fans - 2x45 kW;
- Diverter electric drive 2x40 kW.

Water treatment and treatment facilities

The following water treatment plants and facilities are envisaged:

- WTP with a design capacity of up to 340 m³/h, for feeding HRSGs, starting low-pressure steam boilers, injection of demineralized water into the combustion chamber of gas turbines, feeding a closed cooling circuit and washing gas turbine compressors;
- water treatment plant for the preparation of drinking water with a design capacity of 75 m³/h;
- express laboratory of WTP;
- storage of liquid reagents (acid, alkali, ammonia);
- analytical express laboratory of the main building;
- wastewater neutralizer tanks from chemical washings of boilers;
- treatment facilities for oily wastewater with a capacity of up to 50 m³/h.

In order to meet the regulatory requirements for the quality of feed water for drum HRSGs of two pressures, for the quality of demineralized water for injection into the combustion chambers of gas turbines, etc., a water treatment plant with a design capacity of up to 340 m³/h is provided.

Source water after mechanical filtration on self-cleaning automatic filters undergoes deep clarification at an ultrafiltration unit (UFU). Then the ultrafiltered water is supplied for desalination to the reverse osmosis units of the first and second stages (ROU-I, II). Between ROU-I, II and ROU-II to remove free carbon dioxide, the installation of calciners is provided. Further, the desalinated water enters the electrodeionization units (EDU), designed for deep desalination of water to the required standard indicators for the feed water of the HRSG.

Deeply desalinated (deionized) water after EDU is supplied for feeding steam recovery boilers, low-pressure steam boilers, injection of desalted water into the combustion chamber of gas turbines, feeding a closed cooling circuit and washing gas turbine compressors.

Adjustment of the pH of deeply demineralized water with an ammonia solution is provided.

To reduce the consumption of initial water and the output of effluents, reuse of the ROU-II concentrate and EDU is provided.

For the preparation of drinking water, a water treatment plant for the preparation of drinking water with a design capacity of 75 m³/h is provided.

Source water after mechanical filtration on self-cleaning automatic filters undergoes deep clarification on UFU. Then the ultra-filtered water enters for demineralization at the ROU in one stage. Demineralized water after ROU is sent to ultraviolet sterilization units for final disinfection of water and then supplied for drinking needs.

Chemical cleaning stations are provided for periodic treatment of the modules of the UFU, ROU and EDU units with detergents designed to remove accumulated deposits, maintain the declared characteristics of the units and extend the service life.

To neutralize waste water from the WTP, it is planned to install neutralizing tanks.

The proposed water treatment technologies are based on the use of integrated membrane technologies (ultrafiltration, reverse osmosis, electrodeionization).

Provision is made for corrective treatment of feed water of HRSGs with an ammonia solution, and boiler water with a phosphate solution.

For HRSGs, pre-start alkalizing and operational (if necessary) chemical washings are provided.

To control the operation of the water treatment plant, an express laboratory of the WTP is provided.

For carrying out periodic and operational chemical control of the water-chemical regime of the power plant, an analytical express laboratory is provided in the main building.

To ensure the water-chemical regime of the circulatory system, it is provided (methods for maintaining the necessary water-chemical mode of the circulatory system are specified at the next design stage):

- stabilization treatment of circulating water with sulfuric acid;
- purge of the circulatory system.

To receive and neutralize wastewater from chemical washings and conservation of HRSGs, installation of neutralizing tanks is provided.

For the treatment of oily effluents from power plant equipment (leakage of seals in pumping equipment, washing floors, etc.), oily effluent treatment facilities with a design capacity of up to 50 m³/h are provided.

Fuel facilities and auxiliary buildings

Gas from the main gas pipelines is used as the main fuel at the Facility, diesel fuel is used as an emergency fuel.

The following fuel facilities and auxiliary buildings are provided at the power plant site:

- Gas treatment plant;
- Diesel fuel warehouse with pumping station;
- Oil apparatus with oil storage;
- Air compressor station.

Gas supply on site

From the station fence to the gas treatment plant, a gas pipeline is being laid, on which an electrified disconnecting valve is placed with a shelter from weather conditions and a service platform.

The gas treatment plant includes block-complete equipment, consisting of the following technological units of full factory readiness:

- gas consumption metering station (GCMS): a measurement unit with flow meters and filters for cleaning all gas entering the station;
- block gas control point (GCP): a reduction unit for supplying gas to start-up boilers via a medium-pressure gas pipeline;
- container for collecting high pressure condensate;
- nitrogen ramp in cover.

Air compressor station

- To provide compressed air to control systems for pneumatic actuators of protective, shut-off and control valves of gas equipment installed at the Facility, it is planned to install a container-type block compressor station.
- The block compressor station houses two compressor units: one is in operation, the second is in reserve. The equipment of compressor units is designed for the production of class 1.1.2 instrument air according to ГОСТ Р 8573-1-2016 (GOST R 8573-1-2016).

Diesel facilities

Diesel fuel grade Z-0.2-40 according to ГОСТ 305-82 (GOST 305-82) is used as emergency fuel for gas turbines at the station.

The following buildings and structures are envisaged:

- diesel fuel warehouse;
- diesel fuel pumping station.

The capacity of the diesel fuel storage used as a reserve is taken into account the operation of all GT at their rated load for 3 days. For the storage of diesel fuel, ground steel tanks should be provided, the capacity to be determined by the project.

A diesel fuel pumping station is provided for receiving and pumping diesel fuel. The following equipment is located in the pump room:



- main (fuel) pumps - working and reserve, supplying diesel fuel from storage tanks to the CCGT building;
 - pumps for pumping diesel fuel from the drain device to the warehouse - working and standby;
 - pumping pumps from drainage pits - working and reserve;
 - pump for pumping fuel from the bottom layers of tanks.
- Filters are provided on the suction pipelines of the main and unloading pumps.

Oil facilities

To receive, store and supply oil to the main building of the CCGT, an oil equipment room and an oil warehouse are provided at the site.

The oil pumping station is located in the same building as the diesel fuel pumping station. The following equipment is installed in the oil chamber:

- oil intake pump with a coarse filter installed in front of it;
- transfer pump (main);
- mobile installation for cleaning and regeneration of oil;
- drainage pump.

Six tanks for used, fresh and regenerated oil, with a capacity of 70m³ each, are installed in the oil storage in the dike.

Two fresh oil storage tanks are designed to receive marketable oil from the transport tank and store the necessary supply of fresh oil.

Two storage tanks for regenerated (purified) oil are designed to receive and store oil after it has been processed in the oil equipment room in order to prepare it for filling into equipment. These tanks can be used as a backup for receiving and storing fresh oil, as well as for receiving operating oil drained from the equipment during repairs, which can be refilled into the equipment.

Two waste oil storage tanks are designed to collect and store oil sent for regeneration or disposal.

Mechanical workshop

To carry out current repair work at the power plant, a mechanical workshop with an appropriate set of machine tools and a lifting mechanism is provided.

2.8. Electrotechnical solutions

2.8.1. Scheme for issuing electrical power to the power system

On the territory of the new station, it is planned to build a 220 - 500 kV SG. 220 kV SG according to the scheme «two working sectioned bus systems with a bypass bus system» for 18 cells is required. Communication between the 220 kV SG and the power system is provided by six 220 kV power lines and one 500 kV overhead transmission line (OTL). The 500 kV SG is designed according to the one and a half scheme (2 connections for 3 circuit breakers), which allows reducing the number of switches in the 500 kV SG without compromising the ease of control and reliability of its operation.

2.8.2. Main electrical scheme

To deliver power, it is necessary to build a new 500/220/10 kV substation of a new maneuverable power plant.

The substation provides for the installation of one group of single-phase autotransformers 500/220/10 kV with a capacity of not less than 167 MVA with the installation of a reserve (back-up) phase.

To supply power from the generators of the designed CCGTs, it is planned to install step-up three-phase transformers in accordance with nominal output voltages of the generator.

10 kV SG is provided from metal cabinets complete SG (CSG) type with vacuum circuit breakers. The project provides for the installation of 3(three) CSG -10 kV cabinets, including:

- 1 - introductory;
- 1 - for power supply of the auxiliary transformer;
- 1 - for connecting a voltage transformer;

A block-modular building is provided for the placement of CSG-10kV cabinets.

A block-modular building is provided for the placement of low-voltage complete devices 0.4 kV, Relay protection and automation (RPA), Automated control system (ACS), Monitoring and control system (MCS) and Dispatching and technological control system (DTCS) cabinets. The modules are transportable prefabricated blocks with equipment, apparatus and devices of the main and auxiliary circuits, fully assembled inside, with lighting, heating, ventilation and air conditioning systems, with fire and burglar alarms, and grounding. Blocks are mounted on the foundation.

To supply auxiliary needs of 6 kV, in the taps of blocks switched to 220/500 kV, transformers of the working power supply are installed, with a capacity of not less than 16000 kVA. ASG 6 kV of the main building consists of six aggregate sections. The work of the sections is provided according to the «explicit reserve» scheme. The backup power supply of the sections is provided from two backup transformers with a capacity of not less than 16,000 kVA and a voltage of 220/6.3 kV. Backup transformers are connected to different sections of 220kV SG.

2.8.2.1. Generators and auxiliary equipment

The main building of the station provides for the installation of generators of gas and steam turbines. The GT delivery set includes static excitation equipment (SEE), with an automatic excitation controller (AEC).

2.8.2.2. Generator terminals and generator circuit breaker

The implementation of the generator circuit within the main building, as well as between the main building and the block transformer, is provided for by complete current conductors of the generator voltages in the range of 10-30 kV.

The block circuit provides for the installation of a generator switch.

As a generator circuit breaker, it is planned to use a gas-insulated distribution generator device. The composition of the sulphur hexafluoride gas distribution generator set includes:

- sulphur hexafluoride circuit breaker;
- disconnector;
- current transformers;
- voltage transformers;
- capacitors for surge protection;
- ground electrodes;
- control cabinet

Block transformers

To connect generators with block transformers and tap-off transformers, shielded complete current ducts are provided in a single-phase version at generator voltage. Connection of block transformers with SG 500/220kV is provided by flexible connection.

2.8.3. Direct current installation

The delivery set of the GT includes a direct current installation provided for each unit. The direct current installation includes the following equipment:

- switchboard located in the container of the GT control unit;
- battery in a separate container.

To power relay protection control circuits, drives of high-voltage switches, APCS, signaling devices, communication devices, drives of automatic input and sectional switches of ASG 6 and 0.4 kV sections, electric motors of standby oil pumps of the turbine lubrication system and emergency lighting, installation of batteries with DC shields is provided. current in the electrical rooms of the main building and the relay board of the SG 220 kV.

2.8.4. Means of dispatching and technological control

Control of the main elements of the electrical connection diagram of the station is provided for:

- from the Main control panel (MCP);
- from the Block control panel (BCP);
- from GT, ST control units;

- from local technological boards of auxiliary buildings and structures (or local cabinets).

In accordance with the requirements of the «Electrical installation rules» and other applicable regulatory materials provides for the following types of automation:

- automatic regulation of generator voltage;
- automatic generator synchronization;
- automatic damping of the generator rotor field;
- automatic generator frequency control
- automatic regulation of voltage on buses 6kV for own needs;
- automatic input of reserve power supply of sections 6kV, 400V for own needs;
- automatic regulation of voltage on the buses 220V, 125V of the operating direct current shield;
- automatic switching of emergency lighting;
- automatic switching on of outdoor lighting;
- automatic activation of fire extinguishing systems.

2.8.4.1. Main control panel (MCP)

In terms of the MCP of the station, taking into account modern requirements for the introduction of automated production control systems, an automated control system (ACS-MCP) is provided. ACS-MCP is built on the basis of a hardware and software complex (HSC). The HSC of ACS-MCP system meets the requirements of technical safety at power plants and is designed in such a way that erroneous actions of operational personnel and failures of technical means do not lead to situations dangerous to life and health of people. The system is implemented on the basis of programmable logic controllers. In the MCP room, two automated workstations (AWS) of shift-supervisor of the power plant (SSPP) and shift-supervisor of the electrical workshop SSEW are installed to control electrical equipment and an automated workstation of a relay engineer in the relay team room.

The ACS-MCP system is designed to monitor, control, display, signal and record physical and electrical parameters and discrete signals the main electrical equipment of the station. The system is a functionally and geographically distributed structure that uses several independent local boards or nodes containing control processors, power supplies, data transmission systems. Local boards are able to work autonomously and perform data collection, calculation, control functions of switching devices. Data transmission between nodes and at workstations of operators and engineers located in the main control room is provided through specially designed, redundant high-speed data transmission lines.

ACS-MCP provides the following functions:

- Control of generator-transformer units;
- Element Management 220kV;
- Condition control by generators;
- Spinning reserve monitoring;
- Load distribution;
- Load shedding;
- Registration of alarms and events;
- Registration and analysis of archival data;
- DC installations.

The MCP provides for manual, automatic synchronization for 220kV circuit breakers.

2.8.4.2. Block control panels (BCP)

The BCP is provided with an automated process control system for the APCS of BCP. With the BCP, in addition to controlling the elements of technological equipment of turbo-boiler units, the electrical part provides for control, signaling and control of the elements of the electrical circuit supplied complete with gas turbine, steam turbines:

- generators (excitation circuits and generator switches);
- SG 6kV, 0.4 kV, provided for the distribution and supply of consumers of their own needs of turbo-boiler units;
- direct current installations.

The APCS of BCP is an integrated system, because it integrates into a common system all local control systems for the main and auxiliary equipment, and stand-alone units (GT, HRSG, SB control panels). The BCP provides for accurate, automatic and manual synchronization of generators with the system, as well as semi-automatic self-synchronization.

A separate autonomous control system is provided for the control of gas turbines (GT), heat recovery steam generators (HRSG) and steam boilers (SB).

To provide for the possibility of controlling the gas turbine unit and the auxiliary system of the GT, HRSG and SB with MCP and control panels of the GT, HRSG and SB. Communication between the automated control system of the of the BCP and control panels is carried out via digital communication channels using a standard protocol.

2.8.4.3. Local technology panels

The control of technological and electrical equipment of auxiliary workshops is provided from local technological boards.

For those workshops where permanent staff on duty is provided, control is carried out from the local technological board. Signaling, measurements and control of equipment operation are carried out on the same board.

To call personnel in emergency situations, a call signaling system is provided. The signal is decoded at the local shield.

Depending on the purpose of the structure, the ringing signal is sent either to the MCP or to the technological control panel.

2.8.5. Relay protection and automation

Relay protection is provided in the scope corresponding to the current « Rules for the installation of electrical installations» (RIEI) and «Guidelines for relay protection». Relay protection of the main circuit elements is provided using microprocessor devices.

Relay protection of the main equipment (generator-transformer units, reserve transformers, 220kV connections) is provided in the form of two mutually redundant autonomous protection systems («Main» and «Reserve» protection). The composition of the protections is determined by the requirements of the «Electrical Installation Rules» and the electrical equipment developer. The distribution of protection by sets is carried out on the basis of the principle of interchangeability of protection systems with redundancy for the main most probable and most severe damage. For each set of protections, individual measuring transformers, separate circuits for direct operating current, separate input and output circuits, as well as signaling circuits are provided. Each of the sets of protections is made on the basis of digital protection, implemented on the microprocessor principle. Each microprocessor terminal provides the possibility of indicating electrical quantities, setting parameters and entering protection settings, with the indicated values displayed on the monitor screen of the external human-machine interface or on the liquid crystal display of the terminal. The terminals provide the ability to communicate with external digital devices via independent, galvanically isolated channels.

The ASG 6kV part provides for the installation of CSG with microprocessor relay blocks. The ASG 6kV is equipped with fast-acting protection against arc faults inside ASG, logical protection of busbars and operational blocking, preventing the possibility of erroneous operations with disconnectors, grounding knives, withdrawable carts of PSG.

Relay board 500/220kV with MCP

A new relay board building is envisaged. The relay board building has three floors. On the ground floor there is a 0.4 kV ASG, a storage battery, a direct current shield room and other rooms.

2.8.6. Emergency automation

Emergency automation (EA) is provided on the basis of the principle of allocation (in case of dangerous frequency reduction or dangerous voltage drops) to a balanced load, in order to maintain the stability of the generators, and the operation of the EA system should also be aimed at maintaining the stability of the Turkestan energy center.



2.8.7. Emergency event recorder

Emergency event recorder with reference to astronomical time with an accuracy of 1 ms.

An automated system for registering emergency events (REE) is provided. Electrical process recorder cabinets are designed for:

- measurement of voltage and strength of alternating current;
- registration, storage and analysis of information about stationary and transient processes preceding and accompanying emergency deviations of parameters in electrical networks and machines;
- monitoring the status of on-off devices, input-output modes of generators at power plants;
- registration of short circuits and determination of the location of damage on the line; records of instantaneous values of measured quantities (including in the form of sets);
- the operation of the registrar is carried out around the clock;
- the registrar provides the possibility of remote access and differentiation of user rights and access restrictions.

The Registrar consists of:

- registration block;
- control units;
- power supplies;
- alarm block;
- other equipment - in accordance with the delivery set.

2.8.8. Automated system for the commercial metering of electricity (ASCME)

An automated system for the commercial metering of electricity (ASCME) is being created to organize automated measurement of the parameters of generation, consumption and distribution of electricity, to improve the accuracy of electricity measurements. The work provides for the installation of commercial accounting points in accordance with the RIEI of the RK, approved by the Decree of the Government of the Republic of Kazakhstan No. 1355 dated October 24, 2015 and other regulatory documents of the Republic of Kazakhstan on the following connections:

- generators « G1»÷« G6»;
- on the side of 500/220 kV block transformers « GT1»÷«GT6»;
- on the side of 220 kV standby transformers «OVT10», «OVT20»;
- power lines 500/220 kV;
- bypass switches 220kV.

ASCME is a complex of software and hardware, consisting of:

1. Measuring complex for electricity metering, which includes:
 - primary converters - measuring current and voltage transformers;
 - secondary circuits between measuring transformers and electricity meters;
 - primary means of accounting - digital electricity meters;
2. Complex of software and hardware:
 - system time synchronization devices (STSD);
 - automated workstations of ASCME operators;
 - software tools for meters, AWS.

The current and voltage transformers provided for by the work have the required accuracy class and are selected in accordance with the requirements of the GR (Grid regulations) of the RK.

Microprocessor-based meters measuring active and reactive electricity are used as commercial meters. Commercial accounting meters meet the requirements of regulatory documents of the RK, are included in the Register of the State System for Ensuring the Uniformity of Measurements of the Republic of Kazakhstan and are equipped with:

- a digital interface board for connecting to communication equipment and transferring data to higher levels;
- an optical port for reading data from the meter autonomously.

To collect information about energy consumption from metering meters, it is planned to install a data collection and transmission device (DCTD) cabinets. Communication of DCTD cabinets with the ASCME server is carried out via digital communication channels using a standard protocol.

2.8.9. Power quality control system

The power quality control system (PQCS), in accordance with the requirements of ГОСТ 32144-2013 «Нормы качества электрической энергии в системах электроснабжения общего назначения» (GOST 32144-2013 «Power quality standards in general-purpose power supply systems»), must measure the power quality parameters (PQP) at the interfaces of the balance sheet. This work provides for the measurement of power quality parameters to be carried out at the 220kV connections of the new station.

The complex of PQCS equipment, which includes power quality analyzers and the PQCS server with installed specialized software, should provide the following functions:

- Collection of PQP measurements from controlled connections using quality analyzers;
- Formation of the PQP database;
- Analysis and provision of objective information to the plant personnel about the violation of power quality parameters

The source of power quality control information should be microprocessor-based quality analyzers of the appropriate accuracy class, with long-term memory and having a data transfer port. Quality analyzers must be included in the State Register of Measuring Instruments.

2.8.10. Energy saving measures

This paper takes into account the main directions for energy saving:

- Optimization of modes of production, distribution and consumption of energy;
- Implementation of projects for the introduction of energy-efficient equipment and advanced technologies;
- Implementation of technical, technological, economic measures.

In order to improve the reliability of transmission and distribution of electrical energy, sulphur hexafluoride circuit breakers are provided, which differ from air and oil circuit breakers in higher switching capacity, reliability, durability, low weight, and reduced operating costs. The power transformers are adopted with an efficient cooling system with oil circulation and low no-load losses.

The use of modern supervisory control and data acquisition (SCADA) systems, ASCME and telecommunications equipment in accordance with existing regulatory requirements contribute to:

- Increasing the speed and accuracy of personnel actions by providing them with more complete, reliable and timely information about the operating modes and the state of the main and auxiliary equipment, including for operational management and introduction of modes;
- Increasing the level of control and management of technological processes in normal and emergency modes;
- More accurate accounting and support of the specified parameters of electricity;
- Development of dispatching and technological communications based on corporate networks;
- Ensuring high reliability and safety of equipment operation.

The economic effect is achieved by reducing the number and duration of power outages, reducing damage from downtime, and reducing damage.

2.8.11. Cable facilities

The main cable flows in the building run in cable ducts and on open cable structures. Single cables are laid openly with their protection in places of possible mechanical damage with metal pipes.

In SG 500/220, cables are laid in reinforced concrete trays. Single cables are laid in the ground and protected by pipes.

On the territory of the power plant, the main cable flows are laid on cable and technological racks. Provides for the protection of cables from solar radiation when laying them openly on overpasses. Single cables are laid in the ground in pipes.

Cables with copper conductors are used, with a sheath that does not spread combustion. Cable passages through walls and ceilings, partitions are sealed with fireproof materials.

2.8.12. Grounding devices, lightning protection

Grounding devices and lightning protection are carried out on the basis of the «Rules for the installation of electrical installations of the Republic of Kazakhstan», СП РК 2.04-103-2013 – «Устройство молниезащиты зданий и сооружений» (SP RK 2.04-103-2013 - «Lightning protection device for buildings and structures»).

For grounding electrical equipment installed in buildings, an internal grounding network is provided, made with a steel strip with a section of $40 \times 4 \text{ mm}^2$, and branches to the grounded electrical equipment with a strip with a section of $25 \times 4 \text{ mm}^2$. It also provides for the use of steel building structures for grounding, reinforcement of foundations and cable structures connected to the ground network.

The external ground loop is made with a copper strip with a cross section of $30 \times 4 \text{ mm}^2$, laid in the ground at a depth of 0.5-0.7 m, to which ground electrodes 12-16 mm 3-5 m long are connected.

For the external ground loop, guides for fastening cable structures, spans and other elements of cable and technological racks are also used.

Lightning protection of individual buildings and structures is made in accordance with the category for lightning protection device, according to СП РК 2.04-103-2013 «Устройство молниезащиты зданий и сооружений» (SP RK 2.04-103-2013 «Lightning protection device for buildings and structures»).

Protection against direct lightning strikes is carried out by lightning rods (chimneys) installed separately on structures, as well as by grounding metal parts of structures, metal roofing of buildings and sheds.

If buildings and structures have straight and vented pipes for free removal of gases and vapors of explosive concentration into the atmosphere, the area of protection of lightning rods should include the space above the edge of the pipes, in accordance with Section 9 of СП РК 2.04-103-2013 (SP RK 2.04-103-2013).

Lightning protection of SG-500/220 kV and power transformers is made by rod lightning rods installed on portals and rod lightning rods installed on searchlight masts. According to the RIEI of the RK, the racks of the switchgear structures, on which lightning rods are installed, should be connected to the grounding device. At the connection point, install one or two vertical electrodes 3 m long at a distance not less than the length of the electrode from the post on which the lightning rod is installed.

Measures are envisaged to prevent the removal of dangerous potential from electrical installations with a high earth fault current outside buildings and external fences, as well as to equalize the electrical potential of the earth in the territory of switchgears and approaches to them.

2.8.13. Electric lighting

Electrical lighting is provided in accordance with current regulations and guidelines.

The illumination of individual premises, buildings and the entire territory is taken in accordance with the provisions of СП РК 2.04-104-2012 «Естественное и искусственное освещение» (SP RK 2.04-104-2012 «Natural and artificial lighting»).

The following types of lighting network are provided:

- work lighting for 220 V AC;
- emergency lighting for voltage 220 V;
- evacuation lighting for voltage 220 V;
- outdoor lighting of roads, driveways, passages for a voltage of 220 V alternating current;
- repair lighting on a voltage of 42 V and 12 V.

In industrial premises, explosion-proof, energy-saving lamps with light-emitting diodes lamps are used for lighting.

Outdoor lighting is carried out by lamps with light-emitting diodes lamps, installed throughout the territory on lighting poles along the roads. Lighting of SG-500/220 kV is carried out by light-emitting diodes spotlights installed on portals and free-standing lightning rods.

Work lighting control is provided for manual centralized with automatic activation of the emergency lighting network.

An emergency lighting switch panel is provided to supply emergency and evacuation lighting. In the event of a loss of AC voltage, the grid will automatically switch to battery power.

Outdoor lighting control is provided both manually from the control panel and automatically using photothyristors. The network of outdoor lighting is powered separately from the internal lighting of buildings and structures.

In fire-explosive rooms and zones, lamps in the appropriate design are used. In rooms and areas with an increased risk of electric shock, a voltage of 42 V is used, and for portable lamps - 12 V.

2.9 Video surveillance systems

To monitor the perimeter fences and gates, a video surveillance system (CCTV) should be provided. Fixed-position cameras with fixed focal length lenses and pan-tilt zoom (PTZ) cameras should be used to provide adequate coverage. Remote control of PTZ cameras should be provided. Camera systems must be digital systems that must be user-configurable, scale, use digital storage devices, and be distributed over large networks.

It must be connected to an intrusion detection system and provide automatic calling of cameras in response to an alarm.

The Contractor shall be responsible for designing and placing CCTV cameras at strategic points, mounted on buildings or using masts, as required, to achieve full coverage of the required areas.

The video surveillance system must be a high quality, high performance networked digital video surveillance system and must be designed so that people can be clearly seen anywhere within the coverage area. CCTV images displayed on monitors or retrieved from storage must be clear enough to allow clear and unambiguous identification of people and vehicles.

Solid-state surveillance cameras using charge-coupled devices (CCDs) capable of providing maximum resolution, high-definition color video images should be provided. Cameras must be wide dynamic range (WDR). Cameras installed in locations subject to changes in light levels should have lenses equipped with automatic iris adjustment.

A digital storage system shall be provided capable of storing video surveillance images in a compressed time-lapse format or in real-time as needed. Images should usually be stored in timelapse mode. The storage system must provide redundant storage elements to ensure that data is not lost in the event of a storage element failure. Storage capacity must be at least 72 hours, with some cameras running in slow motion mode and the rest of the cameras running continuously. An alarm should sound when the storage capacity reaches 80 percent. The video surveillance system must comply with the requirements: Decree of the Government of the Republic of Kazakhstan dated May 6, 2021 No. 305. «On approval of requirements for the organization of anti-terrorist protection of objects vulnerable to terrorists». In case of failure of the storage element, the transition to the backup storage element should be automatic. The storage system must have the means to overlay the camera number and date/time onto the images.

Color television monitors capable of displaying any of the system's cameras should be provided. Monitors and controllers will be located both in the CCR and in the security service premises.

2.10. Automation of technological processes

2.10.1. Brief description of automation objects

When designing hardware and selecting software and hardware for an automated control system (ACS), compliance with the tasks to be solved and the requirements of regulatory documents should be taken into account.

When designing, it is necessary to take into account, but not be limited to, the requirements for ACS in accordance with ГОСТ 24.104-85 «Автоматизированные системы управления. Общие требования» (GOST 24.104-85 «Automated control systems. General requirements»), ВНТП 81 «Нормы технологического проектирования тепловых электрических станций» (VNTP 81 «Norms for technological design of thermal power plants»).

The main objectives of the APCS are:

- ensure safe start-up, synchronization, loading, on-load operation, shutdown, emergency shutdown with control and monitoring of all main areas of the installation from the CCR and local control room, if applicable;
- to ensure the optimal and efficient use of all production areas;

- maximum availability throughout the Facility;
- ensure the availability and maintainability of the Instrumentation and Automation (I&A) of the station;
- provide means for integrated monitoring, reporting, storage and presentation of information on the state of the plant;
- provide facilities for comprehensive testing and reporting of system and station performance information;
- provide facilities for comprehensive storage and retrieval of information about the plant, conditions and process parameters;
- for centralization, to centralize the means of monitoring and managing the enterprise in the central control room or control panel;
- to minimize the time of the sequence of starts and stops with minimal mechanical and thermal loads on the installation;
- to reduce the time of «return to work» of the units after the shutdown of the station to a minimum;
- provide fault-tolerant systems;
- provision of selected data and information about the enterprise to the management of the enterprise and support staff;
- provision of prompt service and non-exhaustive diagnostics for all systems;
- provide data for the billing system of the Customer.

To ensure that plant instrumentation meets operational safety and resiliency requirements, the design must include a full HAZOP analysis.

In addition, it is necessary to conduct a SIL study and SIL verification in accordance with IEC 61508/61511 and ГОСТ Р МЭК 61508-2-2007 (GOST R IEC 61508-2-2007), ГОСТ Р МЭК 61508-3-2012 (GOST R IEC 61508-3-2012), ГОСТ Р МЭК 61508-4-2012 (GOST R IEC 61508-4-2012), , ГОСТ Р МЭК 61508-5-2012 (GOST R IEC 61508- 5-2012), ГОСТ Р МЭК 61508-6-2007 (GOST R IEC 61508-6-2007), ГОСТ Р МЭК 61508-7-2012 «Функциональная безопасность систем электрических, электронных, программируемых электронных, связанных с безопасностью. Требования к системам» (GOST R IEC 61508-7-2012 «Functional safety of electrical, electronic, programmable electronic systems related to safety. Requirements for systems»), ГОСТ Р МЭК 61511-1-2018 «Безопасность функциональная. Системы безопасности приборные для промышленных процессов. Часть 1. Термины, определения и технические требования» (GOST R IEC 61511-1-2018 «Functional safety. Instrumental security systems for industrial processes. Part 1. Terms, definitions and technical requirements»).

All decisions made during the design process must be consistent with the objectives of the project.

If it is impossible to use the regulatory documentation in accordance with the requirements of the legislation of the Republic of Kazakhstan, develop special technical conditions within the framework of this project.

The new designed automated control system (ACS) is designed to control the technological processes of the following facilities and structures, including, but not limited to:

- GT, HRSG, SB;
- flue bypass pipes of CCGT (installation of gas analytical systems);
- auxiliary steam boiler for own needs;
- gas treatment plant, including a gas consumption metering station, a block gas control station, nine gas booster compressor stations;
- air compressor station;
- underground water intake with pumps;
- source water storage tanks with a pumping station;
- dry cooling towers;
- circulation pumping station with water conduits;
- water treatment for make-up of HRSGs and auxiliary systems, water treatment for drinking water, treatment facilities for oily wastewater;

- diesel fuel warehouse;
- oil warehouse;
- diesel fuel and oil pumping station;
- two supply pipelines on the site,
- overpasses of technological pipelines and thermal outlets at the site;
- fire extinguishing pumping station and fire-fighting water tanks;
- pumping stations for household and industrial sewerage.

2.10.2. Main technical solutions for equipment automation

The scope of supply described shall be such as is necessary to achieve the required objectives of the project and shall include, but not be limited to, the following:

- integrated control, protection and monitoring systems for the supplied plant, including a gas turbine generator (GTG), a steam turbine generator (STG), a heat recovery steam generator (HRSG), an emergency diesel generator, a fuel gas system, electrical systems of the station, cooling of water supply systems, a station water treatment, general services and all ancillary equipment. (add a set of equipment for oil supply systems for gas turbines and generators);

- APCS, including all hardware, software, operating licenses and related equipment to provide integrated management, monitoring, alarm handling and data processing of the entire enterprise from the central control room;

- interfaces between APCS, field equipment, integrated control systems, power systems and control systems, instrumentation, drives, engine control centers and all related equipment and devices;

- emergency stop systems for safe operation, shutdown and protection of the plant;

- equipment for monitoring the condition of the main and auxiliary equipment;

- instrumentation with an accuracy suitable for tariff accounting of fuel supplies and generated electricity with display, trend analysis and reports on current and aggregate values;

- equipment of the CCR, including operator tables, chairs, furniture, large screen displays, printers and tables;

- I&A cables and cable systems, including all special cables;

- equipment for monitoring atmospheric emissions;

- equipment for control of emissions into water;

- local instrumentation, including pressure, temperature, level and flow sensors, switches, transducers and transmitters, as well as all relevant taps, pipelines, valves, manifolds, thermowells, local cabinets and racks;

- information and telecommunication systems to cover all areas of the enterprise, including a digital private automatic branch exchange (DPABX), a public address system (an enterprise local area network (LAN) system);

- an integrated security system, including a video surveillance system (CCTV), an intrusion detection system and a plant entrance control system;

- heating, ventilation and air conditioning (HVAC) control systems for all HVAC installations;

- meteorostation;

- interface equipment for data/signal exchange with the National Dispatching Center, including a remote terminal for interfacing with the SCADA system of the electrical network;

- fire detection and warning system and fire extinguishing means;

- gas detection and alarm systems;

- engineer's workplace for program development and maintenance;

- uninterruptible power supplies;

- network time protocol (NTP) unified time subsystem;

- control and testing of works;

- installation, installation, testing and operation on site;

- documentation and drawings;

- operation and maintenance manuals, including executive information;

- training of operators and maintenance personnel;

- spare parts (spare parts for commissioning and, as an option, spare parts for 2 years of operation);
- special tools (only provided for installation, others as an option);
- a complete set of all enterprise application software packages and any enterprise base configurations provided by vendors, if applicable.

2.10.3. Organizational and functional structure of equipment management

The APCS is an independent control system with access to the general station level of the APCS.

The stages and methods of organizing interface channels for receiving and transmitting data and tag signals for information and control «control loops» of the designed APCS system, the Contractor must coordinate with the Customer.

A feature of the automated control system (APCS) is that most of the main and auxiliary equipment is supplied by factories with complete control and management systems, which are combined into a common system at the block or station control level.

Of the main equipment, a gas turbine, a steam turbine with a generator are equipped with complete systems, from the auxiliary equipment - a gas treatment plant, a gas flow metering point, a booster compressor room, an air compressor, and microcooling towers.

ACS of HRSGs is designed according to the instructions of manufacturers.

When designing, take into account, but not be limited to the requirements of ВНТП 81 «Нормы технологического проектирования тепловых электрических станций» (VNTP 81 «Norms for the technological design of thermal power plants»), СН РК 4.02-03-2012 «Системы автоматизации» (SN RK 4.02-03-2012 «Automation systems»), СП РК 4.02-103-2012 «Системы автоматизации» (SP RK 4.02-103-2012 «Automation systems»), СН РК 4.04-10-2013 «Электростанции тепловые» (SN RK 4.04-10-2013 «Thermal power plants»), СП РК 4.04-110-2013 «Электростанции тепловые» (SP RK 4.04-110-2013 «Thermal power plants»).

Automated process control system (APCS)

The following hierarchical levels are distinguished vertically in the technical structure of the APCS:

The upper level is the level of visualization and operational management (SCADA - Supervisory Control And Data Acquisition) - the level of information presentation, operational control and management, analysis of current and retrospective information. It is located in the MCP and consists of workstations - servers for processing and storing information, equipment for network components, AWS of operational and maintenance personnel, and provides an interface for interaction between personnel and the control system. The AWS of the boiler room operator uses a SCADA system. The AWS provides convenient and effective control by the operator over the performance of all information, control and service functions of the ACS.

The operator's AWS is designed to visualize process parameters, remotely control actuators, enter tasks for regulators, view individual protocols, reports and summaries, enable or disable control subsystems (autoregulation, AWS, switchable interlocks) and perform other functions. On the computer of the operator's AWS, the programs of the application server operate, which is the main connecting element of information flows between the software of the upper and lower (Software controller) levels of the system.

Top-level Software system includes:

- SCADA package software;
- The software below for each component of the circuit gives a brief description of the characteristics and purpose.

The middle level - the level of control and management (Control Level, PLC (PLC)) - includes control systems for the main and auxiliary equipment based on a redundant software and hardware automation complex with the possibility of hot replacement, as well as automatic control systems (ACS) supplied complete with process equipment, performs algorithmic processing of information from sensors and primary signal converters and the formation of control actions on actuators and regulatory bodies, as well as diagnostics of peripheral equipment.

The controller is an element of the system that provides connection of cable connections from sensors and conversion of analog signals from sensors into digital form (ADC code) necessary for further software processing by the system. The digitized information from the sensors undergoes primary processing inside the controllers for:

- conversion to engineering units;
- calculation of derived quantities (pressure, temperature, flow rates, etc.);
- analysis of the reliability of the measured values;
- formation of an Instantaneous Database corresponding to the current values of the measured values, which are available at any time for controller programs;
- updating values for upper-level subsystems - sending special telegrams about changes in values, achievement of warning, emergency and other settings, and fulfillment of other criteria for updating information of the «upper» level.

The lower level is the input-output level of information (Input/Output Level) - remote data collection modules (devices for communication with the object - DCO modules) from process parameters sensors, primary converters, actuators, regulatory bodies, provides the main functions of the system for collecting information and issuance of control actions on actuators and consists of signal input / output stations, cabinets of industrial intermediate relay and terminal blocks.

Field level - sensors, primary converters, actuators, regulators, supplied complete with equipment or ordered according to the developed project documentation.

Technological equipment sensors are used to convert the measured physical quantities into an electrical analog signal, which is fed into the system through cable connections.

The plant control system APCS should be the main and, if possible, the only (basic) control system used throughout the enterprise. However, it is recognized that certain areas of the plant may be provided by sub-suppliers who provide their own control systems. In such circumstances, these subsystems must be connected through redundant interfaces so that the operators of the MCP are fully informed and can control the entire installation, all installations and objects.

The overall design of the system should be provided by a fully integrated plant management system to control, manage, display, signal and record selected physical and electrical parameters associated with all areas of the plant.

The APCS system will have a functionally and geographically distributed architecture using a number of independent BCPs containing control processors, power supplies, data transmission systems, cross-connect and terminal devices. The BCP must be capable of autonomous operation and perform the functions of data acquisition, calculation, open loop and feedback control. Communication between nodes located in different areas of the station and operator or engineer workstations located in central control rooms and local control rooms should be carried out over properly designed redundant high-speed serial data lines. The communication system must be designed so that rate variables are updated and control commands are issued without compromising system performance. The communication system must include all control equipment, error correction and cabling.

The normal operation of the station, including start-up, load changes, shutdown, monitoring, etc., must be carried out from the MCP through the operator's workstations.

The APCS should be designed to provide maximum availability by including built-in redundancy in both hardware and software. This should include redundant control processors, a data highway, input/output modules as needed, and power supplies with automatic failover to a standby unit when a fault or failure of the operating unit is detected. Redundancy must be provided for input/output modules used for safety-related control or protection functions.

The failure of any individual element in the APCS will not affect the operation of any element of a running plant. In particular, the redundant control processors will be able to failover without changing station state, sequence state, or control actions at any point in the station's operation.

Comprehensive system diagnostics should be included in the APCS to assist with maintenance and troubleshooting. These diagnostic systems should, at a minimum, cover memory, control processors, input/output cards, data highway, and storage devices. System maintenance software must be provided.

The APCS must be connected to a common NTP time synchronization station. This will allow synchronizing the APCS and other monitoring/control systems with a common NTP universal time

synchronization station. The APCS must be capable of automatic time shifting without interrupting operation or data collection according to UTC time/local time shift.

All necessary equipment for sorting and terminating the incoming and outgoing cable network of the Facility must be provided. This should include gland plates, glands, terminal blocks, IS barriers, isolation devices, markings, and wiring as required. The equipment must be designed to operate in hazardous areas where required.

The MCP will be the control center of the Site, so it must be ergonomic and aesthetically pleasing to ensure that the operator's working conditions are of a high standard, providing a safe and comfortable working environment for station personnel and allowing them to work efficiently. The design of the MCP must also ensure that consistent designs are used for all tables, furniture, partitions, walls, ceilings and floors, taking into account aspects such as the type and location of equipment, the distribution of light and shadow, and the coordination of details of color, material and texture.

To provide easy access to cables, moving equipment, etc., a raised floor should be provided for the MCP.

The Contractor shall provide with the application a drawing showing the proposed layout and arrangement of equipment in the MCP and any associated equipment rooms.

Two operator workstations (AWS) are to be provided in the MCP to enable monitoring, control and adjustment of plant conditions using high-resolution Windows-style graphic displays. All visual display device (VDD), keyboards, printers, peripherals and associated control panels must be supplied.

Each operator workstation should consist of two high-resolution 24-inch color displays, one keyboard, and a cursor positioner. The workstations should be the primary means of monitoring and controlling the entire installation, and should also be configured as engineer workstations.

Engineer's workstation (EWS)

One engineer's workstation (EWS) is provided for troubleshooting and programming of the APCS software. All necessary hardware and software must be provided to enable engineering personnel to monitor system and application software failures, create control strategies, display screens, and perform all other maintenance activities. Configuration and modification of the software must be possible both online and offline, as well as the possibility of testing before downloading to the control processors.

The EWS shall consist of at least one high-resolution 24» color display, keyboard input device (e.g. mouse, trackball, etc.) and data storage and loading system mounted on a suitable table that matches the appearance of the operator workstation. The EWS must be connected to the plant management system via a data highway to provide access to system and application software, as well as to the plant's operational data. Access must be password protected.

Printers shall be provided for printing automatically or upon request of the operator of APCS alarms, events, reports and trends. All printers must be state of the art and must be networked so that multiple users can use them. Printing must not interfere with any other activity of the operator or engineer at the workstations.

Control panels, panel designs and consoles

Control panels, design of panels and consoles in the MCP, at least two control panels (one operator panel and one dispatcher panel) must be provided. Desks should be of a consistent rigid steel or steel and wood construction to provide suitable surfaces for mounting displays, keypads, telephones, and other devices. The Contractor shall be responsible for coordinating the design and ergonomics of desks, work stations, panels, furniture, lighting and room lamps/equipment/furnishings of the MCP to ensure that design uniformity is maintained and that the aesthetic appearance of the MCP reflects its status in the Plant.

Consoles and control panels should be ergonomically located, suitable for operation with a minimum number of operators, and in accordance with the operating principles of the station. However, access should be possible for more than one operator for special operations and training purposes.

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Consideration should be given to the use by operators of the operating manuals and the updating of the logbook when observing the installation.

Control desks should include means at the top of the table and at the base of the units for cable routing from computers, displays, keyboards, telephones, hands-free controls, etc. A lockable drawer/storage area should be provided, leveling and securing devices into design. Sufficient space must be provided for freely moving devices such as keyboards, etc., to prevent excessive movement of devices during normal operation of the installation, and sufficient protection must be provided to prevent stress on connections or cables. The height of the tables should be determined by the need for the operator to look down on the control panels, etc. when he is in his normal position at the table. Each workstation should have one fully adjustable swivel chair plus one extra chair.

Panels must be located so that operators have easy access to them.

Panels should be arranged in such a way that the arrangement of control buttons, lights, indicators and other sources of information does not confuse the operator. If the shutdown switches for different process modules or equipment look the same and are located together, it is possible that the wrong equipment will stop if the operator is stressed in an emergency. Stop switches must be physically separated and labeled with the name of the function. There should be a means of testing all light bulbs. Must comply with ГОСТ Р МЭК 61511-2-2011 (GOST R IEC 61511-2-2011).

Local control rooms should be provided primarily to house the command and control equipment associated with the gas turbine generator and its ancillaries and, if necessary, provide a local operating point for each machine. An operator workstation, including flat screen displays, keypads, cursor controls, etc., should be installed in each local control room to enable start, stop and emergency shutdown of the gas turbine and auxiliary equipment. At local control points, it should be possible to select local control or control from the MCP through the APCS.

The plant control system shall be shipped with all hardware, system and application software (VDD displays, alarms, etc.) fully developed, tested, debugged and installed for all analog and digital inputs/outputs. An additional 20 percent expansion per node must be provided, covering input/output cards, processors, and media, fully installed, wired, and completed. The software must be designed to ensure that the intended performance is not degraded when the system is fully loaded, including spare capacity.

The maximum CPU load in all operating modes should not exceed 80 percent.

The system shall provide for the installation of temporary operator workstations for the purpose of testing the APCS interface/installation during the commissioning program.

APCS software

The latest version of the manufacturer's operating system software must be fully developed, tested, installed and in English (the choice of language is determined by the rules of the Customer's dispatch service). The software must be designed to ensure that the intended performance is not degraded when the system is fully loaded, including spare capacity.

Application software and associated displays, reports, databases, etc. must be configured and fully tested when installed on site.

One set of all station application software packages and any basic station configurations shall be provided.

All required software licenses must be included. This will result in Customer's unrestricted use of all installed software packages.

Access to software and applications at various levels should be possible for operational or engineering personnel using passwords. This will protect the software from unauthorized access. Any changes to the software, whether application and/or operational, must be fully documented in a logical and structured manner so that the history and nature of the changes can be easily identified.

The Contractor must include cyber security measures (including equipment) in the system.

Operator interface software

The operator interface software must allow the operator to perform the necessary actions in a safe and efficient manner. In particular, at least the following features should be enabled, but not limited to:



- the software must be structured in such a way as to provide a hierarchy of control from automatic sequential start of the installation to manual control of a single element.
- with manual intervention by the operator (sequence initiation and control of individual devices under normal and failure conditions), all derivative programs, hardware-programmed permissions and overrides should be visible to the operator.
- regardless of the logical control device, there must be manual means (for example, an emergency stop button) to influence the process through the actuating elements of the safety instrumented systems SIS or other means established in the safety requirements specification SRS must comply with the requirements of ГОСТ Р МЭК 61511-1-2018 (GOST R IEC 61511-1- 2018).
- if process measurements have been duplicated to improve reliability, the signals should automatically switch to a good signal in the event of a fault.

A philosophy should be developed to define the operation of the sequencing logic in the event of a stage failure, operator intervention, or initiation of a sequencing when all station conditions are not met. It should be possible to bypass the required station state, but this action should be forewarned and logged. The principle of operation must be agreed with the Customer.

Engineering interface software

The engineering interface software must allow it to perform all necessary installations, updates and checks, monitor hardware and system software for failures, etc. in a safe and efficient manner. In particular, at a minimum, the following features must be enabled:

- the software must be built in such a way as to provide a hierarchy of control systems (from automatic sequential start of the installation to manual control of a single element), which, if necessary, can be changed by an engineer. This must be password protected.
- examine all operations of the hardware (from operator terminals to input/output signals) and operating software (sequencing initiation and control of individual devices under normal and fault conditions, organize all means of trend monitoring and recording). Tools for viewing all software that is initially installed and received later. Implement software and hardware permissions and overrides, and make sure the relevant information is visible to the operator. All these objects must be completed online without interfering with the work of the plant.
- tools for setting the measurement ranges of the process, etc. provide duplication of critical signals in the APCS to improve reliability. If necessary, ensure that these signals can be averaged for monitoring purposes, including quality control, which should be used to automatically switch to a good signal in the event of an interruption.
- the means to meet any additional requirements that the operator or engineering administration may require for operation and maintenance purposes.
- facilities for installing new software, and for recording, displaying and printing all installed information, as well as any changes and updates.

The engineering philosophy shall be developed from previously developed standard arrangements and shall be submitted to the Customer for approval early in the contract.

Operator workstation graphics

A set of graphic displays at the workstations of the APCS operators is the main means of presenting information and managing the Facility. Graphic displays should provide operators with dynamic information about the process and provide a picture of the plant based on piping and IA.

Graphic displays should form an associated hierarchical structure consisting of a number of different graphic display formats, as follows:

- plant overview - displays the overall schedule of the plant, showing production, consumption, and the main parameters of the plant.
- system overview - to display detailed information about the system or systems, as well as to display the status of the main equipment.
- system area - to display all installation elements and process parameters for the area/process in question.
- plant elements - to display all the details of the plant element process (can be displayed in «pop-up windows»).

- faceplates - these graphics shall display the control faceplates for a plant element, system or group of systems on the same graphics.
- trends - to display historical data graphs.
- bar graph - these displays show data values relative to each other (for example, generator stator temperatures).
- menu graphics - they should display a list of system or systems graphics.
- text graphics - these should display ready-to-run lists or help screens.

Graphics should contain only those details that are necessary for the operator to determine the condition of the equipment at a glance.

Graphic displays must be connected through «hotspots» that connect the graphics as a continuation of the process. Hot spots should be highlighted with text describing the continuation of the process.

The design of graphic displays should be concise, uniform in structure, unambiguous, and use reports.

The APCS must be capable of presenting operator-defined and user-configurable reports to the operator. The information in reports can be either in tabular form or in pre-formatted report form and can be automatic at specific times or on demand. Reports should include, at a minimum, periodic logs, daily logs and group logs, alarm summary log, sequence of events log, post-shutdown analysis logs, system status/failures, energy balance and efficiency log, etc. comply with the requirements of the Competent authority and environmental permit. All reports must be configured and report formats must be approved by the Customer. It should also be possible for the operator to generate special reports as needed.

Data logging and trends

The APCS should include long-term data logging and archiving for all analog, digital, serial data, operator actions and other internal points. All signals should either be periodically scanned or reported as exceptions and updated values or digital status changes should be saved.

The operator must be able to display real-time data, call up data stored in the data logger memory or from external archives, specifying the time period of interest. The requested information can be displayed on the Operator's workstations either in tabular form or as selected variables on the trend display.

The data from the data logger must be available to the station's local area network and be available to operators, supervisors, and plant operational control systems. The access level must be password protected.

The format and content of reports on data sampling from the information management system are subject to approval by the Customer.

Sequence of events

A sequence of events recording system should be provided to allow analysis of the causes of plant outages or disturbances.

The number of inputs in the sequence of events system from the unit and the general plant should be finally determined at the detailed design stage.

The status of each point must be time stamped at the source and stored in the database along with other relevant information for a specified period of time. Data must be continuously removed from the database on a first-in, first-out basis after a specified period of time. Means should be provided for archiving data for later use, as well as for easy retrieval for viewing and analysis.

When a scheduled event occurs (for example, a turbine shutdown) or a manual start from the APCS, all data for a certain period of time before the event should be stored and continue to be recorded during the period after the event.

Measurement

Operational and tariff accounting should be provided for process control, electricity quantity, performance control and emissions control. The measured data should be used for billing, process control and monitoring purposes as required and described below. The data must be obtained from factory instruments designed and installed to measure actual values and, if necessary, must be corrected for changes in pressure, temperature, density, etc.

All measurements must be displayed locally and tracked, displayed, recorded by the APCS, and also be available for inclusion in management reports generated by the APCS. The APCS must also provide means for integrating instantaneous values over a predetermined period of time.

The measurement must be supplied to include but not be limited to the following:

- volumetric, mass and energy flow of fuel gas for each main consumer (for example, gas turbines, etc.) for operational purposes;
- volumetric, mass and energy consumption of fuel gas for the installation for fiscal (billing) purposes;
- raw water, demineralized water and cold water flow for operational purposes;
- electrical power (active and reactive) from each generator;
- electric power (active and reactive) for the block transformer;
- total electrical power (active and reactive), imported and exported from / to the power grid.
- consumption of feed water, steam and condensate for operational needs;
- waste water flow.

Tariff metering accuracy must comply with the requirements of electricity and gas purchase contracts, as well as applicable national and international regulations.

The information-computing complex stations include:

- operator stations located on the BCP;
- engineering stations of the hardware and software complex (HSC) of the APCS;
- a database and archival recording station for accumulating information about the parameters of the technological process, for performing work to ensure the functioning of the APCS and obtaining hard copies of the output forms of information tasks;
- station of anti-virus protection and WEB-monitoring;
- NTP time synchronization station.

The AWSs of the drivers of the GT+HRSG+SB or GT+HRSG units consist of workstations with monitors that provide all the functions of monitoring and controlling the units. The workstations are completely interchangeable and allow you to control the process in all modes, even if one of them fails completely.

Engineering stations consist of one monitor workstation with a printer. Drivers' AWSs and engineering stations are equipped with anti-virus programs, access control, etc. Intelligent APCS devices of the station level together with APCS devices of the block level are united by a single information network. Special-purpose subsystems are integrated with the HSC APCS. Appropriate communication devices are used to connect to a single network and introduce them into a single information space.

In the event of a complete failure of the HSC, each unit is provided with an emergency control panel, through which it is possible to stop the unit without accident.

2.10.4. Complete control systems for the main equipment

The main equipment is completed with monitoring and control systems:

- gas turbine (GT);
- steam turbine with generator;

Auxiliary equipment is completed with monitoring and control systems:

- gas treatment plant;
- gas consumption metering point;
- booster compressor room;
- air compressor;
- micro cooling towers.

According to the instructions of manufacturers, the APCS for HRSGs is designed.



2.10.5. Gas turbine's automated control system (GT ACS)

Power units created on the basis of gas turbines as part of power plants ensure the generation of electrical energy in the required volumes and with normalized parameters.

GT ACS is responsible for:

- automatic start and stop of the turbine in a sparing mode on the selected fuel;
- synchronization with the power system;
- adjustment of the load on the turbines;
- frequency stabilization during primary regulation;
- the ability to manage load shedding and meet plant load requirements;
- implementation of the transition from one type of fuel to another;
- prevention of gas turbine overload;
- compressor overload warning;
- protection and signaling in case of deviation of technological parameters.

The system makes it possible to control the gas turbine unit both from the MCP and from the local control unit. To implement the above-mentioned processes, the GT ACS uses a process control system.

2.10.6. Steam turbine's automated control system (ST ACS)

The special turbine control system (TCS) is designed for steam turbines for the energy industries. The architecture of the system consists of decentralized data collection, centralized data processing, and organization of communication and control of the turbine and generator.

The TCS is designed to work through the AWS human machine interface (HMI) system or software connection with the APCS.

The main turbine control program provides fully automatic control of the turbine during operation in all modes.

2.10.7. Tasks performed by the TCS

Turbine control system (TCS) performs the following tasks: control, protection and automation of gas and steam turbines, includes protection against failures according to the safety integrity level 3 (SIL3) standard.

2.10.8. Hardware

To work with the TCS, an application server and AWS of the steam turbine operator are supplied, with the help of which the monitoring and operation of the turbine is carried out.

The TCS is delivered as a solution intended for use via AWS (HMI) or software communication with a distributed control system (DCS) / APCS.

2.10.9. Turbine control in closed circuit section

Closed-loop turbine controls are implemented by a dedicated PLC. Closed loop control functions (steam speed and pressure, and valve position) are implemented through selection logic on the slave valve position controllers and further to the load amplifiers for the actuators. Electro-hydraulic actuators increase the forces for hydraulic positioning of the control valve.

2.10.10. Means of regulating the turbine in the part of the open circuit

Means of regulating the turbine in terms of an open loop and controlling the auxiliary equipment of the steam turbine generator is carried out by a specialized PLC, which includes:

- processing of measured values;
- generation of alarms and shutdowns;
- management of the lubricating oil system;
- control of the turning mechanism of the gearbox and the condensate system.

2.10.11. Turbine protection

When designing, it is necessary to take into account the requirements for turbine protection in accordance with the Guidelines for the implementation of technological protection schemes for thermal power equipment of thermal power plants (Appendix 41 to the order of the Minister of Energy of the Republic of Kazakhstan dated December 30, 2016 No. 580).

ВНТП 81 Нормы технологического проектирования тепловых электрических станций (VNTP 81 Norms of technological design of thermal power plants).

Rules for the technical operation of power plants and networks, approved by order of the Minister of Energy of the Republic of Kazakhstan dated March 30, 2015 No. 247 (registered in the Register of State Registration of Regulatory Legal Acts under No. 11066).

СН РК 4.02-03-2012 «Системы автоматизации» (SN RK 4.02-03-2012 «Automation systems»).

The primary function of the turbine protection system is to detect turbine disturbances at an early stage and initiate appropriate safe shutdown operations to limit damage, repair costs and downtime for generator sets. It is active all the time and cannot depend on operator intervention. Shutdown protection is triggered by the following violations (the list of protections is not limited to this list):

- overspeed;
- excess pressure in the turbine condenser;
- overpressure, last stage;
- high level of condensate, last stage;
- temperature rise of bearings (plain bearings and thrust bearings);
- low lubricating oil pressure;
- excess temperature of the lubricating oil;
- low lube oil level;
- vibrations (turbine shaft or bearing frame);
- axial displacement of the shaft;
- fire protection (fire protection button) + external fire protection;
- shutdown of the turbine through the shutdown button and external shutdown;
- overpressure, main supply steam and hot superheat;
- increased / decreased frequency of the rotor;
- excess seal steam pressure;
- steam temperature excess on seals;
- shaft or housing vibrations;
- temperature rise of bearings;
- cooling water leak, generator - air cooler;
- temperature rise of the generator stator winding;
- vibration of the generator shaft or housing;
- temperature rise of generator bearings.

2.10.12. Fault-tolerance tools built into the TCS

Fault-tolerance system built into the TCS. The protection system uses a fail-safe PLC, special fail-safe input/output boards, and an independent over-speed protection system, combined with SIL-certified current collectors. Its dedicated output terminals actuate the turbine shutdown circuit (direct operation of the solenoids in the electrical or hydraulic shutdown circuit).

2.10.13. Turbomachinery remote monitoring system

Operating modes Remote monitoring systems for the use of turbomachines.

Data collection and analysis: the TCS system automatically records and saves the technological data of the turbine unit. The stored data will be transferred via a secure data connection to the supplier's back-end system.

Existing network access: HSC software company can set up a «virtual private network» (VPN) to provide a fully secure and encrypted transmission of data from the network over the Internet to the hardware supplier's secure server.

Access via DSL (broadband) connection: The equipment supplier company can set up a secure VPN connection through an independent digital subscriber line in accordance with the goals of the project.

2.10.14. Communication system of the TCS

Input/output modules are installed in the local TCS cabinet or in the input/output cabinet of the TCS and are connected to the TCS via data transfer protocols (The communication protocol will be determined in accordance with the decision made).

All decisions made during the design process must be consistent with the objectives of the project. If it is impossible to use the regulatory documentation in accordance with the requirements of the legislation of the Republic of Kazakhstan, develop special technical conditions within the framework of this project.

The interface for generator protection, synchronization, ARI and turbine sub-distribution is implemented via a data communication protocol. The interface provides full-scale operation and visualization of the generator.

The signal exchange between the low voltage secondary distribution system (control of the energy consumer by the steam turbine generator) and the TCS is carried out using remote inputs/outputs and a data communication protocol connection. Depending on the scope of supply, the remote input/output modules are installed in the secondary direct current distribution system or in the central cabinet of the TCS.

The connection between the TCS and the APCS is implemented using the OPC protocol (linking and embedding objects for process control) via the Ethernet bus connection of the TCP/IP protocol. Data transfer is carried out in accordance with the principle server – client. Security software is included in the TCS to protect data. In addition, a separate hardware firewall is included to provide authentication and secure communication over the network.

The physical connections between the OPC server, hardware firewall and DCS are implemented as an Ethernet connection over TCP/IP protocol as part of a copper bus channel. Alternatively, the OPC connection can be implemented as a backup. This connection includes the following:

- transmission of measured values for the control of the steam turbine generator;
- operation of the turbine (controller selection and setpoint adjustment) during normal operation;
- determination of the volume of the signal;
- interface definition (physical, protocol).

2.10.15. Automation of emission (flue gas) monitoring systems

When designing an automated monitoring system (AMS) for emissions (outgoing gases), the following should be taken into account:

Code of the Republic of Kazakhstan dated January 2, 2021 No. 400-VI «Environmental Code of the Republic of Kazakhstan»;

Order of the Minister of Ecology, Geology and Natural Resources of the Republic of Kazakhstan dated June 22, 2021 No. 208. On approval of the Rules for maintaining an automated system for monitoring emissions into the environment during industrial environmental control.

As automatic gas analysis systems for flue gases, stationary gas analysis systems are provided, which are supplied on a turnkey basis by specialized organizations.

The environmental monitoring system consists of the following gas analysis complexes:

1. Complexes that perform the tasks of optimizing the combustion mode of fuel in the boiler furnace and conducting continuous environmental monitoring of harmful emissions into the atmosphere with outgoing flue gases. For each boiler, a separate gas analysis complex is provided, including:

- stationary gas analysis complex for measurement in each flue of the boiler;
- measuring system of flue gas volumetric flow;
- gas analyzer of oxygen in flue gases at the outlet of the boiler furnace;
- calculation of the flow of flue gases; calculation of the amount of nitrogen oxides NO_x;

AWS of the flue gas analytical system, common at the control room.

Sampling connection for each boiler is provided to the main module of the gas analysis complex, which is installed in the gas analysis cabinet located either in the container or in the main building. From

the gas analytical complexes, the signals are transmitted to the automated workstation of the flue gas analytical system of the boilers, installed on the BCP.

2. The complex, which performs the tasks of continuous environmental monitoring of harmful emissions from chimneys, includes for each chimney:

- gas analytical complex;
- humidity measuring device;
- mass gas flow meter;
- flue gas temperature meter;
- flue gas pressure meter;
- controller cabinet;
- block-modular building;
- common chimney equipment:
- meteostation;
- ecologist's AWS.

The gas analysis system performs the following functions:

- issuance of reports on emissions of harmful substances using a data collection and processing system;
- performing calculations (hourly, monthly, quarterly, annual) of flue gas emissions into the atmosphere;
- transfer of data on the results of measurements from the AWS of the gas analytical system to the HSC of the APCs of the corresponding boiler;
- transmission of data from the gas analysis system to remote computers of the enterprise (laboratory of environmental protection).

It is envisaged that periodic monitoring of emissions by the environmental protection laboratory using portable gas analyzers is envisaged.

2.10.16. Location and requirements for panel rooms

The rooms of the central (main), block and group control panels, as well as rooms for computer equipment, are made with soundproofing and air conditioning. There are two exits from the shield premises.

Premises of the central (main), block and group control panels with permanently present personnel, it is recommended to take the permissible sound pressure levels and sound level at permanent workplaces and at control and control panels in accordance with CH PK 2.04-02-2011 (SN RK 2.04-02-2011).

The ceiling of the shield room must be waterproofed.

The height of the central part of the room (MCP, BCP, CP and GCP), in which the operational circuit is located, is assumed to be 4 m in accordance with ВНТП 81 Нормы технологического проектирования тепловых электрических станций (VNTP 81 of the Norms for the Technological Design of Thermal Power Plants).

The interior of the shield is made according to a special project. In the case of installing a relay or other equipment of the control system outside the BCP in separate isolated rooms, the latter are ventilated. When designing heating and ventilation for switchboard rooms, it is recommended to be guided by CH PK 4.02-01-2011 (SN RK 4.02-01-2011) and ЦИ ПК 4.02-101-2012 Отопление, вентиляция и кондиционирование воздуха (SP RK 4.02-101-2012 Heating, ventilation and air conditioning).

2.10.17. Power supply of devices and automation equipment

When designing, it is necessary to be guided by, but not limited to, the following standards:

Order of the Minister of Energy of the Republic of Kazakhstan dated March 20, 2015 No. 230 «On approval of the Rules for the installation of electrical installations» ;

СП РК 4.04-108-2014 «Проектирование электроснабжения промышленных предприятий» (SP RK 4.04-108-2014 in English - Design of power supply for industrial enterprises).

The power supply of the control system must be carried out according to consumer groups:

- technological protection and their sensors;

- remote control and blocking devices;
- technological control devices and their sensors;
- emergency warning signaling devices;
- fire detection and extinguishing systems;
- means of autoregulation;
- computer facilities and their sensors.

Consumers of all groups, except for computer equipment, should be divided into subgroups according to the technological principle: for the boiler and turbine departments. Distribution into subgroups, groups should be carried out through independent protection devices that provide selective shutdown of damaged areas and repair of power supply network elements without stopping the main equipment.

The power supply of all devices of the hardware and software complex (HSC) must be provided from their own power sources (modules).

The primary power sources of the HSC can be two independent networks, each of which is a three-phase AC network 380/220 V, frequency 50 ± 1 Hz. Characteristics of primary power supply networks:

- rated linear voltage - 380 V (+10, -15%);
- rated phase voltage - 220 V (+10, ~15%);
- number of phases - 3.

The primary power sources of the HSC can also be two independent networks, one of which is a three-phase alternating current network with a voltage of 380/220 V, a frequency of (50 ± 1) Hz, and the other is a direct current network with a voltage of 220 V. Power supplies for devices of the HSC lower level (for example, controller cabinets) can be designed to receive power from two independent networks (according to one of the above options), or from only one network. In this case, the devices of the lower level of the HSC should be powered by uninterruptible power supplies (UPS). The UPS must be powered from two independent networks (two alternating current networks or one alternating current network, the other direct current). It is preferable to include UPS in the delivery of HSC.

Technical means must remain operational when:

- independent or simultaneous changes in the voltage of the alternating current and direct current networks by $\pm 25\%$ for up to 100 ms when the HSC is powered from the alternating current and direct current mains;
- during long interruptions in the power supply in one of the alternating current or direct current networks when the HSC is powered from the alternating current and direct current networks;
- during long interruptions in the power supply in one of the two alternating current networks when the HSC is powered from two alternating current networks;
- during simultaneous interruptions of power supply with a duration of not more than 20 ms in two networks.

The HSC devices must be protected against reverse polarity direct current voltage supply. The HSC devices should not be damaged or falsely triggered when one of the two primary power supply networks is connected and (or) disconnected.

The power supply of the HSC devices, which implement the functions of technological protection, must be carried out with the highest reliability from an alternating current source with a voltage of 380/220 V, a frequency of (50 ± 1) Hz, with a battery backup. The operability of devices that implement the functions of technological protection must be ensured in the presence of a voltage of the specified quality at least on one of the two inputs, as well as in case of short-term (up to 5.0 s) deviations of the supply voltage within (+15, -30%) and frequency up to ± 5 Hz. In case of ARI of the supply voltage with a voltage loss for a time of at least 0.5 s, false alarms of the protections should not occur.

The power supply of duplicated HSC devices must be provided from independent sources.

The power supply of technical means of the upper level of the HSC, sufficient for a trouble-free shutdown of the equipment, must be provided from uninterruptible power supplies (UPS) with internal battery support (up to 30 minutes). These UPSs can be powered from a 220 V alternating current source, 50 Hz uninterruptible power supply (via ARI), or from two independent alternating current 220 V, 50 Hz networks. Uninterruptible power supply devices should be included in the scope of delivery of the HSC. The UPS power supply of operator stations included in one AWS and reserving each other must

be provided from independent sources of primary power supply. In this case, it is allowed to power the UPS from one source of primary power supply with a voltage of 220 V, a frequency of 50 Hz.

2.10.18. Subsystem of common time

The hardware and software complex (HSC) should include a unified time subsystem designed to synchronize the timers of all computing facilities of the complex. The unified time subsystem must be adjusted according to the exact time signals (for example, from the power system). Software and hardware complexes must have hardware and software that ensure the reception of signals from standard devices that generate accurate time signals based on signals from an external source (for example, from a radio network). These standard timekeeping devices should be included in the scope of delivery of the HSC upon request of the Customer. The unified time subsystem should provide automatic synchronization of the timers of all HSC devices, including the timers of intelligent devices for communication with an object (DCO).

The error of binding the system time of the HSC as part of the local APCS of an outdoor/indoor switchgear (SG) to astronomical time must be consistent with in accordance with the goals of the project, but not more than ± 0.5 ms.

When designing, it is necessary to take into account the requirements of ГОСТ 34045-2017 «Электроэнергетические системы. Оперативно-диспетчерское управление. Автоматическое противоаварийное управление режимами энергосистем. Противоаварийная автоматика энергосистем. Нормы и требования» (GOST 34045-2017 «Electric power systems. Operational dispatch control. Automatic emergency control of power system modes. Emergency automatics of power systems. Norms and requirements»).

2.10.19. Saving information in case of accidents

The function «Registration of emergency situations» (RES) should be provided for the accumulation and presentation on screens and (or) printing of data on the process of occurrence, development and elimination of emergency situations with reference to astronomical time with an accuracy of 1 ms. The RES function should ensure the registration of, as a rule, reliable technological data for the period preceding the accident and after its occurrence, on the operation of the main and auxiliary equipment, the operation of protections, interlocks, automatic control devices and personnel.

All information involved in the RES is conditionally divided into three groups:

- A - analog and discrete signals characterizing the state of control objects, the registration cycle of which corresponds to the cycle of updating information on monitor screens;
- B - analog and discrete signals characterizing the state of control objects that require registration with high resolution;
- C - initiative signals for the operation of electrical and technological protections, event signals associated with the impact of personnel on control objects, which are also subject to the actions of technological and electrical protections, event signals associated with the output of analog parameters beyond the alarm settings or operation of protections, and also the values of analog parameters for electrical equipment that require registration with super high resolution.

It is necessary to have recorders of different time characteristics for: - thermal power equipment; — electrical equipment; - electrical equipment associated with the receipt of signals from AEA.

It is advisable to have several (the number corresponds to the purpose of the project) independent registrars of emergency situations (ES) in the HSC for units and assemblies. The recorders can work simultaneously and independently and record arbitrary sets of parameters specified at the stages of development and operation of the system.

The following enlarged RES algorithm should be provided. Prior to the arrival of a signal about the beginning of an emergency, there should be a constant sliding accumulation of data in the pre-emergency interval. When an initiating signal characterizing an accident appears, all the accumulated data are «frozen» and the process of registering the development and localization of the accident in the post-accident interval begins.

The RES function should not take into account the real states of the protections (enabled, enabled in the information mode - signaled). Recorders in the mode of post-emergency recording of information should not be switched if the triggered protection is signaled.

Handwritten signature and initials.

The imitation of the beginning of the ES is provided to check the functionality of the function when testing the equipment.

Information about the accident is recorded in the archive.

2.10.20. Safety system

To ensure fire and explosion safety, there must be:

- measures were taken to minimize the risk of fires and an effective system for their detection and extinguishing was organized.
- ACS requirements have been established and implemented to avoid hazardous conditions (in terms of rotational speeds, temperatures, vibrations, etc.).
- an alarm system is organized to alert about dangerous, unsafe conditions and an emergency shutdown of the GT to protect personnel and (or) equipment.
- the placement of units and systems in accordance with a certain category of production in separate compartments with equipment for the corresponding purpose was carried out;
- door closers and a system for monitoring the open position of doors in the GT shelter were installed.

The requirements for explosion and fire safety in accordance with ГОСТ 12.1.010 (GOST 12.1.010) and ГОСТ 12.1.004 (GOST 12.1.004) must be met

The electrical equipment of the GT for electrical safety must comply with ГОСТ 12.1.019 (GOST 12.1.019), ГОСТ 12.1.030 (GOST 12.1.030), ГОСТ 12.1.038 (GOST 12.1.038) and ГОСТ Р 51330.0 (GOST R 51330.0).

The fire protection system shall include, as a minimum, a fire extinguishing system, fire detection and combustible gas detection systems.

When a fire or the presence of fuel gas in the air is detected, the emergency protection system should automatically turn off the GT by direct action on the shut-off fuel valve.

The fire extinguishing system must be switched on automatically. Switching on is duplicated manually by buttons located on the outside of both sides of the casing of the units controlled by the fire alarm.

Do not use halogens or other ozone-destroying gases for extinguishing.

The inlets and outlets in the ventilation and sealing system must be equipped with valves. Their closure during fires keeps the medium supplied for extinguishing inside the shelter.

If the GT is installed under the shell and operates on gaseous fuel or in gas hazardous areas, the sequence of automatically performed operations at start-up includes purging and checking the safety (presence of gas) under the shell. Only after purging and signaling that there is no fuel gas under the casing, electrical equipment located in the hazardous area can be energized. In the presence of continuous purges and control of at least 10 air exchanges per minute, the explosive category of the volume under the skin can be reduced by 1.

2.10.21. Cyber security system from unauthorized access

To protect information from unauthorized access, it is necessary to take into account СТ РК ИСО/МЭК 13335-1-2008 «Информационная технология. Методы и средства обеспечения безопасности. Управление защитой информационных и коммуникационных технологий. Часть 1. Общие понятия и модели для управления защитой информационных и коммуникационных технологий» (ST RK ISO/IEC 13335-1-2008 «Information technology. Methods and means of ensuring security. Management of protection of information and communication technologies. Part 1. General concepts and models for security management of information and communication technologies»).

To protect the information support of the HSC (as well as the APCS as a whole), the following principles should be laid down:

as a rule, single entry and multiple use of information within the system;
protection from inaccurate and unauthorized information, as well as protection of individual users from redundant information;

noise-immune coding and protection against destruction and unauthorized access.

The Facility's control systems must be designed with adequate software and hardware protection against the threats of cyber attacks and industrial sabotage. Control systems in this case refer to electronic

or microprocessor systems that use software-based programs for their main operation, including, but not limited to, APCS, PLCs, and proprietary control systems.

Attacks are often carried out through malicious software and firmware (malware) introduced in a variety of ways, including:

- Direct download through infected equipment introduced into the system.
- On the Internet.
- Via the network to which the control system is connected.

Control systems should include measures to detect and protect against malware introduced by any means, including the above, to prevent system failures, corruption of data stored in systems, and subsequent damage to the installation and physical harm to personnel through infected elements of the installation. Particular attention should be paid to preventing the introduction of malware through third-party systems.

The protection provided must be an integral part of the system design and conform to international standards, best practices and manufacturing procedures and policies, where applicable.

Security methods should be enabled, including anti-virus software, industrial firewalls, and zoning within the system to limit communications to a minimum. Continuous management and continuous improvement must be applied to ensure ongoing protection, with threat awareness and response being important factors.

Malware cyber protection systems must fully comply with all applicable international standards and guidelines, including ISO/IEC 27001, 27002, ST RK ISO/IEC 27001-2015, ST RK ISO/IEC 27002-2015.

Personnel must be trained in best practices for the operation and maintenance of systems designed to prevent the introduction of malware, including awareness of any health and safety risks associated with industrial sabotage.

The design of the technical means installed openly in the engine room must provide protection against unauthorized interference in their work by unauthorized persons.

2.10.22. Emergency stop system

Technological protections form control actions on the executive bodies (actuators and switching devices) in order to protect personnel, prevent damage to equipment and localize the consequences of accidents. Protections have the highest priority in relation to other discrete actions.

The shutdown of the active action of the protections and their output «on/off the signal» are carried out by the maintenance personnel with the approval of the management by non-operational means separately for each protection.

In the event of simultaneous operation of protections that cause a different degree of unloading of the equipment, priority is given to protection of a stronger action (with deeper unloading).

The action of the protections is maintained for a time sufficient to perform the longest control operation.

Protections are entered (and removed) automatically when the specified conditions appear (disappear). After entering the protection work continuously. The fact of input/output protection is signaled and logged. Information on protections that have not been put into operation is also logged and issued for signaling. An authorized manual input-output of protections is provided at the request of operational personnel, and separately for each protection, with the obligatory fixation of this event in the archive.

The action of the protections is unilateral: changing the composition of the operating equipment, the state of the control objects and putting the equipment into operation after it is turned off by the protection is carried out by the operating personnel after the elimination of the causes that caused the protection to operate.

The operation of the protections is accompanied by light and sound alarms and the activation of RES. Protections operating with a time delay of more than 5 s have an indication of the start of work. HSC provides an opportunity to test each of the protections on the stopped equipment. Changes in the technological protection settings are made by service personnel with access authorization, recorded in the archive and accompanied by mandatory logging.



The time for the protection command to reach the executive body does not exceed 0.02 s. The duration of the impulse command varies discretely within 1-2s with a step of 0.1s. Technological locks provide:

- automatic actions (reactions) of the system caused by deviations in the course of the technological process. A special case of such reactions is ARI.
- prohibition of unacceptable actions of the operator during the technological process.

The appointment of working and backup mechanisms, the commissioning and shutdown of the ARI is carried out by the process operator with automatic logging of his actions.

2.10.23. Explosion protection system

For the safe operation of the power plant equipment, a fire and gas detection system (F&G) is provided, which is equipped with means of communication with the APCS.

The choice of instrumentation is made taking into account the class of explosive zones of buildings and premises. Explosion-proof devices are installed in buildings and structures classified as explosive.

All sites and buildings (containers) where the formation of a gaseous mixture is possible are equipped with explosive concentration sensors. Installation of gas detectors is provided for:

- inside the turbine compartments of the GT;
- in the block-boxes of the gas treatment plant;
- in booster compressor containers.

Logic decision devices are provided as part of programmable logic commands of devices (PLC).

The F&G system (fire and gas alarm) is designed to initiate executive actions, the F&G system is equipped with means for an independent executive action, such as the release of fire extinguishing / fire suppression equipment, blocking of the HVAC systems, turning on fire pumps, opening deluge valves, etc.

The P&G system has means for transmitting data to the APCS so that the operator always has at his disposal current information about the state on the territory of the CCGT. The positions of the P&G detectors are displayed on the APCS graphs with dynamic state updates. The power supply of the P&G system is provided from the system 230 V, 50 Hz. In the event of a 220V power failure, the power supply automatically switches to 24V battery backup via the converter. All I&A equipment that is energized must be grounded in accordance with the requirements of the Rules of Electrical Installation of the Republic of Kazakhstan.

2.10.24. Measuring devices

All instrumentation as units of measurement in the territories of the Republic of Kazakhstan must use units of the International System of Units (SI) and units of measurement approved for use by national legislation in the field of ensuring the uniformity of measurements based on ISO 31:1992, ISO 1000:1992, ГОСТ 30957.1-2002 (GOST standards 30957.1-2002), ГОСТ 30957.2-2002 (GOST 30957.2-2002).

All control and measuring instruments must be certified by authorized state bodies in accordance with the Law of the Republic of Kazakhstan dated June 7, 2000 No. 53-II «On Ensuring the Uniformity of Measurements» and have:

- certificates of approval of the type of measuring instruments in accordance with СТ РК 2.21-2019 Порядок проведения испытаний и утверждения типа средств измерений (ST RK 2.21-2019 Procedure for testing and approval of the type of measuring instruments);

- certificates of approval of the type of information-measuring system СТ РК 2.81-2004 «Государственная система обеспечения единства измерений. Системы информационно-измерительные. Порядок проведения испытаний для целей утверждения типа» (ST RK 2.81-2004 «State system for ensuring the uniformity of measurements. Information-measuring systems. The procedure for conducting tests for the purposes of type approval»);

- certificates of approval of the type of measuring controllers in accordance with СТ РК 2.88-2005 «Государственная система обеспечения единства измерений Республики Казахстан. Контроллеры измерительные. Типовая программа испытаний для целей утверждения типа» (ST

RK 2.88-2005 «State system for ensuring the uniformity of measurements of the Republic of Kazakhstan. Measuring controllers. Standard test program for type approval purposes»)

- certificates of metrological certification of measuring instruments in accordance with СТ РК 2.30-2019 Порядок проведения метрологической аттестации средств измерений (ST RK 2.30-2019 Procedure for metrological certification of measuring instruments).

- certificate of metrological certification of information and measuring systems in accordance with СТ РК 2.82-2004 «Государственная система обеспечения единства измерений. Системы информационно-измерительные. Порядок проведения метрологической аттестации» (ST RK 2.82-2004 «State system for ensuring the uniformity of measurements. Information-measuring systems. The procedure for metrological certification»

The Contractor shall ensure that the control and measuring equipment, its installation, commissioning and operation are designed and manufactured in accordance with applicable and relevant internationally recognized standards and norms, the standards of the Republic of Kazakhstan. The list of standards that should be applied to the systems and equipment of APCS is given in the section «Standards and norms».

Contradictions, gaps, conflicts between the standards, norms and laws of the Republic of Kazakhstan and international standards and/or intentions of the Contractor must be brought to the attention of the Customer for resolution/approval. (Contractor's intentions must not conflict with the Project objectives).

In particular, I&A design will be required to be carried out in full compliance with relevant international standards, laws, regulations, codes and statutory guidelines in relation to health, safety and the environment. Control systems and instrumentation for safety use throughout the plant shall comply with the requirements of IEC 61508-1:2010, IEC 61508-2:2010, IEC 61508-3:2010, IEC 61511 and ГОСТ Р МЭК 61508-2-2007 (GOST R IEC 61508-2-2007), ГОСТ Р МЭК 61508-3-2012 (GOST R IEC 61508-3-2012), ГОСТ Р МЭК 61508-4-2012 (GOST R IEC 61508-4-2012), ГОСТ Р МЭК 61508-5-2012 (GOST R IEC 61508-5-2012), ГОСТ Р МЭК 61508-6-2007 (GOST R IEC 61508-6-2007), ГОСТ Р МЭК 61508-7-2012 (GOST R IEC 61508-7-2012), ГОСТ Р МЭК 61511-1-2018 (GOST R IEC 61511-1-2018), ГОСТ Р МЭК 61511-2-2011 (GOST R IEC 61511-2-2011).

2.10.25. Flow measurements

Flow meters operating on the principles of differential pressure must be designed in accordance with the requirements: ISO 5167, СТ РК 2.69-2004 (ST RK 2.69-2004), СТ РК 2.131-2013 (ST RK 2.131-2013), ГОСТ 8.586.1-2005 (GOST 8.586.1-2005), ГОСТ 8.586.2-2005 (GOST 8.586.2-2005), ГОСТ 8.586.3-2005 (GOST 8.586.3-2005), ГОСТ 8.586.4-2005 (GOST 8.586.4-2005), ГОСТ 8.586.5-2005 (GOST 8.586.5-2005),

- turbine flow meters - in accordance with ISO 9951, СТ РК 2547-2014 (ST RK 2547-2014),
- ultrasonic flow meters - in accordance with ISO 17089, СТ РК 2549-2014 (ST RK 2549-2014).

For high pressure applications above 25 barg. The primary device holder must be made of a material similar to that of the pipeline in which it is to be installed and must be butt-welded to the pipeline. In other cases, flange connections should be used whenever possible. For line sizes less than 50mm, a precision pipe assembly with a minimum line size of 50mm should be used. All differential pressure devices must be fitted with draw-off shut-off valves and drain valves or two-way valve manifolds, if required. All differential pressure devices must be supplied with individual calibration data.

Multiple measurements of the same flow may be taken from the same orifice or venturi, but must be made through separate sampling points, parallel connections from the same sampling point are not allowed.

The flowmeters must in all cases comply with the process conditions applicable to the place of measurement and the environment in which they are installed. Meters used for energy, tariff, or metering purposes must be supplied with adjacent piping where necessary, with a calibration certificate from an approved laboratory. Piston flowmeters must be accurate within 0.25%.

If density, temperature, or pressure corrections are required for tariff or capacity control signals, then such measurements must be made near the location of the flowmeter.

Flow measurements of compressible liquids (steam and gas) at variable temperature and pressure must be provided by automatic correction of the flow signal in the control system. Appropriate temperature and pressure measurements must be provided for this purpose.

Flowmeters using alternative measurement principles may be used for the respective applications subject to the Customer's approval.

2.10.26. Level measurements

Level measurement devices can be of direct, differential pressure, or electrical/electronic type, depending on the application. Direct measuring instruments shall be used for local level indication. It shall be possible to dismantle the equipment for level measurement without decommissioning the vessel or other instruments. Level gauges must be designed in accordance with the requirements of ГOCT 28725-90 (GOST 28725-90).

Gauges shall be of the reflex type, constructed of a stainless steel stem, fitted with tempered borosilicate glass, and shall be marked with safe working pressure and temperature, except for low temperature and low pressure applications where clear types may be used.

Gauge glasses for boilers must comply, at a minimum, with accepted norms for boilers. Each pressure gauge must be fitted with top and bottom shutoff valves with a full bore drain valve at the bottom and a plugged vent at the top. Condensation chambers should be used at the top connections of all steam/water level systems. They must be the correct size, securely placed, and the entire installation must be compliant. The sensors must be positioned so that the apparent length exceeds the maximum operating range.

Transmitting guided wave radar transmitters should preferably be used in all but liquid services.

Piston type instruments and switches should be installed in external cages with flange connections rated the same as the vessel. This type of instrument should not be used for applications involving viscous, corrosive or effervescent liquids.

Float switches must be glandless with a magnetic coupling. Internal float devices may only be used in non-critical conditions on atmospheric ships or vessels that can be taken out of service without shutting down the station. Switches installed in chambers located outside the main vessel must be fitted with suitable valves to allow the chamber to be periodically purged to remove sediment.

Direct level measurement with internally mounted floats, etc., should only be used if the switch point is either well defined in advance or if it is adjustable during operation, if the vessel can be emptied and/or depressurized to remove the switch, not affecting the normal operation of the installation, or where there is a possibility of clogging of long connections.

2.10.27. Pressure measurement

Pressure measuring devices must be designed and installed in accordance with applicable international standards and manufacturer's recommendations. As a general rule, local pressure gauges should operate on the Bourdon tube principle and should comply with BS EN 837 or other applicable international standards. Low pressure gauges may use gauges or barometers to indicate a small gauge, differential pressure, or vacuum.

Pressure transducers must be of the capacitive or piezo type, and the sensing element materials must be selected to be resistant to corrosion by the process media. If necessary, the pressure transducers must be compensated by the static liquid head between the device and the sampling point.

In addition to the basic equipment of GT and ST, where pressure gauges or transmitters are used in systems containing viscous liquids or corrosive liquids, a diaphragm must be installed at the process tap and the process pipeline filled with glycerin. Suitable valves must be installed to service the system.

When pressure gauges or transmitters are to be used for applications where the process temperature exceeds 70°C, siphon loops should be used.

Gauges measuring low ranges that can be exceeded at higher containment pressures must be fitted with overrange protection. Pressure gauges must be equipped with bleed/test valves. For high pressure applications, purge devices must be provided. Gauges must be filled with glycerine to dampen any vibrations transmitted in the process.



The pressure switches shall be such that, under normal process operating conditions, the contact in the switch is closed, unless otherwise approved by the Customer. Switches should have adjustable setpoints and adjustable hysteresis if required.

Pressure connections below 40 bar must be equipped with at least one local manual shut-off valve. Double manual shut-off valves above 40 bar must be provided. Manifold valves are not acceptable as check valves.

All ancillary equipment such as containment chambers, dampening means, purge equipment and piping must be provided.

Differential pressure devices must be capable of withstanding the maximum pressure rating acting on one side only.

2.10.28. Temperature measurements

The temperature measurement method used must be selected for the specific application, taking into account the requirements for accuracy and reliability.

Sensitive elements for measuring the temperature of air, inert gases and radiation must be equipped with an appropriate protective sheath. For steam, water and hazardous applications, thermowells must be used. If temperature sensors are to be installed in jacketed pipes and apparatuses, extensions must be fitted with protective covers to allow access to the sensor's connection head and outside the jacket. The connecting heads of the sensitive elements must be located at an ambient temperature not exceeding 50°C and be waterproof.

The thermowells must be suitable for the pipeline in which they are installed and for the physical conditions of the parameter to be measured at the place of installation. Thermowells for steam and feed water must be weldable in place. Test thermowells shall be provided with a cover.

Resistance thermometer detectors (RTDs) should only be used if they can be easily replaced without loss of plant performance. RTDs must comply with IEC 60751, ГOCT 13384-93 (GOST 13384-93) using three-wire connections for applications up to 450°C. Resistive sensors shall be single element or dual element as specified in accordance with the Customer's requirements, Part 4, Tab 34 - Mechanical Diagrams. Resistive sensors must be fitted with waterproof heads with terminal blocks designed to prevent the ingress of all foreign matter and liquids. When measuring temperature difference, use an RTD in accordance with IEC 60751 Grade I, ГOCT 6651-2009 (GOST 6651-2009), ГOCT 13384-93 (GOST 13384-93).

Locally installed signal converters must be provided for all RTD and thermocouple measuring devices. Preferably, they are located at the head of the measuring device or, alternatively, in a local junction box, if required by local environmental conditions. The transmitters must provide a 4-20 mA linear output for further transmission to the control system or monitoring equipment. If necessary, linearization and cold junction compensation must be enabled.

Thermocouples must be used in accordance with the requirements of IEC 60584, ГOCT 6616-94 (GOST 6616-94), ГOCT 6651-2009 (GOST 6651-2009). Special attention must be paid to the design and installation of thermocouple systems to ensure that inaccuracies are not introduced into measurements due to the use of unsuitable materials or incorrect installation. Cold junction compensation must be provided, as well as compensation cables of appropriate size and material where necessary. In addition to GT and ST, thermocouples must be installed on the basic equipment in such a way that they can be replaced during the operation of the plant.

Local temperature indicators should be provided with a thermowell and should preferably be bimetal and polygonal type.

Standard OEM package equipment can have different types of thermocouples according to the application requirements of the equipment.

All resistance thermometers and thermocouples must be monitored for failure and such failures must be signaled through the APCS.

No temperature measurement system should use mercury as a sensitive medium.

Sensing elements should be located in places that are easily accessible for maintenance. Where it is safe and practicable, it should be possible to remove elements while the plant is in operation.

Temperature switches may be used if the application dictates such a need and be such that, under normal process operating conditions, the contacts in the switch must be closed, unless otherwise



approved by the Customer. Please note that in most cases it is expected that the temperature status will be measured by a thermocouple or resistance thermometer detectors connected to the APCS and the alarm/shutdown levels will be determined electronically.

2.10.29. Position measurement

Position measuring devices or switches (e.g. limit switches, proximity switches, etc.) must either be located away from sources of heat or process gases or liquids, or must be appropriately classified for this environment. Where vibration-free locations are not possible, the mounting device must provide flexibility in the connection between the device, the measuring point and the transmitter mountings.

Position switches associated with the GT control system must have two-pole single-position contacts. All other position switches shall have, as a minimum, isolated two-pole, two-position contacts and be such that, under normal process operating conditions, the contacts in the switch will close, unless otherwise approved by the Customer. If necessary, the switches should have adjustable settings.

For encoders in high temperature environments, calibration of encoders and setting of limit switches should not be affected by high/low operating temperatures during startup and shutdown.

As a general rule, the switch contacts must be fully protected from adverse environmental influences, such as dust, moisture, corrosive materials, etc., and the contact closure must be instantaneous by clicking. Switches must be of a robust design to withstand vibration and shock due to the movement of actuators, etc., and wire-break monitoring must be provided.

As an alternative to limit switches, solid-state proximity switches are acceptable.

In addition to the GT and ST on the base equipment, valves and dampers with position feedback should be used for position feedback RVDTs (Rotary Variable Differential Transformers) or LVDTs (Linear Variable Differential Transformers). It must be possible to adjust the 0 percent valve or damper position and adjust the RVDT/LVDT range so that a 100 percent RVDT/LVDT position corresponds to a 100 percent valve or damper position. The RVDT/LVDT output must be 4-20mA or a digital signal for 0-100% travel to connect to a control system.

2.10.30. Quality measurement

Quality instruments that monitor the chemical or physical properties of process fluids and substances can either be installed directly in the process line or vessel, or use a sampling system. Sensors should be of robust construction and not require frequent maintenance or recalibration. If the sensors require the use of calibration solutions or gases, a minimum of one year's supply must be maintained under normal use. If sensors need to be replaced or removed for recalibration or upgrade at regular intervals, sufficient replacement sensors should be supplied for one year of normal use.

Quality measuring instruments should be selected for continuous, infrequent or continuously discontinuous operation as appropriate for the measurement and intended application.

Analyzers should be selected with regard to the accuracy, reliability and maintenance frequency of the individual devices. Wherever possible, analyzers should use a sensor for on-line analysis.

Sampling systems should include, as appropriate, the sensors, valves, filters, coolers, pressure regulators, coalescers, flowmeters, sampling devices, pipe fittings, and pumps necessary to provide analyzers with a representative and properly conditioned sample. The time delay between taking a sample from a process and making the result available to the analyzer should be appropriate for the measurement and control requirements of the particular process. The preferred method of keeping sampling delays within acceptable limits is to place the analyzer close to the sampling point. If this is not practicable, the lag of the sampling system from the speed of the distance should be reduced by increasing the speed of the sample. The sample must be returned to the process or disposed of in an economical and safe manner.

Where practicable, prefabricated analyzer racks should be provided and should include all necessary sample coolers, analyzers, air, electrical and cooling water distribution systems, and pre-assembled accessories. Such devices should allow testing of the entire assembly prior to shipment to the site and should be capable of being conveniently disconnected from real-time operation to facilitate scheduled maintenance and calibration.

Arrangements should also include the collection of used samples and their safe disposal at appropriate disposal points. Emissions of flammable vapors, including sampling lines, must be vented

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to an atmospheric valve independent of any process valve. The vent must be installed at a safe height and in a suitable location.

The analyzers must be installed in an air conditioning area and are subject to customer approval.

2.10.31. Vibration measurements

The vibration control and protection system of GT should be based on seismic and non-contact sensors according to the manufacturer's standard.

All other vibration measurements should be performed on all large rotating or reciprocating machines for protection and condition monitoring. Processing equipment must be provided to power, detect and display vibration data for each measurement point and must be available to the APCS. Each measurement must be accompanied by an appropriate alarm if high levels of vibration could cause damage or affect the safety of the installation.

Vibration monitoring systems must comply with the requirements of ISO 7919 and ISO 10816, СТ РК ГОСТ Р ИСО 13373-1-2011 (ST RK GOST R ISO 13373-1-2011), ГОСТ ISO 20806-2013 (GOST ISO 20806-2013). Where possible, standardization and interchangeability of components should be implemented.

Vibration measuring devices should be of the non-contact transducer type and should be designed and selected to operate under the specific intended conditions. Vibration sensors should be located on or near each machine bearing, mounted in threaded holes in the bearing housing or on a rigid bracket.

Relative vibration should be measured with non-contact sensors (non-contact type) to measure the relative motion between the shaft and the bearing housing. To measure the relative vibration of a shaft, two sensors mounted radially in the same transverse plane, perpendicular to the shaft axis and at an angle of 90° to each other, should be used. A key phase shifter (sensor of one pulse per revolution) must also be provided for each shaft.

Absolute vibration should be measured with non-contact sensors measuring relative shaft vibration in combination with a seismic sensor (velocity sensor or accelerometer) measuring support vibration. Both transducers must be mounted close to each other so that they make the same absolute movement in the measurement direction. Their conditioned outputs must be vector summed to provide a measurement of the absolute motion of the shaft.

Measurement of both relative and absolute shaft vibration must be made on the board to adequately cover the frequency spectrum of the machine.

Vibration switches must be sensitive to low-frequency vibrations associated with unbalance in the mass of machines. Switches must be insensitive to high frequency vibration, external shocks, etc. acting on the machine. The switch must respond to vibrations from any direction. Switch action must be latched and must be reset manually or, if indicated, electrically by means of a solenoid. The setting must be adjustable. Switches should be set to allow easy access for reset and calibration.

2.10.32. Drives

Valves and dampers that must operate during start-up, shutdown and normal operation of the plant must be actuated.

Actuators for control valves and dampers should normally be of the pneumatic, electric or hydraulic type. Actuators for ON/OFF, shutdown or manual positioning of units should normally be electrical. Other types of drives for specific applications in accordance with safety requirements or other technological requirements must be approved by the Customer.

The failure response of all actuators in the event of loss of the prime mover (air pressure, oil pressure, power supply) must be in a safe/predetermined position, which must be indicated on the piping and instrumentation diagrams (piping diagram), schedule valve, etc. d.

If the actuator is required to control position, an actuator position feedback and indication must be provided that is independent of the position request signal. The open and closed positions of all other valves controlled by the actuator must be controlled through the APCS for indication to the CCR.

Where the control requirement requires split range or non-linear operation, the split or non-linear range calculation must be performed external to the drive in such a way that standard drive input signals are supported.



In addition to the basic GT and ST equipment, all actuators must have a manual control device (to operate the valve in the event of a failure of the prime mover) and a mechanically operated indicator showing the true position of the valve.

2.10.33. Control drives

Control valves must be supplied with appropriate actuators that meet the operational and environmental requirements of the plant. All actuators must be provided with local controls that can be used without the use of tools. The operator must be warned of the loss of remote/automatic control of the actuator valve.

All control drives must have local and remote (in the CCR via APCS) indication of the actual position of the valve. For remote control or indication purposes, an appropriate position transmitter must be provided (eg based on a differential transducer with a linear variable characteristic). Where it is necessary to provide only a limited number of discrete position indications, position switches should be used. Switch settings must be adjustable.

All drives must be designed for heavy-duty operation and withstand the maximum drive capability. Actuators must be equipped with stroke and torque limiting switches for blocking and control functions, as well as to provide remote indication to the APCS. Actuators may have electronic torque control instead of torque switches. Actuators that will not be damaged when full power is applied in a permanently locked position do not require torque switches.

2.10.34. Electric actuators

Valves with electric actuator shall consist of self-contained actuator units with motor, gearbox, contactor, controls, limit/torque switches and all ancillary equipment installed in each actuator housing. If necessary, anti-condensation heaters should be provided. Drives using frequency converters and power electronics instead of contactors are also permitted. A lockable local/remote switch with open/close controls must be located on the actuator. With the exception of the basic GT and ST equipment, the actuators must be designed so that they can be disconnected from the valve in the event of a malfunction and the valve operates normally with the handwheel.

The diameter of the flywheel and gear should be such that it can be operated by one person with minimal effort. A lockable mechanical clutch mechanism shall be provided to prevent power control of the actuators while operating the helm.

Each end of the drive stroke must be equipped with suitable limit and torque switches to prevent excessive stroke and subsequent damage to the valve or actuator. Limit switches must also be installed to provide remote (in the CCR via APCS) indication of valve position. The actuator must be provided with a local valve position indication. The relationship between the command signal and the actuator position must be controlled and any discrepancy must be signaled to the APCS.

The drive motors must be suitable for use on a 400 V, 3 phase, 50 Hz network and, as far as possible, must be obtained from a single manufacturer, to be agreed with the Customer.

The drive motors must be equipped, if necessary, with a thermal protection device. Torque switches or electronic torque control shall be provided to control the motor in the fully closed position and, if necessary, to protect the motor in the open position. Torque switches must be independently adjustable and adjustable for each position. Electric actuators must be provided with means of local manual control.

2.10.35. Pneumatic actuators/positioners

Pneumatic actuators/positioners must be supplied in accordance with the operational and environmental requirements of the valve or actuator assembly. Pneumatic actuators/positioners must be capable of actuating the valve or driven unit through its full stroke at a signal between 0.2 and 1 bar using any of the following:

- pneumatic signal directly to the drive;
- pneumatic signal directly to the positioner;
- electrical signal to the electro-pneumatic converter;
- electrical signal to the solenoid control valve.

Pneumatic actuators and actuator units shall automatically return to a fail-safe/predetermined position upon signal and/or air failure, unless process or specification or detailed engineering requirements call for a «stay in place» response.

Where necessary, «lock-out» devices shall be provided to ensure that the actuator remains in the operating position prevailing just prior to the air failure.

All pneumatic control equipment, control actuators and control valves should operate satisfactorily at a normal air supply pressure of approximately 7 barg, but not less than 5.5 barg.

Volume boosters should be provided on hinges where quick response (eg pressure) is required to improve stability.

All positioners and electropneumatic converters must be equipped with three pressure gauges, air supply, signal input and control air output, and an air filter-regulator kit with lubricating oil.

For GT, ST and their auxiliaries, the valve position must be provided by the LVDT, all other position sensors and limit switches must be provided for remote indication of the valve positions in the CCR through the APCS. Position measuring systems using potentiometers must not be used. All feedback switching signals must come from voltage-free contacts.

If an electropneumatic converter precedes or is an integral part of the positioner, measures shall be provided for connecting portable test equipment to provide electrical references or pulses to enable the actuator to stroke for testing purposes.

2.10.36. Electro-hydraulic actuators

As a general rule, these actuators should only be used where high force combined with short run times is required and application requirements cannot be met by the standard range of actuators used in the remainder of the contract.

The units offered are to be self-contained, fully sealed, allowing the entire unit to be moved to a clean room for any maintenance requirement.

Devices accepting either 4-20 mA signals or pulse input signals are allowed.

Failure of the internal hydraulic supply should cause the actuator to lock in its last operating position or move to a fail-safe position and give the appropriate alarm to the CCR.

Position sensors and limit switches should be provided for remote indication of valve positions in the CCR through the APCS.

2.10.37. Solenoid valves

Solenoid valves must be selected based on the type of process fluid, the temperature of the environment in which the valve operates, and the frequency of operation. For applications other than air use, the solenoid must be designed to be protected from the process environment.

Solenoid housings must be waterproof. The solenoid coils must be insulated class H.

The open/closed state of the solenoid valve must be transferred to the APCS.

Solenoids should normally be operated at 24V of direct current and 48V of direct current. However, the Contractor may offer 110 V of direct current in places where voltage drops are expected due to longer connection cables.

2.10.38. Manual valves

Critical manual valves important to the process must have their open/closed state monitored and displayed to the CCR via the APCS. In such cases, limit switches must be provided at each end of the stroke.

2.10.39. I&A cables

All control and I&A cables supplied under this contract shall comply with the latest applicable IEC standards. I&A cables (other than fiber optic cables) and electrical wiring must have stranded copper conductors, thermoplastic insulation, and a thermoplastic outer sheath. Protection against pests and insects must be provided. When laying outside buildings and fences and in the absence of general mechanical protection, the cables must be armored from galvanized steel wire with a waterproof thermoplastic sheath. All methods of mechanical protection, except for steel wire armor, belong to the Customer. Cables laid in cable trays, ladders, troughs and conduits do not require mechanical protection.

All cables must be of a flame retardant type that does not emit halogen gas when burned and produces a minimum amount of smoke. Cables must be designed and installed in such a way as to minimize the risk of fire and damage.

Stranded control cables should generally comply with IEC 60502, with thermoplastic insulation, overall thermoplastic sheath and overall metal shield. Multicore control cables should have approximately 20% spare cores, but at least two cores for future use.

Field multi-pair cables shall have twisted pairs of stranded copper conductors, thermoplastic insulation and an overall thermoplastic sheath. These cables must have an individual insulated metal shield applied to each pair and an overall metal shield. Must have at least 20% spare pairs, but at least 1 pair for future use.

Where, in practice, cables must be laid on racks, trays or in ducts/ducts, burying the cables should be avoided. Cables must be marked at each end of the run with a marker consisting of a label attached to the cable with two straps.

Sufficient sag must be provided at termination points to allow for future damage to the cable and re-termination of the strands.

The design of the equipment shall be such that the cable shields and cable armor remain electrically separated throughout the installation of the equipment. This separation must be maintained by installing a suitable earthing system.

If necessary, separate shielded pairs or stranded cables with an overall shield should be provided to provide additional protection against electrical interference.

If necessary, fiber optic cables must be provided, which must comply with the requirements of IEC 60794, CT PK 3407-2019 (ST RK 3407-2019), CT PK 3408-2019 (ST RK 3408-2019). The cables must operate continuously at temperatures between -10°C and +60°C. Fiber optic cables must be installed in accordance with the manufacturer's installation procedures. In the absence of general mechanical protection, fiber optic cables must be armored with galvanized steel wire with a waterproof thermoplastic sheath.

Fiber optic cables must have at least four fibers. There must be at least 20% spare fibers, but at least two fibers for future use. All fibers must be connected to approved terminal boxes.

2.10.40. Separation

Avoid laying I&A cables along the same route as power cables. If this is not possible, the cables should be insulated as much as possible. Do not run power cables and I&A cables in the same tray/rack/duct/conduit.

Avoid laying I&A cables in trenches with pipelines. If this is not possible, the cables must be completely mechanically separated from the pipeline.

The I&A cables must be laid on special trays with a minimum distance between the I&A trays and all other trays.

If I&A cable trays are to be installed on the same routes as the power cable trays, then I&A cable trays should be installed at a lower level than the power cable trays to ensure that the heat generated by the power cables does not pass through the I&A cables.

2.10.41. Laying and termination

The I&A cables must be fully supported either in conduits, trays, racks, or other supporting systems subject to customer approval. Cables must be properly secured to prevent excessive slack. The cable routes must be such that the cables are not exposed to extreme temperatures from industrial processes and prolonged exposure to direct sunlight without some form of thermal protection.

The installation of the cable core/pair inside the installation equipment, adaptable boxes, junction boxes and junction boxes must be such that the core/pair can be moved from one end of the terminal rail to the other end. In addition, the core/pair must be reconnectable at least three times.

All spare cores/pairs must be connected in such a way that they can be terminated as above.

2.10.42. Identification

The station reference system shall apply to all I&A systems, their software and its configuration, associated documentation and equipment markings.



Each piece of equipment must be labeled in a cohesive, durable and legible manner, indicating the purpose, the identification number of the installation and, if necessary, the functions of any workplace. Labels must be on the front and rear entrance doors of all cells. Labels should also be provided inside cabinets to help identify equipment and terminals. Instruments, valves, actuators, and all other field-installed equipment must be similarly labeled.

All label material and dimensions, legend and printing method must be approved. Surfaces of plates for cabinets and instrumentation should be matte or satin to avoid glare. Colors must be durable and not fade. The labels must be secured with stainless steel screws. Any alternative fastening method is subject to approval.

Hazard signs must have red letters on a white background and comply with BS EN 60073 and ГОСТ 12.4.026-2015 (GOST 12.4.026-2015).

Cables, conductors and pairs must be uniquely identified using a consistent system.

Each cable must be marked with a cable number approved by the Customer.

Each core of the cable must be uniquely identified using plastic lock ring lugs or similar means. «Clip on» or «c» type tips should not be used. The main identifier must be approved by the Customer.

2.10.43. grounding

All de-energized metal parts of the plant, equipment, components, enclosures, racks, etc. must be earthed by connecting to the main earth bus in an approved manner, and all earth terminals and connections must be properly sized.

Grounding must comply with the relevant sections of BS 6739, BS 7671, IEEE 80, IEEE 665 and IEEE 1050 or equivalent international standards and norms, the requirement of the Republic of Kazakhstan ГОСТ 12.1.030-81 (GOST 12.1.030-81), Order of the Minister of Energy of the Republic of Kazakhstan dated March 20, 2015 No. 230 «On approval of the Rules for the installation of electrical installations» (RIEI).

The Contractor shall prepare a document on the principles of I&A grounding prior to commencement of the grounding design. This earthing concept is subject to customer approval.

Protective screens around insulated cables must be isolated from field devices connected through any field junction boxes and be continuous to an earth point in the cubicle or destination panel. If continuous shields are required to provide protection against interference, precautions must be taken to ensure that potentially severe ground faults cannot damage the cable or equipment. The cable armor should only be earthed for electrical safety and not for noise reduction. The cable armor must be insulated from the cable shields and must be connected to site ground at one point.

If the Contractor's project requires a separate and special earthing device (for example, a «clean» earth for electronic equipment), then the Contractor shall provide such means of earthing as necessary. If such earthing systems require a single connection of the I&A equipment to the plant's main earth bus, then a detachable jumper shall be provided to connect the equipment's earth to the plant's main earth bus for testing purposes. This detachable link must be clearly marked with a warning symbol. All clean earth rods of the instrument must also be appropriately marked.

Any separate ground bar for measuring circuits should be located at the bottom of racks or equipment panels and consist of a solid copper rod 6 mm thick and 25 mm wide (length according to the size of the panel), with holes and threads for screw terminals and wire lugs. This ground bus must be isolated from the chassis and equipment ground.

All panels, cabinet doors or swing frames must use a flexible cable or braid to ground these elements; door hinges should not be used as a means of grounding this piece of equipment.

2.10.44. Safety system

An integrated security system for the entire plant shall be installed to control and monitor all access to and from the Site. The security system should consist of video surveillance, intrusion detection and access control systems. These systems should provide security personnel at the gate and operator personnel in the central control room with the means to monitor and control activities within the plant boundaries, as well as to view the plant's external fencing, gates and critical areas of the facility from control desks and color television monitors.



2.10.45. Video surveillance system

To monitor the perimeter fences and gates, a video surveillance system (CCTV) should be provided. Fixed-position cameras with fixed focal length lenses and pan-tilt zoom (PTZ) cameras should be used to provide adequate coverage. Remote control of PTZ cameras should be provided. Camera systems must be digital systems that must be user-configurable, scale, use digital storage devices, and be distributed over large networks.

It must be connected to an intrusion detection system and provide automatic calling of cameras in response to an alarm.

The Contractor shall be responsible for designing and placing CCTV cameras at strategic points, mounted on buildings or using masts, as required, to achieve full coverage of the required areas.

The video surveillance system must be a high quality, high performance networked digital video surveillance system and must be designed so that people can be clearly seen anywhere within the coverage area. CCTV images displayed on monitors or retrieved from storage must be clear enough to allow clear and unambiguous identification of people and vehicles.

Solid-state surveillance cameras using charge-coupled devices (CCDs) capable of providing maximum resolution, high-definition color video images should be provided. Cameras must be wide dynamic range (WDR). Cameras installed in locations subject to changes in light levels should have lenses equipped with automatic iris adjustment.

A digital storage system shall be provided capable of storing video surveillance images in a compressed time-lapse format or in real-time as needed. Images should usually be stored in timelapse mode. The storage system must provide redundant storage elements to ensure that data is not lost in the event of a storage element failure. Storage capacity must be at least 72 hours, with some cameras running in slow motion mode and the rest of the cameras running continuously. An alarm should sound when the storage capacity reaches 80 percent. The video surveillance system must comply with the requirements: Decree of the Government of the Republic of Kazakhstan dated May 6, 2021 No. 305. «On approval of requirements for the organization of anti-terrorist protection of objects vulnerable to terrorists.» In case of failure of the storage element, the transition to the backup storage element should be automatic. The storage system must have the means to overlay the camera number and date/time onto the images.

Color television monitors capable of displaying any of the system's cameras should be provided. Monitors and controllers will be located both in the CCR and in the security service premises.

2.10.46. Intrusion detection system

An electronic intrusion detection system is to be supplied as part of the Contract for perimeter fencing and entry gates to the Site. The detection system must detect and alarm when any perimeter fence is tampered with or entered. When an alarm condition occurs, security lighting in the area and surrounding areas should be turned on, and camera images in the area and surrounding areas should be displayed and recorded automatically. The intrusion detection system should notify security personnel and operatives of an attempt to enter the station through a protective fence.

2.10.47. Entrance security control system

An access control system to the facility should be provided that covers all access points to the facility. The access control system to the site should consist of automatic gates, an electronic access control system, an intercom between the gate and the CCR/gatehouse, as well as gate controls from the CCR and gatehouse. CCTV cameras should be located at all access points to ensure clear identification of people and vehicles.

2.10.48. Meteostation

The meteostation should be permanently installed with the range and accuracy of the instruments necessary to measure meteorological data for both routine performance testing and the ongoing operation of the station. At a minimum, the meteostation should include instrumentation to provide the APCS with the following measurements:

- Ambient temperature.
- Atmospheric humidity.

- Atmosphere pressure.
- Wind speed.
- Direction of the wind.
- Sun indicator.

The meteostation must be provided with all necessary housing and ancillary equipment and must be able to withstand the full range of climatic conditions encountered on the site. The meteostation equipment should be located in an open area away from any structures that may affect the accuracy of measurements, which should be determined at the detailed design stage of the project.

2.10.49. Monitoring of water emissions

At the end points of discharge at the facility, special systems for continuous monitoring of wastewater should be provided. There should be an indication and recording of all parameters required by the relevant authorities, which should include, as a minimum:

- pH.
- Dissolved O₂.
- Chemical oxygen demand (COD).
- Temperature.
- Residual chlorine content.
- Oil/lubricants, hydrocarbons.

The equipment must transmit instantaneous measurement values to the operator in the CCR through the APCs. A local indication shall also be provided for each measurement on the corresponding measuring instrument. Fault alarms for each instrument must be raised to the APCs.

Monitoring equipment must be of proven reliability and must be approved by the Customer. All equipment necessary for the installation, commissioning, calibration (including any calibration materials), operation and maintenance of the control equipment must be supplied.

A self-contained system shall be provided which shall be capable of performing hourly and daily averages, evaluating the degree of compliance and providing early warning of approaching alert levels and violations of statutory levels for all discharges.

2.10.50. Digital private automatic branch exchange (DPABX)

The power plant must have an integrated services digital network (ISDN) type PABX approved for connection to the public switched telephone network (PSTN). The DPABX is to be equipped with 10 public telephone lines and 100 extensions and be expandable in the future. The Contractor shall specify the initial installed capacity, including the number of exchange lines, connecting lines and additional lines.

All necessary equipment and facilities for cabling and connecting the DPABX to the PSTN switch must be provided. The Contractor shall be responsible for all other internal telephone connections within the boundaries of the power plant, including cabling to and within the administration, control and turbine areas.

The telephone system must be powered by the UPS.

The DPABX system must include self-monitoring facilities to monitor the status of the DPABX equipment, and appropriate alarms must be generated automatically to alert the operator when a malfunction is detected in the system.

2.10.51. Public address system

A Public Address (PA) system should be provided to broadcast announcements throughout the station. The station should be divided into zones, which the user can specify for voicing their messages. The loudspeakers must be connected zone by zone to the call station and the zone selector located in the CCR. The PA system must be integrated with the DPABX system to work through any of its password protected extensions.

The equipment should be modular to allow flexibility for system reconfiguration or expansion, and should include an operator's desktop microphone and a zone select box. The system must have at least six zones, plus the ability to select all zones for addressing at the same time.

Loudspeakers must be supplied according to the environmental conditions and must be positioned and oriented to provide the required sound coverage.

Power amplifiers must have sufficient power to power all loudspeakers under normal operating conditions plus a 20 percent overload margin.

The public address system shall include self-monitoring system to monitor the condition of the public address equipment, including loudspeakers. Appropriate alarms should be given automatically to alert the operator when a malfunction is detected in the system.

The public address system must be powered by the UPS.

The facility's emergency evacuation alarm system should be integrated into the public address system to provide audible and visual indication throughout the facility and should block any other use of the system.

2.10.52. Siren system

A factory siren/strobe system should be provided to alert CCGT personnel of an incident and report to predetermined locations. The siren system must be integrated with the public address system. When a siren is triggered, a certain tone should be played in the loudspeakers and beacons should be turned on to alert personnel in all rooms and areas with a high level of noise in the enterprise. Sirens should have two tones, one for «alarm» and one for «all clear».

The siren/strobe system must be supplied and installed in accordance with NFPA and other applicable international standards. The Contractor shall provide sufficient sirens and strobe lights to cover the entire CCGT site, as well as all associated cabling, hardware and software.

It is the Contractor's responsibility to ensure that the sirens and strobes supplied are suitable for environmental conditions, and that the sirens are positioned and oriented to provide the required sound coverage. The number and location of sirens must be such that the entire Facility is covered and approved by the Customer.

In places with a high level of ambient noise, a stroboscopic lighting system should be provided.

The system shall include a self-monitoring facility to monitor the condition of equipment, including sirens. Appropriate alarms should be given automatically to alert the operator when a malfunction is detected in the system.

2.10.53. IT LAN system

An IT system shall be installed at the Facility to provide data communication for administrative purposes at the Facility, as well as communication between the Facility and the Customer's headquarters, and other locations. The Contractor is responsible for providing cable infrastructure, data and power sockets in designated rooms and production areas, and all necessary network switches. The location of data and power outlets, as well as cable routes, must be approved by the Customer. The Internet connection must be secured through a hardware firewall or similar security device to protect the site network from unauthorized access.

A one-way Ethernet TCP/IP interface must be provided with the APCS so that users of the station's LAN can view and access the process and operating data of the station from the APCS. This interface will be fully firewalled from unauthorized access and to ensure complete security of station data on the APCS via PI or HTML.

2.10.54. Monitoring system of building structures

In order to prevent and eliminate the consequences of destabilizing factors in real time, the forecast and the fact of the occurrence of emergencies at the power plant, it is necessary to provide an automatic system for monitoring the building structures of all buildings, buildings, structures and tanks at the Facility.

The main tasks of the system are:

- 1) Monitoring of the stress-strain state of structures and displacement of element points.
- 2) Monitoring of the roll of the building and structure.
- 3) Monitoring of natural oscillation frequency and damping decrement of the object.
- 4) Monitoring of foundation subsidence settlement.
- 5) Monitoring of the level of seismic impact.



6) Monitoring of the consequences of seismic impact.

2.11. Years of production of equipment and systems

The scope of delivery should include equipment and systems that were not in operation with a manufacturing time agreed with the Customer prior to its delivery to the construction site, with a set of technical documentation in Russian.

2.12. Parts and service supplied during the warranty period

The Contractor shall indicate in the price structure of the Facility all necessary spare parts and related maintenance necessary to put the equipment into service before acceptance, as well as parts and necessary scheduled maintenance and inspection during the warranty period (24 months from the date of commissioning).

2.13. Availability and reliability factors

The power plant maintenance concept and equipment selection should reflect the need for high availability plants. The Contractor must demonstrate that his technical documentation provides for the construction of just such a power plant. To do this, the Contractor shall provide detailed information and references to past experience that would confirm the operational reliability (factors) of key equipment and key systems provided for at the power plant.

For the power plant, a repair cycle (inspections; current, medium and major repairs) should be developed, if necessary, linked to an equivalent operating time.

The Contractor must also include in the technical documentation a description of the ability of the power plant to operate in flexible dispatch mode.

Reliability indicators:

- launch reliability factor - not less than 0.97;
- readiness factor - not less than 0.97.

2.14. Service life of the CCGT

The service life of the electrical power plant should be not less than 30 years.

2.15. Environmental impact assessment

The power plant must comply with the requirements for emissions of harmful substances into the atmosphere when burning two types of fuel and the requirements for equipment efficiency, in accordance with the requirements of the new Environmental Code of the Republic of Kazakhstan.

The following main activities should be mandatory:

- Automated emission control system (AECS);
- Evaporation field equipment for sewage disposal with impervious screen.

The design and operational characteristics of the power plant must ensure that any emissions from the power plant, including noise, flue gases must not exceed permissible limits, regardless of the operating mode of the installation.

The control of emissions (flue gases, etc.) must comply with the requirements of the electrical power plant by the relevant authorities or local laws in force.

Table 2.15.1. Emission requirements for combustion plants

Type of fuel burning installations	BAT emission levels (mg/Nm ³)				
	Pollutant	Average daily value (existing requirements)		Average daily value (possible tightening of requirements)	
		Natural gas	Liquid fuel	Natural gas	Liquid fuel
Open cycle gas turbines ≥ 50 MW _{el}	NO ₂	50	50	25–50	145-200 ¹⁾
	CO	100	100	5-40 ¹⁾	2)

Combined cycle gas turbines 50-600 MWt	SO ₂	-	-	-	50-66
	Suspended particles	-	-	-	2-10
	NO ₂	50	50	15-40	145-200 ¹⁾
	CO	100	100	5-30 ¹⁾	2)
	SO ₂	-	-	-	50-66
	Suspended particles	-	-	-	2-10

1) Average annual value.

2) Not established, but please provide, taking into account the tightening of requirements in the future.

2.16. Noise

Sound pressure levels at a distance of 1 m from the equipment and at a height of 1.5 m above floor (ground) level in the work area must comply with the values indicated below. Verification measurements are carried out in accordance with ISO 11204 standards. The average sound pressure level LpA during normal operation of the power plant with a load of 10% to 100% of the installed power must not exceed 80 dB(A). LpA - average sound pressure level (averaged value for the microphones used in the working area).

The equivalent sound level affecting the operating personnel should not exceed the established values of ГОСТ 12.1.003 «Шум. Общие требования безопасности» (GOST 12.1.003 «Noise. General safety requirements»), no more than 80 dB.

The equivalent sound level at the border of the sanitary protection zone should not exceed 55 dB.

The sound level at a distance of 100 m from the intake or exhaust of the GT should not exceed 45 dBA.

2.17. Classification of hazardous areas

The Contractor classifies hazardous work areas and complies with all mandatory requirements of norms and standards. The Contractor shall submit drawings and lists of premises/sites, giving a classification of the power plant and areas near the equipment. When classifying sites, all necessary information should be indicated on:

temperature (environment, surface, etc.);

condition of dust and water;

fire hazard;

explosiveness;

requirements for electrical equipment.

In this case, both ordinary and exceptional conditions are considered.

2.18. Materials

The materials selected by the Contractor shall be suitable for the respective application and shall last for the entire service life of the power plant.

When choosing materials, the main consideration should be the compatibility of materials and operating conditions. An important selection criterion should be the mechanical characteristics as well as the characteristics of the materials related to workmanship and service (eg strength and weldability). The selected materials must be standardized and have convincingly proven their quality in the past.

The materials that make up those power plant components that will be in contact with water or subject to mechanical wear, including heat transfer surfaces, must be highly resistant to corrosion/erosion.

Inappropriate material combinations (eg carbon steel/stainless steel) should be avoided to prevent galvanic corrosion.

Asbestos and polychlorinated diphenyl (PCD) containing materials and persistent organic pollutants (POPs) are not allowed to be used in the manufacture of the power plant.

2.19. Standards, rules and regulations

When implementing the project, the Contractor must comply with the requirements listed in the rules and regulations in force on the territory of the Republic of Kazakhstan.

All electrical equipment supplied must comply with the standards of the Republic of Kazakhstan in force for the period of execution of the Contract, unless the Customer has approved something else. Such approval will be given only if the Contractor can convincingly demonstrate that the equipment complies with other generally accepted international standards and its quality is equivalent to the quality approved by the local standard. Installation must be carried out in accordance with appropriate standards and manufacturer's recommendations.

Electrical installation work must comply with the requirements of the relevant standards of the Republic of Kazakhstan, as well as the electrical equipment used in these areas.

The Contractor is responsible for providing the Customer with information that will be subject to regulatory review and assisting in obtaining the necessary licenses and approvals from the authorities for its scope of work, if such licenses and approvals are obtained on behalf of the Customer.

With regard to the main and auxiliary equipment operating under pressure, the regulatory documentation in force on the territory of the Republic of Kazakhstan is applied.

When preparing, organizing and conducting a competition, regulating its consequences, it is necessary to be guided by the system of regulatory legal acts and regulatory and methodological documents in force in the Republic of Kazakhstan:

- Decrees of the President of the Republic of Kazakhstan.
- Laws of the Republic of Kazakhstan.
- Decrees and Orders of the Government of the Republic of Kazakhstan.
- State standards of the Republic of Kazakhstan (GOST).
- Building codes and regulations, codes of practice and others.

The list of norms and standards in force at the time of signing the Contract is applied.

In the event that any norms, standards or rules contradict each other, the most stringent of them apply.

2.20. Certification

At the time of delivery, all equipment, systems, products, materials subject to certification in accordance with the legislation of the Republic of Kazakhstan must be certified in full to the extent necessary for the industrial operation of the equipment.

In cases established by the legislation of the Republic of Kazakhstan, the presence of a declaration of conformity or a certificate of conformity of goods with the requirements of the Technical Regulations of the Customs Union (TR CU) is mandatory. The need to submit this declaration or certificate before the delivery of the goods is mandatory.

The Contractor shall provide copies of the certificate (declaration) of compliance of the equipment with the requirements of TP TC 010/2011 «О безопасности машин и оборудования» (TR CU 010/2011 «On the safety of machinery and equipment»), TP TC 004/2011 «О безопасности низковольтного оборудования» (TR TS 004/2011 «On the safety of low-voltage equipment»), TP TC 020/2011 «Электромагнитная совместимость технических средств» (TR CU 020/2011 «Electromagnetic compatibility of technical means»), TP TC 032/2013 «О безопасности оборудования, работающего под избыточным давлением» (TR CU 032 /2013 «On the safety of equipment operating under excessive pressure»), and other TR TS, or a written obligation to submit these certificates (declarations) before the delivery of the equipment.

All measuring instruments included in the supplied equipment for which certification is required (instrument transformers, instruments, sensors, meters, software, etc.) must be in the Register of the State Measurement System of the Republic of Kazakhstan, or a written obligation to add measuring instruments must be provided applied at the facility to the Register until the delivery of the equipment.

2.21. Access organization criteria for maintenance

It is necessary to provide that all mechanical, electrical and control equipment of the power plant, etc. provide free access for periodic maintenance throughout the life of the power plant.

When installing, both individual devices and devices related to equipment systems, it is necessary to be guided by the following main criteria:

- One worker in conditions of free access can lift a weight of no more than 30 kg.
- Two workers, each with free access, can lift a total of about 45 kg.

If access is difficult or weights exceed the above values, then provision must be made for lifting with a hoist, a forklift with slings or, in special cases, a mobile or overhead crane. All lifting equipment is included in the scope of delivery.

In cases where electrical equipment is used, the Contractor may need to provide trolleys or special chassis, which will need to be included in the scope of work if it is not possible to provide other means.

If it is not clear from the general arrangement diagram which way a piece of equipment should be delivered to the aisle, then this must be indicated separately. If any component is located at a high elevation, then the path of delivery of such component to the hard surface of the passage or to the road must be clearly indicated, if this is not clear from the existing layout.

If it is planned to use an overhead crane for maintenance, this must be specifically provided for in the plan for the use of cranes, indicating the location of the cranes and equipment, indicating the load capacity of the equipment. The use of tooling with obviously higher load capacity for light parts should be avoided due to the difficulty of access.

The clearance to the front work surface at the point of service must be at least 600 mm and preferably greater than 750 mm. In the case of pumps (or other rotating equipment), these clearances must be provided on both sides of the pump center line. The technical documentation should provide for the possibility of unhindered movement from the point of performance of work to the passage.

2.22. Access criteria for operation and troubleshooting

In addition to normal access, which allows routine monitoring of the operating parameters of the equipment and maintenance (including lubrication and filling of chemicals), the following special criteria must be observed:

- all control or remote-operated fittings must be accessible from floor gratings, ladders or permanent service platforms and meet the maintenance criteria specified above. Elevated platforms must be accessible from steps, ladders or vertical ladders (in order of priority).
- all panels of electrical or electronic devices containing relays or solid state circuits that may require replacement or troubleshooting should be protected by visors (with an angle of inclination up to 15 degrees) from exposure to sunlight or rain and provide protection to workers.

The primary control sensors and the location of the measurement points must also meet the above criteria.

In areas with a large number of equipment, it is acceptable to provide ladders on wheels that can be used to access appliances located at a height of less than 5 meters above the permanent service platform or grating for maintenance or operation purposes. Walkways must not restrict the movement of such wheeled ladders.

2.23. Level of automation

The power plant must be designed in such a way that high reliability and high availability are ensured with minimal operator intervention required during normal operation.

The power plant must be equipped with a permanently operating equipment condition monitoring system (ECMS).

2.24. Stability of the power plant operation

The power plant must operate stably and maintain the frequency and voltage within the range allowed by the components and systems of the power plant.

Systems for restoring synchronization with the network are supplied by the Contractor.

In case of failure, after which the power plant is disconnected from the power system of the Republic of Kazakhstan (complete load shedding), the power plant must continue to operate in an autonomous mode (island mode), supplying power to its own needs.

All systems necessary to re-synchronize the power plant from an autonomous state to the network after a complete load shedding shall be included in the Contractor's scope of supply and work.

2.25. Power plant equipment protection system

The technological protection system of the power plant must comply with the requirements of the legislation in the field of industrial safety of the Republic of Kazakhstan.

The technological protection system of the power plant must be fully automatic and prevent the occurrence and development of emergency situations by automatically generating and issuing control actions on actuators and switching devices for emergency transfer of the protected technological equipment to a safe state in case of violations of the normal operating mode.

The technological protection system must have the highest priority in relation to all other control actions.

The technological protection system must be fault-tolerant. Failures of individual elements of the protection system should not lead to emergency shutdowns when the backup channels are operational.

The system must be protected from electromagnetic interference (high voltage equipment, lightning, etc.).

2.26. General requirements for the main equipment of the power plant

The device of power plants must comply with all functional and operational requirements for a power plant.

The nominal capacity of power plants must be specified, including in accordance with the terms of the ISO standard.

Power plants should provide the possibility of independent automatic start-up, synchronization with the network and full load in automatic sequential mode when monitored and controlled from the control panel of the power plant.

Power plants must provide the possibility of independent automatic reduction of full load, disconnection from the network and safe shutdown in automatic mode.

The design of power plants and auxiliary equipment must be confirmed by examples of successful and long-term commercial operation of such equipment.

Power plants must be able to maintain safe, efficient and continuous operation.

Power installations must be fully equipped with protections to prevent equipment from danger and damage.

Power plants must be equipped with a protection system. The main protection of the generator GT, ST, HRSG must have redundancy.

Power plants must be equipped with starting systems, flushing systems, fire protection and housings.

The automatic fire extinguishing system of the power plant must comply with the norms and requirements of the Republic of Kazakhstan.

A sufficient number of fire detectors and spray nozzles must be provided in each compartment of the unit. It is necessary to provide an alarm signal (remotely and locally) that allows the operating personnel to leave the installation before the fire extinguishing system is automatically activated. The system must be equipped with a switch in order to turn off the unit at the right time if necessary.

All compartments of the installations must have adequate lighting, as well as ventilation and heating.

Sufficient access openings, walkways, etc. should be provided. for the possibility of performing maintenance work on the components of power plants.

Particular attention should be paid to the need to ensure that basic maintenance work can be carried out easily. It shall be possible to use lifting equipment for inspection and maintenance work.

It shall be possible to carry out inspections using an endoscope on the power plant units, as well as, if necessary, the use of barring devices and a stepping inductor motor. All equipment for endoscopic examinations and all instruments necessary for opening and closing holes must be included in the delivery.

To determine the performance of power plants under actual external conditions and various loads, the following graphs should be included in the technical documentation:

dependence of power on the outdoor temperature;

dependence of efficiency on power at various outdoor temperatures;

dependence of fuel consumption on power at various outdoor temperatures;
dependence of exhaust gas consumption on power at various outdoor temperatures;
dependence of the coefficient of excess air in the exhaust gases on the power at different temperatures of the outside air;

dependence of exhaust gas temperature on power at different outdoor air temperatures;
dependence of the relative values of power, fuel consumption, temperature and exhaust gas flow on changes in total pressure losses in the inlet device;

dependence of the relative values of power, fuel consumption, temperature and exhaust gas flow on the change in total pressure losses in the gas outlet device.

These graphs must take into account the elevation and other geographical and climatic characteristics of the area.

Gaseous fuel system

Fuel gas preparation equipment must ensure the normal operation of power plants, both as a single unit and as a complete set.

The fuel system must ensure the supply of fuel to the combustion chamber, its dosage according to the signals of the control system, as well as the termination of the fuel supply at the command of the operator or automatically when the protections are triggered. The system must consist of dispensers, shut-off valves, fuel filters, necessary fittings, sensors and pipelines.

Depending on the type of equipment and the gas supply scheme, it may be necessary to build a gas compressor station, consisting of at least 2 gas compressors. In this case, the gas compressor station and all necessary communications to and from it will be included in the Contractor's scope of supplies and work.

Compressors must be located in special ventilated rooms with noise suppressors that comply with noise standards. When performing a gas compressor station, it is necessary to provide all the necessary safety measures. Gas supply control equipment must be installed on the compressors to instantly provide the required level of gas supply to the power plant equipment under conditions of the entire load range.

The compressor system must ensure that no pressure or flow changes are transmitted to the power plant equipment under any transient or steady states.

Compressors and their ancillary equipment must be able to operate in the full range of temperature regimes of gas supply, as well as in the range of pressures and temperatures of the environment in the area of the power plant. For gas supply, it is necessary to provide for the installation of a complete set of instrumentation to ensure complete control of the gas supply from the control panel of the power plant, as well as to ensure appropriate control during start-ups, shutdowns and under normal operating conditions. Safe shutdown systems must provide protection for compressors and auxiliaries. Particular attention should be paid to transients caused by shutdowns of gas-using equipment.

Gas compressors must be equipped with the necessary lifting equipment for installation and scheduled maintenance, and have sufficient space to accommodate the equipment being serviced. Compressors must respond to load shedding without shutting down. The performance of each of the gas compressors must provide 100% of the required gas flow for the operation of the power plant equipment at rated load. It should be ensured that the backup gas compressor is automatically switched on when the operating compressor is stopped at any load without disconnecting the power plant equipment from the network.

Air intake system

The facility should include an air intake system for power plant equipment, fully equipped with the necessary pressure chambers, ducts, compensators, supports, filter modules, ice alarm, intake air silencer and intake air anti-icing system. The air intake of the filters must be equipped with a canopy to protect the filters from snow, water, shutters to block the air duct when the power plant is not working, and mosquito nets to protect against insects.

The air intake system must be designed in such a way that the start-up of the power plant equipment is also possible when the anti-icing system is in operation (for example, when air is released from the compressor or using external steam).

The air intake duct must contain noise suppression devices.

The risk of icing must be taken into account. It must be possible to control the load of the anti-icing system in accordance with the ambient temperature.

The air intake system must be equipped with the necessary filtration system, including pre-filters and fine filters and differential pressure sensors on the pre- and fine filters.

The design of the air filtration system must be such that dripping liquid and other particles can be successfully removed. The efficiency of the filter must comply with the requirements of the Technical Operation Rules (TOR). Filtration efficiency $\geq E=98\%$. The air velocity must be sufficiently low (<5 m/s) and the distance between the pre-filters and the secondary filters sufficiently large to avoid the transfer of water droplets and dissolved particles to the secondary filters. The channel between these filters is equipped with a drain to collect the collected water.

The filter compartment must be equipped with the necessary ladders, service platforms and sufficient lighting, as well as lifting devices for lifting the filter cartridges.

The system must have proper I&A to reliably measure the differential pressure across the filters.

Ducts and chimneys

The removal of exhaust gases from power plants should be carried out in chimneys.

The exhaust gas duct must be made of a material that can withstand the maximum temperature of the gases for a long time in any mode of use of the power plant equipment. The wall thickness of the gas ducts must have a reserve for oxidation during the estimated service life of the power plant equipment. Noise suppression devices and, if necessary, guide vanes should be installed inside the gas duct to reduce aerodynamic resistance and evenly distribute the gas flow. Compensators for thermal expansion of the flue must be provided. The flue must have sound-absorbing thermal insulation. The outlet gas duct must have the necessary slopes and be drained through a drain pipeline with a water seal.

The thickness of the thermal insulation must be selected depending on the required thermal performance. At an ambient temperature of $+25^{\circ}\text{C}$, the temperature on the insulation surface must not exceed $+45^{\circ}\text{C}$.

At higher surface temperatures, at a certain distance from it, in places that are easily accessible during the normal operation of power plants, protective grilles or steel metal fences are installed to protect personnel.

Exhaust gases from power plants must be directed to their own chimneys.

Heights of chimneys are accepted in accordance with the legislation of the Republic of Kazakhstan, providing acceptable standards for emissions of harmful substances for a given area.

The chimneys are equipped with the necessary connections for the emission monitoring system.

Marking paint must be applied to the chimneys.

Chimneys must be equipped with ladders and service platforms.

The Contractor supplies signal beacons to warn aircraft in accordance with the requirements of the regulatory documentation of the Republic of Kazakhstan. Beacons should be signaling devices of a completely enclosed, moisture-proof type, installed so that they can be easily replaced. Provides access for maintenance of beacons.

Chimneys must be equipped with a lightning protection system.

Compressor flushing system (for GT)

The Contractor must supply a set of equipment for cleaning the GT compressor, give a complete description of it, indicate the frequency and technological procedure for cleaning, guarantee the restoration of the original plant parameters after washing.

The scope of delivery should include a fully equipped gas turbine flushing system with all the equipment included in the kit: pumps, fittings, pipes, tanks, etc. The system must be automated to ensure operation with minimal operator intervention.

For an autonomous GT compressor flushing system, a collection tank must be supplied.

Preservation system (for GT)

The Contractor shall ensure that the GT is equipped with a conservation system that prevents corrosion.



Fire protection system (for GT)

The automatic fire protection system of the entire scope of delivery is a battery of CO₂ cylinders mounted on a frame.

Each section of the GT shelter must have a sufficient number of spray nozzles and fire alarm sensors. The system must give warning signals (remotely and locally) in advance so that operating personnel can leave the shelter before the fire extinguishing system automatically activates. The system must be equipped with a switch so that it can be turned off if necessary.

Units and parts of power plants

The equipment of power plants should be made in the form of block structures. Blocks must be ready for installation without disassembly for revision.

The Contractor shall provide detailed technical characteristics of the main parts of power plants, including information on materials, manufacturing technology and measures taken to ensure their thermal stability.

The design of the elements of power plants should be such that there are no dangerous deformations during start-ups and changes in loads.

The combustion system must ensure complete combustion of the fuel and acceptable NO_x emissions. The Contractor shall describe the combustion chamber, burners, structural and technological means of suppressing the generation of NO_x and other harmful emissions, fuel gas supply system, water injection, if provided.

Regulatory and protection system

The fuel supply system of power plants must ensure a complete shutdown of the fuel supply when the controlled operating parameters go beyond the permissible values. Valves must be properly sealed in the closed position. Check valves must be locally and remotely operated.

Protective devices must have priority effect on the control valves when performing partial unloading or complete shutdown, depending on the degree of danger of the conditions created. These valves must also be affected by the emergency overspeed mechanism.

The necessary number of contacts must be provided for remote indicators of the position of the stop valves and for transmitting a signal to disconnect the generator from the network when they are closed.

Power installations must be equipped with protections that immediately turn off the installation in cases stipulated by the technical operating conditions.

The control system of power plants must meet the following requirements:

- sustainably maintain a given electrical load;

- maintain power plants idling at the nominal rotor speed;

- ensure reliable operation of power plants in start-up and shutdown modes, as well as shutdown of the unit in emergency situations;

- ensure, when the load changes, a smooth change in the operating mode of power plants;

- maintain the rotor speed, which does not trigger the safety switch, with an instantaneous reset of the maximum load to zero;

- ensure the stable operation of the compressor;

- have a degree of static non-uniformity of regulation of the rotation frequency of the shaft generator within 2% of the nominal;

- have a degree of insensitivity at any load not more than 0.2% of the rated speed.

The capacity and duration of operation of power plants with deviations from the normal speed should be regulated by the manufacturer's specifications.

Safety breakers must be adjusted to operate when the rotor speed increases by 10-12% above the nominal value or to the value specified in the manufacturer's specifications.

The characteristic of the control system and the description of all devices intended for the protection of power plants in conditions beyond the permissible ones must be detailed and illustrated with drawings, sketches and diagrams.

The regulation system must be accompanied by a manual for the repair and adjustment of equipment, including repair forms, testing and adjustment procedures.

A set of spare parts, tools, accessories and materials (SPTAM) for the control and protection system must be determined and included in the supply of equipment.

Control regulators must automatically regulate the supply of fuel to power plants from a state of no load to a state of maximum power. There should be a mechanism for setting the load speed of power plants.

The control valve must operate in such a way that there is no overspeed and speed fluctuation over the entire load range. The temporary increase in speed due to the instantaneous release of the maximum load shall not be such as to cause the stop-check valves to actuate.

The electronic part of the control and protection system must ensure the exchange of information and the implementation of control actions with the APCS of the power plant using a digital exchange protocol in real time.

The operation of the system must meet the requirements for equipment involved in frequency and power regulation in the power system of the Republic of Kazakhstan.

2.27. Technical documentation of power plant foundations and shafting dynamics

The technical documentation of the foundations of power plants must be carried out in accordance with the norms of the Republic of Kazakhstan. The Contractor submits to the Customer the technical documentation on the foundations with calculations. Grounding schemes in foundations should also be described.

The rotors are balanced in such a way that they meet the requirements for balancing quality.

A dynamic analysis of the entire shafting should be presented.

The procedure for using the above information in calculations should be described.

Critical speeds and modes are given, including frequency and damping. The results of the dynamic analysis should show good shafting stability.

Torsional stress analysis should consider, among other things, terminal short circuits and other extreme loads.

Bearings and vibration monitoring system

Contractor provides bearings locations;

Bearings data:

- bearing type;
- bearing inner diameter;
- bearing length;
- number, location and geometry of pillows;
- preload of each bearing pad;
- brand of antifriction material;
- ISO brand and characteristics of supplied lubricants.

The Contractor determines the operating temperature range of the machine bearings.

The level of absolute vibration of the bearings or stator elements of the unit, characterized by the rms value of the vibration velocity, must, upon acceptance of the unit into operation and during its further operation, comply with the requirements of GOST ISO 10816-4-2002.

All bearings of power plants must be designed in such a way that it is possible to center the bearings without opening the unit. Alignment shims or similar means must be supplied for subsequent adjustment of the horizontal and vertical position of the bearings to eliminate misalignment due to settlement of foundations or for other reasons. A device must be provided to relieve the bearing from the weight of the shaft during the extraction operation. Means must be provided to prevent oil from flowing along the shaft.

Thrust bearings must be segment type. If they are not, then the Contractor shall note this feature and provide evidence that the proposed bearing type performs reliably under comparable loads. The bearing support ring must be in a position that provides maximum relative expansions and clearances in the constituent elements of the unit, and it must be possible to periodically adjust them. Thrust bearings must have both thrust and bearing surfaces. The entire design of the unit must ensure the absence of reversible axial forces in all modes of operation up to the maximum load.

Lubricating oil must be supplied to the bearings at a pressure sufficient to form a hydrodynamic film during the acceleration of the rotor to the rated speed. Pressure gauges must be installed on the distribution manifold of the oil lines. To control the oil temperature, temperature sensors should be provided at the drain from each bearing, and alarms should be provided when this temperature rises. The oil temperature in the distribution manifold downstream of the oil coolers must be controlled, and an increase or decrease in temperature beyond the established limits must actuate the alarm means. Provision must also be made to measure the temperature of each thrust bearing and thrust bearing at a sufficient number of points to accurately monitor their condition.

Downstream of the oil coolers, oil filters with differential pressure sensors for alarms and, in addition, an additional filter for the thrust bearing must be provided. The pressure of the oil entering the bearings must be measured in the common distribution manifold, as well as after the individual thrust bearing filter. All filters must have individual bypasses and be accessible for cleaning while the unit is under load.

To ensure safe and reliable operation, the unit must be equipped with a standard AVCVDS system (automated vibration control and vibration diagnostics system). The system should include control and protection equipment and a computer monitoring system.

The computer monitoring system must ensure the collection and storage of the data archive for at least one year. Transfer current and archived information on controlled parameters to remote workstations via the local information network of the enterprise for diagnostic purposes. The system must ensure the collection and processing of measurement results obtained from all channels in transient modes during the start and stop of the machine (run-out). Have the possibility of automated diagnostics of the technical condition of the unit and warning of personnel about unfavorable trends in the change in the vibration state.

The units must be equipped with the following measuring elements:

Sensors for measuring the absolute vibration of bearings (at least two per support) or stator elements of the machine, the number and location of their installation must comply with ГОСТ ИСО 10816-4-2002 (GOST ISO 10816-4-2002).

Sensors for measuring relative shaft vibration, each radial bearing must be equipped with two sensors and located in accordance with ГОСТ ИСО 7919-1-2002 (GOST ISO 7919-1-2002).

Each thrust bearing must be equipped with sensors to measure the absolute vibration of the bearing in accordance with ГОСТ ИСО 10817-1-2002 (GOST ISO 10817-1-2002).

To measure the relative vibration of the shaft, each radial bearing must be equipped with two sensors according to ГОСТ 27165-97 (GOST 27165-97).

Each thrust bearing must be equipped with three rotor axial position sensors.

The machines must be equipped (if necessary) with speed sensors, expansion sensors (absolute expansion of the turbine housing and relative expansion of the rotor), bearing support slope sensors that provide objective information about the thermal expansion of the unit.

In addition, the unit must (if necessary) be equipped with sensors for measuring the curvature of the unit's rotor shaft.

Engineering units for controlling shaft relative vibration must comply with ГОСТ ИСО 7919-1-2002 (GOST ISO 7919-1-2002).

Lubrication of power plants and oil supply for their control circuits

The lubrication and control system of the unit must ensure long-term reliable operation of all its parts in all operating modes. The bid must contain a detailed description of the oil system, including the oil cooling system and the method of controlling its temperature.

To safely shut down the unit in the event of a power failure, at least one emergency lube oil pump must be powered by DC power.

Standby and emergency oil pumps must be equipped with an automatic start system.

To filter the oil, full-flow filters (2 x 100%) are integrated into the lubrication system. Differential pressure gauges are installed on the line filters, alarm signals from which are given in the control room. Provides the ability to change the filter during operation. A spare set of lubricating oil filters is supplied for the commissioning period.



An oil cleaning unit must be provided. It can be a motor-driven centrifuge or a multi-stage purification system based on sedimentation and filtration. In any case, this unit must be placed on an oil collecting pan and contain filters, controls, control panels, ventilation equipment and other devices necessary for the normal operation of this unit.

The Contractor shall provide for the installation of an emergency oil drain tank of sufficient capacity to receive all the oil when the system is emptied.

The removal of lubricating oil vapors from the oil tank must be carried out using fans and cleaned using oil separator filters. The oil tank must be equipped with graduated level gauges (remotely and locally), oil drain fittings by gravity and an alarm device in case of excessive level decrease and increase. It must be possible to increase the oil temperature after a long period of inactivity.

Air-filled oil return lines must be made of stainless steel.

The oil control system must be fitted with suitable accumulators or equivalent equipment to withstand a transient and power interruption of at least 5.0 seconds.

To filter non-flammable liquids, full-flow filters (2x100%) are built into the oil control system. Differential pressure gauges are installed on the filters, alarm signals from which are given in the control room. Provides the ability to change the filter during operation. A spare set of filters for the control loop is supplied for the commissioning period.

2.28. Safety requirements

The equipment must be designed taking into account the requirements of the Rules for the technical operation of power plants and networks and the existing regulatory – , technical and legislative documentation on safety, explosion and fire safety in force on the territory of the Republic of Kazakhstan.

In the event of an emergency, the safety of the operating personnel must be ensured.

Mechanical safety must be ensured:

- design of power plants, including heat and sound insulating shelters;
- blocking and protective devices that prevent power plants from going beyond the established limit values of parameters;
- the presence of protective covers on the rotating parts of the equipment of the power plant, the design and placement of parts and assembly units, excluding the possibility of injury to the plant operating personnel during operation.

Explosion and fire safety must be ensured:

- the use of non-combustible or hardly combustible structural materials, insulating and paint coatings that do not contribute to the spread of fire;
- adjustment of pipelines of oil, fuel and air systems of power plants from resonant vibrations in the entire range of operating modes;
- exclusion of explosive accumulations of oil-air mixture, gas and liquid fuel in the elements of power plants;
- ventilation of the gas-air path of power plants before launch;
- the presence of fittings for the supply of extinguishing agent from the fire extinguishing system and the installation of fire alarm sensors in the heat and sound insulating shelter of power plants, in the generator and the block of electrical equipment;
- implementation of electrical equipment, taking into account the requirements for fire safety for the premises of the engine room (shelters);
- the location of the units of the oil system of power plants, excluding their unacceptable heating;
- measures in accordance with the operating instructions for power plants and compliance with the requirements of manufacturers.

Thermal and toxicological safety of power plants should be ensured by:

- the presence of thermal insulation, the temperature on the surface, which should not exceed 45 °C at an ambient temperature of plus 25 °C;
 - the presence of a ventilation and cooling system;
- power plant units of materials that do not emit toxic substances in all operating modes and in the non-working state above the maximum permissible concentrations.

The design of power plants and the automatic control system must ensure the normal safe operation of the plant without the constant presence of maintenance personnel.

The Contractor must provide a description of all measures to ensure fire safety in the premises of power plants.

2.29. Ergonomic requirements

The design of power units, installations and the tools and accessories supplied with it must ensure the implementation of all necessary actions, movements and movements associated with maintenance and repair during its operation, with minimal time and cost.

The volume and form of information received by the operator must correspond to the capabilities of a person in terms of its perception, storage and processing.

On the pipelines of power plants, distinctive, warning signs must be applied and distinctive plates must be installed in accordance with the acting normative-technical documentation (NTD).

Distinguishing plates must also be installed on flywheels and valve drives of power plant systems. The coloring of fire pipelines must comply with ГОСТ 12.4.026-2015 (GOST 12.4.026-2015).

The performance culture (quality and thoroughness of constructions, surface finishing, marking, text, quality of protective and decorative coatings) of power plants must meet the requirements of modern technical aesthetics.

The coloring and finishing materials of power plants must be combined in texture, color, physical features and must be resistant to external influences under normal operating conditions.

2.30. Indicators of resistance to external influences

Power plants must remain operational under the following conditions:

- ambient temperature from minus 40 °C to plus 50 °C;
- barometric pressure from 84.0 kPa to 110 kPa;
- relative air humidity up to 98% (at +25 °C).

Protective, decorative and paint coatings should ensure the safety of surfaces and corrosion resistance of parts and assembly units during storage and operation.

2.31. Ease of operation and maintenance indicators

The Contractor must submit a detailed maintenance schedule for all equipment of the power plant, which indicates the types of maintenance for the entire period of use, their frequency and duration, labor intensity in man-hours, the need for spare parts and materials for their implementation.

Inspection of all operating equipment of the power plant should be carried out no more than once per shift.

The design of all equipment should provide access to the main components and parts and ease of maintenance during operation.

Inspection hatches and special devices must be provided for inspection without opening assembly units, mechanisms and equipment that require periodic maintenance, and also, if necessary, for inspection of the GT flow path.

Power plants must be equipped with technical diagnostic tools that allow assessing its technical condition in the conditions of the plant with a minimum opening and inspection with an endoscope in order to determine the required amount of maintenance.

The vibration level in the service area must comply with ГОСТ 12.1.012 «Вибрационная безопасность. Общие требования» (GOST 12.1.012 «Vibration safety. General requirements»).

The design of all power plant equipment should provide for the possibility of replacing parts and assembly units during repairs. Spare parts must be manufactured so that they do not require adjustment on site during installation and allow minimal adjustments.

The design of power plants should ensure its node-by-unit repair by the maintenance personnel of the power plant.



2.32. Additional and special requirements

The Contractor shall provide all equipment, tools and instruments of acceptable accuracy for carrying out acceptance and warranty tests and for proving the guaranteed performance of power plants. The Contractor transfers to the Customer this equipment, tools and instruments for further operation.

2.33. Labeling and packaging requirements

The marking must be applied to plates made in accordance with the requirements of the standards and attached to the cases of all units that make up all the equipment of the power plant.

Parts and assembly units that are spare parts must be tagged with the designation of the drawing, if the marking cannot be performed directly on the part or assembly unit.

Marking of parts and assembly units must be carried out in accordance with the instructions on the drawings.

Shipping containers (boxes) must be labeled in accordance with standards. Transport marking should be applied on one of the side walls of the box with waterproof paint on a stencil.

On goods not packed in shipping containers, the marking should be applied on the cargo itself in the most convenient, well-visible places.

Preservation and packaging of blocks of all equipment of the power plant at the manufacturing plant should ensure the conditions of transportation and storage periods.

All equipment of the power plant must be delivered to the installation site of the plant in the form of transport containers (shipping boxes) with a factory seal and a number according to the coding system.

2.34. Technical documentation provided with the equipment

Complete set of operational documentation.

Complete set of repair documentation.

2.35. Lifting facilities

The scope of delivery should include cranes of adequate capacity for carrying out subsequent maintenance of the power plant equipment and, if necessary, other lifting structures such as elevators, hoists, etc.

Lifting structures must be designed and built in accordance with the requirements of current codes and regulations or equivalent international standards and ensure in all respects their safe and efficient operation.

Cranes, devices for lifting operations must comply with the requirements of the technical regulations of the Customs Union (hereinafter referred to as TR CU), which apply to them, and other LRA (legal and regulatory acts), including TLRA (technical legal and regulatory acts) operating in the Republic of Kazakhstan.

Before being put into circulation, cranes, devices for lifting operations, electric wire rope and chain hoists must, in accordance with the procedure established by TP TC 010/2011 (TR CU 010/2011), as well as other TR CU that apply to them, go through conformity assessment procedures and have documents of the established form.

Cranes must be fully equipped and include access platforms, ladders, supports, guards, motors, a control mechanism with electrical wiring and a safety switch, as well as longitudinal insulated wires, manifolds / supports and crane tracks with mounting bolts and other necessary equipment to ensure safe crane operation.

It should be possible to conduct periodic static and dynamic tests of cranes with a load exceeding their lifting capacity by 25% and 10%, respectively. For these purposes, platforms should be provided for the placement of goods or the installation of hydraulic loaders, which should be included in the scope of delivery.

2.36. Technical solutions for power output

To cover these deficits, it is proposed to build a power plant. If the shortage of base power is due to a delay in the commissioning of the basic energy source, the shortage of maneuverable power is due to the commissioning of power plants using solar and wind energy and is characterized by uneven output



due to seasonal and daily weather and natural conditions, as well as the need to regulate the frequency in the National Power Grid (NPG) and intersystem flows.

Backup channels for ASCME data transmission are organized via satellite communications. The volumes of the necessary RPA and AEA equipment are preliminary and will be specified after receiving the technical conditions for connection.

In addition, it should be noted that, based on the current Rules for maneuverable generation, the control capacity of the designed CCGT-1000 in terms of active power should be $\pm 50\%$, with a decrease to the technological minimum of the plant. According to a preliminary analysis, for stable operation of the power plant, the technological minimum is determined at 150 MW, respectively, the control range of the station will be ± 425 MW (from 150 MW to 1000 MW).

At the same time, in order to ensure stable operation of the designed CCGT and main power lines during transients, it is necessary at the design stages to provide for automatic control of generator excitation in accordance with the current regulatory documents and coordinate with NDC SO JSC «KEGOC».

Section 3. SCOPE OF WORK

3.1. Architectural and construction part

3.1.1. Initial data

Architectural and construction solutions of the Facility to be developed at the design stage, based on the engineering-geological laboratory.

Engineering-geological conditions of the construction site

Based on the engineering-geological laboratory.

3.1.2. Construction solutions for buildings and structures

The task provides for the development of options for power plant sites. All options have the same type of disposable objects.

When designing, it is necessary to take into account the accessibility of the building for groups with limited mobility and the possibility of organizing special jobs for people with limited mobility (according to acceptable decisions on СП РК 3.02-127-2013 (SP RK 3.02-127-2013); СН РК 3.06-01-2011 «Доступность зданий для маломобильных групп» (SN RK 3.06-01-2011 «Accessibility of buildings for groups with limited mobility»); СП РК 3.06-101-2012 «Проектирование зданий и сооружений с учетом доступности для маломобильных групп населения. Общие положения» (SP RK 3.06-101-2012 «Design of buildings and structures, taking into account accessibility for people with limited mobility. General provisions»).

For all considered options, the following list of facilities is provided at the power plant site:

1. The main building of the CCGT.
2. Outdoor installation of transformers.
3. Outdoor/indoor switchgear 220kV (SG 220).
4. Relay board 220kV with MCP.
5. Gas treatment plant with a booster compressor.
6. Air compressor station.
7. Water treatment for make-up of HRSGs and auxiliary systems. Treatment facilities.
8. Diesel fuel warehouse
9. Pumping diesel and oil
10. Oil warehouse
11. Cable management on site
12. Gas pipelines on site
13. Overpasses of technological pipelines and thermal outlets on the site
14. Lightning protection and grounding

15. Administrative building with a checkpoint
16. Service building with a dining room
17. Joint Auxiliary Corps
18. Checkpoint
19. Dining room
20. Fire station
21. Closed reserve parking garage
22. Fire fighting warehouses
23. Domestic and drinking water supply with facilities
24. Industrial and fire water pipeline with facilities
25. Household sewerage with facilities
26. Rain, industrial sewerage with facilities
27. Tank for collecting industrial effluents from the pump
28. Emergency oil drains for turbine oil
29. Emergency oil drains of transformer oil
30. Outdoor and security lighting
31. Industrial site fencing
32. Checkpoint

CCGT main building

Construction class CC-3.

Level of responsibility - (increased).

Functional fire hazard - F5.1.

Structural degree of fire resistance - IIIa.

C degree of fire resistance – II.

Building category - «B1».

In the architectural and planning and space-planning solution, the building appears to be complex, both in form and in volume, consisting of compartments combined into a single production unit.

For equipment maintenance, overhead tracks are provided for electric hoists with a lifting capacity from $Q = 6.3 \text{ t}$ to $Q = 50 \text{ t}$.

Outdoor installation of transformers

Construction class CC-3.

Level of responsibility - (increased).

Production category - «D»;

A complete device, fully prefabricated, installed on a foundation, made of C20 / 25 concrete, reinforced with A400 ГОСТ ГОСТ 34028-2016 (A400 GOST GOST 34028-2016) reinforcement, laid on monolithic concrete preparation (concrete class 7.5). For draining transformer oil, gravel oil dumps with pits and emergency drain tanks for transformer oil are provided.

A fireproof monolithic reinforced concrete partition is arranged between the transformers.

Territory plots with transformers located on them are to be fenced with a height of 2.0 m, from metal mesh panels, along metal racks installed in pre-drilled wells, filled with C 8/10 concrete.

Switchgears 220, 500 kV (SG-220kV, SG -500kV)

Construction class CC-2.

Level of responsibility - (normal).

An open electrical site with load-bearing and supporting structures for the installation of electrical equipment.

Supports are reinforced concrete.

Portals of substation (SS) and RCPST cell, tire and linear, from metal lattice structures and prefabricated reinforced concrete racks with steel lattice traverses.

Supports for switches - metal.



Cable channels underground and ground: trays with removable reinforced concrete floor slabs.
Lightning rods MS31.

The area for the outdoor switchgear constructions, along the perimeter, is to be fenced with metal mesh panels measuring 3000x1200 (h) mm, with racks installed in drilled wells and concreted with C8/10 concrete.

Relay panel 500/220kV with MCP

Construction class CC-2.

Level of responsibility - (normal).

Production category - «D»;

Degree of fire resistance - II.

Gas treatment plant (GTP)

Metal, insulated prefabricated block containers, with block-assembly equipment:

- Gas consumption metering station (GCMS);
- Block gas control point;
- Booster compressor stations (if necessary);
- Block - container of the control system;
- Block - container of the main start-up of installations.
- Nitrogen ramp:

Construction class - CC-2.

Level of responsibility - (normal).

Degree of fire resistance - IIIa.

- Nitrogen ramp, metal, installed in a shelter, made in the form of an open metal canopy.

- Installation of open above-ground, technological equipment (Tanks for collecting condensate) and service platforms.

- Bearing, free-standing supports made of reinforced concrete and overpass, for laying, above-ground gas pipeline networks.

On the gas pipeline networks, platforms for servicing shut-off valves are provided, with stairs made of metal elements, with grating and railings. A metal canopy serves as protection from atmospheric precipitation. Racks of service platforms and sheds are installed on reinforced concrete foundations.

Structure class - CC-3.

Site responsibility level (increased);

Production category «A».

Along the perimeter, the site is fenced with a metal mesh fence, 1.6 m high.

Air compressor

Functional fire hazard class of the building - F5.1;

Construction class - CC-3.

Level of responsibility - (increased).

The air compressor station is provided in a metal heated module, block-container type, complete prefabrication with the appropriate equipment supplied by the manufacturer.

Water treatment and treatment facilities

Class of functional fire hazard of the building - F5.1.

Construction class - CC-2.

Level of responsibility - (normal).

The WTP is a single complex.

Diesel fuel warehouse

Building class - CC-1.

Responsibility level - (Reduced).

The diesel fuel warehouse is an open area.

On these sites there are monolithic foundations of the ring type, arranged on a compacted artificial base, for the installation of above-ground vertical steel tanks with a capacity of 10,000.0 m³.

Along the perimeter of the storage site for a group of tanks, based on the spilled liquid, a closed enclosing reinforced concrete wall (square) is provided.

Over the entire surface of the site, bordered by the enclosing wall, in areas free from foundations, it is planned to backfill with crushed stone, with compaction. Drainage wells are located at the corners of the site. To support pipelines at oil facilities, free-standing supports are provided. To pass through the enclosing wall on opposite sides of the fence, transitional steel stairs-transitions are provided.

Pumping diesel and oil

Construction class - CC-2.

Level of responsibility - (normal).

Degree of fire resistance - II.

Production category -B.

The building provides for such premises as: diesel fuel pumping room, supply and exhaust ventilation chambers, electrical panel, oil control room, control panel, switchgear, sensor room, sanitary facilities for personnel.

Handling and transport equipment is provided for pumping diesel fuels.

Oil warehouse

Construction class - CC-2.

Level of responsibility - (normal).

To accommodate the oil storage, an open area is provided, with the location of solid foundations on it, arranged on a compacted artificial base, for the installation of above-ground vertical steel tanks, with a capacity of 70.0 m³ (6 pcs.). Along the perimeter of the storage site for a group of tanks, based on the spilled liquid, a closed enclosing reinforced concrete wall (square) is provided.

Over the entire surface of the site, bordered by the enclosing wall, in areas free from foundations, it is planned to backfill with crushed stone, with compaction. Drainage wells are located at the corners of the site. To pass through the enclosing wall on opposite sides of the fence, transitional steel stairs-transitions are provided.

Cable management on site

Linear structure, above-ground laying, which is a flyover with supports and service platforms.

Gas pipelines on site

The gas pipeline to buildings and structures is carried out in an above-ground design, on low reinforced concrete supports.

Overpasses of technological pipelines and thermal outlets on the site

Level of responsibility – II.

The degree of fire resistance is not standardized.

Overpasses are made in the form of temperature blocks, consisting of intermediate and anchor supports. Laying of heat supply pipelines above ground and underground.

Lightning protection and grounding

Lightning rod and grounding to be determined by the project.

Administrative building

Construction class CC-2.

Level of responsibility - (normal).

Production category - «D»;

Constructive degree of fire resistance - IIIa;

Degree of fire resistance - II.

Service building

Construction class CC-2.

Level of responsibility - (normal).

Production category - «D»;

Constructive degree of fire resistance - IIIa;

Degree of fire resistance - II.

Joint auxiliary building

Construction class CC-2;

Level of responsibility - (normal);

Production category - «D»;

Constructive degree of fire resistance - IIIa;

Degree of fire resistance - II.

Checkpoint

Construction class CC-2.

Level of responsibility - (normal).

Production category - «D»;

Constructive degree of fire resistance - IIIa;

Degree of fire resistance - II.

Canteen

Construction class CC-2.

Level of responsibility - (normal).

Production category - «D»;

Constructive degree of fire resistance - IIIa;

Degree of fire resistance - II.

Fire station

Construction class CC-2.

Level of responsibility - (normal).

Production category - «B»;

Degree of fire resistance - II.

Civil defense facilities

Perform in accordance with the requirements of the State Institution «Department for Emergency Situations of the Turkestan Region of the Ministry of Emergency Situations of the Republic of Kazakhstan».

Closed parking garage for reserve cars

Construction class CC-2.

Level of responsibility - (normal).

Production category - «B»;

Structural degree of fire resistance - IIIa;

Degree of fire resistance - II.

Warehouses for fire fighting purposes

Construction class CC-2.

Level of responsibility - (normal).

Production category - «B»;

Structural degree of fire resistance - IIIa;

Degree of fire resistance - II.

Industrial and fire-fighting water supply with facilities

Fire tank with a capacity of $V=003d\ 12000m^3 \times 2=003d\ 24000m^3$.

Buried monolithic reinforced concrete tanks equipped with manholes.

Pumping fire fighting

Construction class CC-3.

Responsibility level - (increased).

Production category - «B»;

Structural degree of fire resistance - IIIa;

Degree of fire resistance - II.

Household sewerage with facilities

Factory-installed underground station with a projecting part above ground level.

Rainwater, industrial sewerage with facilities

Construction class CC-3.

Level of responsibility - (increased).

Tank for collecting industrial effluents with a pumping station

Round underground reinforced concrete tank equipped with a manhole.

Emergency oil drains of transformer oil

Tanks for receiving emergency discharge of transformer oil in the station node, is an underground reinforced concrete tank.

Turbine oil emergency drains

Tanks for receiving emergency oil drain are underground reinforced concrete tanks.

Outdoor and security lighting

Monolithic concrete foundations for lighting poles made of C16/20 concrete, arranged in drilled wells.

Industrial site fencing

Along the perimeter, the territory is subject to fencing, from steel mesh panels to a height of 2150mm, along metal poles installed on prefabricated reinforced concrete foundations. On top of the steel mesh fencing panels, an additional spiral barrier device is provided.

Checkpoint

Construction class CC-2.

Level of responsibility - (normal).

Production category - «D»;

Degree of fire resistance - IIIa.

Steel insulated block-box of full factory execution.

3.1.3. Anti-corrosion measures

All external surfaces of concrete and reinforced concrete products and structures in contact with the ground, the surfaces of steel elements of building structures, are subject to mandatory protection against corrosion by corrosion-resistant materials.

Protective coatings are provided taking into account the type and degree of aggressiveness of the environment in the new operating conditions, using advanced protection methods.



3.1.4. Activities on electrical, explosion and fire safety

All load-bearing structural elements of the building, structures, carried out in construction conditions and block-boxes, made in the factory version, external and internal enclosing structures (walls, ceilings, coatings) and others are made of fireproof materials and do not have fire and explosive components. As a heat-insulating filling, in the external enclosing structures, a fireproof mineral material from the basalt fiber of the not-combustible (NC) group is adopted.

The premises of industrial buildings with different categories of production, with different categories for explosive and fire safety, are separated from each other by fire walls made of non-combustible materials throughout the entire height of the building.

Buildings according to their structural characteristics, with a frame structural scheme and frame elements made of unprotected steel structures, having a fire resistance degree of IIIa according to a constructive solution, are subject to mandatory coating with protective compounds that ensure an increase in the fire resistance of the building to a fire resistance degree of II.

3.1.5. Measures to comply with sanitary requirements and environmental protection

The areas of sanitary premises, sets of sanitary and domestic equipment and wardrobes for changing clothes and storing street clothes in dressing rooms are established according to the total payroll of employees, the maximum number of employees per shift, and also, taking into account the group of production processes.

Building materials used for the manufacture of structures and products, in buildings and structures, materials and products for exterior and interior finishing works, waterproofing and anti-corrosion compounds, should not contain harmful components, radioactive and unpleasantly smelling substances, in quantities exceeding maximum permissible concentrations, as well as pathogenic bacteria, viruses and fungi that adversely affect the condition and health of workers, the environment and comply with the requirements of sanitary standards and environmental protection.

3.2. Engineering survey

3.2.1. Survey organization

The Contractor shall conduct engineering surveys: engineering geodetic, engineering geological, engineering hydrological and engineering meteorological surveys to the extent necessary for the development of design estimates in accordance with the requirements of CII PK 1.02-105-2014 «Инженерные изыскания для строительства. Основные положения» (SP RK 1.02-105-2014 «Engineering surveys for construction. Main Provisions»).

Engineering surveys should be carried out in the presence of permits issued by the relevant executive authorities or local governments.

Permission to conduct surveys must be issued by the organization performing the surveys (performing costs must be included).

Measuring instruments used in engineering-geological surveys for construction are subject to state metrological control and supervision carried out by accredited metrological services.

The engineering survey program should be compiled by the survey organization (subdivision) on the basis of the terms of reference of the design organization approved by the Customer in accordance with the requirements of regulatory documents and with the maximum use of materials from previously completed engineering surveys and other information about the natural conditions of the area, site, site, survey route, and also taking into account the results of the field survey of the area, if it was carried out.

The survey program should establish the composition and scope of engineering and survey work based on the terms of reference, and based on the design stage, a detailed design, type of construction, type of buildings and structures, their significance, area of the study area, the degree of its study and the complexity of engineering and geological conditions.

Surveys without a survey program are not allowed.

The survey program is the main document in the conduct of survey work, internal quality control, acceptance of survey materials, as well as in the examination of technical reports.



When carrying out complex survey work, the program of engineering and geological surveys should be linked to programs of other types of surveys (in particular, engineering and environmental surveys) in order to avoid duplication of certain types of work (drilling, sampling, etc.).

When performing surveys, the requirements of regulatory documents on labor protection, conditions for observing fire safety and environmental protection (ГОСТ 12.0.001-82 (GOST 12.0.001-82), CH PK 1.03- 05-2011 (SN RK 1.03-05-2011), etc.) must be observed.

Based on the results of the engineering surveys, it is necessary to draw up a technical report (conclusion), which should contain the data provided for by the Task for the development of the Customer's design and estimate documentation and the survey program, as well as the rationale for the changes made to the program.

The technical report (conclusion) must be submitted to the Customer and, in the prescribed manner, by the authority that issued the permit for the conduct of surveys, carried out their registration and the territorial survey organization.

One copy of the technical report (conclusion) shall be deposited in the archive of the organization that carried out the survey.

Surveys must be carried out in accordance with the requirements for working documentation for construction. Prepare and publish as part of a working draft as a separate book.

The Contractor is fully responsible for calculations, etc., in which the results of engineering surveys were used.

Engineering surveys should provide a comprehensive study of the natural conditions of the area where the route of the planned transmission line and substation, local building materials, water supply sources and obtain the necessary data to develop economically viable and technically sound solutions for the design and construction of the Facility, taking into account the rational use and protection of the natural environment.

During the production of surveys related to the violation of the soil cover, it is necessary to remove, store and after completion of work, lay the fertile soil layer on the disturbed lands, and also prevent air, water and soil pollution.

3.2.2. Composition and general requirements for engineering and geodetic surveys

Engineering and geodetic surveys to perform Works on the Facility should provide topographic and geodetic materials and data on the situation and terrain (including the bottom of watercourses, reservoirs and water areas), existing buildings and structures (ground, underground, aboveground), planning elements (in digital, graphic, photographic and other forms) necessary for the design of the Facility, as well as for other types of engineering surveys.

The composition of engineering and geodetic surveys for the performance of Works under the Contract includes:

- collection and analysis of engineering survey materials of past years, topographic and geodetic, cartographic, aerial photography and other materials and data on the construction area;
- reconnaissance survey of the survey area;
- creation (development) of reference geodetic networks (geodetic networks of 3rd and 4th classes, geodetic network of concentration of 1st and 2nd categories and leveling network of II, III and IV classes), as well as the construction of geodetic networks for special purposes;
- creation of planned high-altitude survey geodetic networks;
- topographic (terrestrial, aerial phototopographic, stereophotogrammetric, etc.) survey on a scale of 1:10000 - 1:500, including survey of underground and above-ground structures;
- updating topographic (engineering topographic) and cadastral plans in graphic, digital, photographic and other forms;
- engineering and hydrographic works;
- geodetic work related to the removal to nature and binding of mine workings, geophysical and other points of engineering surveys;
- removal of the designed structures in nature with the preparation of the relevant act;
- geodetic stationary observations of deformations of the foundations of buildings and structures, the earth's surface and rock masses in areas of development of dangerous natural and techno-natural processes;

- creation (compilation) and publication (reproduction) of engineering and topographic plans, and cadastral and thematic maps, and plans, atlases for special purposes (in graphic, digital and other forms);
- cameral processing of materials;
- drawing up a technical report (explanatory note).

In separate areas with a dense network of underground and above-ground structures, in order to display the situation and communications in more detail, as well as in small areas, it is allowed to carry out topographic surveys at a scale of 1:200 in agreement with the Customer. The technical requirements for its implementation should be drawn up by a geodetic organization (subdivision) on the basis of the terms of reference of the design organization, approved by the Customer in accordance with the requirements of regulatory documents.

3.2.3. Composition and general requirements for engineering and geological surveys

Engineering and geological surveys for the implementation of Works on the Facility should provide a comprehensive study of the engineering and geological conditions of the area (site, section, route) of the planned construction of the Facility, including the relief, geological structure, geomorphological and hydrogeological conditions, composition, condition and properties of soils, geological and engineering - geological processes, changes in the conditions of developed (built-up) territories, forecasting of possible changes in engineering and geological conditions in the field of interaction of the designed objects with the geological environment in order to obtain the necessary and sufficient materials for the design, construction and operation of the Facility.

Engineering and geological surveys should provide a comprehensive study of the engineering and geological conditions of the area (site, section, route) of the designed power transmission line, including relief, geological structure, composition, condition and properties of soils.

The composition of engineering and geological surveys includes:

- collection and processing of materials from surveys and studies of past years;
- decryption of aero- and space materials;
- reconnaissance survey, including aero-visual and route observations;
- tunneling of mine workings;
- geophysical surveys;
- field studies of soils;
- hydrogeological research;
- stationary observations (local monitoring of the components of the geological environment);
- laboratory studies of soils, underground and surface waters;
- survey of soils of the bases of the foundations of existing buildings and structures;
- making a forecast of changes in engineering and geological conditions;
- clarification of the seismicity of individual sites of the territories;
- cameral processing of materials and drawing up a technical report (conclusion).

Engineering-geological surveys should provide a comprehensive study of the current state of the engineering-geological conditions of the territory (region, site, route) planned for construction development, assessment and forecasting of possible changes in these conditions when using it. The survey program should provide for the performance of engineering and geological surveys, including a set of individual types of survey work.

3.2.4. Composition and general requirements for engineering and hydrometeorological surveys

Engineering and hydrometeorological surveys to perform the Works on the Facility should be carried out to provide the design with initial data when solving the following tasks:

- selection of locations for the construction site (route) and its engineering protection from adverse hydrometeorological impacts;
- development of the general plan of the territory;
- selection of structures of structures, determination of their main parameters and organization of construction;

- determining the possibility of meeting the demand for water and organizing various types of water use;
- assessment of the impact of construction projects on the surrounding water and air environment and the development of environmental measures.

Engineering and hydrometeorological surveys are carried out in conjunction with engineering and geological surveys to predict flooding, surveys of water supply sources based on groundwater, geocryological studies, the study of karst, landslides and other dangerous geological processes.

Engineering and hydrometeorological surveys include:

- collection, analysis and generalized materials of hydrometeorological and cartographic knowledge of the territory;
- reconnaissance survey of the engineering survey area;
- observations of the characteristics of the hydrological regime of water bodies and meteorological elements;
- flooding (flooding), hurricane winds, snow avalanches, snowdrifts, ice, channel process, processing of the banks of rivers, lakes, reservoirs and abrasion of sea coasts;
- study of hydrometeorological processes and phenomena;
- office processing of materials with the determination of the calculated hydrological and (or) meteorological characteristics;
- drawing up a technical report.

When preparing, organizing and performing engineering surveys, the Contractor shall be guided by the legislative and other regulatory legal acts of the Republic of Kazakhstan, building codes and regulations, as well as other state and interstate standards in the field of architecture and construction in force on the territory of the Republic of Kazakhstan.

Reasonable deviations from the technical requirements (conditions, restrictions) of state standards are allowed only if there are written approvals (permits) of the Customer from those state bodies that approved and put into effect these standards (or their successors).

3.3. Design

3.3.1. Norms and standards. General provisions

The design should be carried out in accordance with the regulatory documents in force on the territory of the Republic of Kazakhstan for the period of work.

When performing the design, also be guided by, but not limited to, construction norms and construction regulations of the RK.

All decisions made during the design process must be consistent with the objectives of the project.

If it is impossible to use the regulatory documentation in accordance with the requirements of the legislation of the RK, develop special technical conditions within the framework of this project.

The development of project documentation is considered completed from the moment of its approval in the manner prescribed by law.

According to the project documentation, which has passed the procedure of mandatory approvals in the prescribed manner, before its approval, an examination is carried out, unless otherwise provided by the legislation on architectural, urban planning and construction activities.

The examination is carried out in the manner prescribed by the Government of the Republic of Kazakhstan.

At the same time, the mandatory state examination is carried out by a legal entity authorized by the Government of the Republic of Kazakhstan on objects referred to its exclusive competence in accordance with the legislation of the Republic of Kazakhstan.

Design documentation for construction that is subject to mandatory expertise (state expertise), but has not passed it in the prescribed manner, is considered incomplete and is not subject to approval and further implementation.

Design documentation for the construction of facilities must be submitted for approval no later than three months after receiving a positive conclusion from the state expertise.

Developed and approved in accordance with the design assignment, architectural and planning assignment, other source materials and specifications, project documentation on the basis of a positive conclusion of the examination (state examination) of projects is subject to approval in the manner prescribed by law and in accordance with the requirements established by state regulatory documents.

The following must be approved:

- in case of one-stage design - working project (WP);
- in case of two-stage design - a project (P) developed at the first stage. Working documentation (WD), completed at the second stage, is not subject to examination and is approved by the Customer.

The design documentation at the design stage (P) of a production facility (an enterprise consisting of a separate building, structure or their complexes, engineering and transport communications) includes the following sections:

- project passport;
- energy passport of the project (according to the design assignment);
- general explanatory note;
- general plan of the facility and organization of transport;
- engineering protection of the territory;
- technological solutions;
- management of production, enterprise, organization of conditions and labor protection of employees;
- architectural and construction solutions;
- engineering networks, systems and equipment;
- engineering and technical measures for civil defense and measures to prevent natural and man-made emergencies;
- automated system for monitoring buildings and structures in accordance with the mandatory Appendix 1 of СНиП РК 3.02-05-2010 (SNIIP RK 3.02-05-2010);
- a system for ensuring integrated security and anti-terrorist protection for especially important facilities, large-span structures and high-rise buildings, their complexes;
- construction organization project;
- environmental protection;
- estimate documentation, including a summary sheet of material resources and equipment indicating the country of origin for each material according to the database of goods, works, services and their suppliers, formed in accordance with the Rules for the formation and maintenance of a database of goods, works, services and their suppliers, approved by order of the Acting Minister for Investment and Development of the Republic of Kazakhstan dated November 26, 2015 No. 1107.

3.3.2. Special technical conditions (STC)

If necessary, execute (order) the Special technical conditions section (hereinafter STC), from the profile organization, in accordance with СН РК 1.02-03-2022 Порядок разработки, согласования, утверждения и состав проектной документации на строительство (SN RK 1.02-03-2022 Procedure for the development, approval, approval and composition of design documentation for construction).

The basis for the development of STC is the absence of design standards for objects with special unique parameters.

3.3.3. BIM (Building information modeling) design

General

Perform project development using BIM technologies in accordance with the rules and regulations in force on the territory of the Republic of Kazakhstan, including, but not limited to:

СН РК 1.02-03-2022 Порядок разработки, согласования, утверждения и состав проектной документации на строительство (SN RK 1.02-03-2022 The procedure for the development, approval, approval and composition of design documentation for construction).

Order of the Chairman of the Committee for Construction and Housing and Communal Services of the Ministry for Investments and Development of the Republic of Kazakhstan dated December 10, 2018 No. 253-nq «On approval of the guiding document in the construction of the Republic of Kazakhstan 1.02-04-2018 «Information modeling in construction. Main Provisions».

СП РК 1.02-111-2017 «Применение информационного моделирования в проектной организации» (SP RK 1.02-111-2017 «Application of information modeling in a design organization»).

СП РК 1.02-120-2019 «Применение информационного моделирования в строительной организации». (SP RK 1.02-120-2019 « Application of information modeling in a construction organization»).

СП РК 1.02-121-2019 «Применение информационного моделирования в эксплуатирующей организации» (SP RK 1.02-121-2019 «Application of information modeling in the operating organization»).

Primary goal

Development of a project using BIM technologies, which will achieve:

- high quality IM (information model), P, WD;
- efficient and safe solutions;
- accurate counting of materials and equipment;
- reduction of terms of implementation.

The main task is to create an IM (information model) that will provide information and technical support for the life cycle of an object.

Tasks

Tasks at the design stage:

- creation of IM based on the obtained initial data;
- coordination, approval, successful examination at the stage «P»;
- checking for collisions;
- development of the project at the «WD» stage.

Tasks at the construction stage:

- monitoring of construction and installation works, which will allow to identify deviations from design solutions;

- correction of IM based on the data received from monitoring;
- creation of a single summary information model that corresponds to a facility ready for operation.

Tasks at the operational stage:

- use of IM with the actual list of installed equipment;
- maintenance of current repairs;
- assessment of the current and predicted technical condition of elements of building structures and equipment.

Requirements

General requirements for the Contractor

The Contractor must meet the following requirements:

The presence of highly qualified certified specialists (the certificate confirms the completion of training and the availability of competencies);

Specialists must have experience in implementing BIM projects (provide information on completed projects using BIM technologies).

Experience in designing using BIM technologies (provide information on completed projects).

The executors will be managed by the BIM Manager - this is a specialist who will be responsible for the process of implementing the BIM project and communication with the Customer (or Customer's representative).

Software requirements

IM (information model) must be performed using the following licensed software:

- Autodesk Revit,
- Autodesk Robot Structural Analysis Professional,
- Autodesk Civil 3D,
- Autodesk Navisworks,

- Autodesk InRoads,
- Aveva E3D;
- Tekla Structures.

Software versions are specified and agreed with the Customer immediately before the start of work on the creation of an IM (information model).

Collaboration requirements

To organize collaboration, cloud services BIM 360 Team, ERwin, BPwin (or equivalent), Trello are used.

Document management Aconex (and/or others).

Using external links

To use external links, you must create separate files for each section. If necessary, create multiple files for a specific partition. Link all available files to each other by external links in a single accepted coordinate system.

Using components

Use components from open sources, own libraries, software libraries, if necessary, create libraries for the project.

3.3.4. Detail level of the information model LOD (Level of Development)

A more detailed study of this issue is carried out immediately before the start of the creation of IM by persons responsible for the implementation of the BIM project on the part of the Customer and the Contractor.

The level of detail should be performed taking into account the technical and technological complexity of the project, but not less than the level of LOD350. The level of detail should include interfaces to other building components, such as connections and integrations, specifying installation and interactions with other building systems.

It also includes: the level of elaboration of geometry - location in space, shape, length, width, height, etc. the level of elaboration of attribute information (properties, parameters).

3.3.5. Staging

The staging is to be accepted in accordance with СН РК 1.02-03-2022 «Порядок разработки, согласования, утверждения и состав проектной документации на строительство» (SN RK 1.02-03-2022 « The procedure for the development, approval, approval and composition of the project construction documentation»), two-stage:

- the first stage - a project (P), developed, as a rule, on the basis of the conclusions and indicators of pre-project documentation and subject to approval in the manner prescribed by law;
- second stage - working documentation (WD), developed on the basis of the project approved at the first stage.

Order of the Minister of National Economy of the Republic of Kazakhstan dated December 5, 2014 No. 129. Registered with the Ministry of Justice of the Republic of Kazakhstan on December 8, 2014 No. 9938 «On approval of the Rules for the development or adjustment, conducting the necessary examinations of the investment proposal of the state investment project, as well as planning, consideration, selection, monitoring and evaluation of the implementation of budget investments and determining the feasibility of budget lending».

3.3.6. Commissioning works

Commissioning is a set of works performed during the preparation and conduct of individual tests and during the complex testing of equipment. At the same time, the concept of «equipment» covers the entire technological system of the facility, that is, a complex of technological and all other types of equipment and pipelines, electrical, sanitary devices, automation and other systems provided for by the project. Commissioning works carried out during the period of individual testing of equipment ensure the fulfillment of the requirements stipulated by the working documentation, standards and

specifications for individual machines, mechanisms and units, in order to prepare the equipment for acceptance for comprehensive testing. During the period of complex testing of the equipment, checks, adjustments and ensuring the joint interconnected operation of the equipment in the technological process provided for by the project at idle are carried out, followed by transferring the equipment to work under load and bringing it to a stable technological mode provided for by the project.

3.3.7. Requirements for passing a comprehensive non-departmental and state environmental expertise

Passing of the State non-departmental examination of projects

Passing of the «State non-departmental examination of projects» in accordance with the Order of the Minister of National Economy of the Republic of Kazakhstan dated April 1, 2015 No. 299 (with amendments and additions as of October 21, 2021), registered with the Ministry of Justice of the Republic of Kazakhstan on April 15, 2015 No. 10722 «On approval of the Rules for conducting a comprehensive non-departmental examination of feasibility studies and design estimates intended for the construction of new, as well as changes (reconstruction, expansion, technical re-equipment, modernization and overhaul) of existing buildings and structures, their complexes, engineering and transport communications, regardless from funding sources».

The estimated cost of commissioning works are taken into account as a separate line in the summary estimate for the preparation (development) of the construction site and commissioning of the facility, in accordance with the regulatory document for determining the estimated cost of commissioning works when commissioning construction projects in the Republic of Kazakhstan, approved by the Order of the Chairman of the Committee for construction and housing and communal services of the Ministry for Investments and Development of the Republic of Kazakhstan dated November 14, 2017 No. 249-nq.

When commissioning production facilities (including those for which commissioning provides for a trial production), the summary estimate for the preparation (development) of the construction site and putting the facility into operation (including the estimate for commissioning) is subject to passing industry and state expertise with subsequent approval by the sectoral authorized state body as part of a budget investment project or by the Customer.

The estimated cost of commissioning, approved in the prescribed manner, is the limit of the Customer's funds for the entire period of commissioning until the acceptance of the enterprise, building and structure for operation.

Passing of the State environmental expertise

Passing of the «State environmental expertise» in accordance with the Order of acting Minister of Ecology, Geology and Natural Resources of the Republic of Kazakhstan dated August 9, 2021 No. 317, registered with the Ministry of Justice of the Republic of Kazakhstan on August 9, 2021 No. 23918 «On approval of the Rules for conducting state environmental expertise».

3.3.8. Site preparation

Before starting work, remove fertile plant soil from the construction site with a thickness according to engineering and geological surveys and use it in accordance with the requirements of the Land Code of the Republic of Kazakhstan (with amendments and additions as of August 31, 2022). Excess fertile soil removed from the construction site should be stored in a designated place on the territory allocated by the Customer and transferred to the Customer for further storage.

Organization of the vertical layout

The vertical layout of the sites, the floor level of the first floor of buildings, the slopes of the site surface should be applied in accordance with the requirements of CII PK 3.01-103-2012 (SP RK 3.01-103-2012).

Landscaping and gardening

Designed roads should be provided to all buildings and structures adopted in the project.



Provide a transport scheme for on-site roads and railways in accordance with the requirements of the regulatory document СП РК 3.03-122-2013 «Промышленный транспорт» (SP RK 3.03-122-2013 «Industrial transport»), including for the transportation of auxiliary and household goods, for the passage of fire, repair and emergency vehicles.

To determine the categories of the road according to its purpose in accordance with СП РК 3.03-122-2013 «Промышленный транспорт» (SP RK 3.03-122-2013 «Industrial transport»).

To provide for the pavement structure of the roadway in accordance with СТ РК 1225-2019 «Смеси асфальтобетонные дорожные, аэродромные и асфальтобетон» (ST RK 1225-2019 «Asphalt concrete mixes for road, airfield and asphalt concrete»), СТ РК 1284-2004 «Щебень и гравий из плотных горных пород для строительных работ» (ST RK 1284-2004 «Crushed stone and gravel from dense rocks for construction work»); ГОСТ 23735-2014 «Смеси песчано-гравийные для строительных работ» (GOST 23735-2014 «Sand and gravel mixtures for construction work»).

Along driveways, along the edges of sidewalks and platforms, provide for the installation of concrete stones in accordance with ГОСТ 6665-91 (GOST 6665-91).

Provide for landscaping of the territory with trees and lawns in accordance with the requirements of СП РК 3.01-103-2012 «Генеральные планы промышленных предприятий» (SP RK 3.01-103-2012 «General plans for industrial enterprises»).

The width of green planting lanes and their planting in order to protect the enterprise, industrial parks and industrial centers from prevailing winds shall be determined in accordance with СП РК 3.01-103-2012 «Генеральные планы промышленных предприятий» (SP RK 3.01-103-2012 «General plans for industrial enterprises»).

The distance from buildings and structures to trees and shrubs and the distance between trees and shrubs in row planting should be taken in accordance with СП РК 3.01-103-2012 «Генеральные планы промышленных предприятий» (SP RK 3.01-103-2012 «General plans for industrial enterprises»).

The number of entrances to the territory, the width of the gates of automobile and railway entrances, the width of roads and their distance from the building, the category is determined in accordance with the requirements of СП РК 3.01-103-2012 «Генеральные планы промышленных предприятий» (SP RK 3.01-103-2012 «General plans for industrial enterprises») and Fire safety rules for energy enterprises of the Republic of Kazakhstan dated February 20, 2015 year No. 123, Technical regulation «General requirements for fire safety» dated August 17, 2021 No. 405.

Install containers for solid household waste in accordance with the Order of acting Minister of Health of the Republic of Kazakhstan dated December 25, 2020 No. ҚР ДСМ-331/2020 (QR DSM-331/2020). On approval of the Sanitary Rules «Sanitary and Epidemiological Requirements for the Collection, Use, Application, Neutralization, Transportation, Storage and Disposal of Production and Consumption Waste».

The territory of the production facility must have functional zoning in accordance with СП РК 3.01-103-2012 (SP RK 3.01-103-2012).

On the territory of an industrial facility, it is necessary to allocate administrative and auxiliary zones, production and transport and storage, which are separated from each other by gaps in accordance with СП РК 3.01-103-2012 (SP RK 3.01-103-2012).

Parking and storage places for vehicles operating at enterprises, in technology parks, should be located on the territory of the enterprise, technology park, in accordance with the Sanitary Rules «Sanitary and epidemiological requirements for the establishment of a sanitary protection zone of production facilities».

The location of buildings of amenity premises, entrance points, the distance from the entrance points to the entrances to the sanitary premises of the main workshops shall be taken in accordance with СП РК 3.01-103-2012 (SP RK 3.01-103-2012).

Sites in front of the checkpoints and entrances to sanitary facilities, canteens and control buildings should be provided in accordance with СП РК 3.01-103-2012 (SP RK 3.01-103-2012).

Placement of engineering networks

Between adjacent underground engineering networks in case of their parallel placement, horizontal distances (in the light) from the nearest underground engineering networks, with the exception

of combustible gas pipelines, to buildings and structures shall be applied in accordance with CII PK 3.01-103-2012 (SP RK 3.01-103-2012).

Cranes and lifting equipment

Sufficient cranes, winches, monorails, platforms, lattice ladders, ladders, traverses, trolleys, lifting blocks, etc., must be provided at the Site for the maintenance of all equipment.

Each crane must be sized for the heaviest component it must lift and remove for service. Zones and platforms with equipment for testing cranes (hydraulic loaders) should be provided.

The lifting mechanism (or metal structures for its fastening) must be provided if the component weighs more than 25 kg (from the condition of lifting by one person) or 50 kg (if lifted by two employees).

Heating, ventilation and air conditioning (HVAC) systems

All buildings and structures must have complete automated heating, ventilation and air conditioning (HVAC) systems, consisting of the necessary air conditioning units, fans, heaters, control units and other things.

All systems must be designed and installed to provide acceptable environmental conditions for each area. The conditions of temperature, humidity, air movement and air filtration must be controlled in accordance with the requirements in the respective premises.

The HVAC systems should be designed to shut down in the event of a fire alarm, pressurize supply and exhaust ducts as required for clean agent fire extinguishing systems, and start pressurization systems. Ventilation and smoke removal in the event of a fire must be provided in all areas of the installation using roof vents or exhaust fans.

The Contractor shall prepare calculations for sizing and selecting HVAC equipment and components. These calculations should be based on, but not limited to, environmental design requirements. Heating and cooling load calculations must be made in accordance with the requirements of the TNLA of the Republic of Kazakhstan.

The Contractor shall submit to the Customer the final engineering calculations of the system design, approved design drawings (air flow diagrams, equipment drawings indicating connecting dimensions) and equipment certificates.

The cooling load should include heat input from internal electrical and mechanical sources, solar heat input to buildings, and ventilation loads. The design must include at least a 10% capacity margin for heating and cooling loads.

The HVAC equipment must be of industrial quality and designed for round-the-clock operation. All systems and components must be suitable for operation in harsh environments with high temperature and humidity, match the design life of the power plant, be designed for maximum life cycle and complexity in accordance with the functional performance of the power plant, be easy to maintain and reliable.

Belts, pulleys, chains, gears, couplings, and protruding objects such as set screws, keys, and other rotating parts must be completely enclosed and positioned so that personnel in the immediate vicinity are adequately protected.

3.3.9. Estimate documentation

The Estimate documentation for the working draft to be developed in accordance with the «Regulatory document for determining the estimated cost of construction in the Republic of Kazakhstan», approved by the Order of the Chairman of the Committee for Construction, Housing and Communal Services and Land Management of the Ministry of National Economy of the Republic of Kazakhstan dated November 14, 2017 No. 249-nk.

Estimate documentation for the construction of the Facility shall be executed by the resource method using the ABC-4 software package, presented at current prices, in accordance with the regulatory documentation in force in the Republic of Kazakhstan for the current year for the period of work.

When performing, be guided by, but not limited to, the following standards:



- Regulatory document for determining the estimated cost of construction in the Republic of Kazakhstan (Appendix 1 to the order of the Chairman of the Committee for Construction and Housing and Communal Services of the MID RK dated November 14, 2017 No. 249-NK).
 - Regulatory document for determining the amount of overhead costs and estimated profit in construction (Appendix 2 to the order of the Chairman of the Committee for Construction and Housing and Communal Services of the Ministry of Investment and Development of the Republic of Kazakhstan dated November 14, 2017 No. 249-NK);
 - Regulatory document for determining the additional costs associated with the decisions of the construction organization project (Appendix 3 to the order of the Chairman of the Committee for Construction and Housing and Communal Services of the Ministry of Investment and Development of the Republic of Kazakhstan dated November 14, 2017 No. 249-NK).
 - Regulatory document for determining the costs of engineering services (Appendix 4 to the order of the Chairman of the Committee for Construction and Housing and Communal Services of the Ministry of Investment and Development of the Republic of Kazakhstan dated November 14, 2017 No. 249-HK), approved by order of the Chairman of the Committee for Construction and Housing and Communal Services MID RK dated May 11, 2018 No. 102-NK.
 - Amendments and additions to the order of the Chairman of the Committee for Construction, Housing and Communal Services of the MID RK dated November 14, 2017 No. 249-NK ORDER No. 8-NQ dated January 25, 2022.
 - Collection of estimated cost norms for the construction of temporary buildings and structures (HДЗ PK 8.04-05-2015 (NDZ RK 8.04-05-2015), taking into account changes and additions;
 - Collections of estimated prices at the current level for building materials, products and structures (CCII PK 8.04-08-2021 (SSC RK 8.04-08-2021). 2022;
 - Collection of estimated prices at the current level for engineering equipment of construction facilities (CCII PK 8.04-09-2021 (SSC RK 8.04-09-2021). 2022;
 - Collection of estimated prices at the current level for the operation of construction machines and mechanisms (CIQЭM PK 8.04-11-2021 (SCEM RK 8.04-11-2021). 2022;
 - Collection of estimated tariff rates in construction (CTC PK 8.04-07-2021 (STS RK 8.04-07-2021). 2022;
 - Collection of estimated prices at the current level for the transportation of goods for construction. Department 1 Road transportation (CIQIIГ PK 8.04-12-2021 (SCPG RK 8.04-12-2021). 2022;
 - Collection of estimated prices at the current level for the transportation of goods for construction. Department 2 Rail transportation (CIQIIIГ PK 8.04-12-2021 (SCPG RK 8.04-12-2021). 2022;
 - Collections of aggregated indicators of the estimated cost of constructs and types of work (YCH PK 8.02-03-2021 (USN RK 8.02-03-2021). 2022;
 - Collections of aggregated indicators of the estimated cost of construction of buildings and structures. Objects for non-industrial purposes (YCH PK 8.02-04-2021 (USN RK 8.02-04-2021). 2022
 - General provisions on the application of elemental estimated norms for commissioning of the ЭCH PK 8.04-03-2022 (ESN RK 8.04-03-2022)
 - Elemental estimated norms for construction work and installation of equipment, a collection of estimated prices at the current level for the operation of construction machines and mechanisms, collections of aggregated indicators of the estimated cost of constructs and types of work for non-industrial facilities, collections of aggregated indicators for the cost of building buildings and structures non-industrial facilities Changes and additions Issue 25 Unforeseen work and costs are accepted in accordance with the Regulatory document for determining the estimated cost of construction in the Republic of Kazakhstan
- Include others:
- for the construction of temporary buildings and structures in accordance with the HДЗ PK 8.04-05-2015 (NDZ RK 8.04-05-2015)
 - additional costs when performing work in climatic conditions ЭCH PK 8.04-02-2015 (ESN RK 8.04-02-2015) Table G.1 and Table D.
 - costs associated with the performance of work on a rotational basis;

- costs associated with sending workers to perform construction and installation work.

Costs of the Customer for project management, technical supervision and architectural supervision in accordance with the Regulatory document for determining the costs of engineering services in construction (according to Appendix 4 in accordance with Order No. 102 of May 11, 2018, with amendments and additions as of December 14, 2018).

Value added tax (VAT) in accordance with the tax legislation of the Republic of Kazakhstan in the amount of 12%.

All decisions made during the design process must be consistent with the objectives of the project.

If it is impossible to use the regulatory documentation in accordance with the requirements of the legislation of the Republic of Kazakhstan, develop special technical conditions within the framework of this project.

3.4. Construction

Construction and installation works shall be carried out in accordance with the developed design and estimate documentation within the time limits stipulated by the Contract.

3.5. Test and acceptance procedure

This paragraph sets out the principles and procedures to be adopted during the final construction and commissioning stages of the Project prior to the final handover of all sites and systems.

General

This section provides detailed information on the stages and means of control that will be provided by the Contractor at the stages of preparation and carrying out of pre-commissioning works (PCW) to meet the requirements of the Customer. This section provides for the sequence of testing and acceptance of objects, but does not replace the rules for testing and acceptance established with the legislation of the RK.

This section should not contradict the testing and acceptance rules provided for by the legislation of the RK and the terms of the Contract. The Contractor will perform all the necessary testing and acceptance procedures established by the legislation of the RK.

Definitions

«Completion of pre-commissioning works» means the completion of the Facilities/Works (excluding deficiencies in any item of the Defect list of D category) and that the permanent Items meet the requirements of all stages and tests set out in this section, and also meet all legal requirements and requirements for issuance The Act of completion of commissioning works in accordance with the Contract.

«Mechanical readiness» means the completion of the Facilities /Works (excluding deficiencies in any item of the Defect list of C category and on the items of the Defect list of D category), including mechanical readiness checks and tests on the Items to demonstrate that the Facilities /Works comply with the requirements of the Contract and construction documentation, as well as readiness for commissioning in a safe mode.

«Defect list» means a list of works that have not been completed and that must be continuously performed, corrected, adjusted or repaired by the Contractor at its own expense as a condition for obtaining the Acceptance certificate in accordance with the Contract.

«Items of the Defect list» means all items of the Defect list that are part of the Facilities/Works, determined (periodically) during any inspection of the Facilities/Works as not complying with the terms of the Contract and as defined in this section, which are divided into the following categories:

1. «Items of the Defect list of A category» means work that must be corrected prior to hydraulic/electrical testing;

2. «Items of the Defect list of B category» means works that must be corrected to Mechanical readiness;

3. «Items of the Defect list of C category» means works that must be corrected prior to Completion of pre-commissioning works;



4. «Items of the Defect list of D category» means minor outstanding issues that:

- will not affect the reliability, non-failure operation, functionality, safety, compliance with current legislation, as well as the mechanical and structural integrity of objects;
- will not affect the performance of tests upon completion of construction;
- will not affect the use of the facilities for their intended purpose (either until or until these issues are resolved);
- will not interfere (in the future) with obtaining permission (permits) for commissioning and operation of facilities, and will not cause the Customer to violate its obligations in accordance with applicable law, if these issues are not resolved.

«Automated Completion System». Electronic database of equipment numbers, class A, B and C inspections required, inspection date, items of the Defect list, etc. The database should be organized subsystemically, according to the systemic (by functionally nodal) approach will become the main source of registration and reporting on the status of completion of construction and the progress of the PCW.

3.5.1. Inspection, testing and acceptance during construction work

The Contractor will prepare and provide the Customer with a Testing plan and a Working plan (procedures) defining the appropriate points of inspection and testing at all stages of production, construction, installation and completion within 28 (twenty eight) days from the date of commencement of work.

3.5.2. Drawing up a Defect list

The Defect list is an integral part of the completion process and is also a critical step in determining the completion status of the Facilities/Works. The category of items of the Defect list determines the fact that the Facilities / Works are ready to move to the next stage. Items of the Defect list take precedence in the categories as specified in this section.

Twenty-one (21) days prior to the expected date of Mechanical readiness, the Contractor will provide the Customer with notice of the readiness of the facilities for a joint interim inspection (hereinafter referred to as the Interim inspection) and of the planned date of the Mechanical readiness of the facilities. Unless otherwise agreed by the parties, an Interim inspection will take place more than ten (10) days prior to the expected Mechanical readiness date of the facilities. The Contractor undertakes to be present during the Interim inspection at any reasonable time during these 10 (ten) days for the purpose of conducting and completing the Interim inspection.

Twenty-one (21) days before the expected date of Completion of pre-commissioning works, the Contractor will provide the Customer with a notice of the readiness of the facilities for the joint Final inspection (hereinafter referred to as the Final inspection) and of the planned date of Completion of the pre-commissioning works of the facilities. Unless otherwise agreed by the parties, the Final inspection will not take place more than ten (10) days prior to the expected date of Completion of pre-commissioning. The Contractor undertakes to be present during the Final inspection at any reasonable time within these 10 (ten) days for the purpose of conducting and completing the Final inspection.

The Contractor undertakes to take into account any comments, information and notices provided by the Customer in accordance with this section, as well as to make adjustments or otherwise eliminate any inconsistencies or errors to the extent required by the Customer.

Unless otherwise agreed, an Interim inspection and a Final inspection will be carried out and completed within 10 (ten) days following receipt of the Contractor's notice in accordance with the time periods specified above, on the date or days specified by the Customer.

At the time of the Interim inspection on the day or days specified above, the parties shall draw up items of the Defect list, which the Contractor undertakes to correct or amend. The Customer shall act in accordance with the terms of this section and the Contract in determining the category of items of the Defect list (Item of the Defect list of A category, Item of the Defect list of B category, Item of the Defect list of C category and Item of the Defect list of D category), and may change the category of items of the Defect list from time to time.

When conducting a Final inspection, the parties must update the Defect list to reflect completed items of the Defect list and identified items of the Defect list.



The Customer shall have the right, acting within reasonable limits, to require additional items to be added to the relevant Defect list by giving notice to the Contractor if the inspections and/or tests provided for in this section have not been completed by the time of the Interim inspection or Final inspection, or such inspections and/or or tests must be repeated under the Contract and as a result such additional items are found.

If the parties cannot agree on a Defect list by the end of the Interim inspection or Final inspection (as appropriate), then the Defect list shall be determined by the Customer acting reasonably.

The Contractor undertakes to provide the Customer with the Defect list in its final form (depending on the Customer's requirement to add any additional items), immediately after the parties reach an understanding or settle all disputed items by the Customer.

The presence of any items of the Defect list in the Defect list shall not prejudice the responsibility and obligations of the Contractor for the compliance of the Facilities/Works with the requirements of the Contract. Joint Interim inspection, joint Final inspection and Customer's comments, no objection, agreement and correction of deficiencies in accordance with the Defect list shall not change or reduce the Contractor's obligations and responsibilities under the Contract, including the obligation to complete all work and the Customer's right to require the Contractor to complete work in accordance with the Contract.

The Contractor must not:

- proceed with hydraulic testing until all deficiencies are eliminated according to the Items of the Defect list of A category;
- apply for a Mechanical readiness certificate until all deficiencies are eliminated on the Items of the Defect list of A category and on the Items of the Defect list of B category;
- apply for a Pre-commissioning works completion certificate until all deficiencies are eliminated on the Items of the Defect list of A category, Items of the Defect list of B category and Items of the Defect list of C category.

3.5.3. Mechanical readiness

The Mechanical readiness of facilities is considered achieved if:

1. Works are completed in accordance with the Contract, including in accordance with the provisions specified in the terms of the Contract, including defects, with the exception of defects in accordance with Items of the Defect list of C category and Items of the Defect list of D category, specified in the Defect list;
2. Works meet all the requirements of the applicable legislation of the RK;
3. The facilities are built in accordance with the applicable drawings and specifications set out in the Contract;
4. The facilities satisfy the requirements that determine the achievement of Mechanical readiness specified in this section, the related tests and procedures are performed with a satisfactory result in accordance with the additional procedure agreed by the parties;
5. Eliminated all defects according to Items of the Defect list of A category and Items of the Defect list of B category;
6. The Contractor has provided a list of all existing bonds and undertakes to indemnify and hold harmless the Customer from any claims for such bonds.

The list of types of work related to Mechanical readiness to be determined jointly with the Customer, after the approval of the design and estimate documentation.

If the Contractor considers that the requirements that determine the achievement of Mechanical readiness set forth in this section are fulfilled, the Contractor shall notify the Customer in writing and request the Customer to issue a Mechanical readiness certificate.

Within 14 days after receiving a request from the Contractor, the Customer undertakes:

- issue to the Contractor a Mechanical readiness certificate, indicating the date on which the facilities reached Mechanical readiness in accordance with the Contract;
- reject the request, stating the reasons and indicating the work to be performed by the Contractor in order to obtain the Mechanical readiness certificate. Subject to its obligations to carry out such Works, the Contractor shall complete the Works prior to further notice under this Section.

3.5.4. Pre-commissioning works

General provisions

The Contractor is responsible for all engineering and preparatory work, and then all field work under the PCW. The Contractor bears full responsibility for the preparation and development of the necessary documentation and control methods for PCW and subsequent commissioning.

The Contractor will provide a breakdown by systems (functional and technological units) for the fastest and optimal transfer to operation. In agreement with the Customer, the Contractor will also develop a Test matrix with a breakdown by class (A/B/C) depending on the priority of the work.

The Contractor, on the instructions of the Customer, will develop and provide specifications and acceptance criteria for both Mechanical readiness and Pre-commissioning works completion.

The scope of work of the Contractor within the framework of the PWC also includes:

- participation in factory acceptance tests;
- preparation of a list of necessary spare parts and materials for PCW and operation for a two-year period with the provision of price offers from suppliers;
- development of technical regulations and plan for scheduled preventive work with a schedule for the use of spare parts;
- support for the Customer's operating personnel and training;
- interaction with authorized state bodies;
- carrying out the necessary work on conservation of equipment or a written recommendation to the Customer on conservation of equipment;
- organizing and planning the presence of representatives of equipment suppliers and subcontractors;
- preparation and transfer of the entire package of documentation (dossier of the system). Development and submission for agreement with the Customer, the contents of the dossier of the functional unit.
- recycling.

Project of Pre-commissioning works (Plan of PCW)

At the design stage, the Contractor will develop the PCW Plan that describes the general philosophy and methodology for carrying out the PCW. Defines the tasks that must be performed to complete the commissioning of the system. The Plan will ensure the priority of the activities of the works. The Contractor will develop this plan, which will contain, but is not limited to the following:

- list of systems;
- a list of all equipment included in the system;
- ranking of each type of test by class (A/B/C) depending on the priority of the test;
- test matrix with reference to an individual element or the entire system (subsystem) as a whole;
- detailed test schedule;
- the schedule of arrivals of equipment suppliers and other necessary third-party specialists;
- list of consumables;
- list of spare parts and materials;
- procedures (or references to procedures) for carrying out type individual tests;
- procedures (or references to procedures) for conducting complex tests;
- forms of acceptance and approval of tests with the Customer;
- organizational structure of the PCW team;
- matrix of responsibility of all involved parties;
- safety techniques and methods (Including the system of permit orders for simultaneous operations).

Test procedures

The Contractor will prescribe all the necessary procedures for both individual and complex tests. The test procedures detail the sequence of steps that will be followed to complete the commissioning of each system, partial system and subsystem, individual item. Test procedures will describe in detail and clearly the procedure for carrying out certain stages in which increased attention must be paid. The procedure will include, but not be limited to:



- preparatory work and minimum requirements for the start of the test;
- detailed description of works;
- precise criteria for successful completion of the test;
- applicable safety measures;
- forms of the act of acceptance sheet, or test report;
- safety data sheets of chemical products (if necessary);
- a list of equipment subject to mechanical or electrical isolation;
- a list of specific tools and techniques.

It is possible to apply standard procedures when testing on similar equipment and if this testing is routine.

The procedure will comply with all requirements of the equipment manufacturer.

Factory acceptance testing (FAT)

Factory testing focuses on testing equipment components to confirm that the equipment meets specifications.

The Contractor will indicate in the project schedule a list of factory tests that require the presence of a PCW team. Factory acceptance testing (FAT) is the first step in the PCW process. All comments must be indicated in the general table of the Defect list indicating the number of equipment and system according to the previously prepared numbering system, regardless of whether these comments have been eliminated or not.

Ranking of types of inspection by classes A/B/C

Class «A» inspections refer to the stage of Mechanical readiness confirming that all necessary inspections and tests on the mechanical installation have been completed.

Class «B» inspections are applicable for pre-commissioning tests or individual tests.

Class «C» inspections are applicable for functional tests and tests of individual components or automatic control systems. Site acceptance test (SAT). Complex tests.

Inspection notification

The Contractor will send to the Customer a notice of readiness for the inspection at least 1 (one) month before the appointed day.

Issuance of the Certificate of compliance

The Certificate of compliance will be prepared and submitted by the Contractor to the Customer after confirmation of the acceptance of documents for production and quality assurance in accordance with the requirements of the Contract.

3.5.5. Completion of pre-commissioning works

Completion of pre-commissioning works in relation to facilities is considered to be achieved if:

- the work has been completed in accordance with the Contract, including the completion of preparations for pre-commissioning works and the completion of pre-commissioning works in accordance with the project of pre-commissioning works, with the exception of deficiencies in Items of the Defect list of D category, which are available in the List of deficiencies and Operational tests;
- the facilities comply with applicable laws and all relevant permits;
- the Contractor has fully fulfilled its training obligations in accordance with the terms of the Contract;
- the facilities are ready for operation;
- all deficiencies have been eliminated on the Items of the Defect list of A Category, on the Items of the Defect list of B Category and on the Items of the Defect list of C Category;
- the Contractor provided the Customer with complete and final maintenance and operation manuals, technical passports for equipment and construction documentation required in accordance with the Contract;
- the Contractor delivered spare parts to the construction site in accordance with the Contract;

- the entire previously agreed package of documentation was transferred. Contents of the functional node dossier.

The pre-commissioning works are a verification process used to confirm that the facilities are designed, supplied, manufactured, installed, tested and ready for operation in accordance with the drawings and specifications stipulated by the Contract.

In order to obtain permission to conduct pre-commissioning works, the Contractor will develop a procedure for conducting pre-commissioning works, which consists of the following standard activities, but is not limited to:

- development of a system for pre-commissioning works;
- development of a breakdown system (system matrix);
- definition of pre-commissioning works packages;
- schedule of pre-commissioning works;
- budget for pre-commissioning works, including spare parts for pre-commissioning;
- monitoring the readiness of pre-commissioning works.

The pre-commissioning works procedure will include:

1. Tasks. Providing detailed information on the operational parameters to be achieved;
2. Description. A brief description of the systems/subsystems to be commissioned will be provided;
3. Lists of temporary equipment and consumables. A list of all consumables, tools, temporary equipment, and vendor assistance requirements;
4. Health, Environment and Safety. A list of all toxic and polluting liquids and materials will be compiled, describing methods for their handling and disposal. A checklist of all necessary security measures will be drawn up, including work permit requirements. Prior to the connection of electrical equipment, a notice of connection of the power supply is issued in order to inform all parties involved about the upcoming supply of electricity;
5. Conservation. Requirements will be developed for the removal of existing conservation/protection and subsequent new conservation of the system if the system is out of service for a certain period of time;
6. Scope. The scope of work will contain a detailed description of the work to be carried out step by step, including pre-commissioning works, audit trails. Faults and/or errors will be logged. Contractor commissioning procedure will be included. Performance parameters will be included for comparison with the actual result achieved during pre-commissioning works;
7. Planning. Detailed pre-commissioning works plans for systems/subsystems will be developed;
8. Transfer. The transfer of works will be carried out by signing an act or other documents agreed in the commissioning procedure developed by the Contractor.

The standard list of activities for the Completion of pre-commissioning works shall be determined jointly with the Customer, after approval of the design and estimate documentation.

If the Contractor considers that the requirements for the Completion of pre-Commissioning works set forth in this Section are satisfied, the Contractor will notify the Customer in writing and request the Customer to issue a Certificate of completion of pre-commissioning works.

Within 14 days after receiving a request from the Contractor, the Customer undertakes:

- issue to the Contractor the Certificate of completion of pre-commissioning works, indicating the date on which the facilities reached the Completion of pre-commissioning works in accordance with the Contract;
- reject the request, indicating the reasons and indicating the work that must be performed by the Contractor in order to receive the Certificate of completion of pre-commissioning works. Subject to its obligations to perform such work, the Contractor shall complete the work before receiving new notice in accordance with this section.

3.5.6. Operational tests

After the completion of the Pre-commissioning works for all Works and the successful launch of the facilities, and when the facilities reach operational parameters, the Contractor will notify the Customer of the readiness of the facilities for Operational tests, during which the facilities will operate uninterruptedly for 24 (twenty-four) hours.

The Operational tests include, but are not limited to:

- operation of facilities in various conditions to determine the operating range of objects;
- measuring the performance of facilities, including the measurement of the composition of the original product and the trend of pressure, temperature and product flows for their throughput, and quality;
- comparison of actual performance with predicted performance and/or with specifications according to the projects approved by the Customer, and the requirements of the Customer, and confirmation of achievement of target indicators;
- reporting results, identifying corrective actions and responsible party.

In the event that the facilities or any part of them do not pass the Operational tests due to identified defects, the Contractor will eliminate such defects in accordance with the terms of the Contract, and after the expiration of the period for the elimination of defects, a 24-hour Operational test of the items is repeated.

The Operational tests, which are part of the Post-construction tests, will be conducted in accordance with the conditions set forth in this Section and the terms of the Contract.

The Contractor shall be responsible for providing all equipment, assistance, documents and other information, temporary supply of electricity and water, equipment, fuel, supplies, tools, labor, materials and suitably qualified, experienced and competent personnel, as necessary for the effective and proper carrying out these tests. All instruments, equipment and tools must be calibrated in accordance with applicable local and international codes and standards. The Contractor shall provide test equipment calibration certificates prior to testing by the Customer. In addition, prior to commencement of post-construction testing, the construction and maintenance Contractor shall provide the Customer with preliminary Operations and maintenance manuals for the Works or site. If any error or defect in the preliminary operation and maintenance manuals is discovered during the completion tests, the Contractor shall immediately correct the error or defect at his own expense.

Any product produced, as well as any income or other benefit received as a result of the launch, are the property of the Customer. Once the Work has passed each stage of the commissioning process, the Contractor shall submit to the Customer a certified report of the results of these tests. The report must contain the following information:

- tested material/installation or part of the Works;
- location of the material/plant or part of the Works;
- place of testing;
- date and time of testing;
- technical personnel supervising or conducting tests;
- equipment used and test method;
- calibration certificates for used test equipment;
- readings and measurements taken during the tests;
- test results, including any calculations and graphs;
- established acceptance criteria;
- other details specified in the Contract.

The Customer shall review each such report and may send a written notice to the Contractor if the test results do not comply with the terms of the Contract.

3.5.7. Tests carried out by the Contractor for compliance with guaranteed indicators

When planning and conducting tests for compliance with guaranteed indicators, the Contractor shall comply with the requirements described in the form «Guaranteed indicators».

The Contractor allocates for planning and conducting tests for compliance with guaranteed indicators, as well as for the preparation of reports on them by an independent, well-reviewed, qualified company (the «Testing company»). The choice and appointment of the Testing company is approved by the Customer.

During the tests, the power plant is operated by the Customer's personnel under the supervision of the Contractor.



The Contractor describes the methodology for measuring the required parameters, and also provides information on the expected standards and accuracy of instrumentation, expected measurement points and attaches the necessary documentation.

For each case of failure to achieve the guaranteed performance due to the fault of the Contractor (manufacturer of goods), the Contractor shall pay a penalty in the amount specified in the Contract.

The output electrical power is measured in the high voltage cell of the switchgear.

Current and voltage transformers are supplied by the Contractor to measure useful output power, generator power and auxiliary power. They must generally comply with the requirements for equipment, except that the current transformers must have an accuracy class of 0.1, and voltage transformers must have an accuracy class of 0.2 according to IEC 60044-1 and IEC 60044-2. The Contractor submits calibration documents for all transformers.

Power kilowatt-hour meters for own needs for efficiency and performance testing are provided by the Contractor. These meters shall comply with accuracy class 0.5 according to IEC 60521 and accuracy class 0.5 according to IEC 60687. The current and voltage transformers associated with these kilowatt-hour meters are also supplied by the Contractor and comply with the standards specified in the Contract.

On HRSGs, tests are carried out for the purity of steam (water) on them. Tests are carried out at all loads using permanently installed high-speed samplers. Particular attention is paid to situations with a change in load or unstable conditions, when mechanical transfer is most likely.

A sample of the fuel used in the tests is taken at the beginning, during and at the end of the tests for compliance with the guaranteed performance. The necessary equipment for taking and storing fuel samples is supplied by the Contractor. During each sampling period, three separate samples will be taken, i.e. a total of nine samples will be taken. One sample from each set of three samples is fully analyzed by an independent body, the costs of such analyzes are borne by the Contractor. The other two samples from each set are duly labeled and handed over to the Customer.

During the measurements taken during the tests, the records in the logs should include the following information:

- barometric pressure;
- ambient air temperature according to dry and wet thermometers;
- drop in air pressure in the incoming air filters;
- pressure and temperature at the outlet of the compressor;
- fuel consumption, pressure and temperature;
- flow rate, pressure and temperature of the medium at the outlet of the turbine;
- flow rate, pressure and temperature in the gas outlet channel;
- output power, voltage, current, frequency, and power factor of the generator;
- input current and power factor (if applicable) of energy used for own needs;
- consumption of cooling air from the compressor to the turbine;
- turbine rotation speed;
- gas temperature at the inlet to the tube bundle of each HRSG and the gas temperature during the passage of the high, medium and low pressure circuits, then the temperature at the inlet to the chimney;
- temperature and pressure of high, medium and low pressure superheated steam and overheated water;
- high and low pressure in the turbine (if applicable) before and after the valves. Temperatures only up to shut-off valves;
- pressure and temperatures at each point of steam extraction and ejection of the steam turbine;
- pressure, temperature and flow rate of feed water;
- temperature and consumption of cooling water;
- pressure in the steam turbine condenser;
- exhaust temperature in the condenser;
- all pump leakage volumes;
- leakage volumes through the shaft seals in the high pressure cylinder;
- parameters of steam supplied to air ejectors;
- power supplied to the supply pumps;



- power supplied to the high-pressure boiler pumps;
- levels in the drums of the HRSG, in the deaerator and in the condensate collector at the beginning and at the end of the tests;
- electricity supplied to the high-voltage network;

The final list of measurement points according to the PCW project is agreed by the Customer before testing.

3.5.8. The Facility commissioning acceptance certificate

After successful completion of the trial operation period with a duration of ten (10) days from the date of Completion of pre-commissioning works and completion of final Operational tests (in relation to the Works), the Customer shall issue an Facility commissioning acceptance certificate on the basis of the preliminary submission of the Contractor's declaration of conformity, positive and unconditional the conclusion of the organization exercising architectural supervision on the compliance of the Works with the Projects, a positive and unconditional conclusion of the organization exercising technical supervision on the quality of the Works in the form prescribed by the legislation of the RK.

3.6. Requirements for the training of the Customer's personnel

The Contractor shall arrange for on-the-job training for technology transfer to the Customer's supervisory, operational, technical and instructor and inspection personnel in relation to the operation and maintenance of plants and equipment provided under the Contract. The Contractor shall submit to the Customer at least 21 days before the date on which the Contractor intends to commence training, a detailed training program indicating the expected date, place and time of the training.

The Customer will review the proposed training program and send a written notice to the Contractor if it finds non-compliance with the terms of the Contract.

The final operation and maintenance manuals, production instructions, process flow diagrams will be provided by the Contractor in English, Russian languages prior to the issuance the Acceptance certificate. Operation and maintenance includes a stock of spare parts required by the Customer for the future operation and maintenance of the plant.

3.7. Guarantees and suretyships

All materials, equipment and works provided by the Contractor must have a guarantee (guarantee) for at least two years after the date of the Transfer and acceptance act.

The Contractor shall obtain all guarantees for materials and equipment and the guarantee required by the terms of the Contract. A list of all guarantees and suretyships required by the terms of the Contract shall be included in the Application register provided by the Construction project Contractor to the Customer. The Contractor shall provide the Customer with all guarantees and suretyships for materials and equipment before requesting an inspection to obtain a Transfer and acceptance act.

The Contractor shall perform warranty work in accordance with the terms of the Contract. If any work is found to be defective or not in conformity with the terms of the Contract during the warranty period, the Customer will notify the Contractor in writing of its responsibility for performing corrective work in accordance with the Contract.

Two months prior to the expiration of the warranty period, the Customer will conduct (or engage a consultant) a review of the guarantees and suretyships with users and check the work of the Contractor to ensure that the items covered by the guarantees are functioning properly. If deficiencies are identified, the Customer will send a written notice to instruct the Contractor to eliminate the identified deficiencies. The Contractor shall be responsible for the correction/elimination of identified defects at his own expense.

3.8. Defects after acceptance

The Contractor will be responsible for the elimination of all undetected defects issued to the Contractor in the transfer and acceptance acts, as well as any further defects found at his own risk and expense. The Customer will send a correction notice to the Contractor to complete the work in progress and/or remedy the defects.

3.9. Operating permits and certificates

All permits or operating certificates from local authorities required to operate equipment or facilities will be obtained by the Contractor.

All permits and operational certificates will be submitted by the Contractor prior to the Transfer and acceptance act.

List of documentation for the period of operation

The Contractor must provide the following package of documents, but not limited to:

- Official correspondence of the Project Contractor.
- Executive drawings in electronic format.
- Confirmation of the elimination of items in the list of deficiencies.
- Certificate of satisfactory cleaning.
- Operation and maintenance manuals.
- Guarantees and sureties.
- Shipping logs.
- Request for information logs (RFI).
- Daily checks and progress reports.
- Certificate of substantial completion.
- The Act of final inspection, release and acceptance.
- Claims log and files.
- Photo and video of the project.
- Electronic 3D model with BMS.

All other technical documents specified in paragraph 7 - Order of the Minister of Energy of the Republic of Kazakhstan «On approval of the Rules for the technical operation of power plants and networks» dated March 30, 2015 No. 247.

3.10. Occupational health, safety, environmental and social management

General information

All persons associated with this project must work without injury, without the influence of alcohol and drugs, and must comply with the Customer's safety requirements, as well as all applicable codes and standards of the RK.

The Contractor is responsible for initiating, maintaining and overseeing all Occupational Health and Safety (OH&S) and environmental safeguards and programs in connection with the work. The Contractor shall provide a safety program, OH&S policies, an occupational health and safety management system, risk assessment and management adapted to the Project.

The Contractor's health, safety and environmental program shall provide for frequent and regular inspections of work sites, materials and equipment by competent persons in the field of OH&S (capable of identifying existing and predicted hazards in the environment or working conditions that are unsanitary, harmful or hazardous to workers and who are authorized to take prompt corrective action to address them).

The Contractor is responsible for providing and maintaining safe and healthy working conditions for its employees, as well as other project participants and the public. Thus, the responsibility for correcting or, the cost of correcting unsafe practices or conditions remains with the Contractor.

The Contractor shall develop and submit to the Customer for approval its «Specific OH&S plan for the project» which shall include, but not be limited to:

- 1) Work production plan;
- 2) Identified hazards, risk assessment and their management;
- 3) Control over the work;
- 4) Identification of inconsistencies and measures to eliminate them.

Health, safety and environment goals

The objectives of the project in the field of health, safety and environmental protection are:

- Do not directly or indirectly harm people, the environment and/or property.



- Promote and strengthen an ongoing and appropriate culture of OH&S and environmental protection and a proactive attitude towards OH&S and environmental protection.
- Include health, safety and environmental issues on the agenda of meetings.
- Monitor the implementation of the necessary measures to eliminate or reduce the hazards/risks at the site to the level «as low as reasonably practicable» (ALARP). Comply with and monitor compliance with all applicable local/national and international health and safety codes, statutes and regulations by all engineers and Contractors working on the project.
- As early as possible, assess the environmental impact of project activities and the environmental consequences of any significant development and adjust necessary environmental impact mitigation plans accordingly.
- Minimize waste generation during project implementation.
- Minimize the consumption of natural resources.
- Monitor the periodic review and evaluation of the effectiveness of the labor protection, industrial safety and environmental protection management system.
- Take the time you need to safely complete the job.
- Pay due attention to health, safety and environmental indicators when appraising employees, engineers and the Contractor.
- Monitor incident/accident reporting and investigation for prevention purposes.

Roles and responsibilities

The Contractor undertakes:

- Assume primary responsibility for ensuring the health, safety and well-being of people, protecting the environment and property in accordance with the health and safety requirements set out in this document, the general conditions of construction contracts, local laws, regulations and ministerial orders for protection labor, safety and environment and any other requirements.
- Develop and maintain a detailed health and safety plan/policy and environmental and social management plan (ESMP) in accordance with the guidelines set out in this document, general terms and conditions of construction contracts, local laws, regulations and ministerial orders for occupational health and safety, safety and environmental protection.
- Periodically review the Health and safety plan and ESMP and communicate them to all concerned within your organization.
- Ensure that all its employees and others on site strictly comply with the health, safety and environmental requirements of the Project, taking proactive measures at all times.
- Report and investigate all incidents/accidents occurring on the site in accordance with the General conditions of construction Contracts and at the discretion of the Customer.
- Properly and adequately manage the work and behavior of all subcontractors, suppliers and vendors working on the construction site in the field of health, safety and environmental protection at all times.
- Cooperate and coordinate their actions with the Customer.
- Employ suitably qualified/competent occupational health and safety personnel in a number consistent with regulatory requirements and internationally certified in the field of OH&S (NEBOSH), environmental protection personnel in accordance with their legal and contractual obligations.
- Ensure compliance with all health, safety and environmental requirements and measures at the site at all times, including living conditions for workers.
- Conduct qualification testing of personnel knowledge;
- Train personnel on S&LP, industrial and fire safety.
- Train senior and middle managers in international OH&S requirements (IOSH).

Legislation and standards in the field of health, safety and environmental protection




All health, safety and environmental standards and regulations of the Project shall be established in accordance with and in accordance with the general and specific terms of the Contract and specifications equivalent to international standards deemed appropriate and applicable. In addition, the following are some of the resources/links that can be referred to in the development and implementation of the health, safety and environmental programs of the Project:

- Fire safety Rules of the Republic of Kazakhstan dated October 9, 2014 No. 1077 (last amended on December 13, 2019 No. 921)
- «Sanitary and epidemiological requirements for working conditions and consumer services during construction, reconstruction, repair and in water, operational construction facilities» Guidelines approved by order of the Minister of Health of the Republic of Kazakhstan dated June 16, 2021 No. KR DSM-49
- Order of the Minister of Labor and Social Protection of the Population of the Republic of Kazakhstan «Rules for ensuring safety and labor protection when working at height» dated March 31, 2022 No. 109.
- Law of the Republic of Kazakhstan «On Civil Protection» No. 188-V 3PK (with amendments and additions as of 14.07.2022)
- Law of the Republic of Kazakhstan «On Normative Legal Acts» No. 213 dated March 24, 1998, Article 4 «Hierarchy of Normative Legal Acts».
- Order of the Acting Minister of Ecology, Geology and Natural Resources of the Republic of Kazakhstan dated July 27, 2021 No. 271. Registered with the Ministry of Justice of the Republic of Kazakhstan on July 30, 2021 No. 23782
- On approval of the Sanitary Rules «Sanitary and epidemiological requirements for the collection, use, application, disposal, transportation, storage and disposal of production and consumption waste» Order acting. Minister of Health of the Republic of Kazakhstan dated December 25, 2020 No. KR DSM-331/2020.
- On approval of the Rules for developing a program for industrial environmental control of facilities of categories I and II, maintaining internal records, generating and submitting periodic reports based on the results of industrial environmental control. Order of the Minister of Ecology, Geology and Natural Resources of the Republic of Kazakhstan dated July 14, 2021 No. 250.
- On approval of the Rules for the use of water supply and sanitation systems in settlements. Order of the Minister of National Economy of the Republic of Kazakhstan dated February 28, 2015 No. 163.
- On approval of the Sanitary Rules «Sanitary and epidemiological requirements for water sources, places of water intake for domestic and drinking purposes, domestic and drinking water supply and places of cultural and household water use and safety of water bodies». Order of the Minister of National Economy of the Republic of Kazakhstan dated March 16, 2015 No. 209. Registered with the Ministry of Justice of the Republic of Kazakhstan on April 22, 2015 No. 10774.
- Environmental Code of the Republic of Kazakhstan dated January 2, 2021 No. 400- VI ZRK
- СНиП РК 1.03-05-2011 (СНиП РК 1.03-05-2011), СП РК. 1.03-106-2012 «Охрана труда и техника безопасности в строительстве» (SP RK. 1.03-106-2012 «Labor protection and safety in construction»);
- «Land Code of the Republic of Kazakhstan» No. 442-II dated June 20, 2003
- «Water Code of the Republic of Kazakhstan» No. 481-II ZRK dated July 09, 2003
- СТ РК 12.1.013-2002 «Система стандартов безопасности труда. Строительство. Электробезопасность, общие требования» (ST RK 12.1.013-2002 «System of labor safety standards. Construction. Electrical safety, general requirements»).
- СНиП РК 1.03-106-2012 «Охрана труда и предупреждение несчастных случаев в строительстве» (СНиП РК 1.03-106-2012 «Labor protection and prevention of accidents in construction») (with amendments and additions as of December 20, 2020)
- Order of the Minister for Investments and Development of the Republic of Kazakhstan «On approval of the Rules for ensuring industrial safety in the operation of lifting mechanisms» dated December 30, 2014 No. 359



- Order of the Minister of Health and Social Development of the Republic of Kazakhstan «On approval of the Rules and terms for training, briefing and testing knowledge on the safety and labor protection of workers, managers and persons responsible for ensuring labor safety and labor protection» dated December 25, 2015 No. 2019
- Rules for working with personnel in energy organizations of the Republic of Kazakhstan. Order of the Minister of Energy of the Republic of Kazakhstan dated March 26, 2015 No. 234.
- Safety regulations for the operation of electrical installations.
- Order of the Minister of Energy of the Republic of Kazakhstan dated March 31, 2015 No. 253.
- Safety regulations for the operation of consumer electrical installations.
- Order of the Minister of Energy of the Republic of Kazakhstan dated March 19, 2015 No. 222.
- Safety regulations for the operation of thermal mechanical equipment of power plants and heating networks.
- Order of the Minister of Energy of the Republic of Kazakhstan dated February 20, 2015 No. 122.
- Order of the Minister for Emergency Situations of the Republic of Kazakhstan «On approval of the Rules for training employees of organizations and the population in fire safety measures and requirements for the content of fire safety training programs» dated June 9, 2014 No. 276.
- All other applicable norms and standards of the Republic of Kazakhstan
- ISO 45001:2018 Occupational health and safety management systems
- ISO 14001:2015 Environmental Management Systems Standard
- ISO 9001:2019 Quality management system standards
- American National Standards Institute (ANSI).
- Occupational Safety and Health Administration (OSHA)
- National Fire Protection Association (NFPA).

Health and safety management

1. The Contractor is responsible for providing and maintaining a safe, non-hazardous and healthy workplace for its employees, employees of its subcontractors, the Customer, interested parties, visitors and the general public.

The Contractor shall ensure health and safety measures at all stages of the project and comply with all applicable laws, rules, regulations and standards of the Republic of Kazakhstan, as well as international best practice in the field of safety, including:

- a) general medical examination before employment;
- b) special pre-employment medical examination and a check for suitability for a particular position;
- c) periodic health checks of workers;
- d) pre-shift and pre-trip medical examination of employees involved in the maintenance and operation of hazardous technical devices and mechanisms;
- e) vaccination programs;
- f) other types of disease prevention that are global in nature or specific to the area where the project is being implemented.
- g) organizing the provision of first aid and a medical center.

Contractor's Occupational health and safety plan

The Contractor shall develop and submit to the Customer for approval its Occupational health and safety (OH&S) plan for the Project. The Contractor's OH&S plan shall comply with the requirements of the Contract and all applicable the RK rules and regulations, as well as best international practices to be implemented throughout the Project. This Plan shall describe responsibilities and procedures for all aspects of OH&S management and shall be capable of being audited regularly throughout the duration of the Works. No work shall commence until the Customer's approval of the Contractor's health and safety plan.

The Contractor shall ensure that its OH&S plan includes:

- Regulations on the occupational health and safety policy of the project Contractor.
- Policy of the project Contractor regarding alcohol and drugs.




- Description of duties and responsibilities of the Contractor's team.
- Organizational chart showing the reporting relationships between the Contractor's project management, line managers, workers and staff.
- Detailed description of the procedures, precautions, safety checklists, accident records and other safety documents that will be used throughout the Project.
- OH&S training requirements - introductory briefing and hazard recognition training, task-specific training, first aid training, etc.

The OH&S and environmental orientation training provided by the Contractor shall include, as appropriate, the following:

- Information to familiarize the employee with special requirements at the workplace, including safety and traffic rules
 - Description of the nature of the project
 - OH&S policy of the project
 - Alcohol and drugs policy
 - A description of any hazards that the worker may be exposed to during the performance of the work
- Required work practices and behavior in the workplace
- General rules and responsibilities for site safety/critical safety rules
- Legal requirements
- Requirements for personal protective equipment
- Incident reporting procedures
- Reporting unsafe activities or conditions
- Emergency response procedures
- Safety procedures unique to each workplace
- Work permit procedures
- Traffic procedures
- Environmental procedures
- Working around heavy equipment
- Identification of hazards, risks and means of controlling these risks in the workplace.
- Hazard communication standards requirements
- Hazards specific to a particular facility
- A warning that violation of health and safety regulations may result in disciplinary action or dismissal.

This orientation must be recorded in writing.

Hazard identification and risk assessment

Risk assessment is the basis for a successful occupational health and safety management system. The Contractor is required by law to carry out an appropriate and sufficient risk assessment, which must be proportionate to the nature of his scope of work and the level of risks that may be associated with or arise in connection with the work under his obligations.

The Contractor prepares and submits a detailed activity-based risk assessment along with each method statement that it submits to the Customer for review and approval.

The Contractor shall effectively communicate the job hazard analysis (JHA) prior to commencement of work to key Project personnel.

The Customer has the right to stop any activity that could result in personal injury or damage to property until work can be carried out in a safe manner.

Scheduled OH&S audits

OH&S audits shall be conducted on a semi-annual basis or at the discretion of the Customer to review the Contractor's OH&S management systems, identify deficiencies and recommend actions for improvement.

Reporting and investigation of accidents and incidents/close incidents

The Contractor shall immediately provide information to the Customer about all incidents, accidents, injuries, illnesses and cases close to misses.

Legal notification of the incident/accident to local authorities will be made by the Contractor in accordance with the Labor Code of the RK.

In the event of an accident resulting in injuries requiring medical attention beyond the first, the scene must be preserved for a thorough investigation of the scene by any state authorities.

All incidents shall be thoroughly investigated by the Contractor and the relevant parties to determine the root and immediate causes of the incident. All witnesses to the incident must give written statements, and all data, monitoring records, medical reports, etc. relating to the incident or its immediate causes must be taken into account and included in the investigation report. The Contractor shall comply with confidentiality requirements when attaching medical records to investigation reports.

All incidents shall be recorded by the Contractor in the Project incident register, regularly updated and distributed to all interested parties for review and appropriate action.

All accidents/incidents and lessons learned shall be reviewed and published by the Contractor to all interested parties through all available means of communication such as meetings, tool talks, memos, alerts, bulletin boards, etc.

Environmental and social management

Monitoring and management plan in the field of environmental protection and social aspects

The Contractor will be responsible for the design and implementation of the environmental (nature protection) due diligence process of the project in accordance with local codes and standards. The Contractor must conduct an environmental impact assessment prior to commencement in order to obtain an impact permit in accordance with the current norms and rules of the RK.

The Contractor will develop and submit an Environmental and social monitoring and management plan (ESMP) for activities related to the Contractor (engineering, design, procurement and construction) in accordance with the Environmental impact assessment (EIA) and ESMP prepared by an independent consultant in accordance with all relevant local laws and norms, as well as international standards.

The Contractor shall be responsible for the management of the construction site and its activities in accordance with the approved ESMP, as well as the Customer's ESMP.

The ESMP will define the Contractor's personnel with roles and responsibilities, resources mobilized, site procedures, communication, non-compliance management, document and data management, auditing.

Good environmental and social practices to be applied by the Contractor throughout the life of the project to minimize the negative impact of construction activities on the natural environment as well as on the local population, to prevent pollution and to ensure that hazardous materials are properly stored without risk to the environment.

Measures to ensure compliance with the environmental and social requirements of the project

The process of monitoring and analyzing environmental and social indicators

The Contractor will prepare and implement the following plans and procedures under the ESMP:

- Air quality management plan
- Noise and vibration management plan
- Hazardous substances management plan
- Waste management plan
- Water and land management plan
- Traffic management plan
- Health and safety management plan
- Chance finding procedure



- Recruitment and workforce management.

The Contractor will appoint a competent environmental specialist to effectively manage and implement all environmental and social mitigation and monitoring measures as defined in the ESMP and related management plans. If the Contractor engages subcontractors to perform part of the scope of work, the Contractor shall be responsible for ensuring that the subcontractors comply with the environmental and social requirements set out in the specific project's ESMP.

The Contractor will schedule the work in such a way as to avoid/minimize adverse environmental and social impacts and will work in accordance with the management measures listed in the ESMP.

The Contractor will implement environmental monitoring measures in accordance with international best industry practice and local regulations and standards of the RK.

Air quality management

The Contractor will prepare and implement a detailed Air quality management plan as part of the ESMP. The plan will detail all environmental mitigation measures that the Contractor will implement during the construction period to reduce harmful impacts caused by air emissions and dust generation from construction activities, including from traffic on access roads. The plan will also include specific measures to reduce greenhouse gas emissions.

The Contractor will take at least the following measures to control dust and other emissions from construction activities:

- Good maintenance of access roads that will be used for a long time.
- Strict control of dirt roads and other dust-producing areas by spraying with water under dry weather conditions for dust suppression.
- Keep the drop height of potentially dusty materials as low as possible.
- Ensure that a wheel washer is available and used to avoid spreading dust to neighboring roads via construction trucks.
- Apply speed limits in the area.
- Do not allow the engine of construction machines to run in the parking lot and without use.
- Use well maintained equipment and machines in accordance with the methods specified by the manufacturer.
- Ensure regular technical inspection of used construction machines, mechanisms and vehicles for minimum exhaust emissions.
- The use of electricity for the technical needs of construction (heating of materials, heating of water) instead of solid and liquid fuels.
- Use of special vehicles and containers for storage, loading and transportation of bulk, dusty and wet materials.

Noise and vibration control

The Contractor will prepare and implement a detailed Noise and vibration management plan as part of the ESMP. The plan will describe how the Contractor will minimize and manage noise and vibration impacts during construction.

The Contractor will take at least the following measures to mitigate the negative impact of noise and vibration at construction sites:

- Regular inspection and maintenance of all machines and vehicles.
- Installation of mufflers or acoustic guards on machines where applicable, such as fitting suitable mufflers to engine exhaust pipes and compressor components, and use of portable sound barriers (if required).
- Implement speed limits for trucks traveling to and from construction sites.
- Route shortening project traffic through public areas where possible.
- Installation of shock absorbers to dampen vibrations.
- Placement of working machines and mechanisms on the construction site, taking into account mutual protection from noise and natural barriers.
- Installation of silencers on air intakes, vibration isolators and vibration dampers on compressor units.



Noise levels in the air shall be measured and the necessary noise reduction measures shall be taken by the Contractor to reduce the noise level to acceptable limits.

Hazardous substances management

The Contractor shall prepare and implement a detailed Waste management plan (WMP) as part of the ESMP. The plan will define all site-specific measures that the Contractor will implement during the construction phase to identify and control hazardous substances to be used at the construction site.

The Contractor will take at least the following measures to mitigate the adverse effects of handling and storage of hazardous materials on construction sites:

- The storage area will be designed and equipped taking into account the chemical and physical properties of the products, as well as the types of containers stored, the number of people requiring access and the amount of substance used. The amount of storage space will be calculated based on the amount of hazardous material that will be on site during peak construction for the respective activity.
- All areas where hazardous substances are stored must be equipped with a secondary containment to contain any spill or leak from one of the stored containers. The volume of the secondary containment is determined to be at least 110% of the volume of the largest container stored on site, taking into account the volume occupied by the containers stored. The secondary containment must be kept empty of rainwater.
- Material safety data sheets (MSDS) should be available and easily accessible for all stored fuels and chemicals.
- All hazardous substances must be clearly labelled.
- Access to storage areas is limited to trained and authorized persons wearing appropriate personal protective equipment.
- Chemicals that can react with each other must be physically separated.
- Tanks must be equipped with level sensors to prevent overfilling.
- Suitable absorbents and spill kits will be available on site to clean up any spills and leaks. Contaminated soils must be removed and disposed of as hazardous waste.
- Vehicle refueling at the construction site will be carried out using a secondary containment to minimize the risk of leakage and soil contamination. Portable equipment will be systematically used to collect accidental spills.
- Prevention of cases of illegal discharge and disposal of construction waste.

The Contractor will prepare, submit and implement a contingency plan describing the actions to be taken in the event of a spill of a hazardous substance. All spills will be stopped and cleaned up immediately to avoid potential impacts on water and soil quality. All spills will be reported to the Customer.

Waste management plan

The Contractor will prepare and implement a detailed waste management plan (WMP) as part of the ESMP. The plan will describe how the Contractor will identify, collect, transport and handle all waste generated at the work sites of its personnel.

The Contractor will adopt at least the following waste management practices as part of its WMP:

- The Waste management hierarchy (WMP) will be applied throughout the project as per the WMP, waste will be eliminated as the preferred option. The second most desirable option would be to minimize the amount and hazard of waste generated. If these options are not possible and not feasible, the waste will be separated, reused, recovered and recycled to the maximum extent possible as the next option. Waste treatment and safe disposal will be considered as a last resort.
- The Contractor shall arrange for a temporary waste storage site. The Contractor shall provide suitable waste containers, which shall be provided for each type of waste and clearly marked according to the contents of the containers, as well as ensure that the waste is effectively contained and protected from wind, rain, and vermin. Recyclables (plastics, glass, ferrous metals and metals) should have separate storage containers.



- Waste must be transferred to specialized organizations under an agreement, which must be concluded before the commencement of the Works.
- Waste incineration is not allowed.
- Non-hazardous waste must be transferred to specialized organizations under an agreement, which must be concluded before the commencement of the Works.
- Hazardous waste, such as used oil, hydrocarbons, lubricants, paints, solvents and used batteries, must be transferred to specialized organizations, under an agreement that must be concluded prior to the commencement of the Works.
- A waste register will be created and available from the moment the Contractors are mobilized to the site, and records will be maintained until the completion of the Project.

Water management plan

The Contractor will prepare a water management plan to ensure proper environmental management of water abstraction, treatment and discharge in accordance with applicable laws and standards, and to minimize pollution of surface and ground water during all construction activities for the Project.

Traffic management plan

The Contractor will prepare and implement a detailed Traffic management plan (TMP) as part of the ESMP. The plan will define the characteristics of construction vehicles and machinery at the construction site, describe the expected Project traffic (frequency of trips to the construction site, working hours, convoys and detail any site-specific measures) that the Contractor will implement during the construction period to minimize disruption for the neighbors created by its fleet and reduce the risk of accidents.

Health and safety management plan

The Contractor will prepare and implement an Occupational health and safety management plan.

Archaeological finds

The Contractor's procedure regarding chance finds will include reporting to the Customer and notifying the relevant competent authorities of the RK about the objects or places found, preserving and securing the discovered chance finds to avoid any violation, notifying project personnel.

Recruitment and workforce management

The Contractor will develop and implement a personnel management policy and principles of human resources management in accordance with the labor legislation of the RK and the requirements of the International Labor Organization (ILO). This plan will detail labor requirements for the entire construction period, the Contractor's local recruitment process, and the planned approach to maximize local employment opportunities. Human resources policy and principles of human resource management will cover the following issues:

- equal rights, equal pay;
- a ban on all forms of child and forced labor;
- rights and obligations of employees (including the Code of conduct for employees);
- freedom of association in trade unions and the right to conduct collective bargaining;
- informing employees about their rights, duties, responsibilities and privileges;
- the terms of the employment contract, including the terms of employment, hours of work, overtime and related compensation, as well as the right to refuse requests for overtime.

Environmental impact assessment

The Contractor will carry out a suitable and sufficient environmental impact assessment, which should be proportionate to the nature of its scope of work and the level of environmental impact that may be associated with or may result from work within its enterprise.

The Contractor shall prepare and submit a detailed EIA by activity along with each terms of reference to the Customer for review and approval.

Education and information

The Contractor shall ensure that all personnel of the Contractor and subcontractors involved in the work are:

- Under signature informed about the strategy and management plan for environmental and social issues related to the works, as well as about their role and responsibility.
- Briefed on the requirements of environmental legislation, rules, regulations and special permit conditions applicable to the area and work.
- Informed about environmental issues and special conditions at the job site.
- Informed about specific chemical hazards.

Environmental monitoring

Environmental parameters subject to monitoring by the Contractor include the quality of air, water and land resources, noise and vibration levels, volumes of waste generated and transferred for disposal. The Contractor shall mobilize the necessary field equipment for air quality and noise monitoring at the site.

The results of environmental monitoring and reports on the status of environmental non-conformities will be submitted to the Customer on a weekly basis as part of the Weekly report.

The Contractor will summarize on a monthly basis all environmental and social initiatives implemented in connection with the performance of the work, and submit them as part of the Monthly report to the Customer.

Key performance indicators

The Contractor will develop key environmental performance indicators (KPIs) to be agreed by the Customer as part of the final ESMP and will monitor until the end of the project.

Section 4. GENERAL REQUIREMENTS

4.1. Participant requirements

Contractor license requirements:

- availability of the State license for exploration activities;
- availability of the State license of the 1st category for project activities;
- availability of the State license of the 1st category for construction and installation works.

4.2. Deadline

The term for the completion of the Works is from the date of signing the Contract until August 2025.

The main critical points and stages of the project that will affect the completion time of the Works:

Signing of the Contract

Start of the order of the main equipment immediately after the signing of the Contract

Development of project documentation at the «P» stage

Obtaining an environmental expert opinion

Obtaining the conclusion of RSE «State expertise» («Gosexpertiza») and obtaining a building permit

Construction and installation works

Supply and installation of the main equipment.

The Contractor shall develop an enlarged Schedule for the production of Works for the main stages of construction, after the conclusion of the Contract, provide a detailed phased Schedule for the production of Works.

4.3. Temporary objects of the Customer on the site

The Contractor shall provide an office large enough to accommodate the Customer's personnel, including office space. The number of the Customer's personnel (including the Supervisor, technical supervision and architectural supervision) is estimated at 100 people. The Contractor must provide a minimum of 8 sq.m. space for each employee when designing office space on site. The Customer's offices must be fully equipped and furnished.

In addition, offices should be designed and built, including:

- Meeting room for 20 people, equipped and equipped for videoconferencing (including furniture, audio and video equipment, microphone, projection equipment, projection curtain, LED television, internet).

- Two small meeting rooms for 6 people each conference rooms should be equipped for use, including a white board, chairs, a meeting table.

- Kitchenette with the necessary equipment and furniture (cupboards, washbasin, kitchen worktop, refrigerator, microwave oven, coffee machine, electric kettle, water dispenser with hot and cold water functions, household materials for serving the kitchen (dishes set, glasses, coffee mugs) and cups, tea cups, forks, knives, spoons for 24 people, consumables such as kitchen towel, sugar, paper napkins, etc.)

- Dressing room with lockers.

- Toilet and lavatory (separately for men and women).

- Room for printer.

- IT-room for the server (closed and cooled).

- IT support services.

- UPS of sufficient power;

- Wi-Fi network server;

- 5 pieces of iPad Pro 2022 with Fieldwire - Business Licensed Software

- High-speed Internet access (regardless of the Contractor)

- External telephone access with PABX (regardless of the Contractor).

- Color copier for the work center and consumables (laser type, including A3 photocopying, enlargement-reduction, scanning and printing);

- A0 color plotter and consumables.

- All office equipment and supplies

- Sufficient number of parking spaces

- Breakfast, lunch and dinner (quality to be approved by the Customer's representative) for all employees of the Customer's representative;

- 4 off-road 4x4 vehicles and 2 sedan vehicles for use by the Customer not older than 2 years or 5000 km (all insurance, maintenance and fuel to be provided by the Contractor);

- 2 shuttle buses (all insurance, maintenance, drivers' salaries and fuel to be provided by the Contractor).

Facilities security (Temporary fencing around the perimeter of facilities in accordance with local legislation and building requirements, video surveillance system, including around the perimeter of the fence and the perimeter of office and other buildings, controlled access systems for vehicles and pedestrians to the main entrance gate, card entry system, turnstiles and etc.).

Each room should be equipped with windows, mosquito nets and blinds, daylight, air conditioning and heating. All doors must be equipped with locks. The office must be equipped with all necessary security features such as fire detectors, fire fighting equipment, a fire button, or any other required by local law.

The premises must comply with relevant local regulations. The Contractor shall provide all expenses, running costs for operation, maintenance, electricity, communications, heating, cooling, piped and sewer water, etc., necessary for the facilities of the temporary site.

The Contractor shall submit for Customer's approval prior to commencement of construction plans and drawings showing the proposed details and location of Customer's offices.

The Contractor shall also submit for the Customer's approval the details of the proposed furniture and fittings. These items should generally be of the best quality available locally.



4.4. Communications management

General

The purpose of communications management is to establish, streamline and implement clear and effective means of communication that will facilitate the processing of information between all parties associated with the Project. Effective communications management ensures the completeness of documentation for legal, fiscal, administrative and historical purposes (project review, audits, claims, etc.).

Effective communication is a key element to the success of a project, as it ensures that all team members are informed of the decisions made in a timely manner. Communications management ensures that all relevant stakeholders receive the information they need effectively and in a timely manner.

Communication on the project can be carried out in the form of printed and electronic documents, informal and formal meetings:

Provide timely guidance to the parties involved in the project:

- Ensure that project stakeholders are informed and given the opportunity to contribute in a timely manner and agree to project decisions;
- Ensure timely provision of accurate project information to all project stakeholders;
- Monitor and coordinate other communications for internal or external audiences.

The day-to-day exchange of information will take place, as a rule, between the Contractor and the Customer.

Communication plan

The communications management process will be governed by the Communications management plan, which will be prepared by the Contractor and approved by the Customer. This plan will outline how communications will be maintained throughout the life of the project. The communication requirements will be documented in a responsibility matrix to be developed prior to the start of the project.

Language: The language used in all written communications under the Contract relating to all items of service offered by all parties must be written in both Russian and English.

Units of Measure: The units of measure for the project and all related documentation must be in metric units.

Form of communications: Communications include submission of engineering, design, procurement and construction documentation, calculations, reports, letters, accompanying documents, memorandums and minutes of meetings.

Reference Acquaintance: All written correspondence must have a specific reference code, and replies must reference this code. The Gregorian calendar is used in the format day, month, year (either 06.08.2022 or August 06, 2022).

Oral communication: All formal oral communications must be confirmed in writing as soon as possible.

Acknowledgment of receipt: Receipt of all documents and correspondence must be confirmed by signature and seal.

All documents and correspondence should be on standard communication forms to ensure the flow of information is managed. For all forms of communication, logs should be kept indicating the originator, recipient, date of issue, date of response required, days when no response was received, and these logs should be issued and transmitted by the originator.

The aim is to monitor the responsiveness of the parties and avoid or minimize late responses that affect the project.

Correspondence

All correspondence between the Contractor and the relevant authorities, including all submissions, must be copied to the Customer.

Meetings



Various types of weekly and bi-weekly meetings should be held between project team members, who specify for each meeting a date, time, location, attendance, agenda, chairman for each package.

The meeting agenda will be distributed by the Customer two (2) business days prior to the meeting. The first item on the agenda will be a review of the actions taken at the previous meeting.

The minutes of the meeting will be prepared and distributed by the Customer within five (5) business days after the meeting. The minutes of the meeting will include the status of all agenda items as well as new actions.

The table below provides an example of a project communications plan and meeting minutes.

Communication type	Audience	Target	When /Frequency	Commu nicator	Practical results	Type of/ Method(s)
Opening meeting	All stakeholders	Gather information before designing	First before project start date	Customer	Agenda Meeting minutes	Meeting
Project start	Customer Contractor	Report plans, introduce the project team. Review project goals and management approach	On or near the start date of the project	Customer	Agenda Meeting minutes	Meeting
Progress meetings	The entire project team. Individual meetings for subgroups and technical group, as needed	To review overall progress, technical and coordination issues, detailed plans (tasks, assignments and actions)	Regular schedule - weekly or biweekly is recommended for the whole team depending on the situation.	Customer	Agenda Meeting minutes Work schedule	Meeting - If the consultant(s) is/are not available locally, a conference call may be accepted.
Project management meetings	Customer Contractor	Notify the Customer status and discuss important issues. Handle issues and change requests here before escalating them.	Weekly	Customer	Agenda Meeting minutes	Meeting
Design review meetings	Customer Contractor	To review and evaluate design progress, identify deviations from plan, problem areas, problems, and discuss mitigation measures to resolve problems.	Weekly	Customer	Agenda Meeting minutes Design schedule	Meeting
Preparatory meeting	Contractor	The purpose of this meeting is to ensure that the Contractor fully understands all contractual requirements, Customer's expectations and QA/QC standards established by the Customer for full compliance by the Contractor.	Before starting each defined work function.	Customer	Agenda Meeting minutes Method statement ETC Design drawings Work schedule	Meeting
Preparatory meeting	Contractor	The purpose of this meeting is to ensure that the Contractor fully understands all contractual requirements,	Before starting each defined work function.	Customer	Agenda Meeting minutes method statement ETC Design drawings	Meeting

		Customer's expectations and QA/QC standards established by the Customer for full compliance by the Contractor.			Work schedule	
Executive meetings	All parties concerned	View status, problem, and risk reports. To identify and communicate potential risks and issues that could affect schedule, budget, or results. Inform the Customer about the status and discuss important issues. Request approval for changes to the project plan	Not on schedule. As needed when issues cannot be resolved or changes to the project plan need to be made. recommended monthly	Customer	Agenda Meeting minutes Slides updates Project schedule	Meeting
Periodic demonstrations and targeted presentations	Customer	Present the design at the end of each stage to get the approval of the Customer and move on to the next step.	At the stages of the project	Customer Contractor or	Agenda Presentation slides	Presentation/ demonstration
Symposium	Contractor Customer	To obtain information from a construction Contractor, resolve coordination issues, discuss design/critical issues, or make a major decision that requires the involvement of other parties. Keep all consultants updated on project status.	Once the project is sufficient to «show», or the Contractor has to share critical issues (mostly issues related to coordination)	Customer	Agenda Presentation slides	Presentation/ demonstration
Progress reports	Contractor	Inform the Customer about the progress of the project.	Monthly	Customer	Project status report Project schedule	Distribution in hard and soft copies (if possible)
All parties concerned		General communication such as contractual matters, billing, day to day communication (mostly email).	If necessary			in hard and soft copies

Reports

As a minimum, the following reports will be issued by the responsible party of the Project. The table below provides an example of a project communication plan and reports.

Report *	Frequency	Responsibility	Comments
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Monthly progress reports	Monthly	Contractor for the Customer	The format of the report will be provided by the Construction Contractor and agreed with the Customer.
Weekly progress reports	Weekly	Contractor for the Customer	The format of the report will be provided by the Construction Contractor and agreed with the Customer.
Daily activity reports	Daily	Contractor for the Customer	The format of the report will be provided by the Construction Contractor and agreed with the Customer.
Design	Monthly	Contractor for the Customer	The Project status report will be submitted as a sub-item of the Monthly progress report.
Purchasing status report	Monthly	Contractor for the Customer	The Procurement status report will be submitted as a sub-item of the Monthly progress report.
Project cost and budget report	End of each design phase	Contractor for the Customer	The Project budget and cost report will be submitted as a sub-item of the Monthly progress report.
Final report		Contractor for the Customer	End of project

4.5. Project management

Introduction

This section briefly outlines the main project management tools. The project management tools discussed in this section provide the basis for quantifying the scope, schedule, and budget for all phases of a project. These tools will provide timely and accurate information describing the status of Project activities and will be helpful in formulating corrective actions for potential problems.

In this regard, the Contractor shall hire qualified and experienced personnel competent in the preparation of an integrated project program, cost, allocation of resources (manpower, plant/equipment, materials, etc.) and development of a baseline plan for monitoring and measuring performance.

Joint platform

In order to ensure that all parties to the project use the same system, documents will be developed by the Contractor and agreed by the Customer and issued prior to construction or as soon as possible.

- Work distribution structure;
- Document coding system;




- Schedule design guide;
- Reporting requirements;
- Shipping requirements.

The use of these agreed documents by all parties will simplify the management of documents and projects.

Work distribution structure (WDS)

Hierarchical structure of works (hereinafter referred to as the HSW) provides a logical framework for project management planning, reporting and control, and has been prepared to describe the Program with an increasing level of detail for adequate management visibility. The HSW covers all phases of the project and provides a consistent coding structure throughout, accommodating the various requirements of all parts of the Program.

The HSW reflects the set of works to be performed by the Contractor and all third parties that may be involved in the Program. The HSW will be a common identifier between time/cost/documents and will be used to ensure integration of time, cost, document and program resource elements.

The Contractor will install a HSW that will allow structuring information about the project at different levels of detail for different organizations. The HSW will include an alphanumeric numbering system that uniquely identifies each element of the HSW, ensuring transparency in both directions (top to bottom and bottom to top). The Contractor shall submit the Hierarchical structure of works (HSW) to the Customer for approval.

The Contractor will be responsible for including the appropriate WDS number in the master schedule, organizing and grouping works according to the approved structure. The Contracting Surveyor will include the same structure in the Cost Coding System to accurately reflect all associated costs budgeted, accepted and expended. In this way, he can develop cost reports related to budget/cost incurred at each level of the HSW structure, which will now be converted to a cost breakdown structure (CBS). The Contractor warrants that:

- The HSW includes all the tasks that must be completed during the implementation phases of the project. The HSW is developed using the software application and hosted on a server.
- The HSW identifier (alphanumeric) is included in the cost breakdown structure to provide integration between time, cost, and documents.
- The HSW identifier (alphanumeric) is entered into all documents to identify all documents and allow integration with the above parameters (cost, schedule, and documents).

The advantage of having all three control components linked together via a WDS number allows close monitoring of project progress at all times, and easy access and monitoring of emerging issues.

Reporting and updates

- Linking all three (3) components to the HSW identifier project will allow for a better and more closed, controlled method of measuring day-to-day progress as well as emerging issues and impacts that will occur during the course of the work.
- All reports related to Schedule/Cost/Documents must include the HSW identifier.
- Reports should also be generated and grouped using the HSW code for easy access to project/task information reporting issues based on time, cost, or related documents related to any specific HSW code number.

Project budgets

Project budget management consists of developing and maintaining a budget, as well as reporting and controlling costs at each stage of project development in accordance with the budget and cash flow, in order to avoid delays and cost overruns.

The main goal of budgeting is to establish a balance between the developed and approved budget at the tender stage and realized costs in order to effectively manage the design, procurement and execution processes to achieve the set goal.

The Contractor shall structure the project budget with a cost breakdown structure (CBS, in accordance with the project HSW) and submit to the Customer for approval. Such a structured budget

would effectively control the procurement process of each work package, as well as control the project development (project manufacturability and cost-effectiveness, specifications and selection of materials, etc.) associated with the work packages, in order to maintain budget limits and goals.

The budget and cash flows shall be updated and compared with previous and initial versions by the Contractor at each design stage and for each commitment, as well as during the procurement process. To track progress, you can plan actions to address any overflows; The following activities can be directed towards meeting budget objectives.

The set of information and assumptions on which the budget is based should be documented so that any changes in these criteria and assumptions can be tracked along with improvements in the information available. This will allow you to identify deviations in advance and allow you to calculate the cost for other budget items.

Time management (schedule) and control

The Contractor shall prepare a detailed timetable for consideration, comments and approval by the Customer, in which he shall indicate the activities to be followed, the timeframe, milestones and date of completion of work milestones for works related to pre-construction, mobilization and demobilization, temporary works, design, procurement, construction, commissioning and transfer. The Work schedule and subsequent updates shall be in accordance with the requirements of the Contract.

The Contractor must purchase **Primavera (minimum version 6.0) Project Planner** with sufficient licenses for use by his own team, the Customer's team. All schedules will be prepared using **Primavera (minimum version 6.0) Project Planner**.

Preliminary schedule - tender stage

The Contractor will develop and submit a temporary level 2 program which should include key activities and key dates as part of the bidding documents to clarify how the Contractor intends to execute the project (sequence and duration).

Project schedule - baseline

The project schedule is drawn up in accordance with the Preliminary work schedule. The Contractor shall prepare and submit a resource-laden (manpower, equipment, materials, cost) level 3 project program showing the design, procurement, construction, testing and commissioning activities, milestones, critical paths and planned subcontracting activities to the Customer within 28 days after receipt of the Work start notice. The term for consideration of documents will not exceed 21 days from the date of receipt of the project program of the construction Contractor with the form of transfer.

The Customer shall review the Project program and each revised program submitted by the Contractor and may send a Notice to the Contractor stating the extent to which it does not comply with the Contract, or ceases to reflect actual progress, or otherwise does not comply with the Contractor's plan of commitment. If the Customer fails to provide such Notice:

- within 21 days of receiving the initial program; or
- within 14 days of receipt of the revised program, the Customer shall be deemed to have given a No objection notice and the original program or the revised program (as applicable) shall be accepted as the approved Program.

Once the Project schedule has been approved and agreed upon, it replaces and governs the Application to the Contract, the Preliminary work schedule, as the Base schedule. The baseline schedule will contain an integrated detailed design, procurement, construction and testing and commissioning schedule for the Project and will be used to track the progress of the project. The base schedules will be revised (rebased) only if significant changes are made with the approval of the Customer. A Basic level 3 schedule should include the following modules, as well as any individual tasks.

- Design schedule: The design module should be prepared taking into account all engineering, survey, engineering and design work and all project deliverables, including key milestones for all contractual obligations.
- Procurement schedule: The Procurement Module should be formulated considering procurement, fabrication, delivery times, project timeframes for each procurement task, including key



milestones for all contractual obligations. Consideration and approval of the submitted materials/equipment/plan will be linked to the activities.

- Construction schedule: The Construction module shall be developed in detail for all construction works, taking into account the logical sequence and duration of the Works, including requirements for procurement time, quantities, resources, adhering to the established HSW structure by the Contractor at a lower level of detail with key milestones included for all contractual obligations.

- Testing and commissioning schedule: Testing and commissioning shall be formulated for all testing, commissioning and commissioning of plants and equipment, taking into account the sequence and timing of inspections and tests specified in or required by the Contract, including project results and included key milestones in all stages. contractual obligations.

The Contractor shall prepare a 90-day Schedule at level of detail 2 to allow work to proceed until detailed Project schedules are prepared. The schedule shall be distributed to the project participants through the Contractor's document Controller as a controlled document.

Project schedule

The Project schedule should be adapted to the levels of the HSW hierarchy and reporting will be based on these levels.

Once the baseline is adopted, updates and reporting on progress will be based on the agreed schedule.

Schedule control reviews will include:

- Critical path and reserve
- Project performance
- Changes in logic and restrictions
- Deviations
- Approved extensions

Periodic updating of the duration of future works will be carried out by the Contractor on a regular basis. Updating includes estimating the duration of activities, changing planned start and finish dates, modifying and changing the network logic of the project.

The method to be implemented in all schedules shall be controlled by the Critical path method (CPM) and submitted to the engineer and Customer for review and approval. The CPM will cover all stages of the modules described below, from design to procurement, production/supply, construction and commissioning. It will include all activities with the smallest or negative margin.

The project schedule update will be submitted to the Customer as part of the overall monthly progress report as outlined in Section 8.

Additional reports shall be prepared on a regular basis, as required, by the Contractor's project team.

The nature of the reports should cover the following types of reports:

- Updating reports;
- Deviation reports;
- Forecast of progress (3 months ahead).

Failure by the Contractor to submit the Schedule or any weekly or monthly update thereof may result in the withholding of any payments due to the Contractor until the Schedule is submitted or subject to weekly or monthly renewal.

If the Customer rejects the Contractor's Schedule, including any subsequent update or revision, the Contractor shall, within 7 days of receipt of the rejection, revise the Schedule to conform to the Contract Documents and resubmit it to the Customer.

The schedule should be used as the basis for progress reporting, schedule control, and schedule forecasting. It shall also be used by both the Contractor and the Customer as a basis for evaluating changes, claims and requests for payment.

The schedule shall be updated bi-weekly and monthly to reflect the actual progress of the Works.

Making report



All reports must be submitted. The Contractor shall provide the Engineer with six (5) color copies, an electronic copy on the digital platform and a copy of the Monthly progress report via **ACONEX** (Electronic document management system) or an alternative platform.

Daily report

The Contractor's daily report shall contain at least the following information:

1. Information about the reporting period and weather conditions;
2. Detailed workforce;
3. Details about the equipment;
4. Information about activities at the Facility;
5. Information about delivered materials/equipment on site;
6. Other special events and comments;
7. Incidents (near accidents, accidents and other violations) related to OH&S;

The format of the daily report is agreed with the Customer.

Weekly progress report

The Contractor shall prepare and submit a Weekly progress report in a format approved by the Customer. The weekly progress report must contain at least the following information:

1. Design status
2. Current activity at the Facility
 - a) Labor force levels
 - b) Construction schedule for this week
 - c) Labor productivity
 - d) Planned versus actual
3. Purchasing activities
4. Documentation monitoring
5. Health, safety and environmental protection
6. QA/QC activities
7. Activities planned for next week
8. Applications
 - a) HSE details and issues with weekly HSE statistics
 - b) Schedule
 - c) Nonconformance report (NCR)
 - d) Photos of the construction site.

The content of the weekly progress report is agreed with the Customer.

Monthly progress report

This Report is submitted on the last day of each calendar month and reflects all actually performed work from the 26th (twenty-sixth) day of the last month to the 25th (twenty-fifth) day of the month inclusive. Monthly progress reports must be submitted monthly thereafter, each within 7 days of the last day of the period to which it relates.

A monthly progress report must be submitted in a format approved by the Customer.

The Contractor's monthly progress report shall contain sections/subsections on the topics described in this paragraph, but not limited to them.

1. Summary of the section on physical progress
 - a) An introduction that summarizes each action
 - b) Status of work performed, significant achievements (engineering, procurement, project control, documentation control, OHSE, testing and commissioning)
 - c) Simplified information about the presentation of progress, measured as a percentage, compared to the planned percentage, obtained from the Work schedule.
 - d) Areas of concern, including critical items and areas of concern, corrective actions taken or planned, and other related actions.
2. Technical section
 - a) Design section



- Detailed information about the progress of design and drawing,
- Examination status,
- Status Request for information (RFI)
- Problems

a) Budget and cost section

b) Purchasing section

- A summary of all significant purchasing transactions during the month,
- Areas of concern, including details of any pending actions.
- A report listing the main types of equipment and materials to be included in the Works.
 - Purchase order date - planned/actual;
 - Manufacturer/supplier and origin;
 - Date of shipment of the manufacturer/supplier - planned/ actual;
 - Date of arrival at the Facility, Kazakhstan - planned/ actual;
 - Analysis of any delay

«Project control» section

- An updated work schedule prepared by recording actual work completion dates and the percentage of work completed by the twenty-fifth (25th) of the month, along with estimates of remaining duration and expected completion based on current progress. It will be accompanied by a descriptive statement and activity report that explains:
 - approach to meeting the necessary deadlines specified in the Contract.
 - actual performance as reflected on the Critical path in terms of days ahead or behind the allowed dates. The Contractor must determine:
 - deviation from the critical path of the previous month;
 - the reasons for the delays, with an explanation of the affected work and proposed corrective actions to mitigate the delays;
 - activities in progress and activities planned to be completed.
- Performance analysis: The Work schedule update should show:
 - the actual start dates of the activities that started in accordance with the scheduled start date of each activity;
 - actual completion dates for activities that completed according to the planned end date for each activity and time span;
 - forecast and planned start dates of planned activities;
 - forecast and planned completion dates for activities that have begun;
 - percentage of completion of each activity started but not completed;
 - the total reserve of each pending action;
 - any delay in activity.

Purchasing status section:

- Procurement activities for materials and capital equipment completed during the previous calendar month.
- Procurement activities for materials and major equipment expected to be completed during the current calendar month.
- For the production of each major unit of machinery and materials, the name of the manufacturer, place of production, progress percentage and actual or expected dates:
 - start of production
 - Contractor checks
 - factory tests
 - shipment and arrival at the Facility.
- Areas of concern: Details of any delays beyond the contracted delivery date and/or estimated installation date, any impact of delays on the major equipment procurement schedule.

Construction status section:

- Activities implemented during the reporting month
- Forecasts to achieve any missed key dates and those due next month

- Organization and resources of the Contractor
- Details of any other items deemed necessary by the Customer or Contractor.

Testing and commissioning:

- Documentation overview of the commissioning process
- Overview of the commissioning program
- Factory test progress
- Test progress at the construction site
- Course of study
- Final progress of test documentation

Quality section (QA/QC):

- An introduction that summarizes the quality assurance and quality control activities at the facility during the reporting period.
- Project submission status:
 - Method statements / Technology statements
 - Inspection and test plans
 - Results of laboratory control
 - Quality control results
- Report on the status of nonconformities, including information on corrective actions taken by the Contractor to eliminate nonconformities
- Activities planned for next month
- Areas of quality concern

«Document Control» section

«Occupational health and safety»: an overview of all aspects of safety during the month, including reports on all accidents, incidents, a report on key OH&S performance indicators and statistics, actions proposed to prevent further incidents, and safety performance to date, including the results of all audits conducted during the reporting period, the number of inspections, the implementation of inspection measures, an audit in terms of OH&S, and the provision of PPE by employees.

«Environmental and social section»: An overview of all environmental issues in the past month, including all monitoring reports, environmental and social KPIs, mitigation measures taken, and environmental impact control activities, including the results of all audits conducted during the reporting period.

«Risk Management» section: update of the risk register and forecast

Applications

- a) Property photo: Photographs must be identified by a brief description indicating:
 - The specific location and theme of the photo;
 - The date and time of the photo.

On a monthly basis, the Contractor shall propose and the Customer shall confirm the number of photographs to be provided in order to accurately record the progress of all parts of the work.

- b) Main chart (schedule)
- c) Stage schedule
- d) 3-month forecast chart (schedule)
- e) Register of Nonconformance reports
- f) Risk register
- g) Any other required information

4.6. Document flow and control

Introduction

The Document management system (DMS) will be implemented using automated electronic document management tools in combination with paper archives of documents in accordance with the overall management structure of the HSW system. Records Management will allow Program participants to exercise greater control over the production, storage and distribution of documents, improving the ability to reuse information, control the flow of documents and reduce lead times. The full range of

functions that a document management system (DMS) must perform includes document identification, storage and retrieval, tracking, versioning, workflow management, and presentation.

The elements of the DMS that must be implemented for a project include the software, hardware, and procedures to perform all the functions necessary to manage records in projects from initiation, or whenever they are introduced in a project, to completion. Each element is described below.

All data will be stored in the electronic document management system (EDMS) – **Aconex** (and/or others) and will be readily available in real time to the parties involved in the project. All data must not be changed or deleted, and the EDMS server administrator must grant appropriate access rights to each member of the project team.

A common platform for the use and identification of documents and other data should be established by the HSW.

The system will collect data, including documents, at a hierarchy level based on the scope of the project. The system will provide a specific view of the database at a specific time to a specific user.

The Contractor must purchase and maintain an electronic document management system (EDMS) - **Aconex** (and/or others) and a sufficient number of licenses for use by your own team, the Customer's team, as well as other necessary legal entities required by law, throughout the entire duration of the project (technical supervision and architectural supervision, etc.).

Joint platform

To make sure that all project Participants use the same platform, the Customer develops and releases documents «General requirements - Standards» in the shortest possible time.

These documents will include:

- Document coding and numbering system;
- Schedule Development Guide;
- Protocol for creating drawings;
- Reporting requirements;
- Shipping requirements.

Document encoding structure

The Contractor's document control administrator will be responsible for registering all documents prepared by this company in the Document Registry (EDMS) and distributing them to the responsible persons after consultation with the Customer. Each Record keeper is responsible for maintaining a hard copy for access and use on the file system. The Customer's clerk must also be able to report documents from the system for the Monthly Report.

The Customer's clerk will assist project personnel to correctly code documents in accordance with the established coding system described below. Document coding must follow the code numbering system as specified in the Document control procedure.

After signing the contract, the transfer of all types of information between the parties is carried out by correspondence. Information flow other than written correspondence/submissions is considered «informal». The parties shall deliver correspondence and transmissions to the Customer by e-mail until the document management software is installed and put into operation.

Submissions, invoices, reports, schedules and all documents that are Contractual deliverables must be submitted from the Contractor to the Customer via a submission form.

All submitted documents, including design drawings, calculations, etc., must be legible and transmitted in Russian and English or in two languages Russian-English. The submission must be in an electronic copy through the EDMS in PDF format and in printed form.

Technical data sheets for equipment, instruments, cables, etc., as well as bills of materials included in documents, must be in Excel format.

All electronic materials must be printable without the use of special software or significant conversion. In the event of a conflict between the submission of an electronic and hard copy, the hard copy shall prevail.

The Contractor shall arrange for efficient distribution of documentation to ensure timely review of the project by the Customer.



All correspondence is registered by senders and recipients.

All prepared documents must be numbered with a unique number. The number must match the document encoding index listed below:

Correspondence coding sample;

L = letter, N = number

CCP	-	L	L	L	-	L	L	L	-	L	L	L	-	N	N	N	N
Project code		1	2	3		4	5	6		7	8	9		10	11	12	13

Cell	Code type	Definition
13	Letter	Sender
4 - 6	Letter	Document type
7-9	Letter	Recipient
10-13	Number	Subsequence

Document encoding examples:

CODE	DESCRIPTION
ACT	Act
ASB	Executive technical documentation
BVO	Budget Variance Order
COR	Change Order Request
CLC	Calculations
CTF	Certificate
CTR	Contract
DCR	Design Change Request
DEF	Defects List
DWG	Drawing
HSM	Health and Safety Management
INV	Invoice for payment
IPC	Act of Interim Completed Works
LTR	Letter/correspondence
MAF	Material/Equipment Approval Form
M.E.M.	Memorandum
MOM	Minutes of Meeting
NCR	Non-Conformance Report
POR	Purchase Order
PPR	Project Progress Report
PRC	Procedure
PRG	Program
QAM	Quality Assurance Manual
RFI	Request for information
RFP	Request for a price
RPT	Report
RSK	Risk Assessment Table
SAF	Subcontractor Approval Form
SCH	Schedule
SHD	Sketch Drawings
SKA	Sketch
SPC	Specifications
SWM	Site Waste Management Plan
TRM	Transfer-acceptance certificate

Example:

« CCP - EPC - LTR - CCGT -0003» means the outgoing letter and the 3rd letter sent by the Contractor to «CCGT Turkistan» LLP (Customer) as part of the project «Construction of a power plant on the basis of CCGT with a capacity of 1000 MW in the Turkestan region».

CCP : «Construction of a power plant on the basis of a CCGT unit with a capacity of 1000 MW in the Turkestan region»

LTR : Letter

EPC : Contractor

PGU : «CCGT Turkistan» LLP (Customer)

0003 : Sequence of correspondence sent to the recipient company during the Project.

Letters

Letters must be prepared in a legible manner on a computer.

Letterhead: A heading, logo or letterhead indicating the author of the letter should be located at the top and to the right of the letter.

Date: The date must be placed above and to the left of the letter. The format must be DD/MM/YYYY, and the date sent (not prepared) must be in the specified area. A Letter without a date cannot be accepted by other parties.

Outgoing number: The Outgoing number (hereinafter referred to as «Outgoing No.») should be located at the top and on the right side of the letter above the form. The Outgoing numbers are the identification numbers of letters, so each letter should have only a unique outgoing number.

Addressee: This part of the form indicates the name and address of the organization to indicate the addressee.

Subject: Each letter should have only one subject, and for submission purposes, the subject should be written correctly and clearly. The subject should summarize the content of the letter, and no other irrelevant definitions should be written in this part.

Attachment: Appropriate attachments submitted with the Letter should be listed in this section below the Letter.

To: The letter should be addressed to the project director or responsible person. This part should remain unchanged until the end of the project term, unless new responsible persons are appointed for the specified responsibility.

Copy: If the letter is planned to be duplicated to another person/party, this part is filled in with the name of the specified person/party.

Cover letter: All types of documents, files, reports, charts, etc., which must be sent along with the letter, are considered an «attachment». A document designated as an attachment must be attached to both the letter and the copy. The Original attachment is attached to the letter, which remains with the recipient.

Delivery: Letters to be delivered by hand must be prepared as two copies: the original must remain with the recipient and the copy must be signed as «received» along with the date, time and name of the recipient.

Procedure

The Document Management System (DMS) is regulated by the Documentation Control Procedure, which will be developed by the Customer. The procedure will provide information about the workflows used and forms followed by all project participants.

Representation

Applications to be submitted by the Contractor to the Customer will include, but are not limited to, the following types of documents.

List of documents	Deadline from the date of signing the contract
List of technical results (for review)	30 days - draft contract Maximum. 60 days - Final version

Project quality plan (for approval)	30 days
Project HSE Plan (for approval) including the OH&S Policy, Occupational Health and Safety Management System, Hazard Identification and Risk Assessment.	30 days
Project environmental and social management plan (for approval)	30 days
Baseline project schedule (to be approved)	28 days after receipt of the Work Start Notification
Project schedule (for approval)	30 days
Procurement schedule (for approval)	30 days
Forecast chart 3 months ahead (for review)	3 months - look forward to the schedule to be submitted with monthly reports
Mobilization plan (for approval)	14 days
Design drawings and calculations (for approval)	Periodically
General plan - Site plan (for approval)	15 days
Electrical single line diagram (for approval)	30 days
Circuit diagrams (for approval)	75 days
Process control scheme (for approval)	90 days
Bill of materials (for approval)	Periodically
Specifications of the main equipment of the plant and technical specifications (for approval)	20 days before order
List of suppliers (for review)	Periodically
Building standards and norms to be used (for approval)	Periodically
Detailed diagrams and layout drawings (for approval)	Periodically
Connection diagrams (for approval)	Periodically
Method Statements (for approval)	Periodically
Technology Statements (for approval)	Periodically
Inspection and Test Plans (ITP) (for approval)	Periodically
Monthly progress report (for review)	Periodically
Safe work procedures (for approval), including a work plan, organizational and technical measures	Periodically
Register of project risks	30 days

Any document, report, drawing, sketch, calculation, etc. prepared by all parties at any stage of the Project must be submitted using the «Document Submission Form». In order to facilitate and expedite the submission and approval process, each type of submission must be delivered with the appropriate submission form.

Each Submission Form must be completed exactly as defined, and Form No. must be specified exactly as specified in the codes in the appropriate sections of this section. All submissions will be registered and filed by the Customer according to these forms along with their attachments by the parties.

Drawing creation protocol

Criteria for creating drawings

Drawings officially issued by the Contractor shall meet the following criteria:

- Where appropriate, the original size of the drawing sheet should be A0 and reproduced in a reduced A3 size. Where it is not practical to use A0, A1 can be used, but must be playable on A3. The reduced figure must be clear and legible using appropriate line thickness and font.

The drawings must contain the following information:

- Scale in original size, e.g. 1:200 on A0,
- Scale bar.
- Site layout plans should include a key location plan, a key diagram showing where within the project or section boundaries that part of the site or structure is located.
- Location plans should include a point of north (proposed and true north).
- Computer-aided design (CAD) drawings must also include the path to the CAD file and the list must be in **Aconex** (and/or others).
- The drawings must be produced in the following format, regardless of the source CAD program used:
 - Paper copies are in A0 format only unless presented at discussion meetings or specifically requested by the Engineer.
 - Electronic copies - All electronic drawings (Tekla, Revit, AutoCAD) will be in the original software format with duplication in PDF format.
 - All drawings, documents and reports must be registered and submitted to Aconex (and/or others) or other selected program.
 - Direct file transfer between the Contractor and the Engineer can only be in DWG format, but where the Customer's copying is required, then PDF must also be included.

Standard drawing title block

A standard drawing title block will be agreed with the engineer prior to drawing submission.

CAD protocol

All electronic drawings will be in AutoCAD format DWG because DWG files have now become the worldwide standard for transferring CAD files and all CAD programs can import and export to a greater or lesser extent in DWG format. The standard to be adopted for the project will be AutoCAD DWG and all other CAD files must be compatible with this standard.

Line weight and color protocol will be set, as will layering. They will be determined by the Contractor.

Measurement protocol

This project uses the **metric system**, so all drawings must be done in the metric SI system. Dimensions will be as follows:

- All dimensions must be given in millimeters, i.e. 7.2 meters should be expressed as 7200 and not as 7.2 m, 7200 mm or 720 cm.
- All areas must be expressed in square meters, i.e. 7.2 m².
- All point levels, reduced levels, floor levels, etc. must be expressed in meters, i.e. RL 32.525 above zero.

CAD and dimensional protocols must comply with local regulations.

Verification and approval of documents

The term for consideration of documents will not exceed 21 days from the date of receipt of the Contractor's Document and the Contractor's Support. Electronic copies will be sent by email and Aconex (and/or others). The Customer shall make every effort to carry out the verification within less than 21 working days, and in particular if the Contractor indicates which specific documents are critical for early consideration.

The Contractor's Applications with the status «Submitted for approval» will be answered at the above processing time with any of the following statuses; «Approved», «Approved as marked exceptions» or «Rejected».

Approved	A
Exception Approved Marked	B
Revised/Resubmitted	C



The Applications reviewed and answered as B and C will be re-submitted for approval to the Customer after all noted amendments/modifications have been made to the documents.

The Contractor shall bear full responsibility for the time and financial consequences of re-submission of applications for the performance of Works arising from the rejection or re-submission of applications that are incorrect, incomplete or do not comply with the terms of the Contract. If the Contractor's document does not comply with the Contract, the Contractor shall submit to the Customer revised documents in accordance with the above approval procedure.

4.7. Design management

General information

The Contractor will carefully review the tender documents and the Customer's requirements and complete the development of the Project. The design must cover all civil, mechanical, electrical and engineering requirements.

The Contractor shall carry out the design and be responsible for the Project. The Contractor will be required to accept the general specifications as provided in this Technical Specification and refine them piece by piece to obtain complete, thorough, comprehensive and high quality designs, as-built drawings and specifications for the Project.

The project must be prepared by designers:

- a) engineers or other professionals qualified, experienced and competent in the design areas for which they are responsible;
- b) be qualified and entitled in accordance with the current legislation of the RK.

The Contractor will be responsible for ensuring the accuracy, completeness and correctness of the design, estimate and all engineering concepts and details of the work, including the coordination of the various architectural, construction, structural, mechanical, electrical and other departments with each other and with technical specifications.

The Contractor shall provide the Customer for approval with the name, address, details and relevant experience of each proposed designer/design subcontractor.

The Contractor warrants that the Contractor's designers and Design Subcontractors have the experience, capability and competence required for the design. The Contractor undertakes that the designers and Design Subcontractors will take part in discussions with the Customer at any convenient time (on the construction site or outside it) prior to the issuance of the Transfer and acceptance act.

During the project, there will be requirements for project documentation and drawings, which must be submitted for consideration by the Customer.

The Contractor shall prepare all documents under the Contract and any other documents necessary for the completion and implementation of the project during the execution of the Works. The Customer's personnel shall have the right to check the preparation of all these documents (including any studies, simulations and tests), wherever they are prepared.

The project documentation must include the following elements and any other documents that may be requested by the Customer:

a) Design Basics Document: The Contractor will describe the technical approach planned for the Project. The Design Fundamentals document will include conditions, requirements, project needs and performance criteria, including legal requirements. This document should also detail the design parameters for each equipment/material to be used in the project and should include preliminary test results that confirm that the equipment/material meets the proposed design parameters.

b) Detailed design, construction drawings and specifications

- The Contractor shall provide a complete and detailed design of work showing the Contractor's compliance with the Customer's requirements, as well as any other data that the Contractor has viewed or received, including details of any proprietary products to be used.
- The Project should include clear details of any eligibility criteria.
- The Specifications shall describe the materials/equipment and workmanship to be used for the construction of the Facilities.

c) Test and monitoring regime: The Contractor shall provide details of the test regimes that he will implement in order to demonstrate that the work performed is in accordance with the project.



Constructability overview

All project documentation, drawings, selection of equipment, systems and considerations related to the Contract are subject to review and approval by the Customer.

The provision of project documentation is carried out by the Contractor in accordance with the development of the project and in accordance with the agreed schedule.

The term for consideration of the project documentation shall not exceed 21 days, counting from the date of receipt by the Customer of the project documentation from the Contractor.

All submissions must be submitted in soft and hard copy.

The Customer (or on behalf of the Customer a Consulting Engineer) will conduct a high level design review and structural analysis of the critical structure design completed by the Contractor for the construction project. The review will include a detailed analysis of design features, functionality, quality, form and fit, safety, efficiency, interfaces, technical adequacy and solutions, relationships with neighboring systems and structures, cost effectiveness, size, sustainability, etc.

If the Customer determines that the project is defective and deficient and does not meet the goals and objectives of the project, the Customer will require the Contractor to immediately provide a solution and correction plan that will eliminate the deficiencies and achieve the required quality and standard of the project under the Contract.

If the Customer indicates that additional documents from the Contractor are required to demonstrate that the Contractor's design complies with the Contract, the Contractor shall prepare and promptly submit them to the Customer at the expense of the Contractor.

Cost engineering

The Contractor may submit to the Customer a written proposal which (in the opinion of the Contractor) if accepted:

- a) speed up execution;
- b) reduce the cost of the Customer for the performance, maintenance or performance of the Works;
- c) increase the efficiency or value of the completed Works for the Customer;
- d) otherwise benefit the Customer.

The proposal shall be prepared at the expense of the Contractor and shall include the following details:

- a) Description of the various works to be carried out, including details of the resources and methods to be adopted by the Contractor.
- b) The program for its implementation and the General Contractor's proposal for any necessary modifications.

Executive documentation

The Contractor shall prepare and maintain up to date a complete set of performance records for the performance of the Works showing the exact location, dimensions and details of the works performed by the Contractor. As-built documentation will be generated and maintained in accordance with CH PK 1.03-00-2022* «Строительное производство. Организация строительного производства предприятий, зданий и сооружений» (SN RK 1.03-00-2022* «Construction production. Organization of construction production of enterprises, buildings and structures»).

The format, reference system, electronic storage system and other relevant details of executive documents must be agreed with the Customer.

Executive documentation includes (but not limited to):

- Technical documentation of manufacturers (passports, operating and installation instructions, assembly documentation, software for installed equipment and SPTAM).

- As-built production documentation (journals, acts, protocols, executive schemes showing the progress of work, quality and responsible persons).



- Technical reports on commissioning and testing, mainly reflecting the acceptance control of the quality of special installation work, as well as the amount of work on individual testing of equipment.
- Acts of acceptance of the geodetic center base.
- Executive schemes for the location of buildings (structures) on the ground (landing of the building).
- Executive drawings and profiles of engineering networks and underground structures.
- Executive general plans for industrial facilities.
- Executive geodetic schemes of erected structures, elements and parts of buildings and structures.
- General work log, combined work log and special work logs, logs of incoming and operational quality control, filled out during the entire period of construction and installation work , etc.
- Certificates of examination of hidden works.
- Acts of intermediate acceptance of critical structures.
- Acts of acceptance of engineering systems with the attachment of documents on the results of acceptance tests.
- Acts of testing and testing of equipment, systems and technical devices.
- Working drawings for the construction of the facility with inscriptions on the compliance of the work performed in kind with these drawings (taking into account the changes made to them), made by persons responsible for the construction and installation works.
- Other documents reflecting the actual execution of design decisions, at the discretion of the construction participants, taking into account its specifics.

The as-built documentation is transferred by the Contractor to the Customer in the amount of 4 (four) copies on paper and 1 (one) copy in electronic form on a solid-state drive in the formats of the programs in which they were executed, as well as scanned copies in *pdf format through Aconex (and/or others).

Building information modeling (BIM)

The project will be developed in accordance with LOD 350, which includes model details and elements that represent how building elements interact with various systems, and other building elements with graphics and written definitions.

The Contractor shall provide licensed Tekla or Revit software for use by the Customer's team as required.

4.8. Risks management

Concept

The Contractor shall develop and implement a risk management approach in accordance with the international standard ISO 31000:2018 «Risk Management - Principles and Guidelines», as well as other ISO standards that supplement it, including, but not limited to: ISO / TR 31004:2013 «Guidelines for implementation of the ISO 31000 standard», ISO 31010:2019 «Risk assessment methods».

The Contractor's risk management process shall include three interrelated approaches as shown below:

- Systematic and continuous identification of factors (sources, drivers, triggers) of risks, identification of exposure to risk, assessment of the nature, likelihood and potential impact of risks on the Project.
- Planning and organizing the necessary measures for the treatment of risks of the Project: reducing the likelihood of risks, transferring risks, as well as methods of financing risks to minimize damage from risk realization;
- Implementation of risk management methods within the organization at all levels of the Project implementation, as well as outside it: working with insurance organizations, loss containment specialists and other risk financing specialists.

The following table (Table 1. Risk Management Plan) shows examples of the main risk areas and risk management approaches that the Contractor should identify, assess and manage.

Areas of risk	Management approach
Project cost	Cost control program for project management, cost optimization, work schedule; Change Management Process
Work schedule	Analysis of a constructive solution by an engineer; opportunities, previously started tasks, management of key resources, management of deadlines compared to the estimated duration of work
Project quality	Contractor's quality assurance programs; supply quality management, non-compliance act management / corrective and preventive actions
System integration	Integration and optimization of electrical and mechanical systems, consideration of the design solution and design criteria
Worker injury/environmental accident	Contractor's zero accident preventive health and safety program, pre-job hazard analysis, health and safety instructions, training

Risk management plan

The Contractor shall develop and implement a Risk management plan to manage uncertainties and their subsequent potential negative or positive effects in order to achieve project objectives in terms of time, cost and quality. The Risk management plan shall describe the Contractor's methodological approach for identifying, assessing and continuously managing risks, including, but not limited to: informing the Customer, consultations with stakeholders, monitoring, analysis and control of the project life cycle, including risk management in the area OH&S.

Risks assessment

The Contractor shall continuously identify and assess project risks as part of ongoing Project activities.

Each risk must be assessed against criteria in terms of its likelihood of occurrence and its impact on timing, cost, volume, performance/quality. Likelihood and impact shall be assessed to determine the overall risk criticality of the Project.

The Contractor will establish and maintain on an ongoing basis a register and risk matrix that will be used to identify key risks and to analyze and manage potential Project risks.

Risks responsibility

The Contractor will be responsible for managing all risks associated with the project arising from the Contractor's work plan. Updates on risk management, risk register, risk matrix of the Project shall be submitted to the Customer as part of weekly and monthly progress reports on the Project.

The Contractor will be responsible for immediately informing the Customer about the key risks (if any), as well as for the implementation of risk management measures.

4.9. Quality control

General information

In performing the work, the Contractor shall adopt and implement the quality management principles specified in the current version of the ISO 9001 Standard: «Quality Systems - Quality Assurance Model for Design/Development, Manufacturing, Installation and Maintenance».



The Contractor will ensure that the Quality Management System (QMS) is integrated with its other management systems and does not lead to conflicts or duplication. In addition, the Contractor is responsible for integrating its Quality System with the Quality Systems of its subcontractors and suppliers.

The QMS will contain a description of the key processes and procedures. If additional procedures are needed to control the scope of work, they will be described in the project-specific quality plan (PQP).

As a minimum, the Contractor's Quality Plan will contain the following.

a) An organizational chart that includes reporting relationships for those responsible for controlling and executing the scope of work, identifying the responsible party for assurance, and interfaces between the job site and departments.

b) Persons responsible for quality, including their responsibilities and reporting relationships, who will be key personnel. Key QA/QC personnel will include involved inspectors and auditors.

A list of procedures required to perform and control the scope of work, including:

a) Engineering processes, including: technical review and approval processes, peer reviews, and customer specification and requirements verification processes.

b) Nonconformity control and corrective action procedures

c) Documentation and document management

d) Supervision of subcontractor/suppliers

e) Material control - traceability (tracking) – receipt, verification and storage of materials

f) Inspection and test requirements

g) Commissioning and facility transfer management

h) Audits.

Organization and responsibilities

The Contractor is responsible for the quality.

The Contractor has the first responsibility for ensuring that their products and methods meet established quality standards. The Contractor shall have a competent team of managers, engineers, technicians, etc. to perform the work satisfactorily in accordance with the quality requirements of the project and the Customer's quality expectations.

The Contractor shall document and submit for approval all Quality Management System documentation developed to ensure compliance with the specific requirements of the project.

The Contractor is obliged to provide documentary evidence confirming compliance with the required standards. This will require the use of independent laboratories and testing.

Factory control, Factory testing and acceptance

The Contractor shall conduct a program of inspection and testing during the production of his own work and the work of his subcontractors to ensure that the mechanical accuracy of the components, conformity with the drawings, conformity with the functional and performance requirements, acceptability of all materials, parts and equipment, and their conformity with the test certificates. The Contractor shall be responsible for carrying out all tests/inspections necessary to establish that the products/equipment comply with the requirements of the design specifications and relevant codes/standards, in addition to carrying out tests in accordance with the reviewed Inspection and Test Plan.

In addition, the Customer must carry out an inspection of the work during the manufacture of the equipment at the manufacturing plant. Whenever inspections or tests are carried out in the workshop, all material certificates, as well as all other intermediate test certificates in accordance with the agreed test schedule, must be provided to the Customer in a legible copy for his review.

The Contractor shall factory test the equipment in accordance with relevant specifications which require such testing. Factory Acceptance Tests (FAT) shall be carried out to determine the conformity of materials and equipment to the Specification and to provide the necessary performance data. All tests should be organized in such a way as to reflect the working conditions as accurately as possible.

Unless an alternative test location is agreed or specified, testing shall be carried out at the manufacturer's site. As many tests as possible should be organized simultaneously in accordance with the Project Schedule, which should be agreed with the Customer.

The manufacturer's test equipment must be of satisfactory quality and condition and, if necessary, must be properly calibrated in accordance with international standards by an accredited laboratory at the expense of the Contractor.

After the conclusion of the Contract and clarification of the main features of the project, the Contractor will develop a detailed Manufacturer's Inspection and Test Plan, as described earlier in this document, for each item of equipment.

The Contractor shall notify the Customer at least forty-five (45) working days prior to any such testing. The tests must be witnessed and verified by the Customer. All activities for visiting the plant and acceptance tests (travel, reception and hotel accommodation) for the Customer will be organized at his expense at the request of the Customer.

After the inspection, carried out in front of witnesses at the manufacturing plant, all parties must sign the protocol (inspection report) indicating the subject, participants, results and comments and distribute it immediately after the test. A detailed test report with certified copies of all tests, data and results themselves shall be distributed by the Contractor to the Customer.

Under no circumstances should equipment be shipped, or even packaged, prepared for shipment, or dismantled for the purpose of packaging for shipment, unless it has been satisfactorily inspected and approved for shipment, or inspection has been denied in writing.

The following equipment must be subjected to a Factory Acceptance Test (FAT) to prove that reliability, operation and performance meet the requirements of certain specifications and relevant standards:

- Gas turbine (GT)
- Steam turbine (ST)
- HRSG
- Generators
- Transformers
- Lifting mechanism (bridge cranes).

All other factory equipments requested by the Customer.

Inspection and testing

The Contractor is responsible for preparing Inspection and Test Plans (ITPs) for all works at the Site and for each equipment and submitting them to the Customer for review and approval. The Contractor shall ensure that ITP determines critical manufacturing steps, acceptance criteria (codes, standards, drawings, etc.) and specific methods and procedures to be used to meet all project requirements.

The ITP may take any form consistent with the Contractor's system, but must, as a minimum:

- a) Critical stages of construction/installation and every point of inspection and testing.
- b) Characteristics to be checked, examined and tested at each point, as well as reference drawings, procedures and acceptance criteria used.
- c) Test frequency criteria.
- d) Quality «checkpoints» (H, W, R) set by the supplier and Contractor, which require the verification of selected characteristics of a document, product or process before that work can proceed. The Customer may assign additional Quality Control Points (H, W, R) to the Contractor on a case-by-case basis .
- e) Supporting document for use (manufacturer's instructions, material certificates, laboratory test results, etc.) .

The Contractor shall also submit copies of the reference documents/standards/acceptance codes/testing and inspection procedures, etc. referred to in the ITP along with the ITP.

All ITPs will be presented to the Customer 14 days prior to the commencement of field work. Inspection tests will be carried out after mutual agreement between the Customer and the Contractor.

The Contractor shall inform the Customer and the Technical Supervision about inspections/tests 24 hours before the start of work by submitting an application for inspection of the Contractor's work.

The results of all tests, routine tests and site tests shall be recorded in the appropriate test formats. All on-site test formats must be approved by the Customer.

Test instruments and equipment

The instruments and apparatus used for inspection and testing, both on site and in the workshop, shall be of high quality as agreed with the Customer and shall be calibrated to an agreed standard. The Contractor shall provide the tools and equipment necessary for installation and testing, as well as for commissioning. Control and measuring devices must comply with the current standard.

Control and measuring devices must comply with the current standard, are included in the register of measuring instruments of the Republic of Kazakhstan.

Records of calibration certificates will be kept at the Contractor's Production and Technical Department (PTD) and should be available for viewing by the Customer. Instruments must be provided with certified labels indicating the date of certification.

For performance testing, instruments must be listed and have a valid calibration certificate, and calibration records must be made available to the Customer for review.

For on-site inspections, the Contractor shall be responsible and include in his supply all safety measures, such as barriers, warning signs, etc., necessary for inspection and testing during installation.

Examples of equipment to be calibrated are surveying equipment, torque wrenches, load cells, pressure gauges, etc.

- Each instrument must be clearly and permanently numbered. Only qualified and experienced persons directly responsible for the equipment/tools, or personnel under their direction, will work with calibrated and certified equipment/tools.

- Each piece of equipment/tool will be tested for accuracy in accordance with the manufacturer's recommendations and relevant regulations.

- Equipment/instruments that are accidentally dropped, damaged or deemed inaccurate will be immediately taken out of service, calibrated and re-certified.

- Certification of the equipment prior to its delivery to the site will be carried out, and the relevant certification documents must be submitted by the Contractor to the Customer prior to the use of this equipment.

Quality Process Control

The Customer will verify that the Contractor and its suppliers effectively apply documented procedures to ensure that all manufacturing, fabrication, construction, installation, commissioning and turnover are carried out under controlled conditions, which include the following:

- Documented work instructions that define the order of production.
- Use of suitable and properly maintained production and installation equipment.
- Appropriate working environment.
- Compliance with relevant standards, codes and quality assurance/quality control plans.
- Monitoring and control of processes and product characteristics.
- Approval of processes and equipment, if applicable.
- Determination of manufacturing quality criteria.
- Control of «special processes», i.e. processes, the results of which cannot be fully verified by subsequent inspection and testing, and which may cause deficiencies to become apparent only after the product has been used. Processes in this category are subject to ongoing monitoring and/or adherence to documented procedures; personnel are qualified and meet the established requirements; records are kept of qualified processes, equipment and personnel, if necessary.

- Compliance with health, safety and environmental procedures, including safety instructions for each workplace, profession and position, tools and equipment.

The Contractor shall notify the Customer of the readiness of any work before it is concealed or packed for storage or transport. Thereafter, the Customer shall either carry out inspection, inspection, measurement or testing without undue delay, or promptly give notice to the Contractor that the Customer



is not required to do so. If the Contractor fails to give notice, he shall, if and when required by the Customer, open the work and then restore and restore to proper condition, all at the expense of the Contractor.

The Engineer's QA/QC process consists of 5 (five) main steps as listed below:

- Pre-mobilization/commencement processes
- Quality processes of the preparatory stage before installation
- Initial Inspections
- Subsequent Inspections
- Final inspection.

Initial inspection

For a particular Definable Feature of Work (DFOW), a representative sample of jobs should be performed. The initial inspection shall be carried out by the Customer on a representative sample of the work immediately upon receipt of a notice of completion from the Contractor. Breadboards/samples of completed scopes of work may be included in contract work if they fully comply with all relevant standards and requirements. This inspection is a logical continuation of the preparatory meeting held earlier. The main function of this inspection is to ensure that the Contractor starts work in full accordance with the requirements of the Contract, and to establish an acceptable level of work quality, setting a physical benchmark. During the initial inspection on representative samples of work, the inspection team checks and makes sure that the materials, equipment and technologies used comply with all established standards and requirements. The ultimate goal of the initial inspection is to ensure that the team performing the work has all the necessary equipment, skills and knowledge to successfully complete the entire scope of work under the Contract.

The Contractor shall be responsible for sending to the Customer an Inspection Notice or a Construction Inspection Request (CIR) at least twenty-four [24] hours prior to the actual inspection date of any major inspection and testing as required by the approved ITP. The CIR format should be provided to the Customer.

Subsequent inspections

Inspection of work in progress should be carried out on an ongoing basis. Subsequent inspections will be carried out throughout the life of the work to ensure that the work is being carried out satisfactorily and in accordance with all applicable requirements.

Follow-up inspections may be conducted daily, scheduled, predetermined and/or selectively as necessary to ensure compliance with the Contract. Subsequent inspections confirm that the materials and methods demonstrated during the initial inspection continue to be installed and applied to agreed standards. Follow-up inspections are required for all DFOWs and must be conducted prior to the successful completion of the inspected activity. The Customer, together with the relevant representatives on the part of the Contractor, shall follow specific inspection and test plans to be developed by the Contractor and approved by the Engineer and the Customer, where each inspection shall be carried out using the appropriate checklists/forms, and shall also be summarized and set out in the appropriate reporting and registration documentation.

Final inspections

Final inspections will be carried out with the participation of the Contractor, technical supervision, architectural supervision and the Customer. These checks are designed to ensure the final quality of the work and its readiness for acceptance. The following steps must be followed:

- All participants check the work for compliance with plans, specifications, quality, professionalism and completeness. The interaction of electrical circuits, electrical equipment systems, control systems in various modes is adjusted.
- The Customer shall make a list of all items that were found to be incomplete or not in accordance with the contract documentation, specifications and the latest revision of the drawings at the time of inspection. This includes any non-compliance items or action items identified during previous inspections for the type of work.

- The Customer transfers this list to the Contractor indicating the required time to eliminate the deficiencies and instructs the Contractor to document the completion of the listed items of work.
- All participants perform the follow-up steps to verify that each item in the list of corrections is completed and/or corrected and is acceptable, and put their signatures under the control of the Customer.

Additional inspections

Off-site inspections are inspections of the facilities and/or products of manufacturers and suppliers off-site, if considered necessary and/or required by the Contract specifications, to ensure that all requirements are met and observed. Off-site inspections will be carried out by the Customer as required. The Customer's personnel must, at any reasonable time:

- a) have full access to all parts of the Site and to all places where inert materials come from;
- b) during production, manufacture and construction (on the Site and elsewhere) have the right to inspect, inspect, measure and test the materials and workmanship, as well as to check the progress of the Plant and the production and manufacture of the Materials.

A visit to the manufacturer's plant (travel, reception and hotel) will be organized by the Contractor at its own expense at the request of the Customer.

The results of each inspection will be documented in an offsite inspection report. The Contractor must be notified of any deficiencies and must submit a corrective action report. All materials and equipment inspected off site will be properly stored and permanently labeled with the project name and other relevant data to ensure long-term follow-up.

Materials inspections are conducted upon receipt of product delivery of any given facility. The Customer will inspect and supervise the Contractor's physical inspection of the materials to ensure that they have been properly stored and conform to the approved shop drawings and/or submitted product data. This will be documented in the Contractor's daily reports and the relevant project log by using a material receipt record. Equipment or materials that do not pass inspection upon receipt should not be released to production, and rejected items should be separated and included in the non-conformity report.

Non-compliance with requirements

The Customer shall develop a procedure that defines the methodology and the associated roles and responsibilities for the control of nonconformities related to the product and the quality process. The procedure should include a format for documenting nonconformities (Nonconformity Report Form). The Contractor will be required to implement this procedure.

If defects in material, work performed or installation are found, the Customer will notify the Contractor by issuing a Non-Conformity Report (NCR). Upon receipt of such notice, the Contractor shall immediately take steps to prevent the inadvertent use or supply of certain nonconforming material, work or installation.

Upon receipt of the NCR, the Contractor shall immediately investigate the matter and prepare corrective actions in relation to measures to eliminate the identified non-conformities and submit them to the Customer for review and approval.

If the Customer approves the proposed corrective actions, the Contractor shall implement the proposed measures and upon completion resubmit the NCR to the Customer for closure.

The Contractor and suppliers effectively apply documented procedures that ensure that any non-conforming product is prevented from unintended use or installation.

The Contractor shall maintain a Register of Nonconformities to indicate the status of all nonconformities that have been identified by the Customer and the Contractor. The Customer reserves the right to inspect and audit these documents at any time.

Technical documents

The Contractor is responsible for submitting all prepared technical documentation, such as design documentation, calculations, sketches, reports, material / product specifications, equipment specifications, guidelines, inspection and test plans, working drawings, condition drawings, samples,

certificates, laboratory results tests, mock-ups, operation and maintenance manual, etc. and the Customer for review and approval.

Quality records

The Contractor shall maintain complete and accurate records of all inspections and tests carried out, providing appropriate logs, reports, inspection checklists, material/equipment certificates and all other quality records in a timely manner to the Customer, where the process will be monitored, reviewed and supervised and ensured that strict compliance with this requirement. These records will serve as factual evidence that inspections and tests have been carried out and that the work has been performed in accordance with contractual, legal and industry best practices. Such records will fully cover and reflect both relevant and non-compliant characteristics.

Project of Work production

The Contractor shall submit separate individual Project of Work Production (PWP) detailing the sequence and methods of work that he intends to apply to the various elements of the work. The PWP must contain detailed information on the construction sequence and technical execution methodology and must comply with any additional construction requirements of the specification and drawings.

The PWP should contain detailed information on the materials, equipment and installations used, labor requirements, timing and schedule of work, including preconditions, details and procedures for each operation, safety measures, environmental controls and any other relevant technical aspects, including activities in area of OH&S.

The PWP must comply with the requirements of the Quality Plan of a particular project of the Contractor and include a description of the frequency of tests and inspections, records, reporting channels. The relevant ITPs for the relevant activities should be attached to the PWP for ease of reference.

Quality audits

The Contractor shall conduct internal audits of the QC system on a regular basis, at least once every 6 months. The Contractor shall submit to the Customer a report listing the results of each internal audit within 7 days of completion. Each report shall include, as appropriate, proposed measures to improve and/or correct the QC System and/or its implementation.

In addition, the Customer has the right to audit any aspect of the quality management system (QMS) of the Contractor's project, in accordance with the requirements of ISO 19011:2011. These will include audits of quality system elements, audits of products and services, and audits of processes and technical procedures.

The Customer can conduct scheduled and unscheduled inspections and quality audits. When conducting scheduled audits, the Contractor will be notified of the quality audit at least five (05) days in advance.

The purpose of quality audits is to test the effectiveness of the project's quality management system and the Contractor's quality assurance process to ensure compliance with project requirements, relevant codes and standards, as well as the Customer's quality expectations, and to identify areas for continual improvement.

The status of the audits and the corresponding conclusions, as well as the required corrective actions for improvement within a certain time frame, should be summarized and issued by the Customer to the Contractor.



Part 5. ABBREVIATION

Abbreviation	Description
ACS	automated control system
ADC	analog-to-digital converter
AEA	anti-emergency automatics
AEC	automatic excitation controller
AECS	automated emission control system
AFPCS	automatic frequency and power control system
AGDS	automated gas distribution station
ALARP	as low as reasonably practicable
ANSI	American National Standards Institute
APCS	automated process control system
ARI	automatic reserve input
ASCME	automated system for the commercial metering of electricity
ASG	auxiliary switchgear
AVCVDS	automated vibration control and vibration diagnostics system
AWP	automated workstation
BAT	best available technologies
BCP	block control panel
BGCP	block gas control point
BIM	Building information modeling
BMS	building modelling software
CAD	computer-aided design
CBS	a cost breakdown structure
CCD	charge-coupled devices
CCGT	combined-cycle gas turbine
CCR	central control room
CCTV	video surveillance system (closed-circuit television)
CDLU	cleaning device launch unit
CDRU	cleaning device receiving unit
CIR	construction inspection request
COD	chemical oxygen demand
CP	control panel
CPM	critical path method
CSG	complete switchgear
DCO	devices for communication with the object
DCTD	data collection and transmission device
DFOW	definable feature of work
DMS	document management system
DPABX	digital private automatic branch exchange
DTCS	dispatching and technological control system
DTCS	dispatching and technological control system
DPT	dew point temperatures
EA	emergency automation
EDMS	electronic document management system

EDMS	electronic document management system
EDU	electrodeionization unit
EF	efficiency
EF of GT	efficiency factor of the gas turbine
EIA	environmental impact assessment
ES	emergency situation
ESMP	environmental and social management plan
EWS	engineer's workstation
F&G	fire and gas detection
FAT	factory acceptance testing
GCMS	gas consumption metering station
GCMS	gas consumption metering station
GIS	gas-insulated switchgear
GOST	State standards of the Republic of Kazakhstan
GT	gas turbine
GTG	gas turbine generator
GTP	gas treatment plant
HAZOP	hazard and operability study
HRSG	heat recovery steam generators
HSC	hardware and software complex
HSE	Health, Safety and Environment
HSW	hierarchical structure of works
HTML	hypertext markup language
HVAC	heating, ventilation and air conditioning
HVAC	heating, ventilation and air conditioning
I&A	instrumentation and automation
ILO	International Labor Organization
IM	information model
IOSH	Institution of Occupational Safety and Health
ISDN	integrated services digital network
ISO	International Organization for Standardization
ITP	inspection and test plans
JHA	job hazard analysis
LAN	local area network
LOD	level of development
LRA	legal and regulatory acts
LTSA	long-term service agreement
LVDT	linear variable differential transformers
MCP	main control panel
MCS	monitoring and control system
MGP	main gas pipeline
MSDS	material safety data sheets
NC	non-combustible group of materials
NCR	non-conformance report
NCR	non-conformity report

NDC SO JSC «KEGOC»	National Dispatch Center of the System Operator JSC «KEGOC»
NEBOSH	National Examination Board in Occupational Safety and Health
NFPA	National Fire Protection Association
NFPA	National Fire Protection Association
NPG	national power grid
NTD	normative-technical documentation
NTP	network time protocol
OEM	original equipment manufacturing
OH&S	occupational health and safety
OSHA	Occupational Safety and Health Administration
OTL	overhead transmission line
P	Project
PA	public address
PABX	private automatic branch exchange
PCD	polychlorinated diphenyl
PCW	pre-commissioning works
PLC	programmable logic control
POP	persistent organic pollutant
PQCS	power quality control system
PQP	power quality parameters
PQP	project-specific quality plan
PTD	Production and Technical Department
PTZ	pan-tilt zoom
PWP	Project of Work Production
QA/QC	Quality Assurance/Quality Control
QMS	quality management system
RCPST	reinforced concrete portal with steel traverses
REE	registering emergency events
RES	registration of emergency situations
RFI	request for information
RK	The Republic of Kazakhstan
ROU	reverse osmosis unit
RPA	relay protection and automation
RTD	resistance thermometer detector
RVDT	rotary variable differential transformers
SAT	site acceptance test
SB	steam boilers
SCADA	supervisory control and data acquisition
SCU	security crane units
SEE	static excitation equipment
SG	switchgear
SI	International system of units
SIS	safety instrumented systems
SN RK	civil normatives of the Republic of Kazakhstan
SPTAM	spare parts, tools, accessories and materials

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SS	substation
SSEW	shift supervisor of the electrical workshop
SSPP	shift supervisor of the power plant
ST	steam turbine
STC	special technical conditions
STG	steam turbine generator
STSD	system time synchronization devices
TCS	turbine control system
TLRA	technical legal and regulatory acts
TMP	traffic management plan
TOR	Technical Operation Rules
TR CU	technical regulations of the Customs Union
UFU	ultrafiltration unit
UPS	uninterruptible power supplies
WD	working documentation
WDR	wide dynamic range
WDS	work distribution structure
WMP	waste management plan
WP	working project
WTP	water treatment plant

SIGNED

In the name of and on behalf of the Employer:

In the name of CCGT TURKISTAN LLC

Kusainov Askar Aitbolatovich
General Director acting on the basis of the Charter



SIGNED

**In the name of and on behalf of the Contractor:
Consortium composed of:**

**In the name of Doosan Enerbility Co. Ltd.
(Consortium Leader)**

Iljoon Won
Deputy General Manager acting on the basis of
Power of Attorney dated 9 March 2023

(signature)
/stamp/



**In the name of Bazis Construction Limited
Liability Partnership**

Uali Bauyrzhan Kudaibergenuly
President, acting on the basis of the Charter

(signature)
/stamp/

